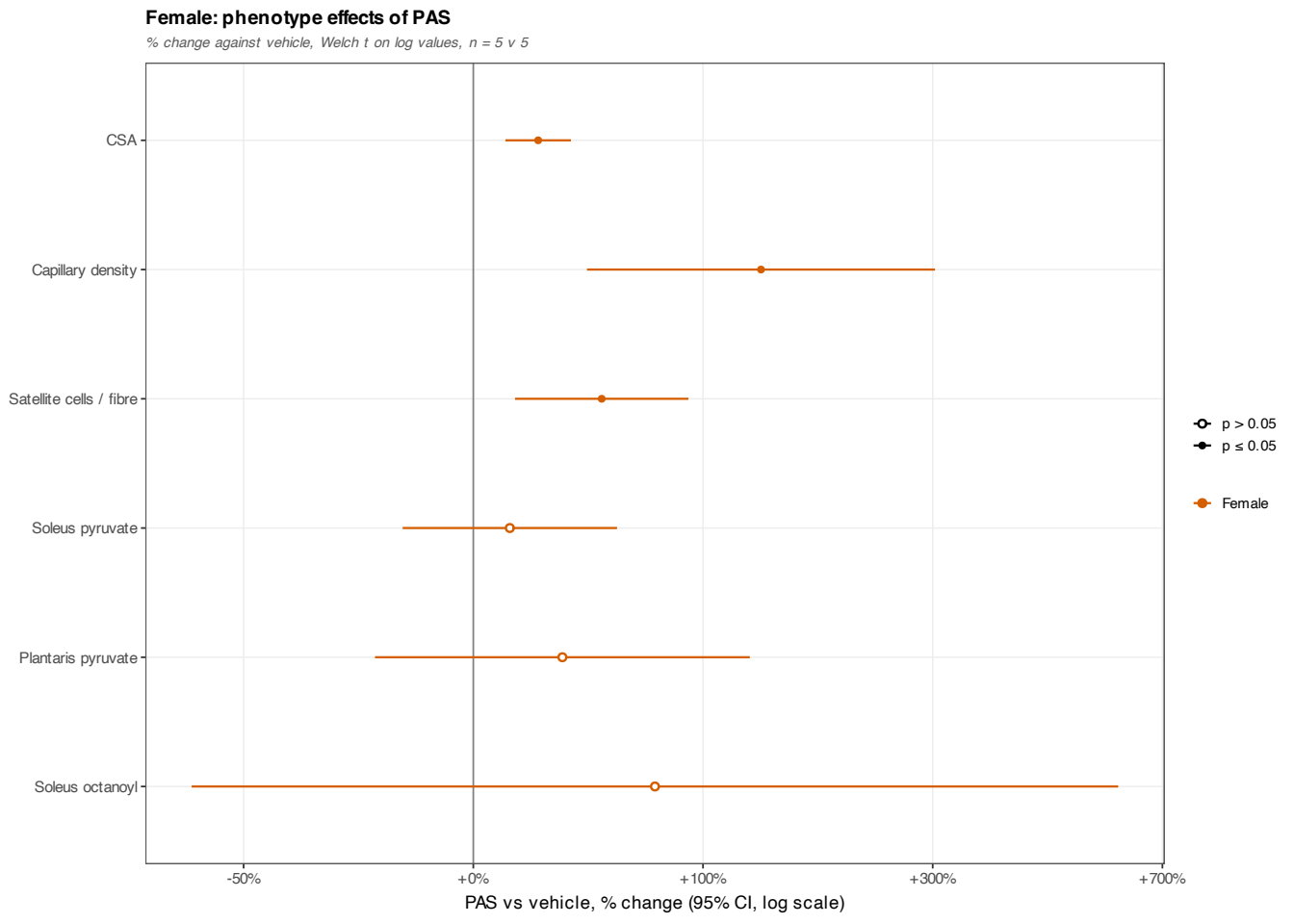
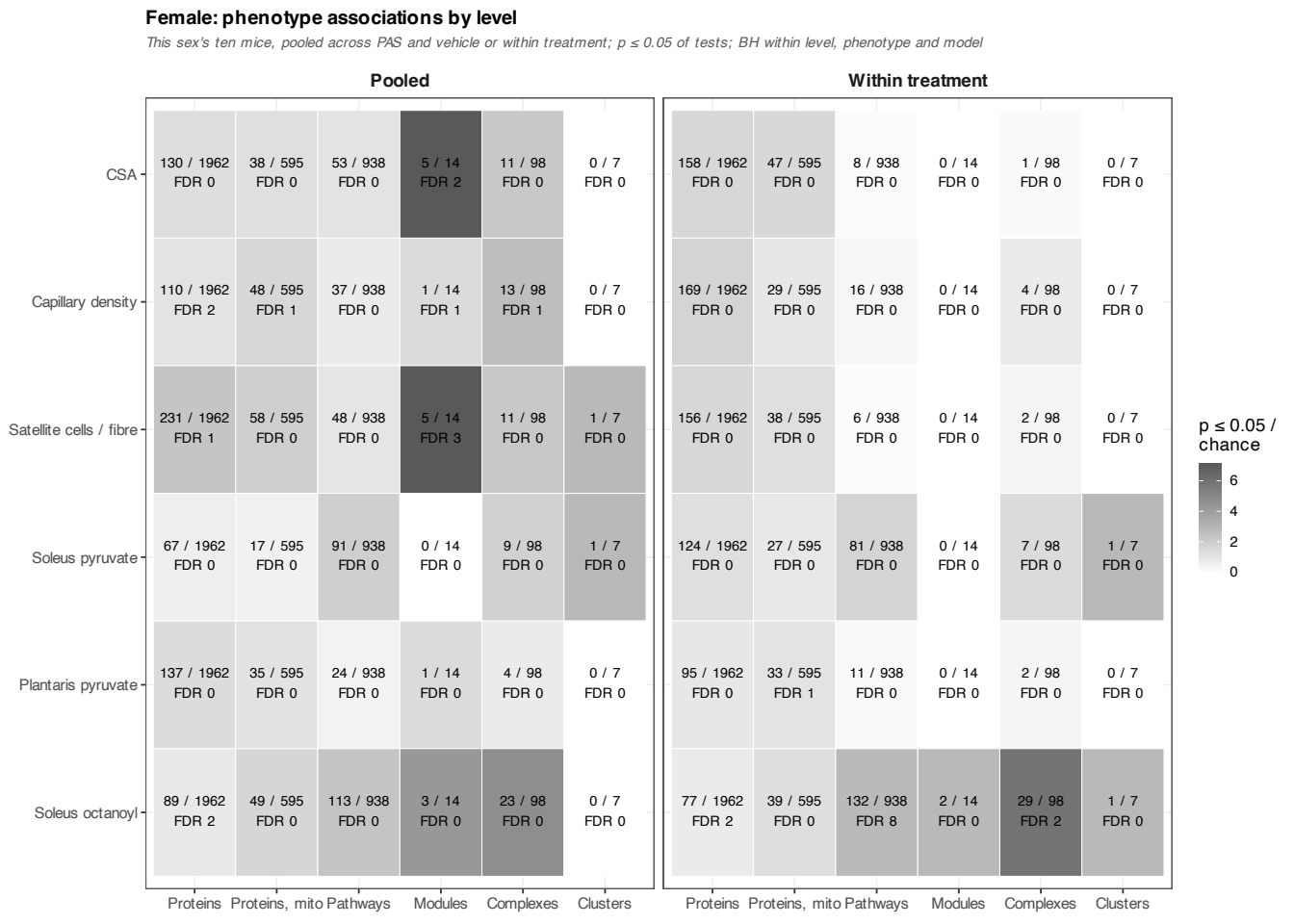
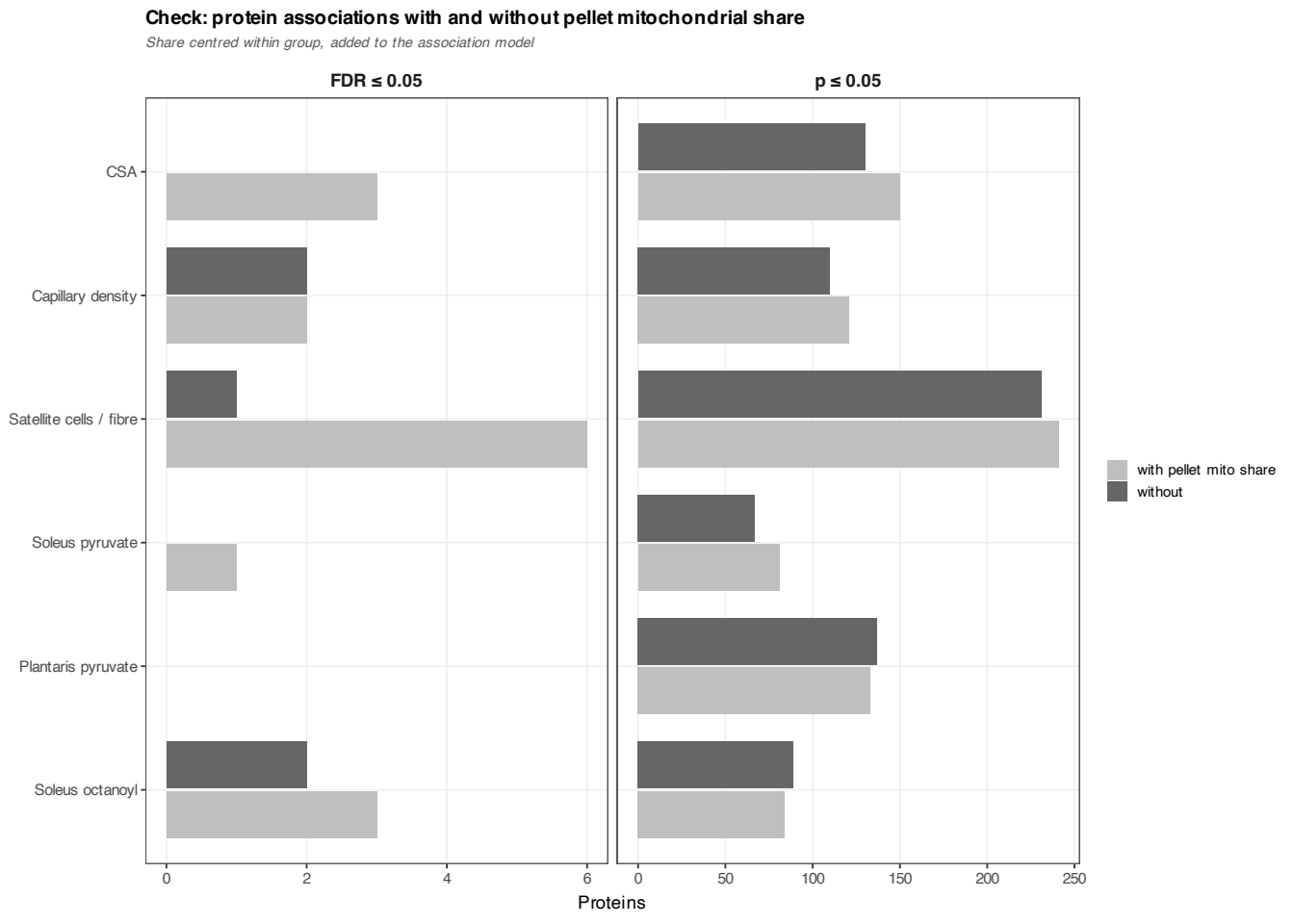
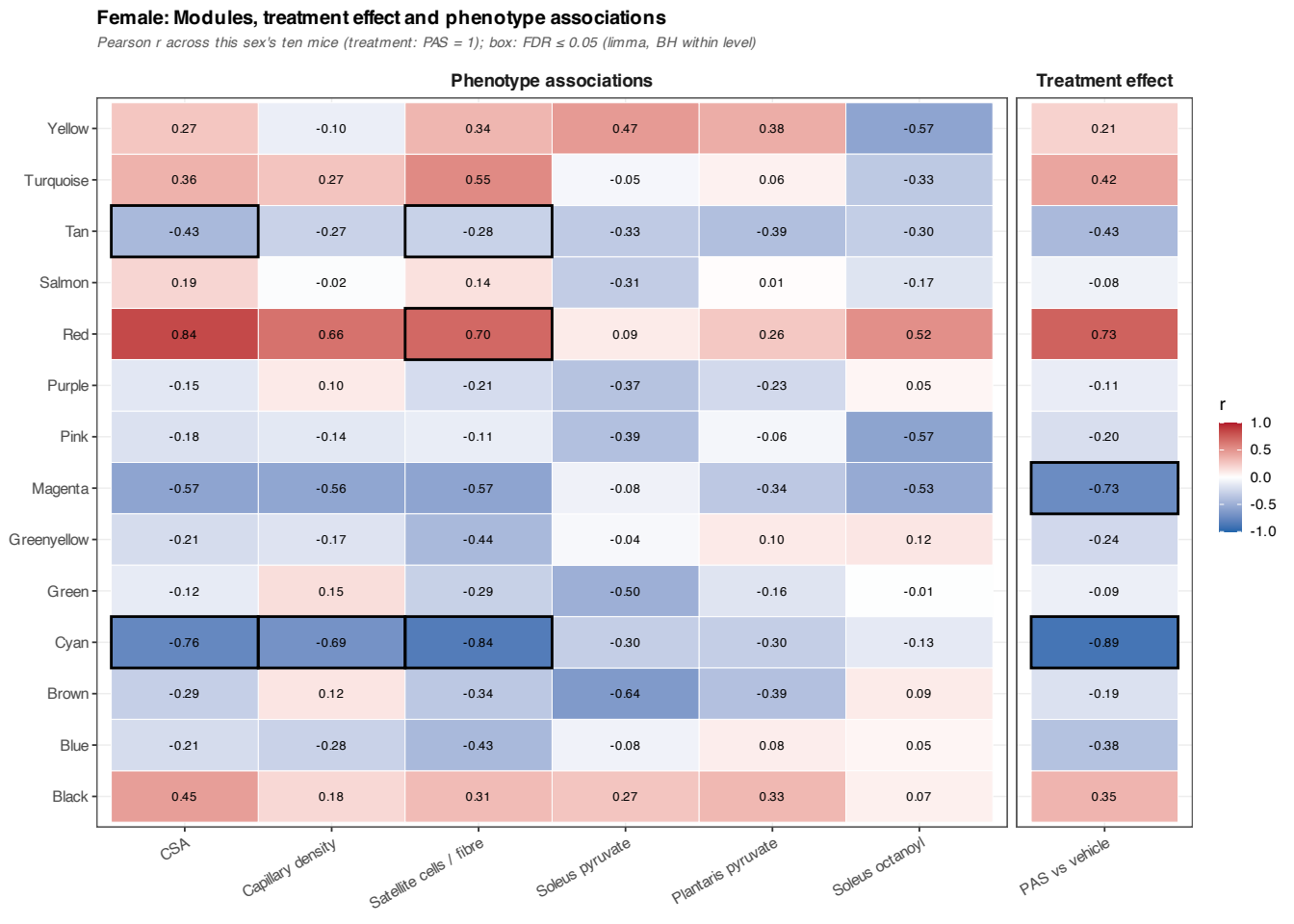
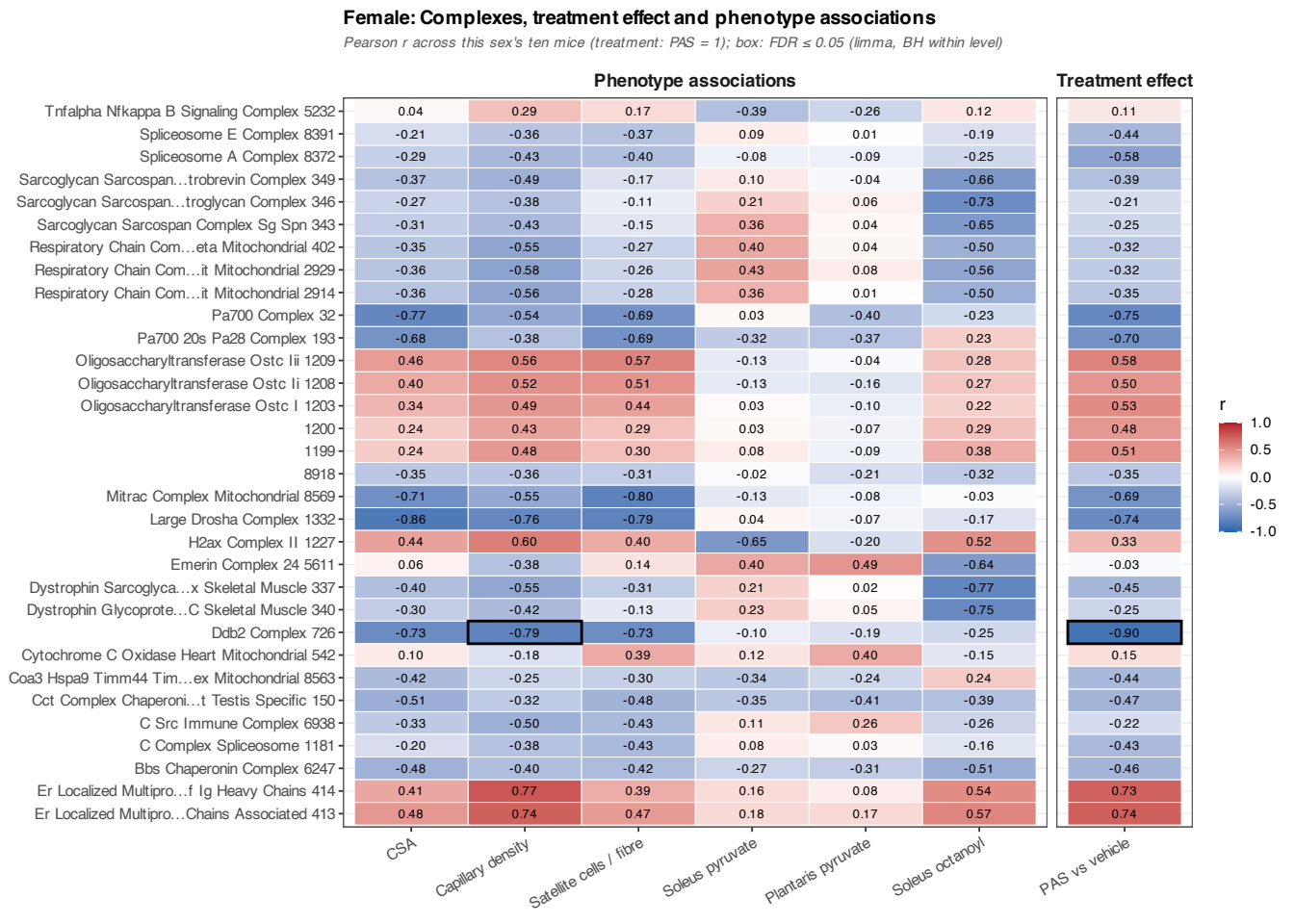
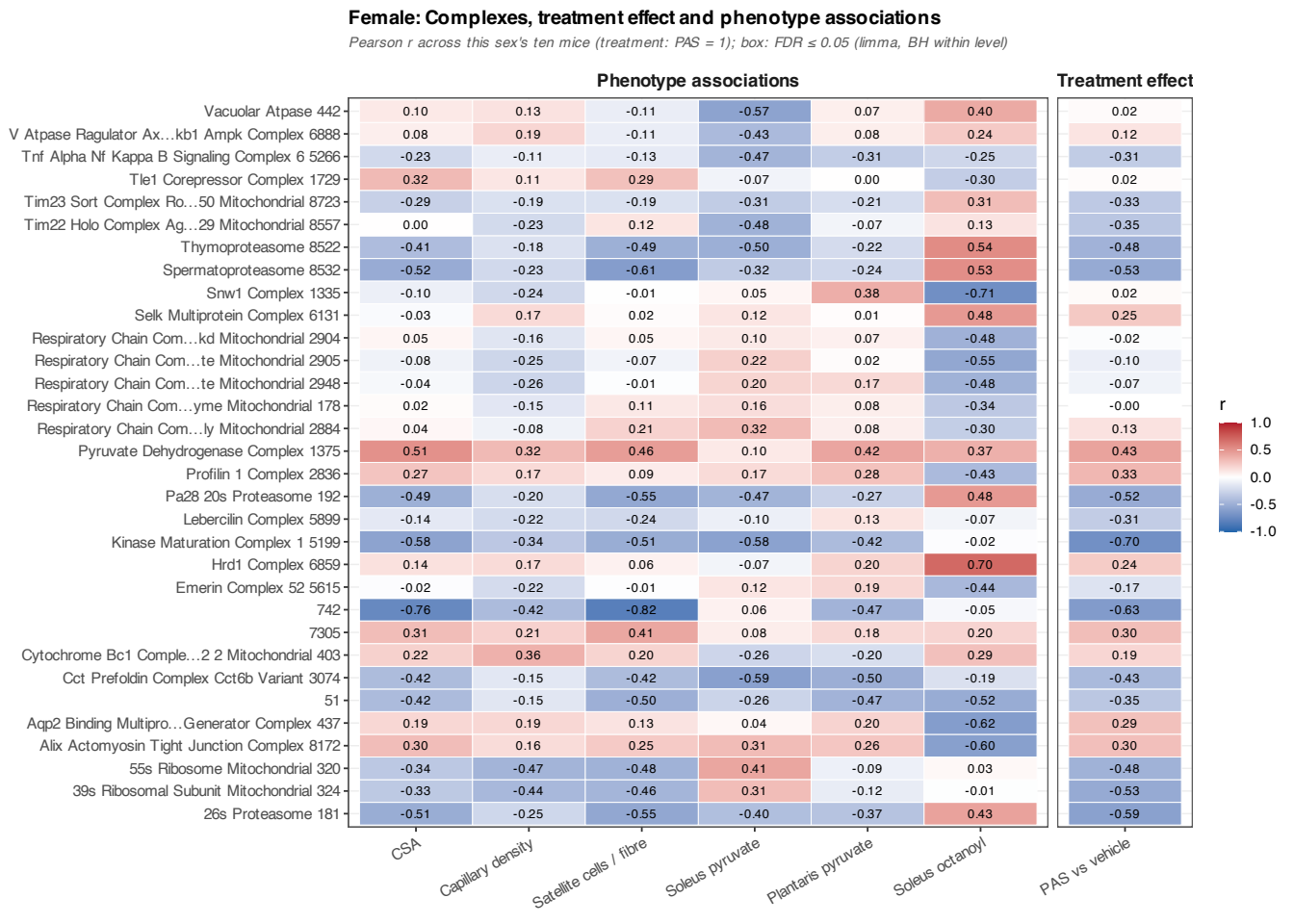

Female: phenotype effects of PAS
% change against vehicle, Welch t on log values, n = 5 v 5
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
p > 0.05
p ≤ 0.05
Female
-50%
+0%
+100%
+300%
+700%
PAS vs vehicle, % change (95% CI, log scale)

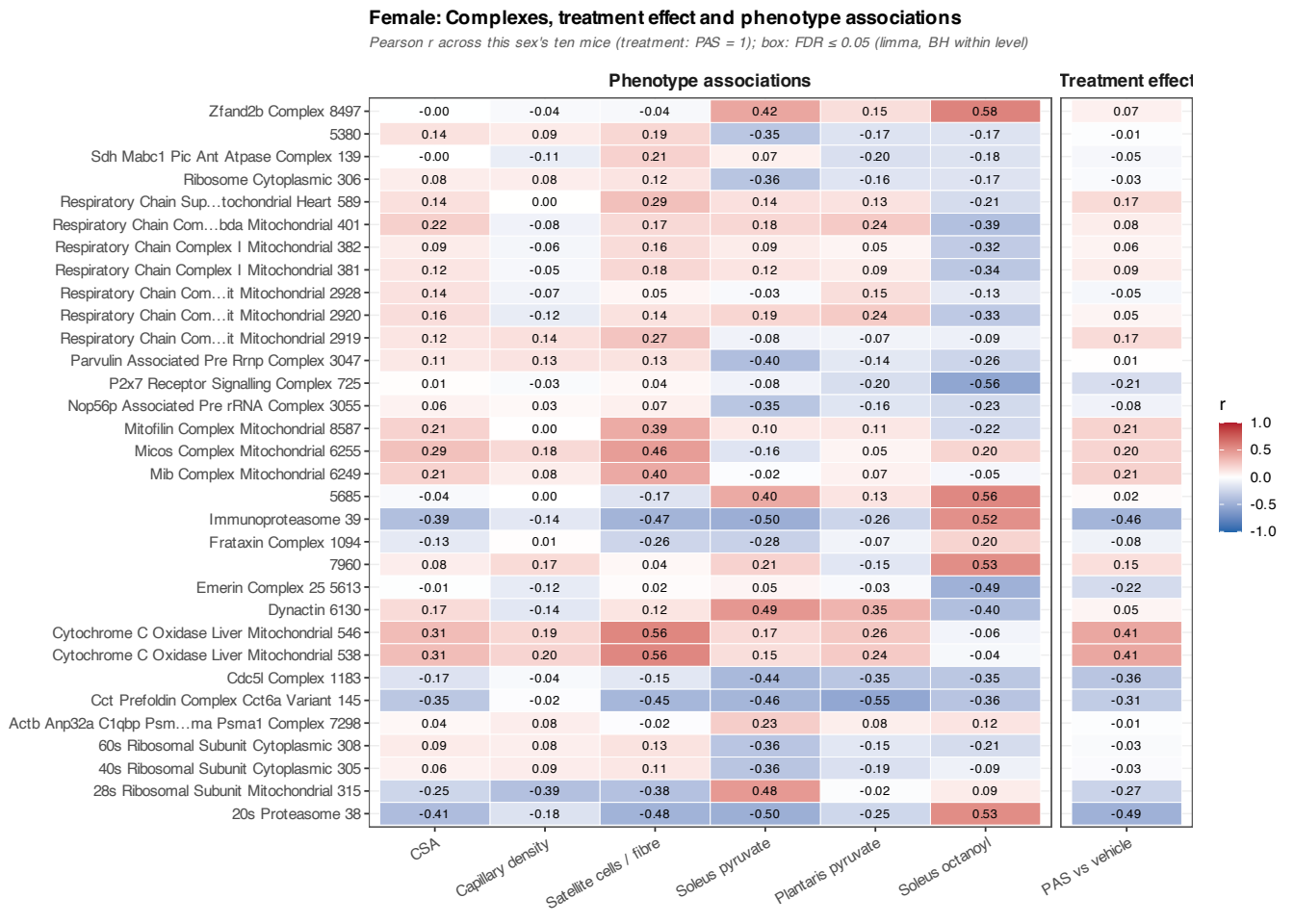

Female: phenotype associations by level
This sex's ten mice, pooled across PAS and vehicle or within treatment; p ≤ 0.05 of tests; BH within level, phenotype and model
Pooled
Within treatment
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
130 / 1962 FDR 0
38 / 595 FDR 0
53 / 938 FDR 0
5 / 14 FDR 2
11 / 98 FDR 0
0 / 7 FDR 0
158 / 1962 FDR 0
47 / 595 FDR 0
8 / 938 FDR 0
0 / 14 FDR 0
1 / 98 FDR 0
0 / 7 FDR 0
110 / 1962 FDR 2
48 / 595 FDR 1
37 / 938 FDR 0
1 / 14 FDR 1
13 / 98 FDR 1
0 / 7 FDR 0
169 / 1962 FDR 0
29 / 595 FDR 0
16 / 938 FDR 0
0 / 14 FDR 0
4 / 98 FDR 0
0 / 7 FDR 0
231 / 1962 FDR 1
58 / 595 FDR 0
48 / 938 FDR 0
5 / 14 FDR 3
11 / 98 FDR 0
1 / 7 FDR 0
156 / 1962 FDR 0
38 / 595 FDR 0
6 / 938 FDR 0
0 / 14 FDR 0
2 / 98 FDR 0
0 / 7 FDR 0
67 / 1962 FDR 0
17 / 595 FDR 0
91 / 938 FDR 0
0 / 14 FDR 0
9 / 98 FDR 0
1 / 7 FDR 0
124 / 1962 FDR 0
27 / 595 FDR 0
81 / 938 FDR 0
0 / 14 FDR 0
7 / 98 FDR 0
1 / 7 FDR 0
137 / 1962 FDR 0
35 / 595 FDR 0
24 / 938 FDR 0
1 / 14 FDR 0
4 / 98 FDR 0
0 / 7 FDR 0
95 / 1962 FDR 0
33 / 595 FDR 1
11 / 938 FDR 0
0 / 14 FDR 0
2 / 98 FDR 0
0 / 7 FDR 0
89 / 1962 FDR 2
49 / 595 FDR 0
113 / 938 FDR 0
3 / 14 FDR 0
23 / 98 FDR 0
0 / 7 FDR 0
77 / 1962 FDR 2
39 / 595 FDR 0
132 / 938 FDR 8
2 / 14 FDR 0
29 / 98 FDR 2
1 / 7 FDR 0
Proteins
Proteins, mito
Pathways
Modules
Complexes
Clusters
p ≤ 0.05 / chance
6
4
2
0

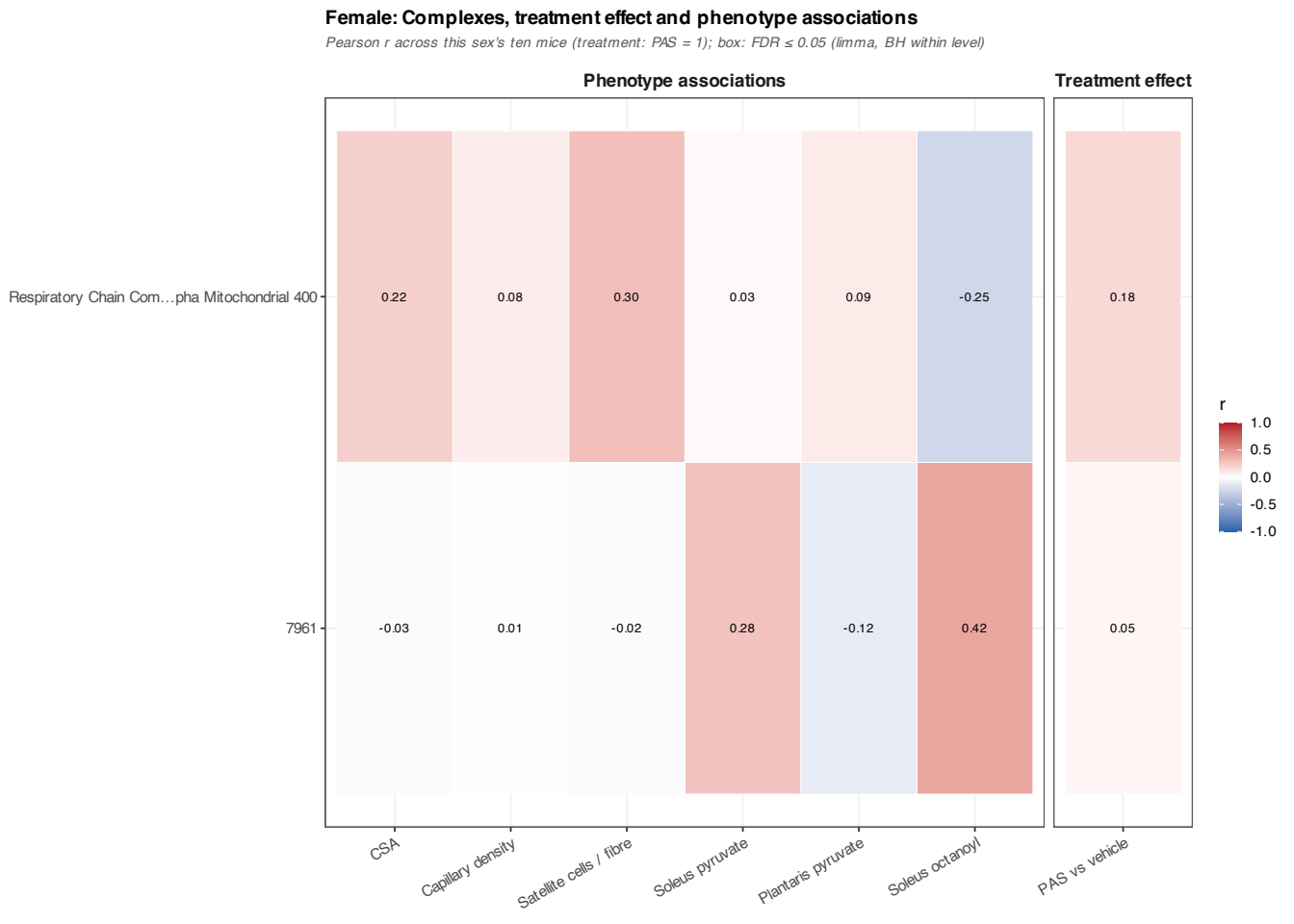

Check: protein associations with and without pellet mitochondrial share
Share centred within group, added to the association model
FDR ≤ 0.05
p ≤ 0.05
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
0
2
4
6
0
50
100
150
200
250
Proteins
with pellet mito share
without

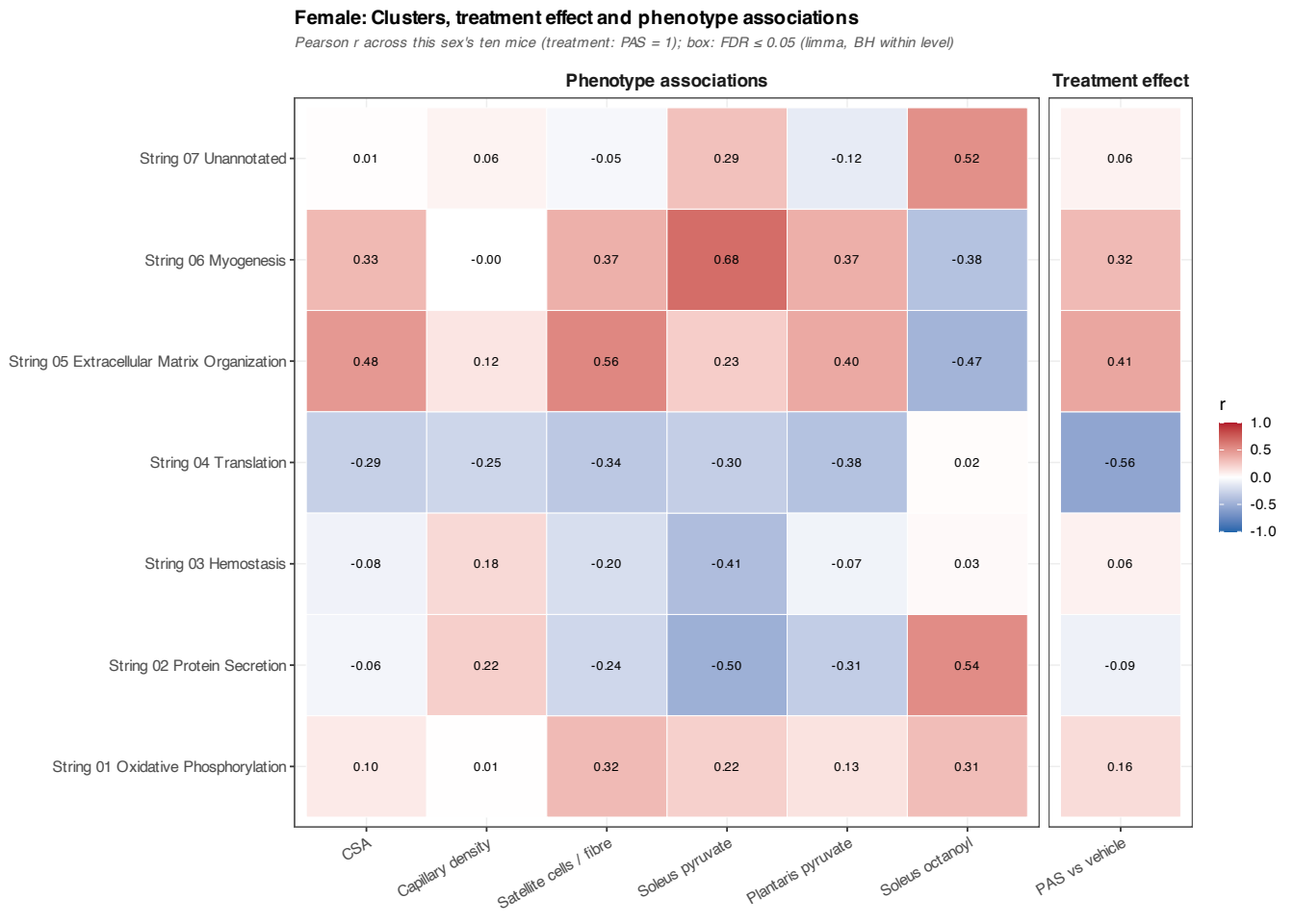

Female: Modules, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Yellow 0.27 -0.10 0.34 0.47 0.38 -0.57 0.21
Turquoise 0.36 0.27 0.55 -0.05 0.06 -0.33 0.42
Tan -0.43 -0.27 -0.28 -0.33 -0.39 -0.30 -0.43
Salmon 0.19 -0.02 0.14 -0.31 0.01 -0.17 -0.08
Red 0.84 0.66 0.70 0.09 0.26 0.52 0.73
Purple -0.15 0.10 -0.21 -0.37 -0.23 0.05 -0.11
Pink -0.18 -0.14 -0.11 -0.39 -0.06 -0.57 -0.20
Magenta -0.57 -0.56 -0.57 -0.08 -0.34 -0.53 -0.73
Greenyellow -0.21 -0.17 -0.44 -0.04 0.10 0.12 -0.24
Green -0.12 0.15 -0.29 -0.50 -0.16 -0.01 -0.09
Cyan -0.76 -0.69 -0.84 -0.30 -0.30 -0.13 -0.89
Brown -0.29 0.12 -0.34 -0.64 -0.39 0.09 -0.19
Blue -0.21 -0.28 -0.43 -0.08 0.08 0.05 -0.38
Black 0.45 0.18 0.31 0.27 0.33 0.07 0.35
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Female: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Tnfalpha Nfkappa B Signaling Complex 5232 0.04 0.29 0.17 -0.39 -0.26 0.12 0.11
Spliceosome E Complex 8391 -0.21 -0.36 -0.37 0.09 0.01 -0.19 -0.44
Spliceosome A Complex 8372 -0.29 -0.43 -0.40 -0.08 -0.09 -0.25 -0.58
Sarcoglycan Sarcospan…trobrevin Complex 349 -0.37 -0.49 -0.17 0.10 -0.04 -0.66 -0.39
Sarcoglycan Sarcospan…troglycan Complex 346 -0.27 -0.38 -0.11 0.21 0.06 -0.73 -0.21
Sarcoglycan Sarcospan Complex Sg Spn 343 -0.31 -0.43 -0.15 0.36 0.04 -0.65 -0.25
Respiratory Chain Com…eta Mitochondrial 402 -0.35 -0.55 -0.27 0.40 0.04 -0.50 -0.32
Respiratory Chain Com…it Mitochondrial 2929 -0.36 -0.58 -0.26 0.43 0.08 -0.56 -0.32
Respiratory Chain Com…it Mitochondrial 2914 -0.36 -0.56 -0.28 0.36 0.01 -0.50 -0.35
Pa700 Complex 32 -0.77 -0.54 -0.69 0.03 -0.40 -0.23 -0.75
Pa700 20s Pa28 Complex 193 -0.68 -0.38 -0.69 -0.32 -0.37 0.23 -0.70
Oligosaccharyltransferase Ostc Iii 1209 0.46 0.56 0.57 -0.13 -0.04 0.28 0.58
Oligosaccharyltransferase Ostc Ii 1208 0.40 0.52 0.51 -0.13 -0.16 0.27 0.50
Oligosaccharyltransferase Ostc I 1203 0.34 0.49 0.44 0.03 -0.10 0.22 0.53
1200 0.24 0.43 0.29 0.03 -0.07 0.29 0.48
1199 0.24 0.48 0.30 0.08 -0.09 0.38 0.51
8918 -0.35 -0.36 -0.31 -0.02 -0.21 -0.32 -0.35
Mitrac Complex Mitochondrial 8569 -0.71 -0.55 -0.80 -0.13 -0.08 -0.03 -0.69
Large Drosha Complex 1332 -0.86 -0.76 -0.79 0.04 -0.07 -0.17 -0.74
H2ax Complex II 1227 0.44 0.60 0.40 -0.65 -0.20 0.52 0.33
Emerin Complex 24 5611 0.06 -0.38 0.14 0.40 0.49 -0.64 -0.03
Dystrophin Sarcoglyca…x Skeletal Muscle 337 -0.40 -0.55 -0.31 0.21 0.02 -0.77 -0.45
Dystrophin Glycoprote…C Skeletal Muscle 340 -0.30 -0.42 -0.13 0.23 0.05 -0.75 -0.25
Ddb2 Complex 726 -0.73 -0.79 -0.73 -0.10 -0.19 -0.25 -0.90
Cytochrome C Oxidase Heart Mitochondrial 542 0.10 -0.18 0.39 0.12 0.40 -0.15 0.15
Coa3 Hspa9 Timm44 Tim…ex Mitochondrial 8563 -0.42 -0.25 -0.30 -0.34 -0.24 0.24 -0.44
Cct Complex Chaperoni…t Testis Specific 150 -0.51 -0.32 -0.48 -0.35 -0.41 -0.39 -0.47
C Src Immune Complex 6938 -0.33 -0.50 -0.43 0.11 0.26 -0.26 -0.22
C Complex Spliceosome 1181 -0.20 -0.38 -0.43 0.08 0.03 -0.16 -0.43
Bbs Chaperonin Complex 6247 -0.48 -0.40 -0.42 -0.27 -0.31 -0.51 -0.46
Er Localized Multipro…f Ig Heavy Chains 414 0.41 0.77 0.39 0.16 0.08 0.54 0.73
Er Localized Multipro…Chains Associated 413 0.48 0.74 0.47 0.18 0.17 0.57 0.74
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Female: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Vacuolar Atpase 442 0.10 0.13 -0.11 -0.57 0.07 0.40 0.02
V Atpase Ragulator Ax…kb1 Ampk Complex 6888 0.08 0.19 -0.11 -0.43 0.08 0.24 0.12
Tnf Alpha Nf Kappa B Signaling Complex 6 5266 -0.23 -0.11 -0.13 -0.47 -0.31 -0.25 -0.31
Tle1 Corepressor Complex 1729 0.32 0.11 0.29 -0.07 0.00 -0.30 0.02
Tim23 Sort Complex Ro…50 Mitochondrial 8723 -0.29 -0.19 -0.19 -0.31 -0.21 0.31 -0.33
Tim22 Holo Complex Ag…29 Mitochondrial 8557 0.00 -0.23 0.12 -0.48 -0.07 0.13 -0.35
Thymoproteasome 8522 -0.41 -0.18 -0.49 -0.50 -0.22 0.54 -0.48
Spermatoproteasome 8532 -0.52 -0.23 -0.61 -0.32 -0.24 0.53 -0.53
Snw1 Complex 1335 -0.10 -0.24 -0.01 0.05 0.38 -0.71 0.02
Selk Multiprotein Complex 6131 -0.03 0.17 0.02 0.12 0.01 0.48 0.25
Respiratory Chain Com…kd Mitochondrial 2904 0.05 -0.16 0.05 0.10 0.07 -0.48 -0.02
Respiratory Chain Com…te Mitochondrial 2905 -0.08 -0.25 -0.07 0.22 0.02 -0.55 -0.10
Respiratory Chain Com…te Mitochondrial 2948 -0.04 -0.26 -0.01 0.20 0.17 -0.48 -0.07
Respiratory Chain Com…yme Mitochondrial 178 0.02 -0.15 0.11 0.16 0.08 -0.34 -0.00
Respiratory Chain Com…ly Mitochondrial 2884 0.04 -0.08 0.21 0.32 0.08 -0.30 0.13
Pyruvate Dehydrogenase Complex 1375 0.51 0.32 0.46 0.10 0.42 0.37 0.43
Profilin 1 Complex 2836 0.27 0.17 0.09 0.17 0.28 -0.43 0.33
Pa28 20s Proteasome 192 -0.49 -0.20 -0.55 -0.47 -0.27 0.48 -0.52
Lebercilin Complex 5899 -0.14 -0.22 -0.24 -0.10 0.13 -0.07 -0.31
Kinase Maturation Complex 1 5199 -0.58 -0.34 -0.51 -0.58 -0.42 -0.02 -0.70
Hrd1 Complex 6859 0.14 0.17 0.06 -0.07 0.20 0.70 0.24
Emerin Complex 52 5615 -0.02 -0.22 -0.01 0.12 0.19 -0.44 -0.17
742 -0.76 -0.42 -0.82 0.06 -0.47 -0.05 -0.63
7305 0.31 0.21 0.41 0.08 0.18 0.20 0.30
Cytochrome Bc1 Comple…2 2 Mitochondrial 403 0.22 0.36 0.20 -0.26 -0.20 0.29 0.19
Cct Prefoldin Complex Cct6b Variant 3074 -0.42 -0.15 -0.42 -0.59 -0.50 -0.19 -0.43
51 -0.42 -0.15 -0.50 -0.26 -0.47 -0.52 -0.35
Aqp2 Binding Multipro…Generator Complex 437 0.19 0.19 0.13 0.04 0.20 -0.62 0.29
Alix Actomyosin Tight Junction Complex 8172 0.30 0.16 0.25 0.31 0.26 -0.60 0.30
55s Ribosome Mitochondrial 320 -0.34 -0.47 -0.48 0.41 -0.09 0.03 -0.48
39s Ribosomal Subunit Mitochondrial 324 -0.33 -0.44 -0.46 0.31 -0.12 -0.01 -0.53
26s Proteasome 181 -0.51 -0.25 -0.55 -0.40 -0.37 0.43 -0.59
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0

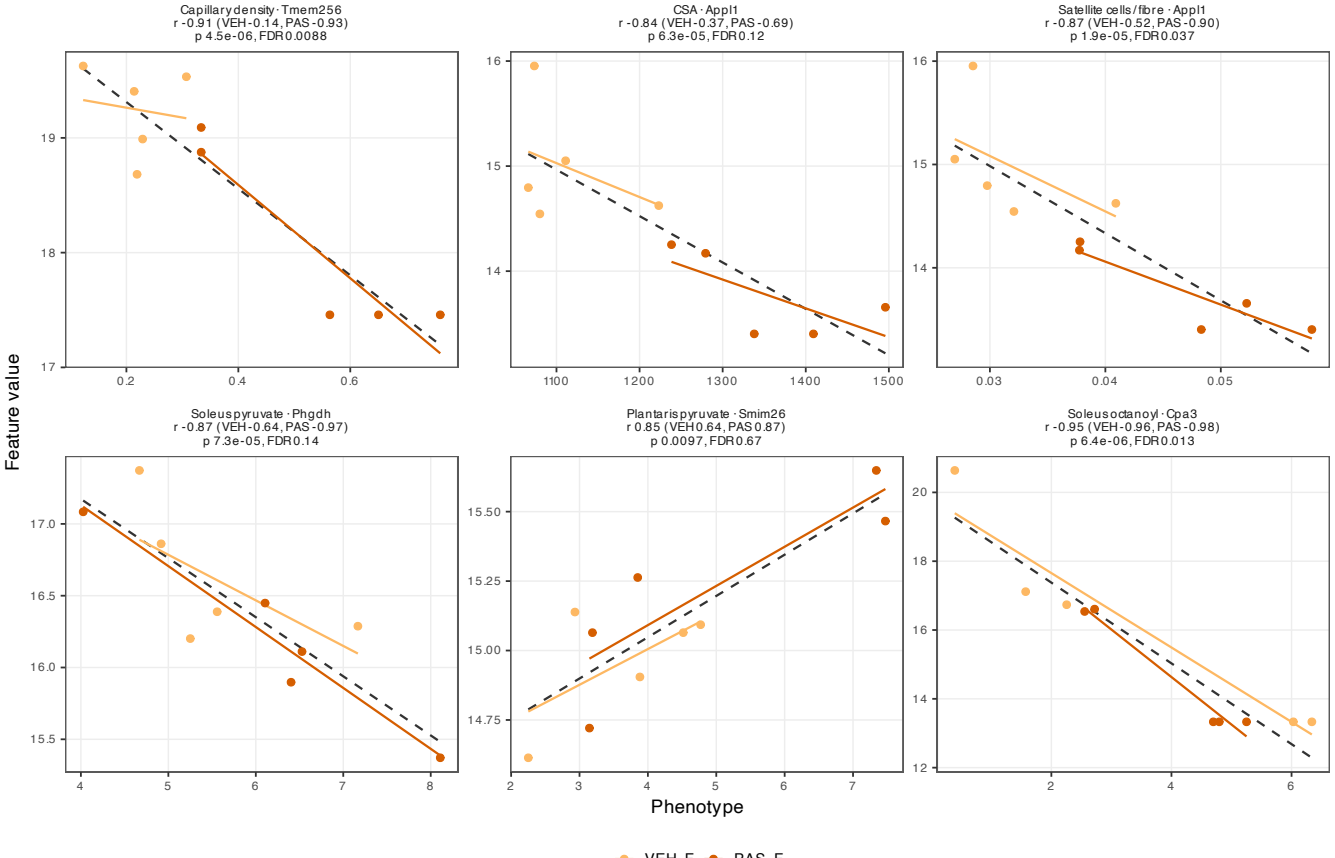

Female: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Zfand2b Complex 8497 -0.00 -0.04 -0.04 0.42 0.15 0.58 0.07
5380 0.14 0.09 0.19 -0.35 -0.17 -0.17 -0.01
Sdh Mabc1 Pic Ant Atpase Complex 139 -0.00 -0.11 0.21 0.07 -0.20 -0.18 -0.05
Ribosome Cytoplasmic 306 0.08 0.08 0.12 -0.36 -0.16 -0.17 -0.03
Respiratory Chain Sup…tochondrial Heart 589 0.14 0.00 0.29 0.14 0.13 -0.21 0.17
Respiratory Chain Com…bda Mitochondrial 401 0.22 -0.08 0.17 0.18 0.24 -0.39 0.08
Respiratory Chain Complex I Mitochondrial 382 0.09 -0.06 0.16 0.09 0.05 -0.32 0.06
Respiratory Chain Complex I Mitochondrial 381 0.12 -0.05 0.18 0.12 0.09 -0.34 0.09
Respiratory Chain Com…it Mitochondrial 2928 0.14 -0.07 0.05 -0.03 0.15 -0.13 -0.05
Respiratory Chain Com…it Mitochondrial 2920 0.16 -0.12 0.14 0.19 0.24 -0.33 0.05
Respiratory Chain Com…it Mitochondrial 2919 0.12 0.14 0.27 -0.08 -0.07 -0.09 0.17
Parvulin Associated Pre Rrnp Complex 3047 0.11 0.13 0.13 -0.40 -0.14 -0.26 0.01
P2x7 Receptor Signalling Complex 725 0.01 -0.03 0.04 -0.08 -0.20 -0.56 -0.21
Nop56p Associated Pre rRNA Complex 3055 0.06 0.03 0.07 -0.35 -0.16 -0.23 -0.08
Mitofilin Complex Mitochondrial 8587 0.21 0.00 0.39 0.10 0.11 -0.22 0.21
Micos Complex Mitochondrial 6255 0.29 0.18 0.46 -0.16 0.05 0.20 0.20
Mib Complex Mitochondrial 6249 0.21 0.08 0.40 -0.02 0.07 -0.05 0.21
5685 -0.04 0.00 -0.17 0.40 0.13 0.56 0.02
Immunoproteasome 39 -0.39 -0.14 -0.47 -0.50 -0.26 0.52 -0.46
Frataxin Complex 1094 -0.13 0.01 -0.26 -0.28 -0.07 0.20 -0.08
7960 0.08 0.17 0.04 0.21 -0.15 0.53 0.15
Emerin Complex 25 5613 -0.01 -0.12 0.02 0.05 -0.03 -0.49 -0.22
Dynactin 6130 0.17 -0.14 0.12 0.49 0.35 -0.40 0.05
Cytochrome C Oxidase Liver Mitochondrial 546 0.31 0.19 0.56 0.17 0.26 -0.06 0.41
Cytochrome C Oxidase Liver Mitochondrial 538 0.31 0.20 0.56 0.15 0.24 -0.04 0.41
Cdc5l Complex 1183 -0.17 -0.04 -0.15 -0.44 -0.35 -0.35 -0.36
Cct Prefoldin Complex Cct6a Variant 145 -0.35 -0.02 -0.45 -0.46 -0.55 -0.36 -0.31
Actb Anp32a C1qbp Psm…ma Psma1 Complex 7298 0.04 0.08 -0.02 0.23 0.08 0.12 -0.01
60s Ribosomal Subunit Cytoplasmic 308 0.09 0.08 0.13 -0.36 -0.15 -0.21 -0.03
40s Ribosomal Subunit Cytoplasmic 305 0.06 0.09 0.11 -0.36 -0.19 -0.09 -0.03
28s Ribosomal Subunit Mitochondrial 315 -0.25 -0.39 -0.38 0.48 -0.02 0.09 -0.27
20s Proteasome 38 -0.41 -0.18 -0.48 -0.50 -0.25 0.53 -0.49
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Female: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Respiratory Chain Com…pha Mitochondrial 400
7961
0.22
0.08
0.30
0.03
0.09
-0.25
0.18
-0.03
0.01
-0.02
0.28
-0.12
0.42
0.05
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Female: Clusters, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
String 07 Unannotated
String 06 Myogenesis
String 05 Extracellular Matrix Organization
String 04 Translation
String 03 Hemostasis
String 02 Protein Secretion
String 01 Oxidative Phosphorylation
0.01 0.06 -0.05 0.29 -0.12 0.52 0.06
0.33 -0.00 0.37 0.68 0.37 -0.38 0.32
0.48 0.12 0.56 0.23 0.40 -0.47 0.41
-0.29 -0.25 -0.34 -0.30 -0.38 0.02 -0.56
-0.08 0.18 -0.20 -0.41 -0.07 0.03 0.06
-0.06 0.22 -0.24 -0.50 -0.31 0.54 -0.09
0.10 0.01 0.32 0.22 0.13 0.31 0.16
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0

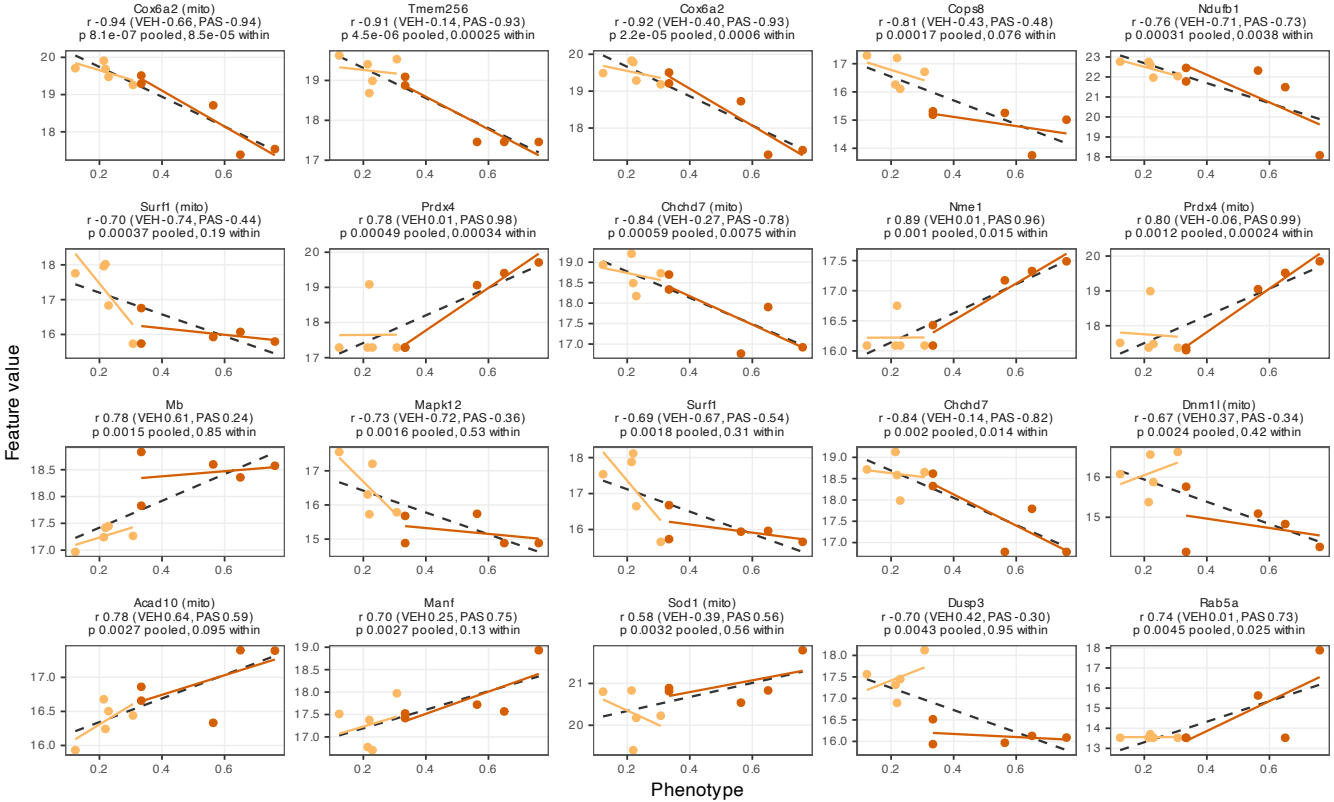
Female: strongest robust protein per phenotype

Smallest association p that keeps its sign and p ≤ 0.05 with any one mouse left out



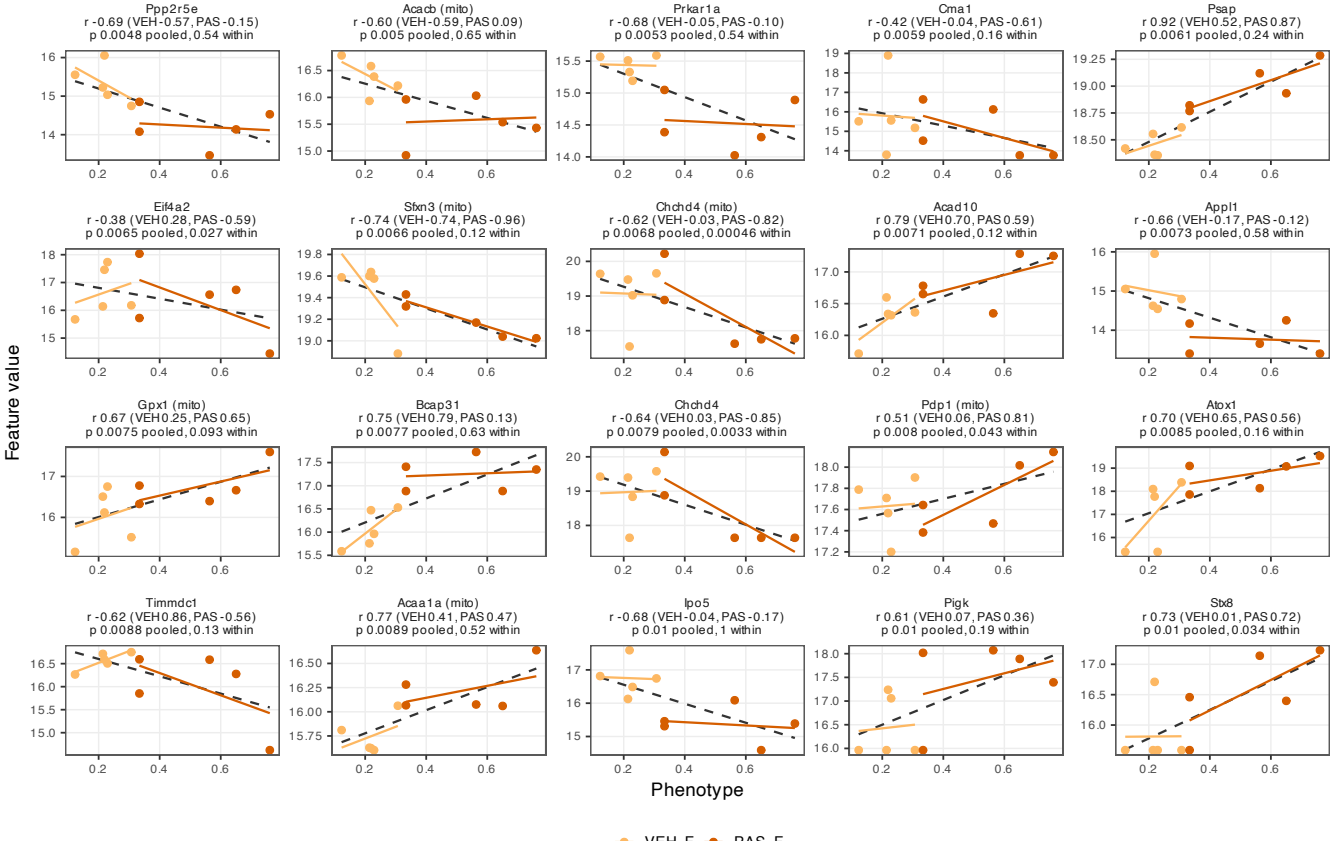
Female: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 320)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



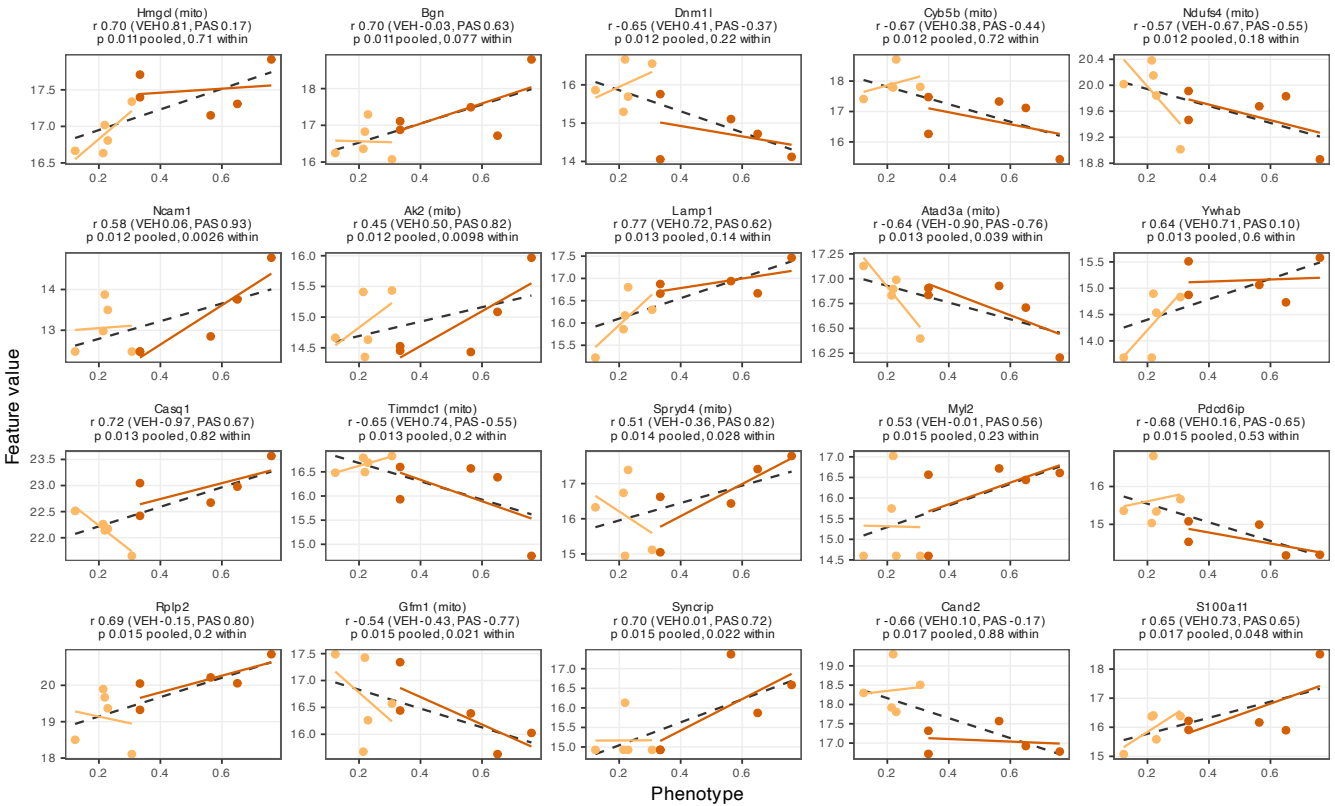
Female: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 320)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



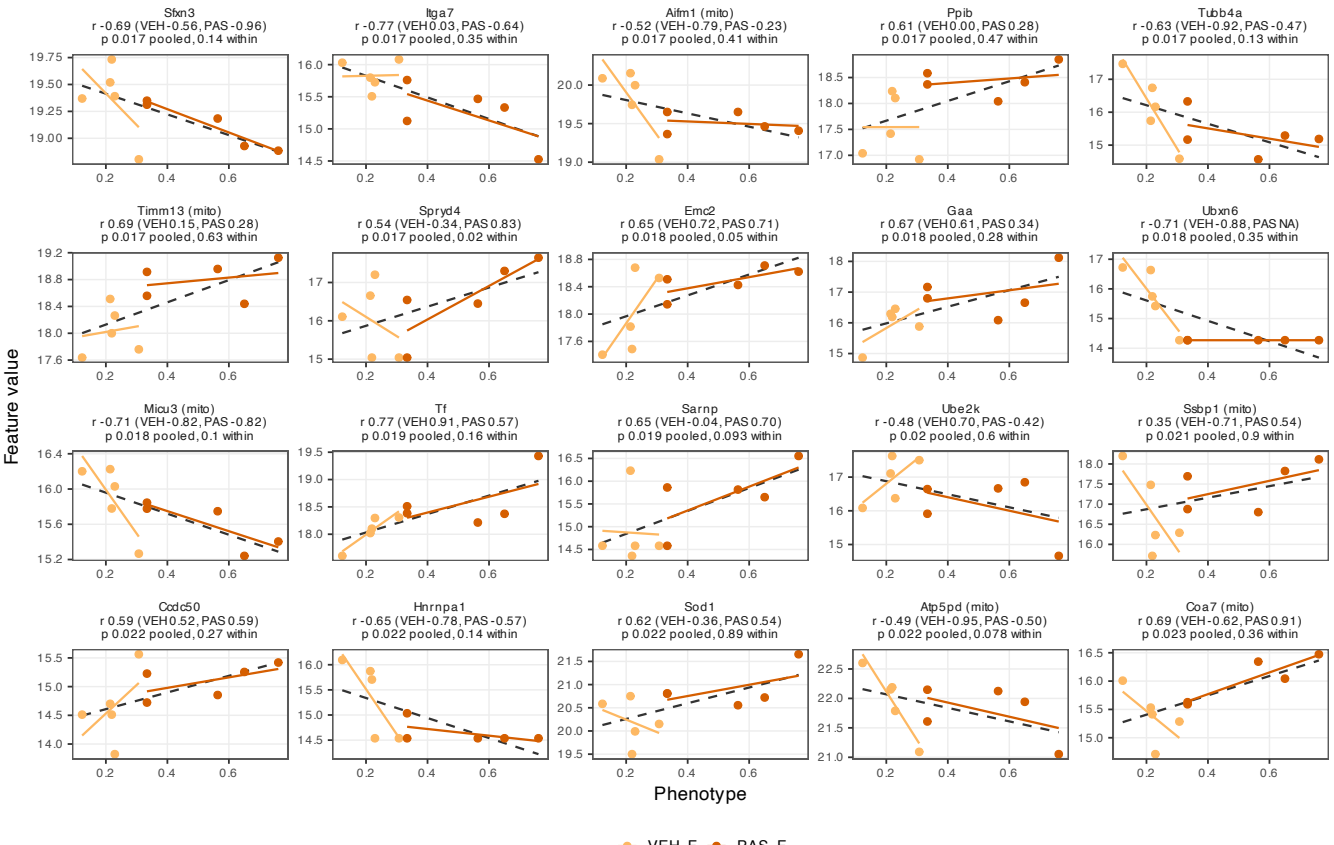
Female: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 320)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



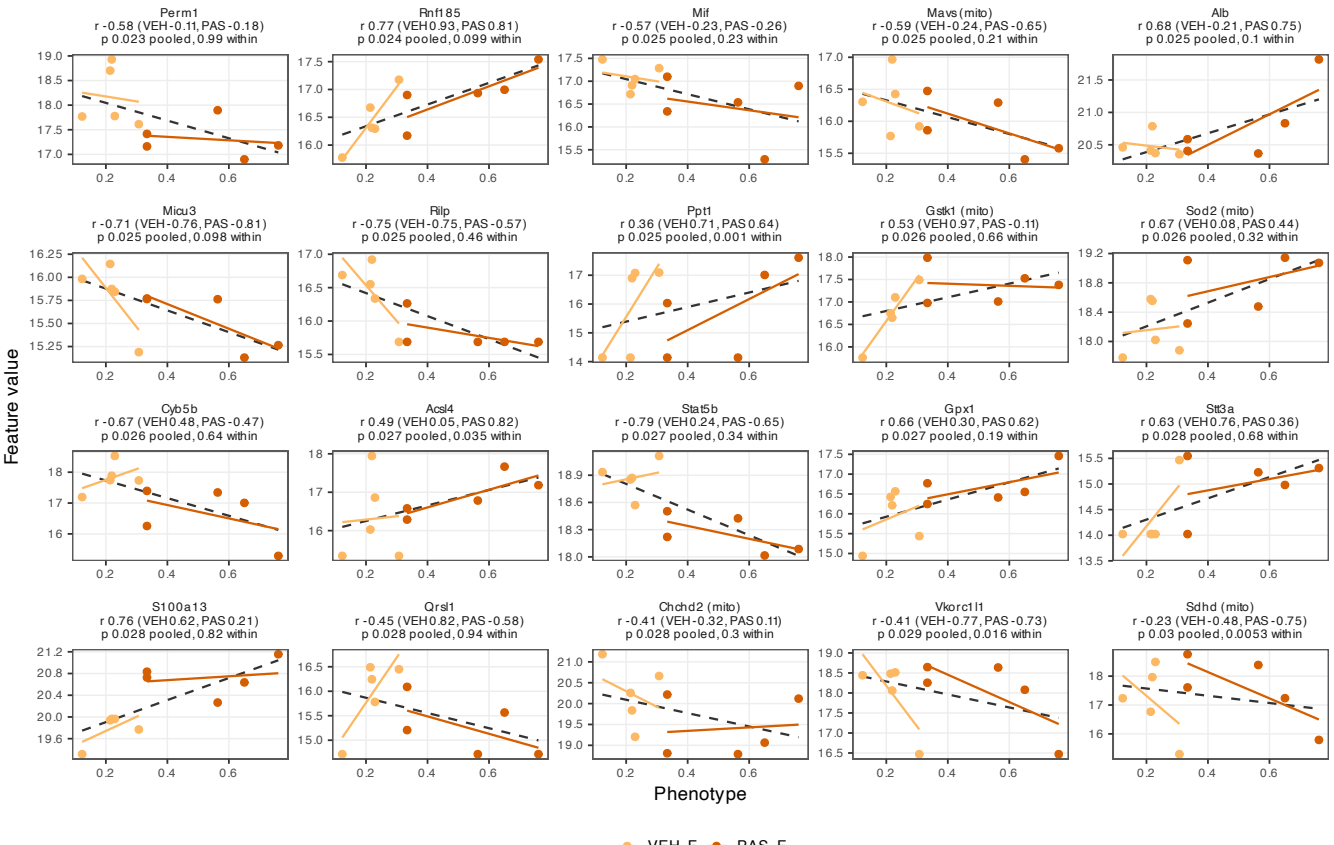
Female: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 320)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



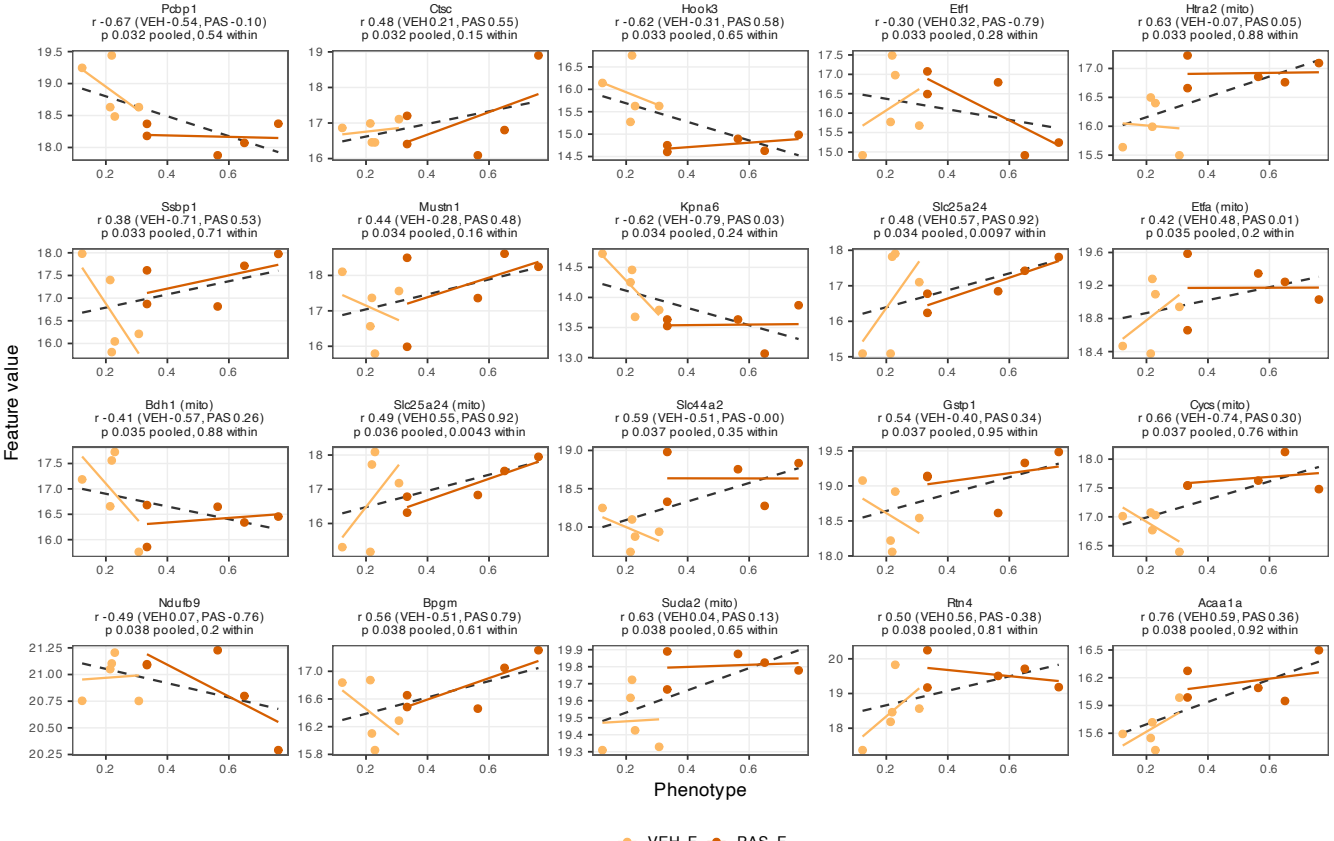
Female: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 320)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



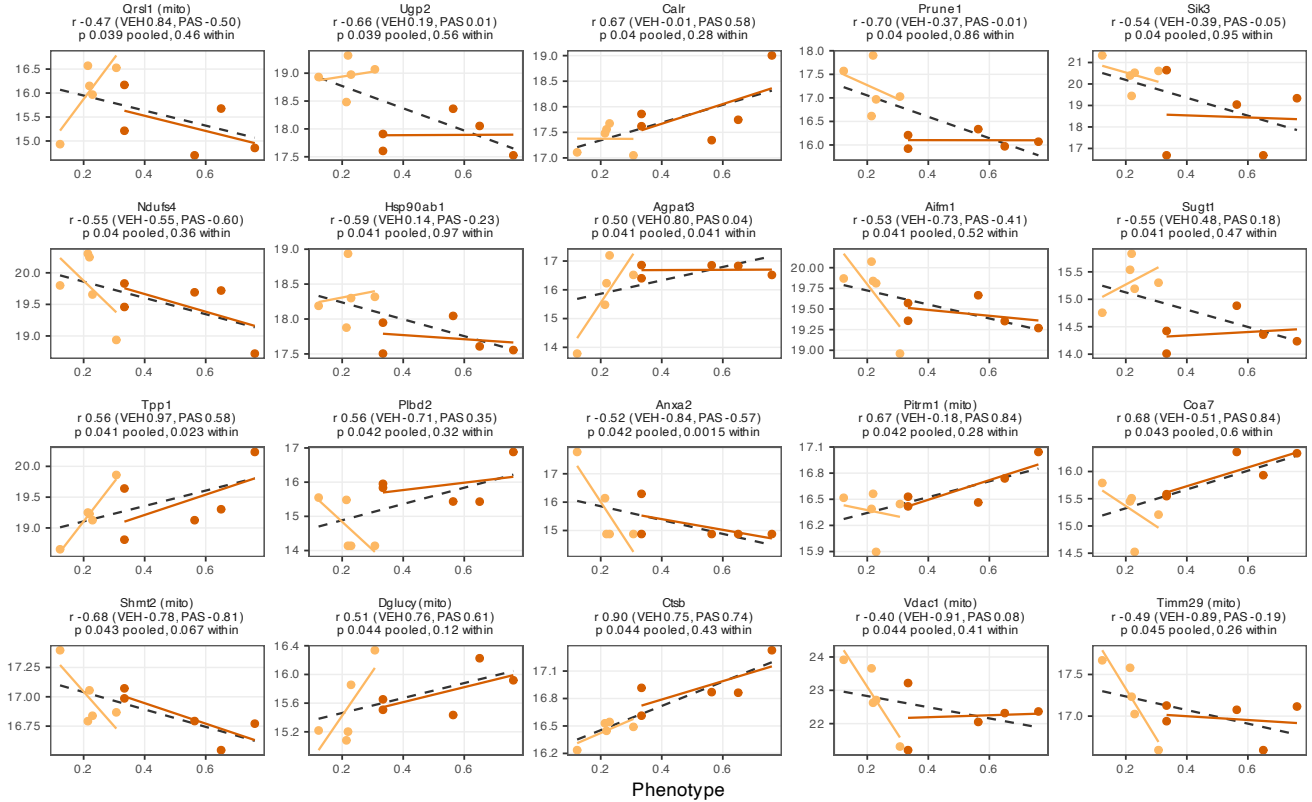
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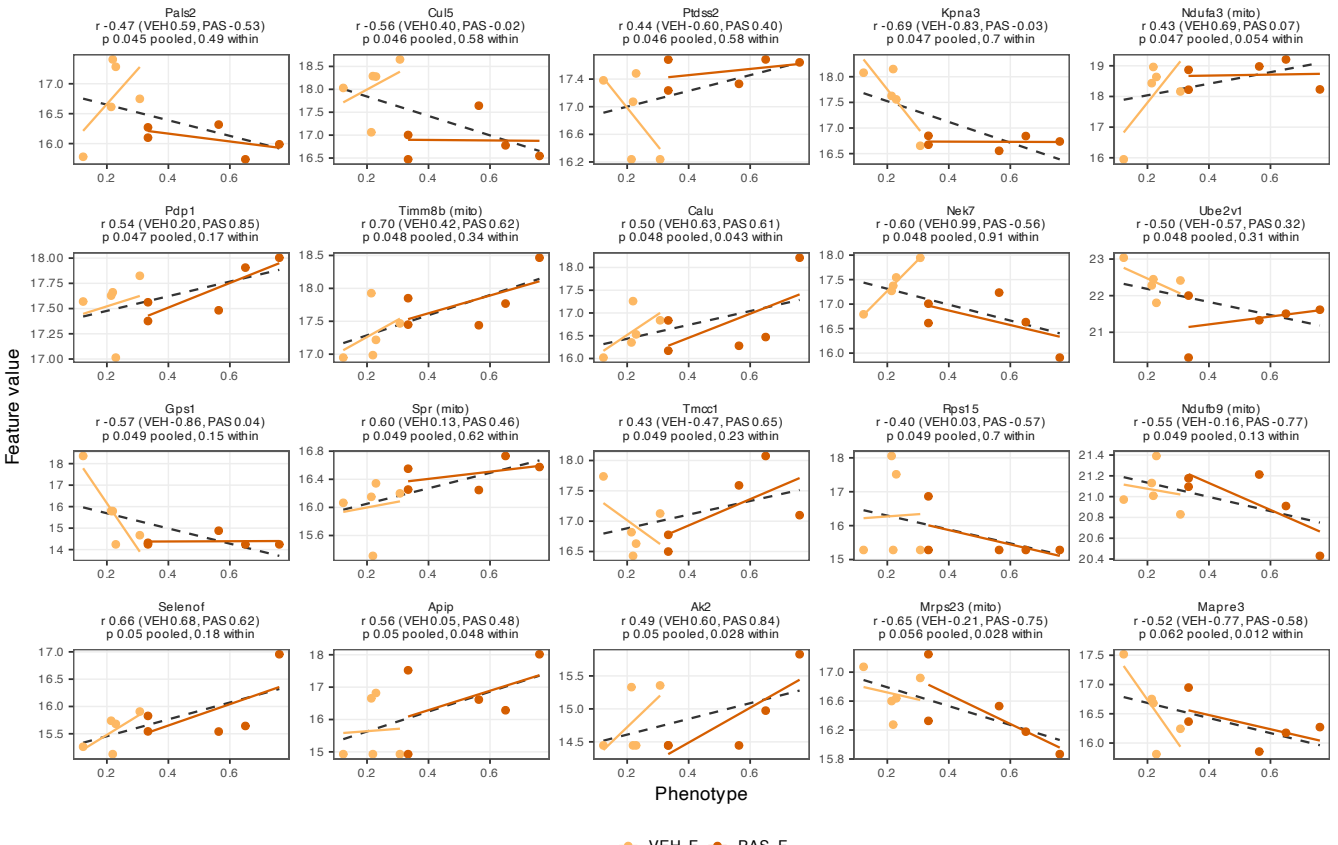
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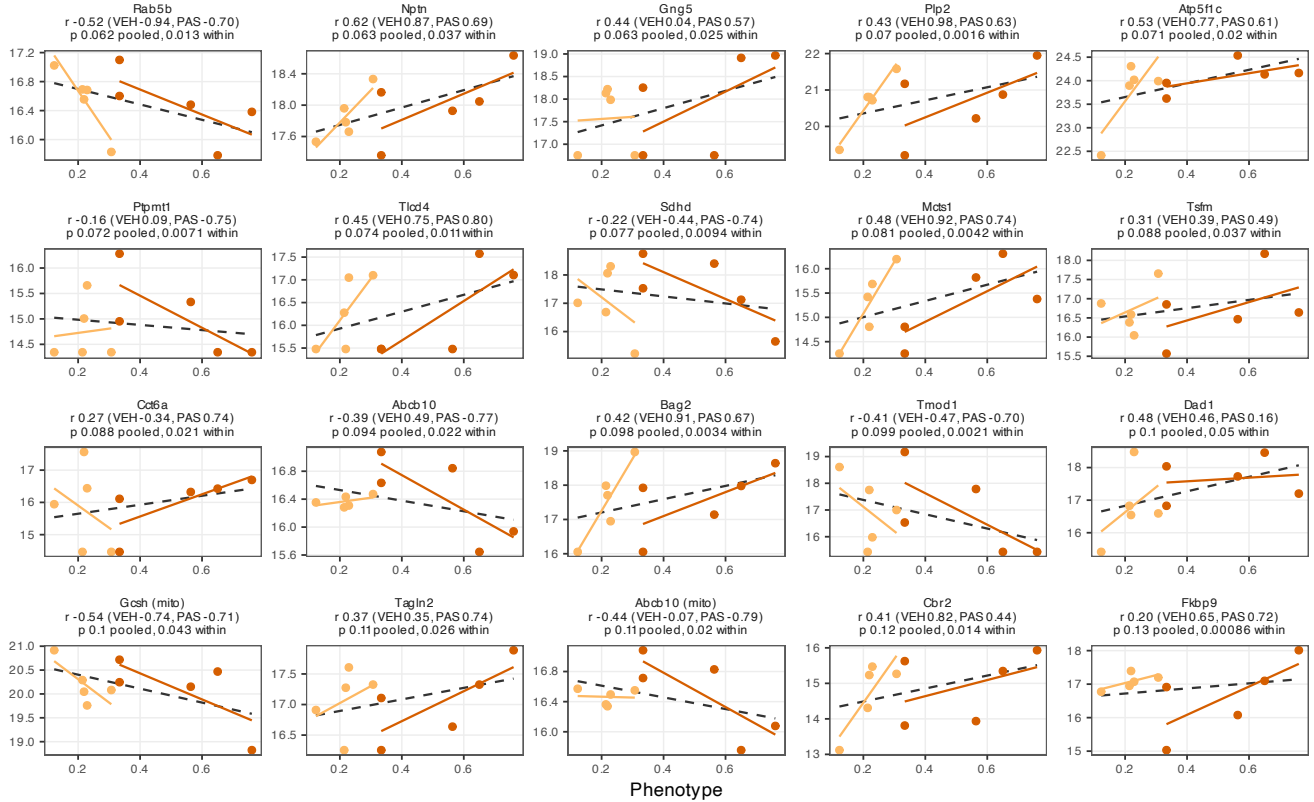
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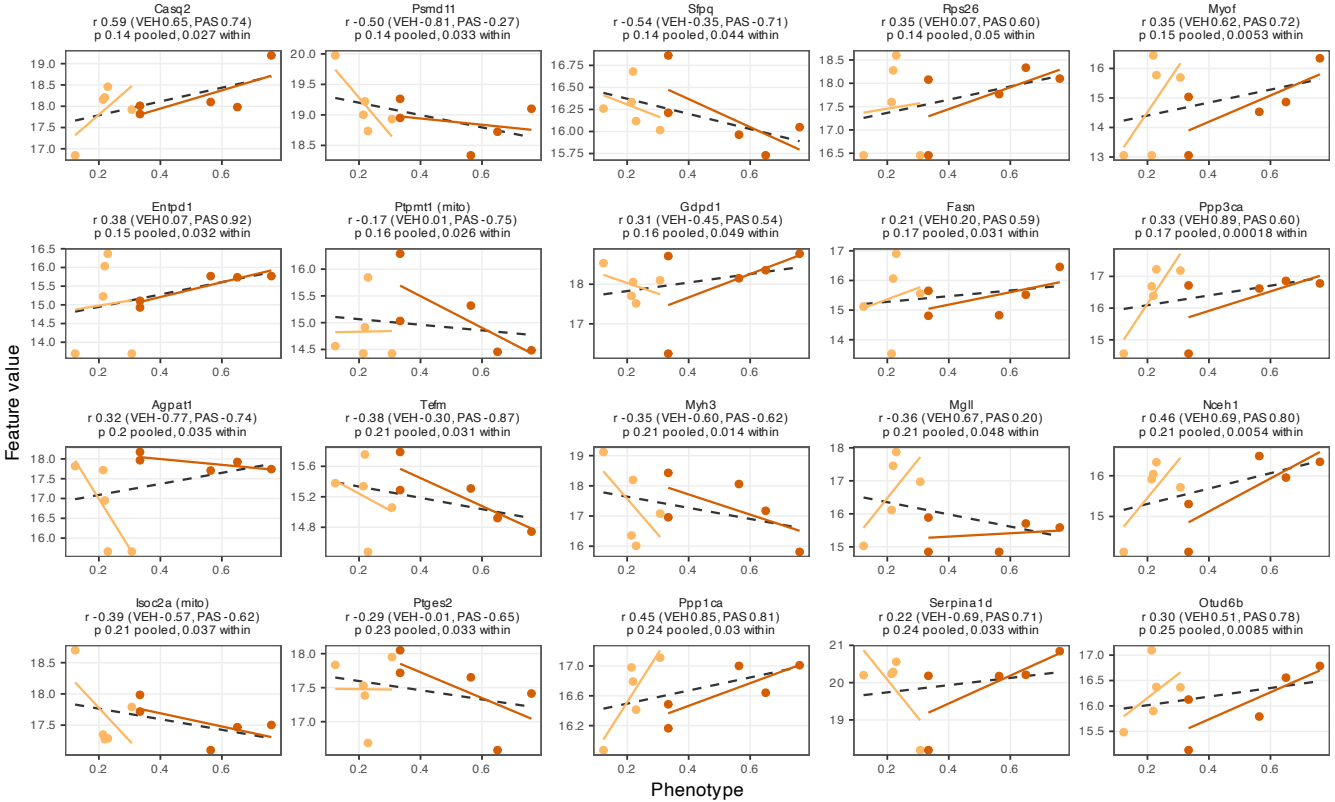
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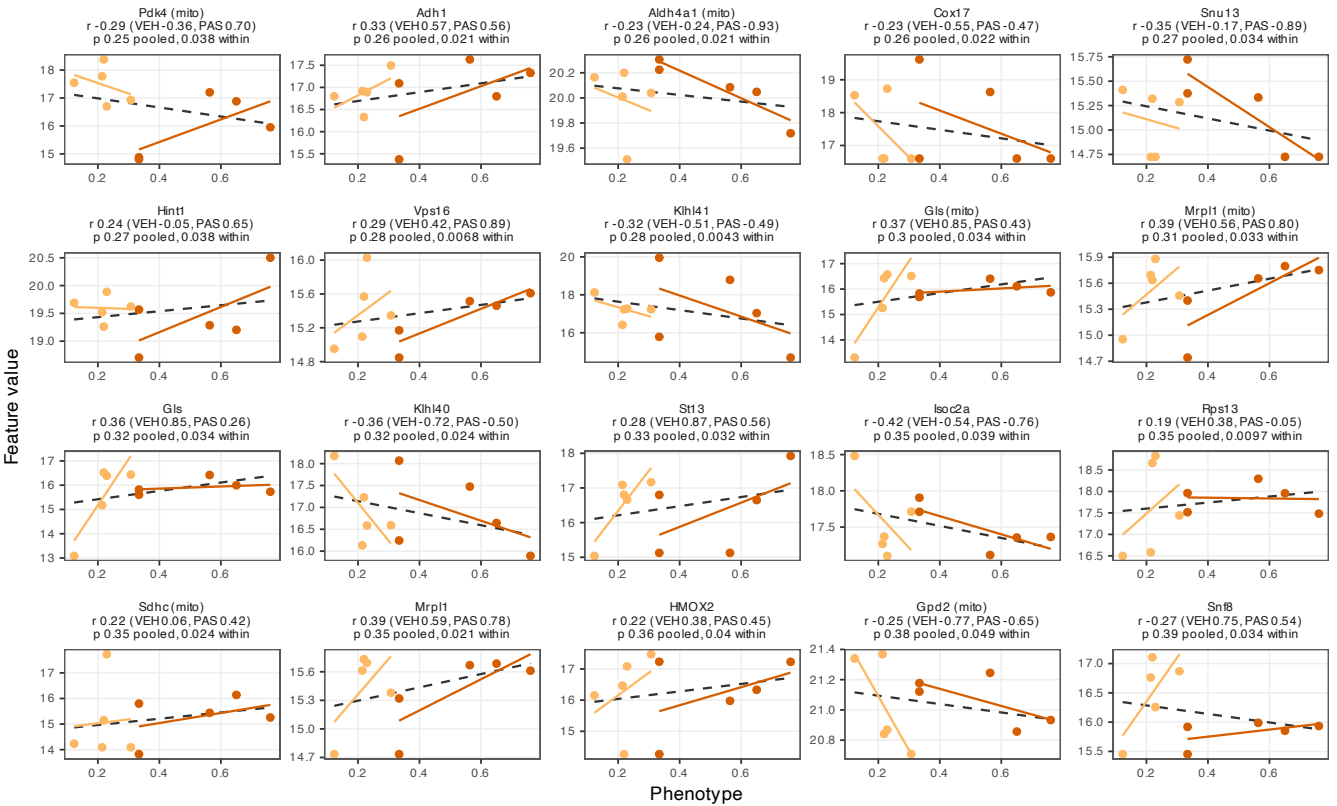
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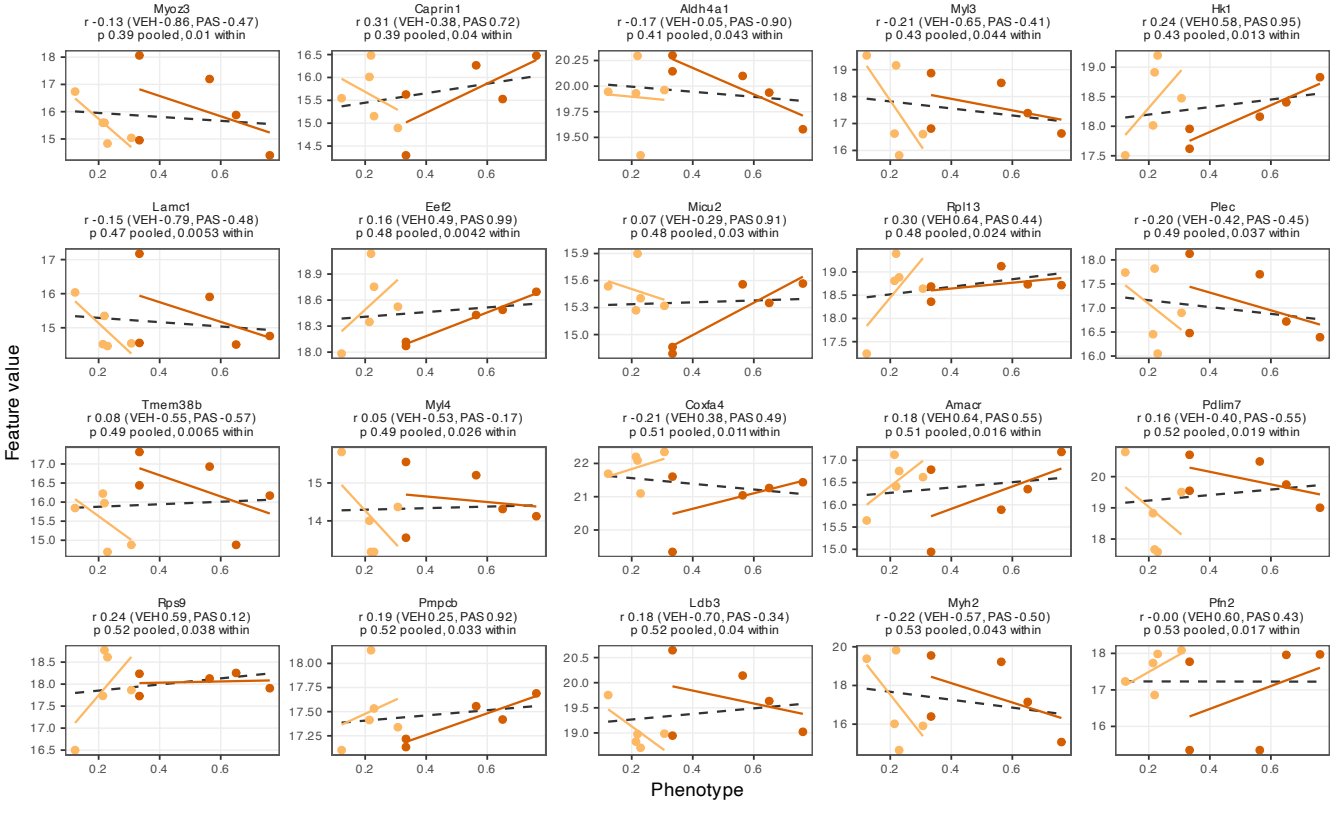
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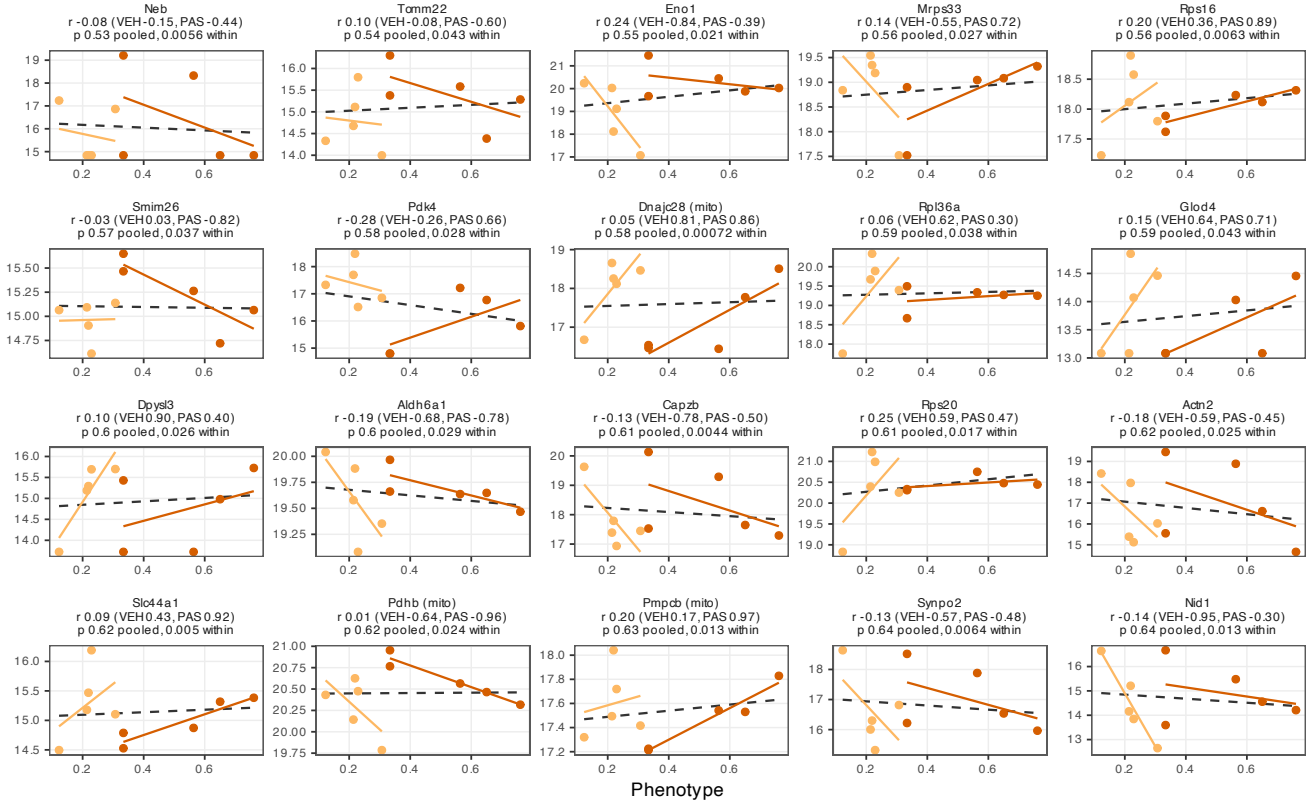
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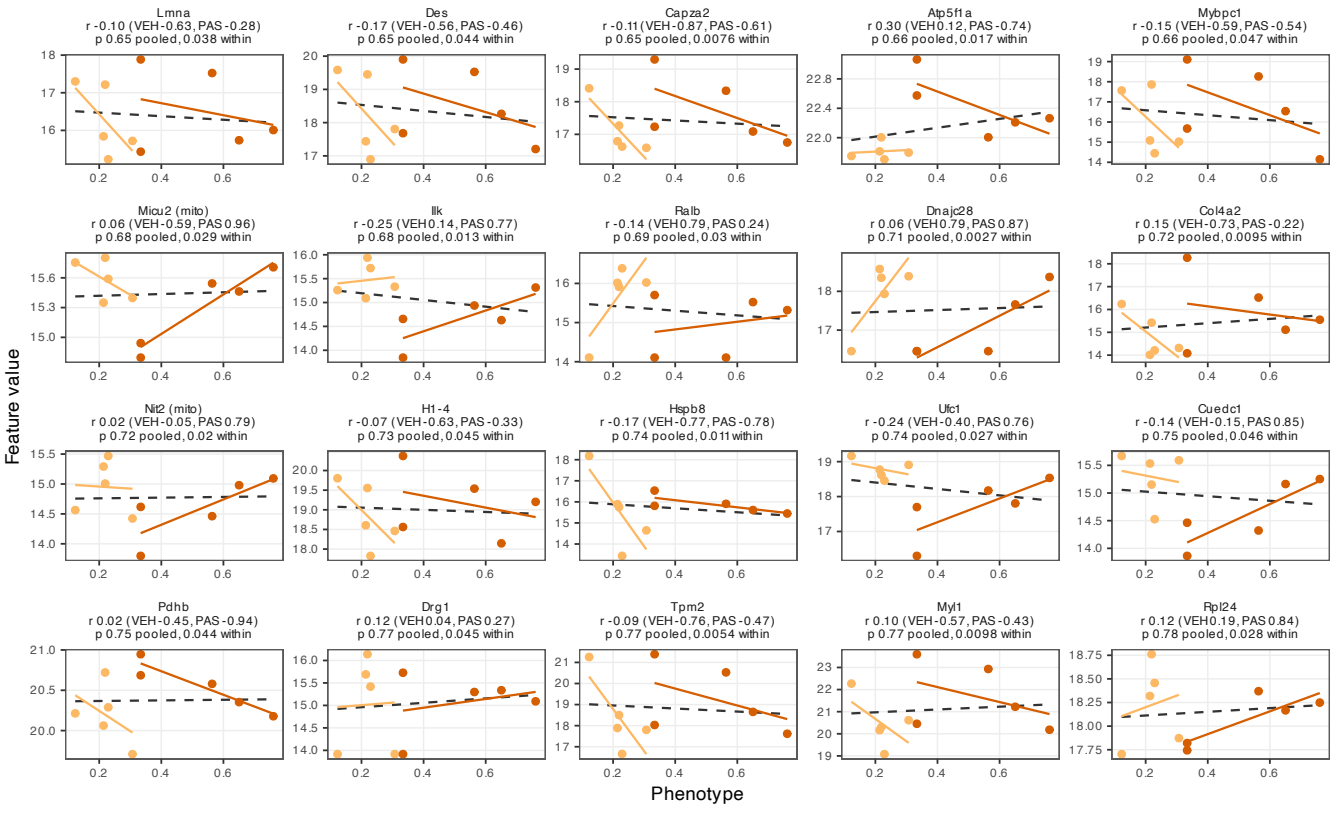
Female: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 320)

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Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



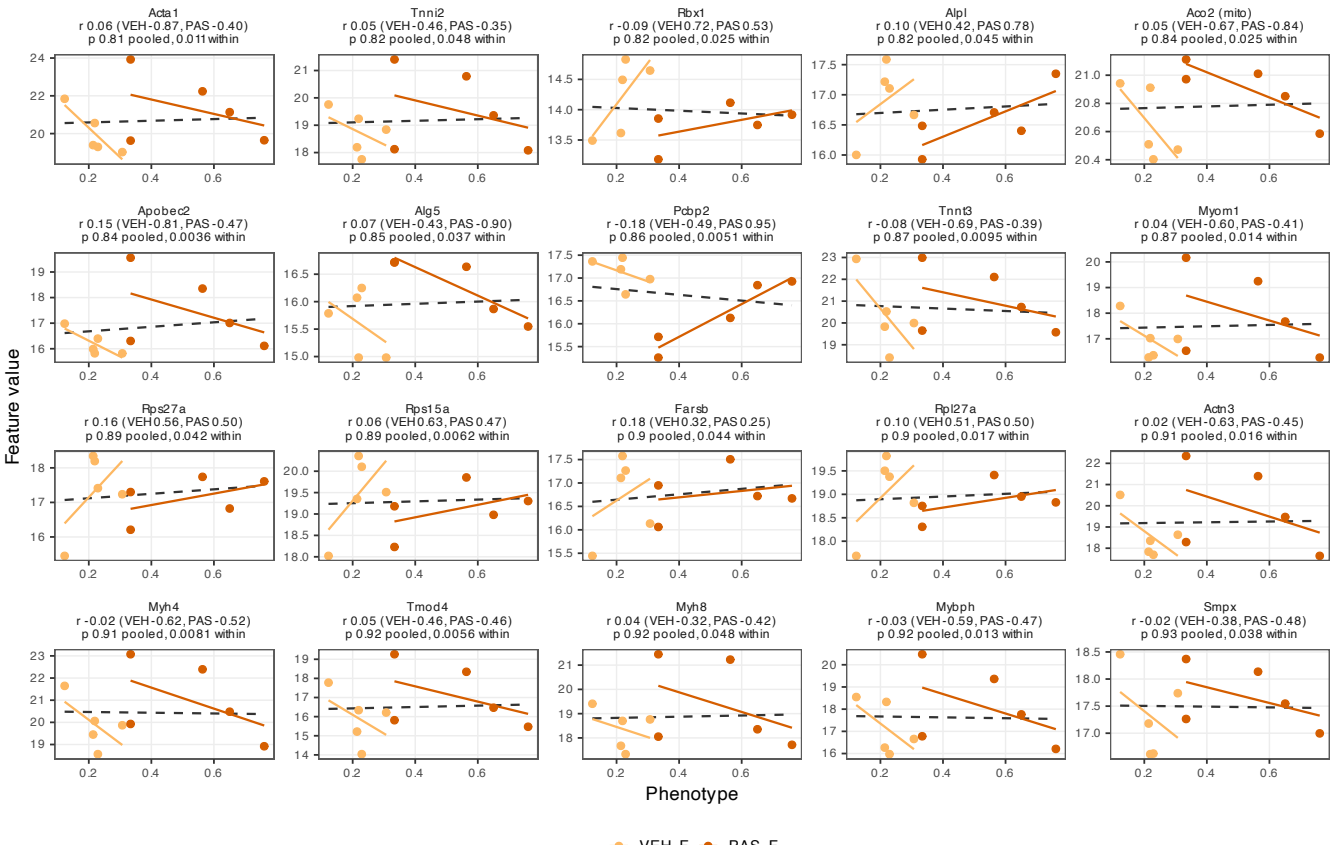
Female: Proteins associated with Capillary density, p ≤ 0.05 in either model (n = 320)

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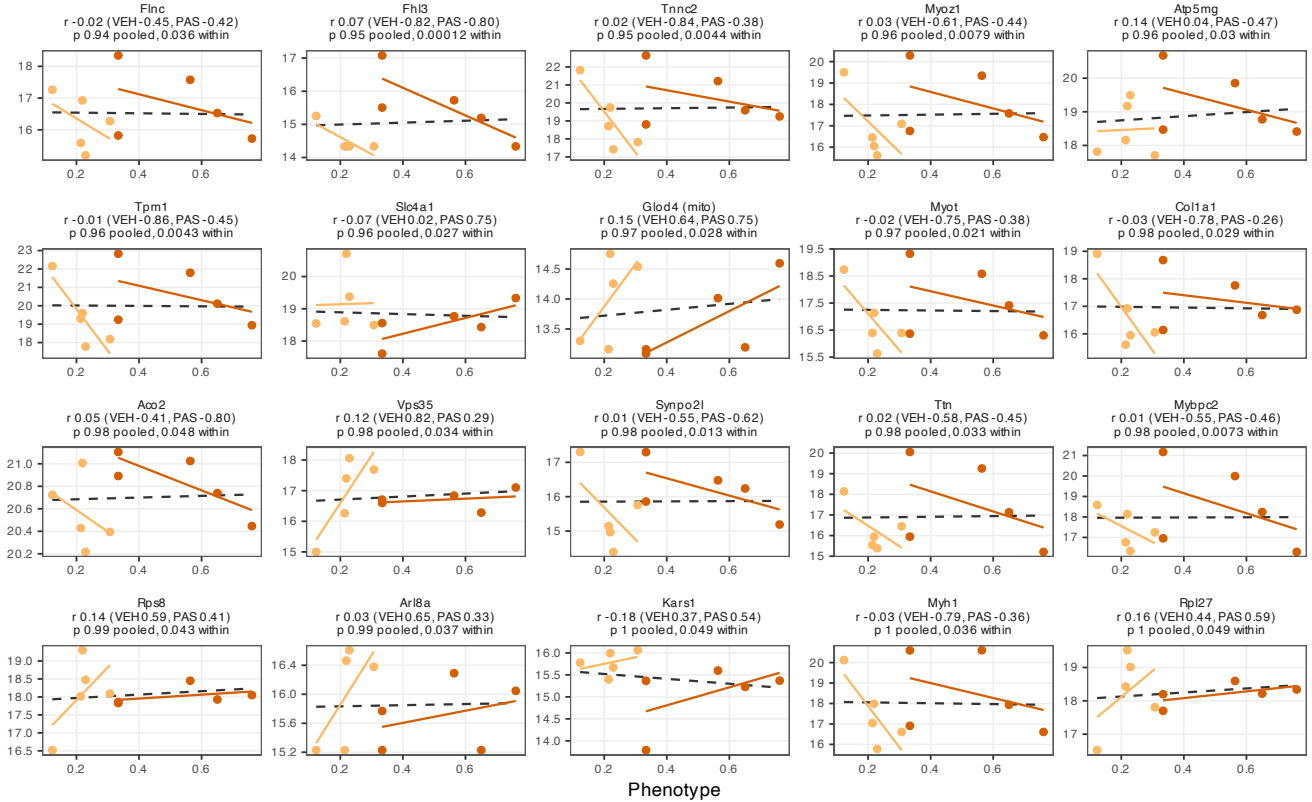
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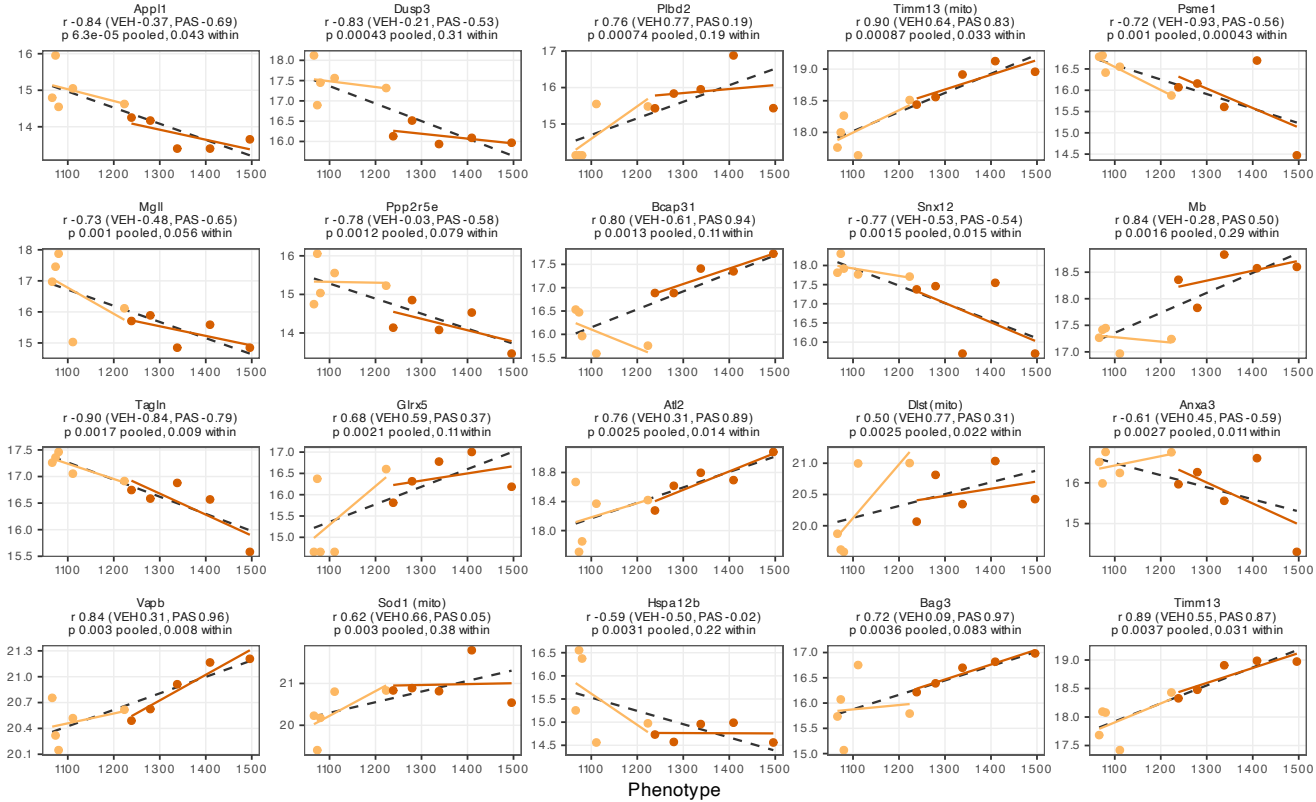
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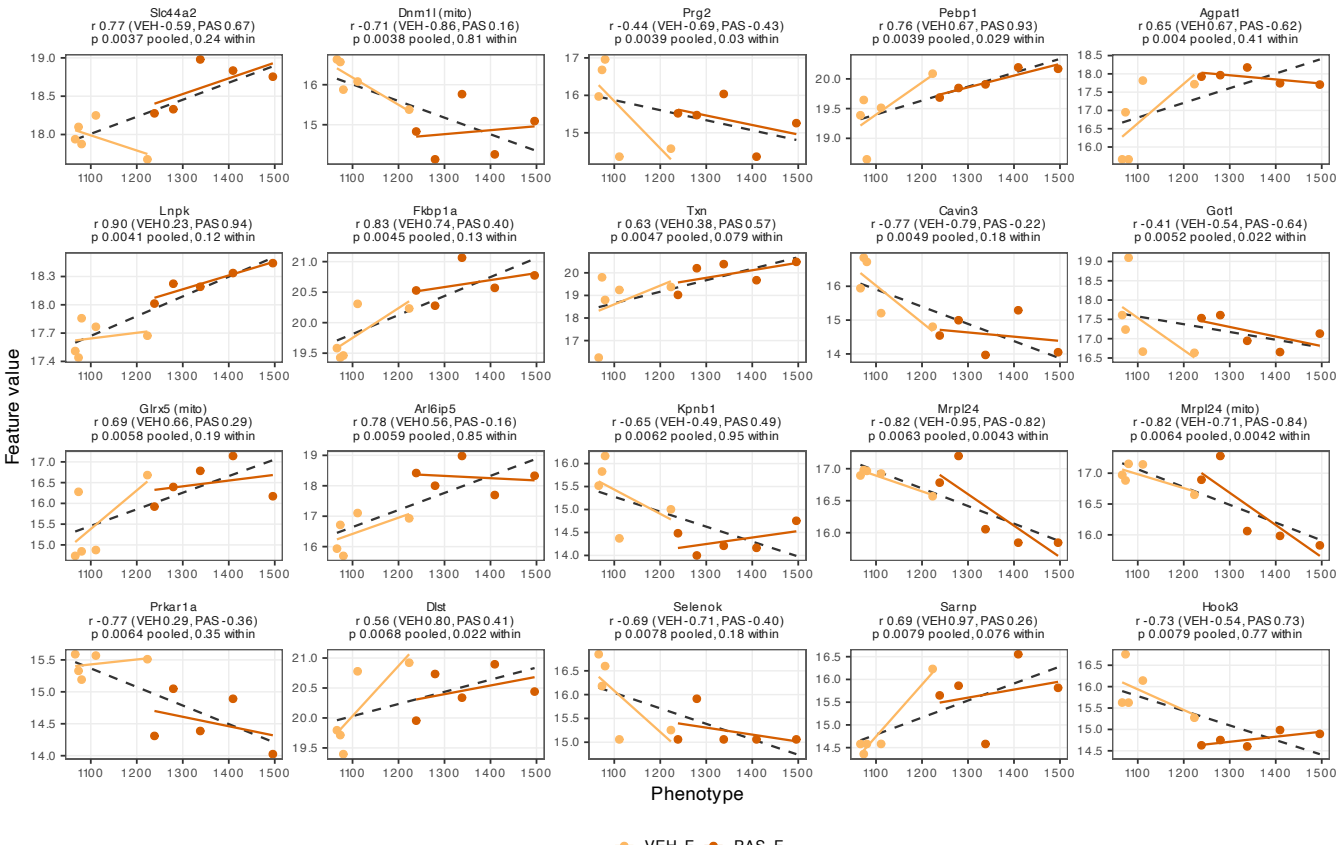
Female: Proteins associated with CSA, p ≤ 0.05 in either model (n = 328)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



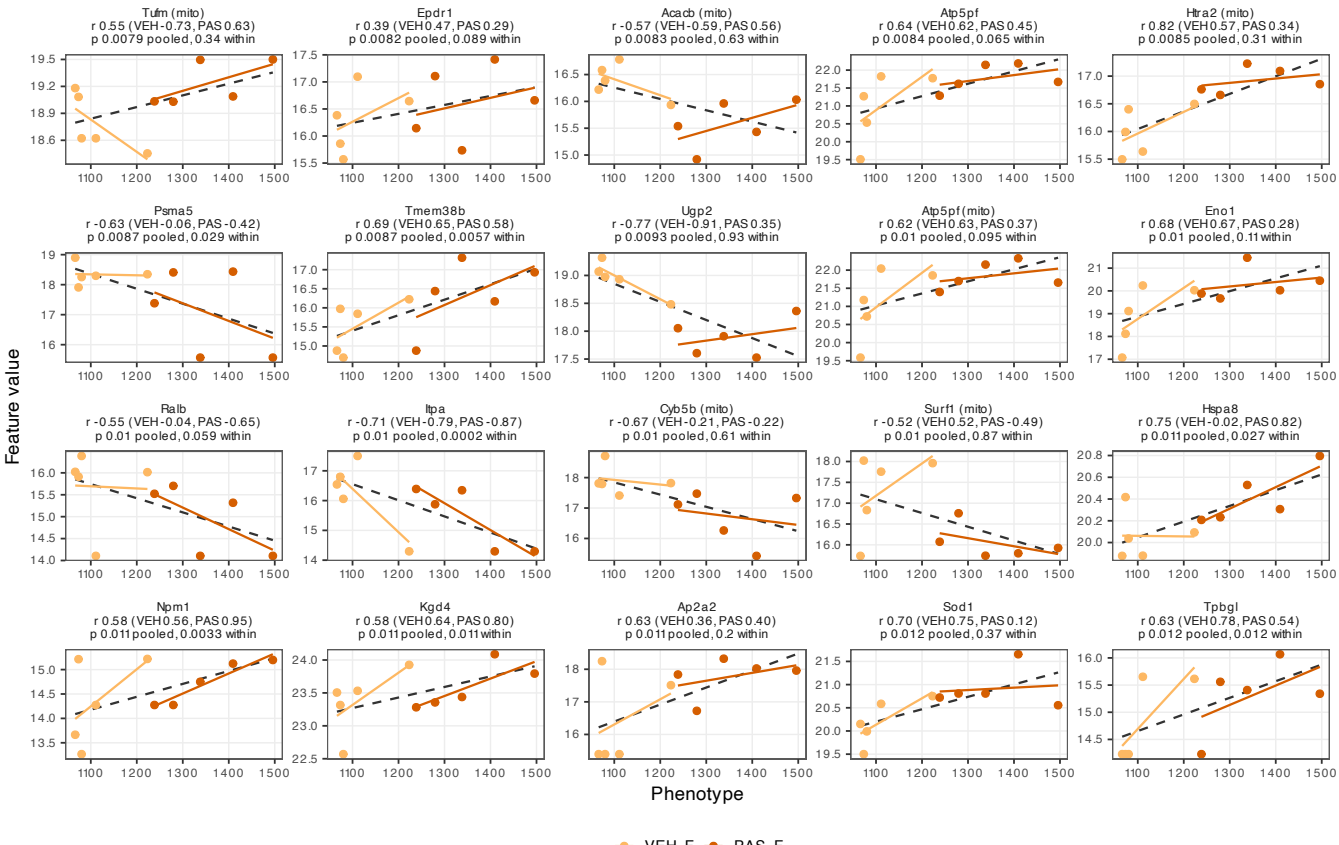
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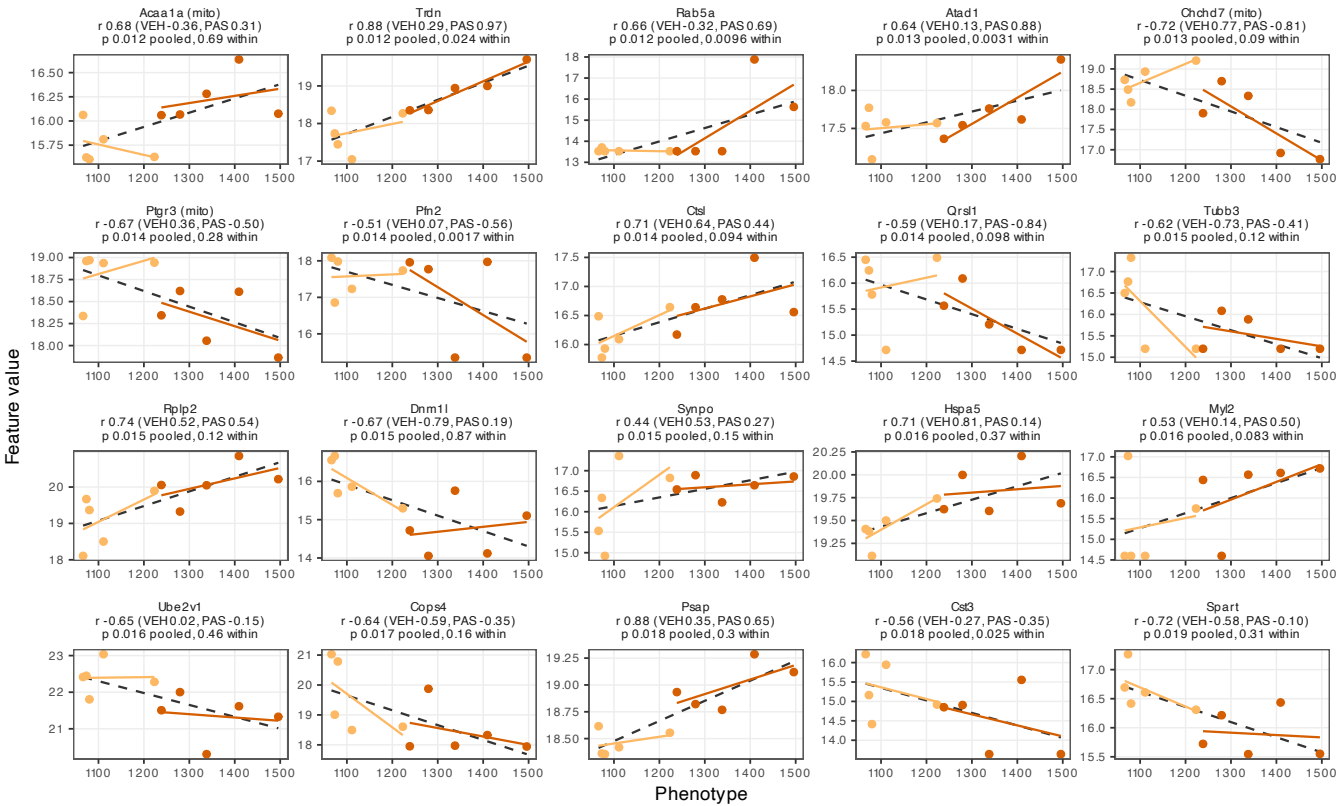
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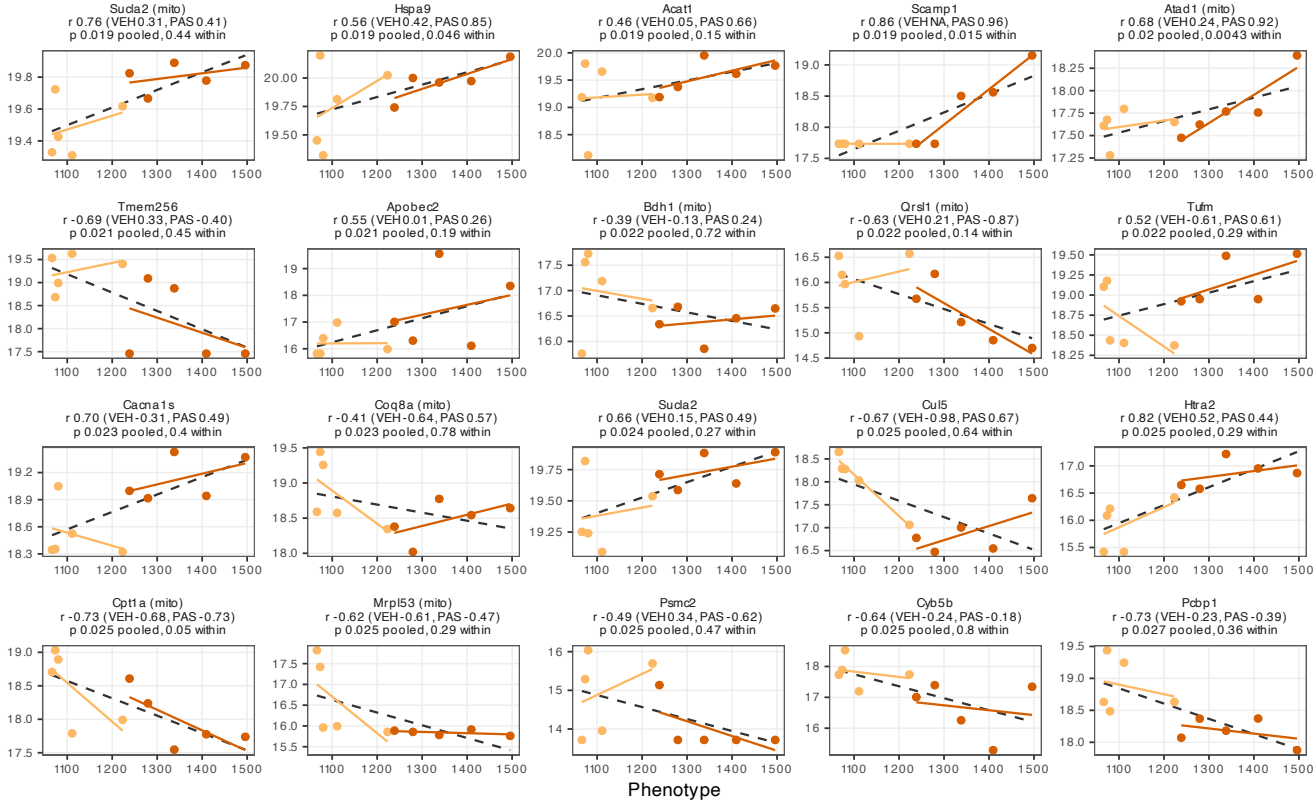
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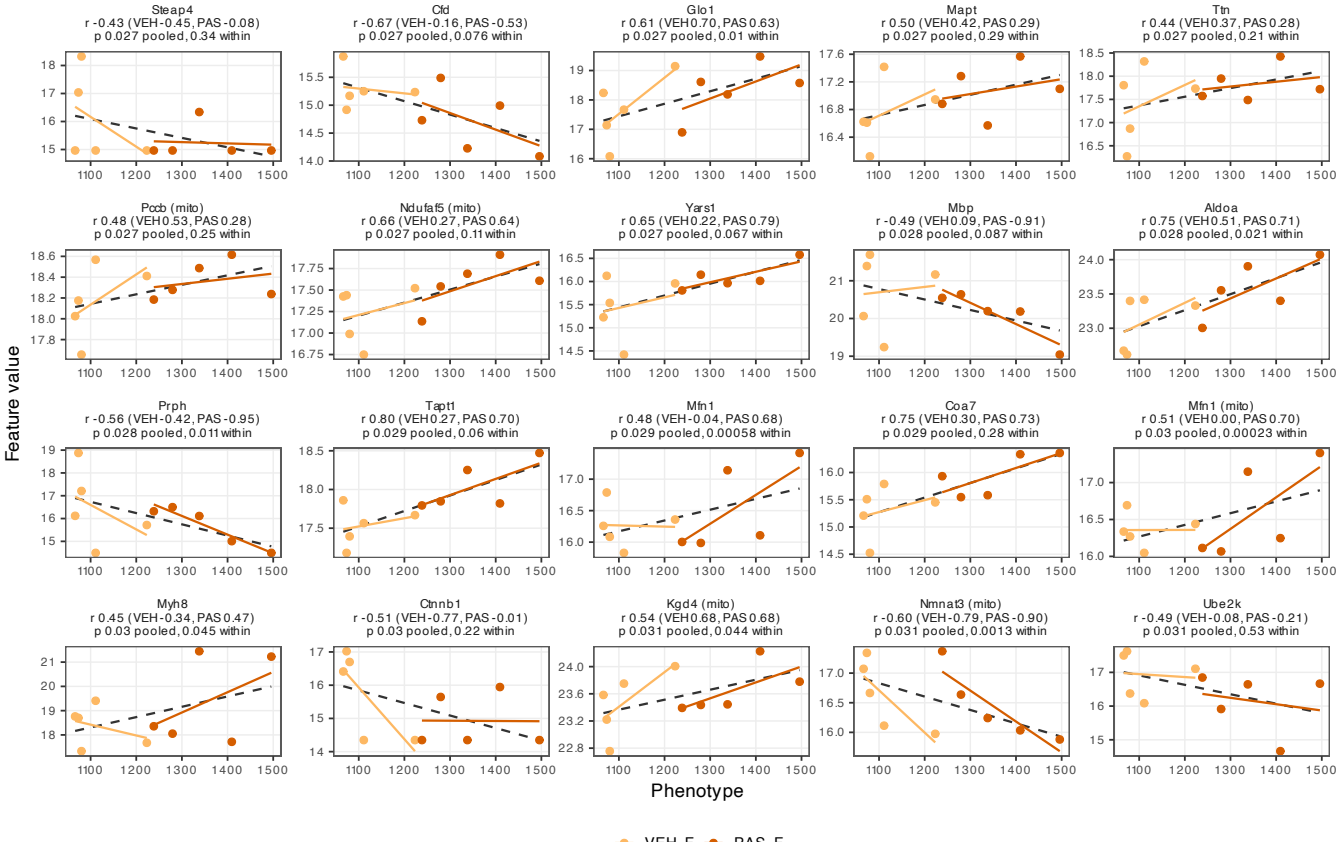
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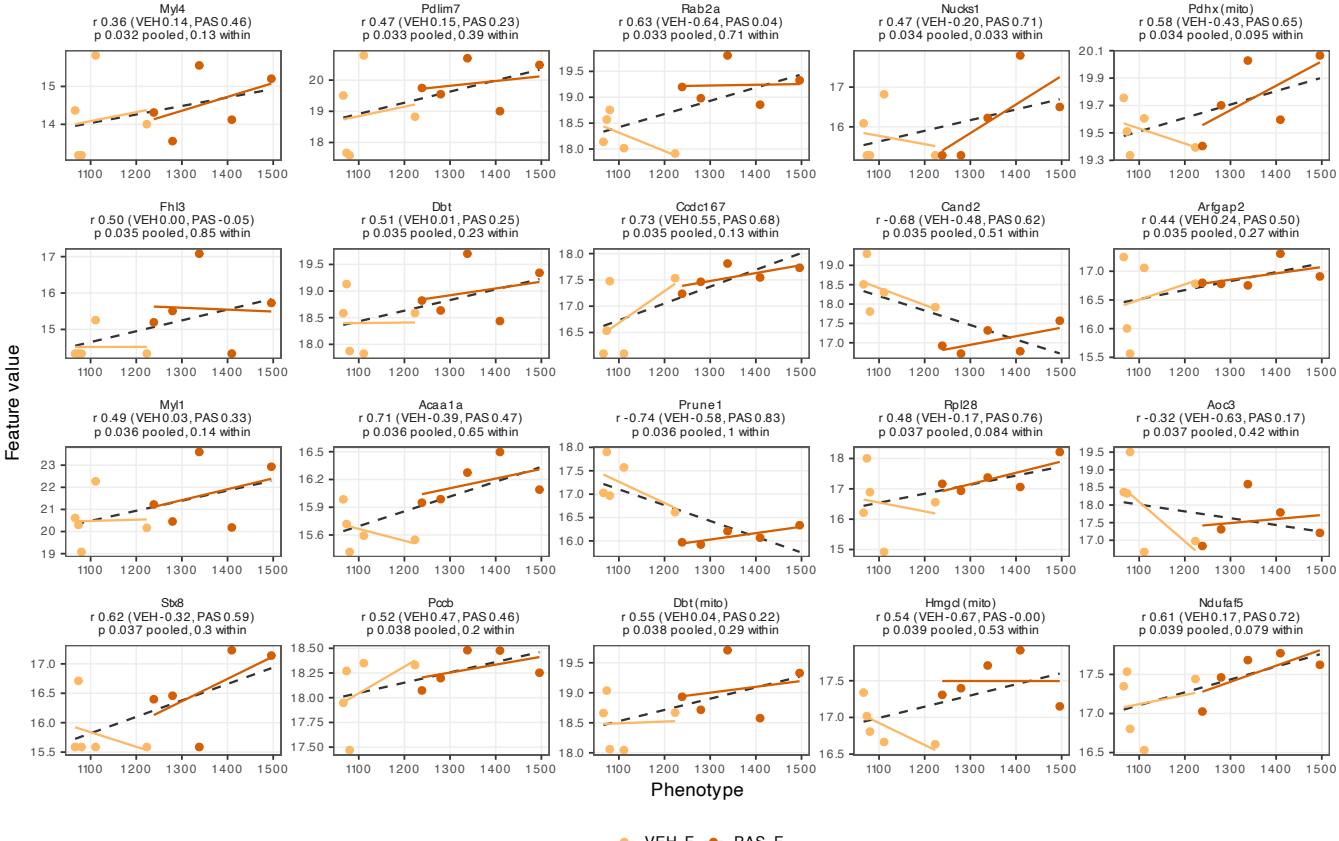
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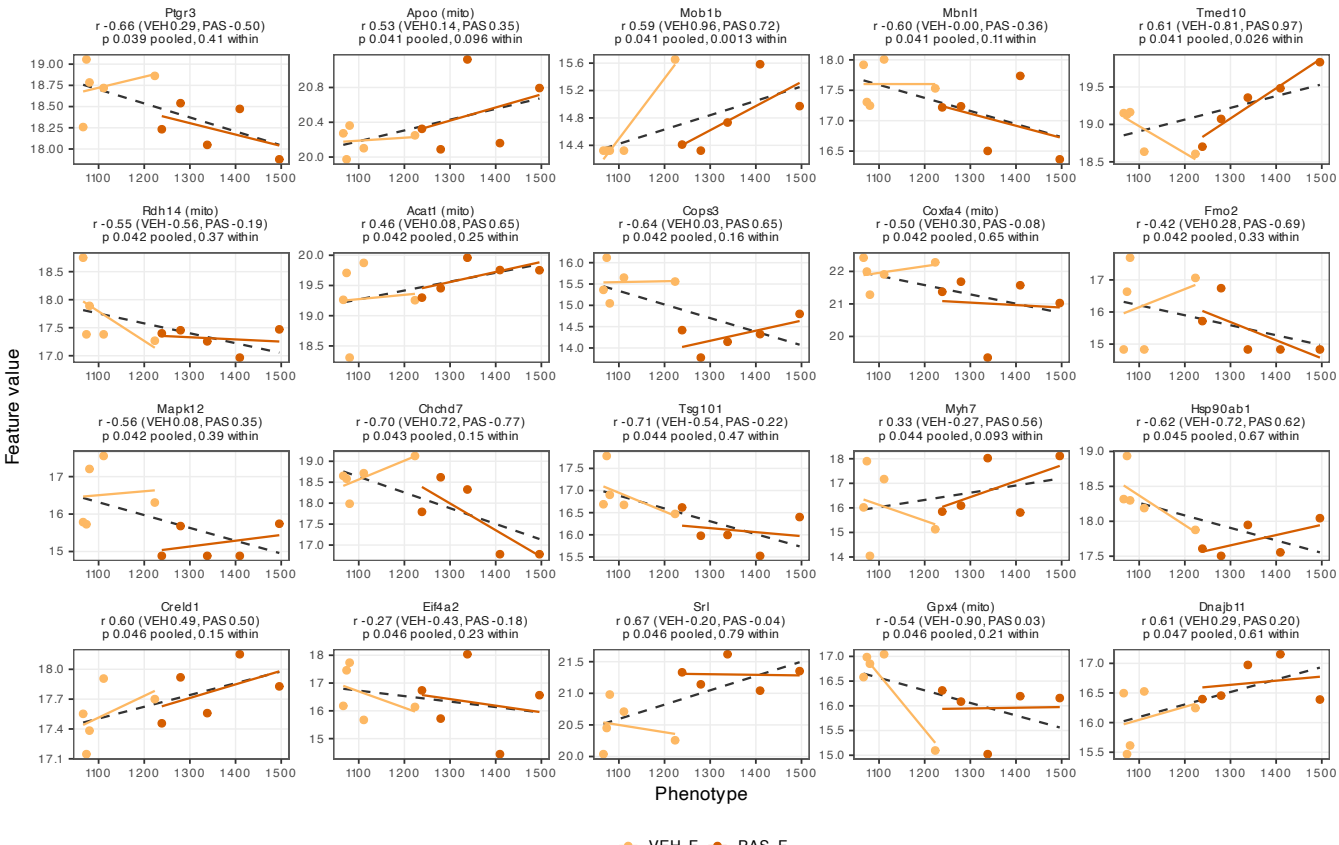
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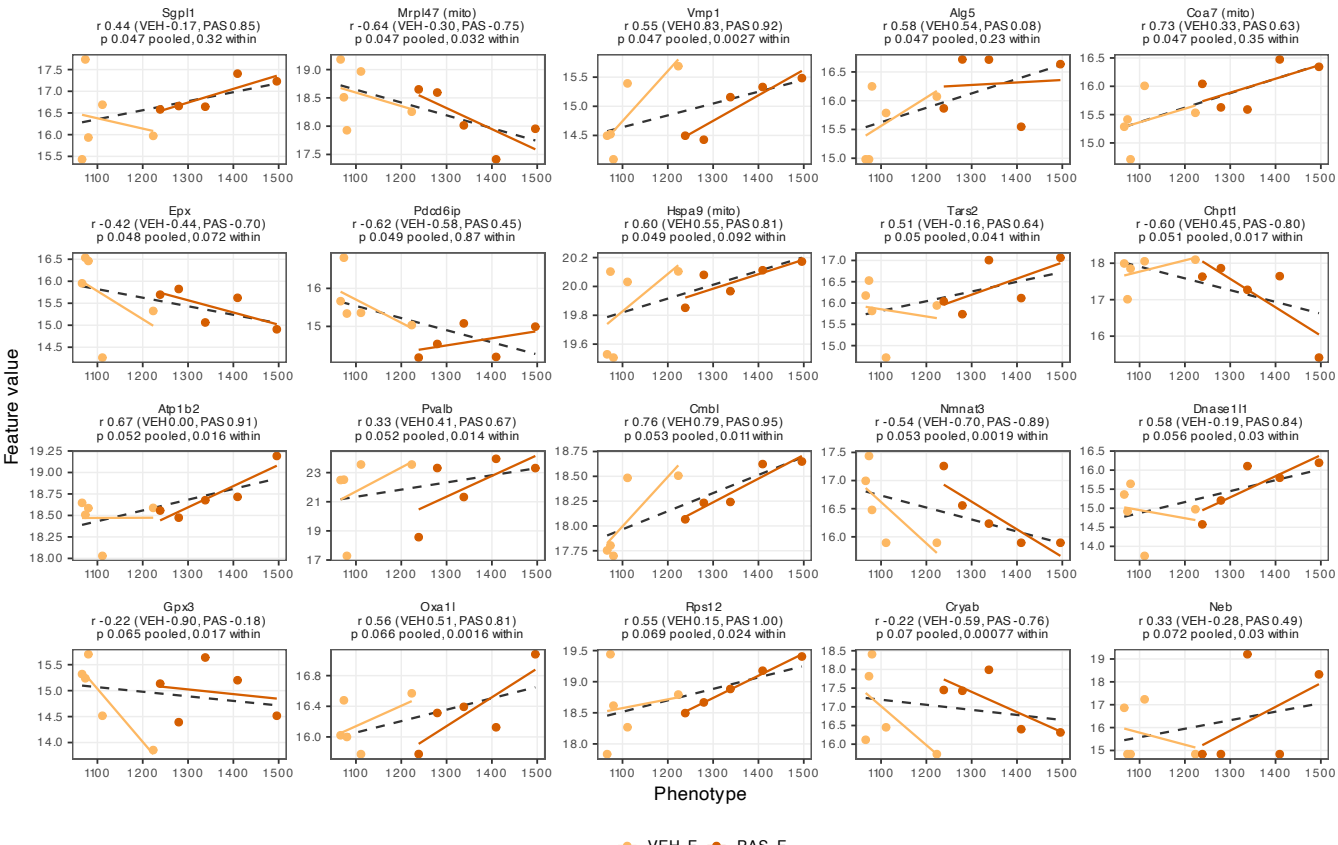
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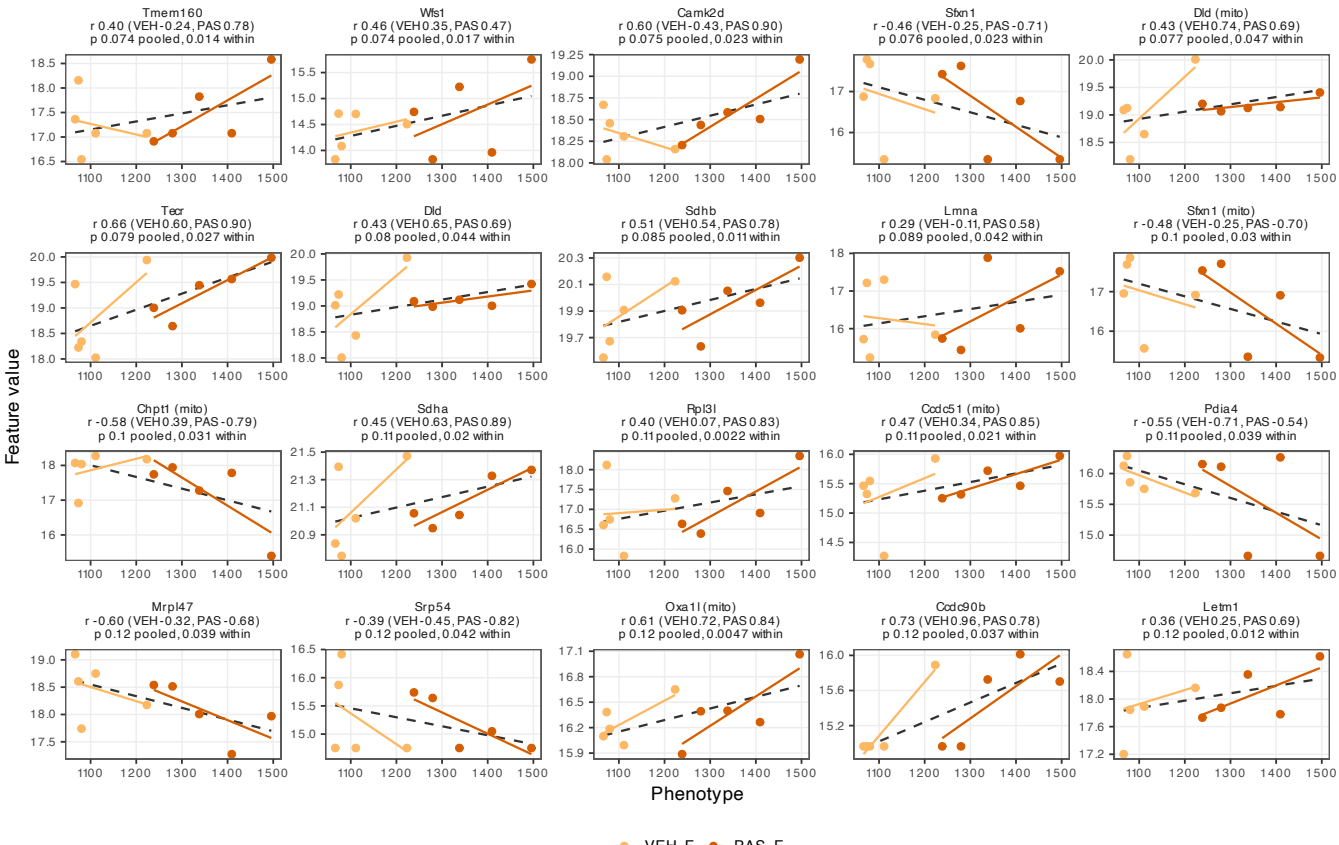
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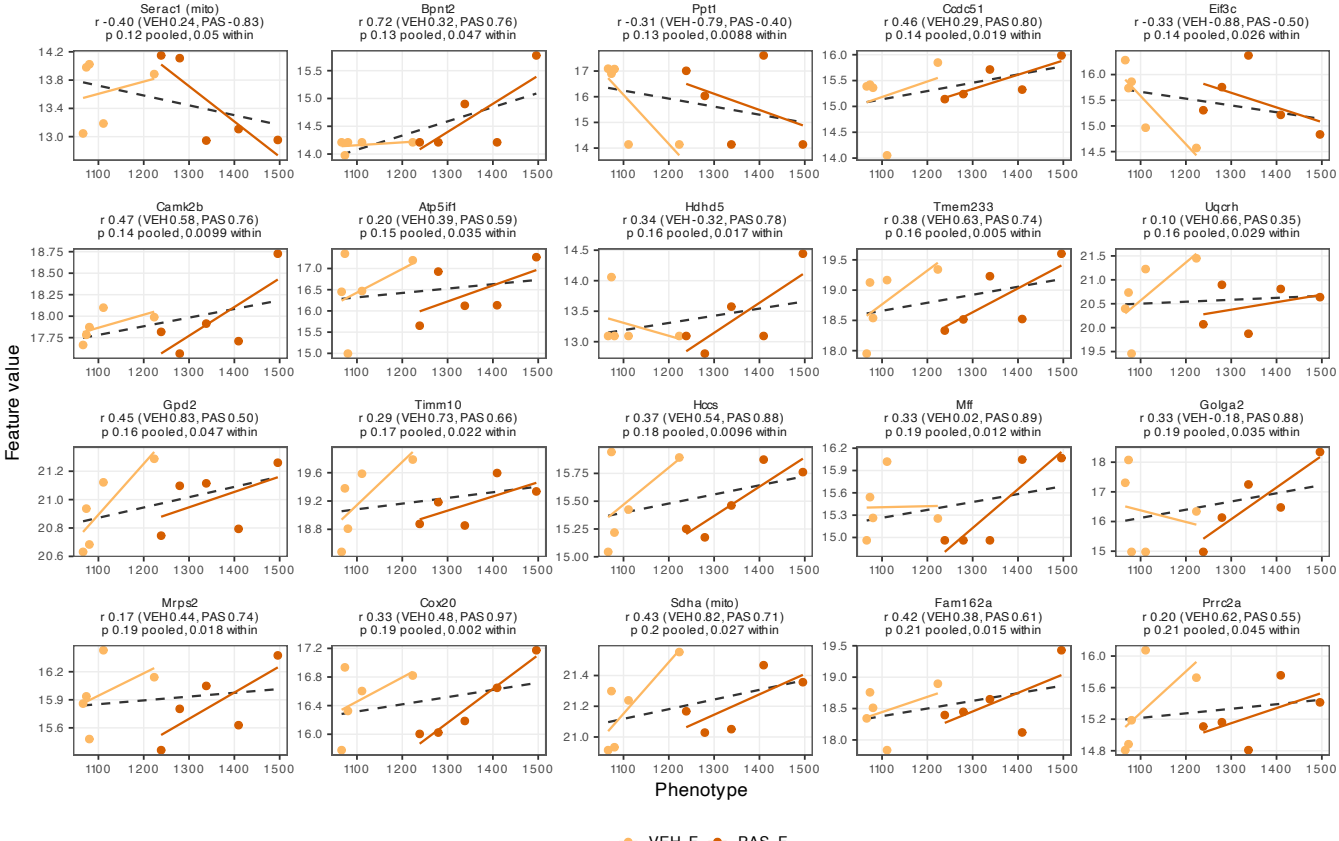
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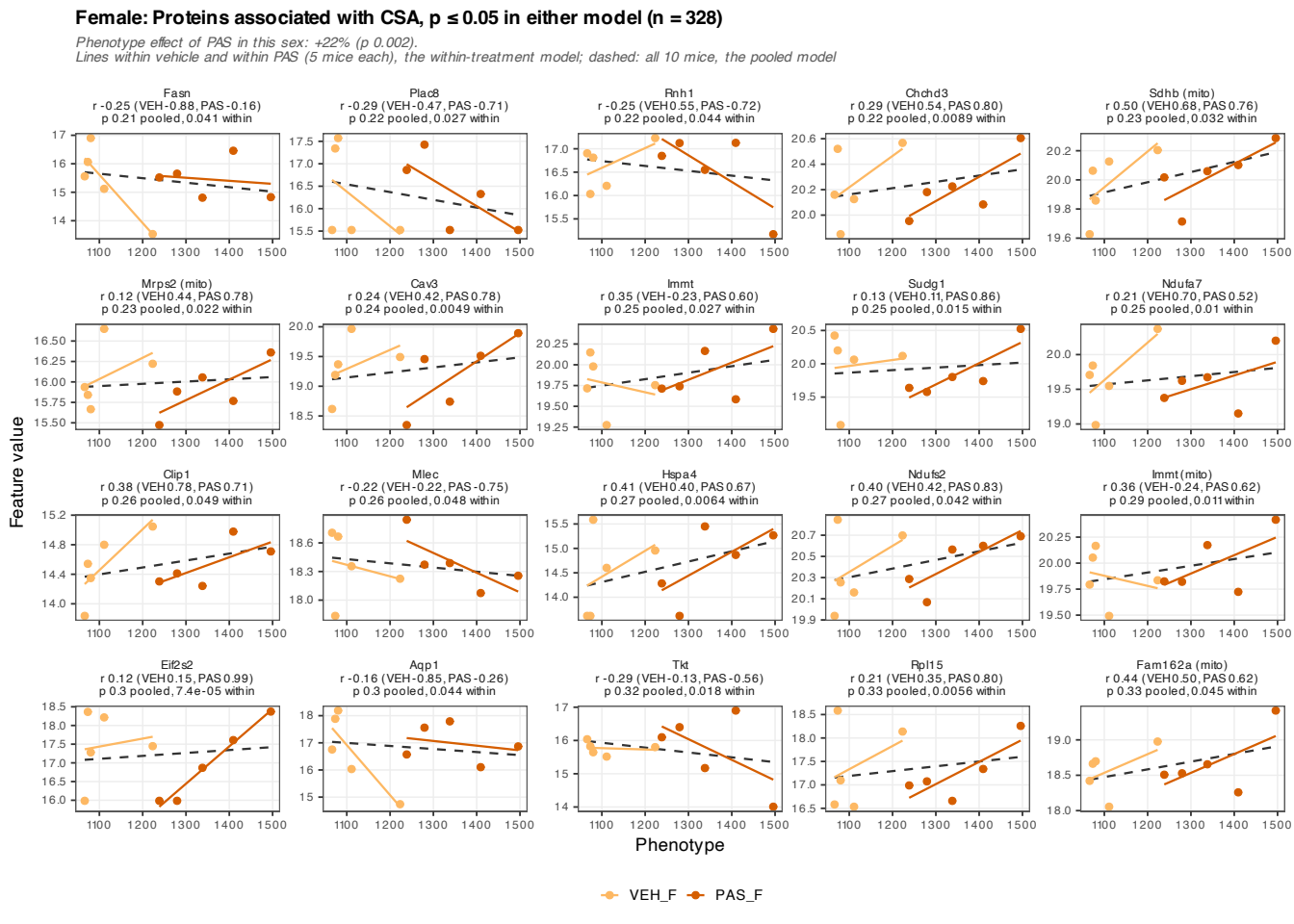
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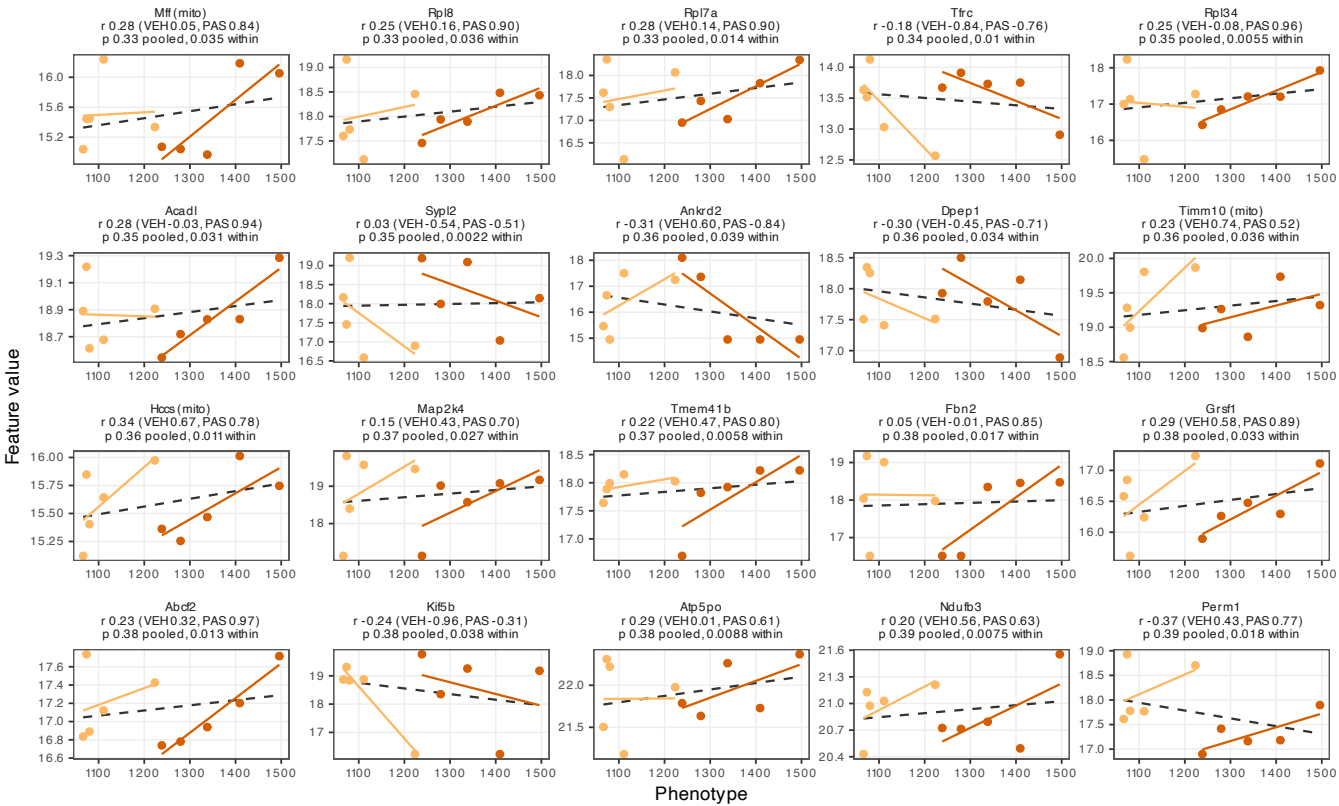
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Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model
Fasn
r -0.25 (VEH -0.88, PAS -0.16)
p 0.21 pooled, 0.041 within
Plac8
r -0.29 (VEH -0.47, PAS -0.71)
p 0.22 pooled, 0.027 within
Rnh1
r -0.25 (VEH 0.55, PAS -0.72)
p 0.22 pooled, 0.044 within
Chchd3
r 0.29 (VEH 0.54, PAS 0.80)
p 0.22 pooled, 0.0089 within
Sdhb (mito)
r 0.50 (VEH 0.68, PAS 0.76)
p 0.23 pooled, 0.032 within
Mrps2 (mito)
r 0.12 (VEH 0.44, PAS 0.78)
p 0.23 pooled, 0.022 within
Cav3
r 0.24 (VEH 0.42, PAS 0.78)
p 0.24 pooled, 0.0049 within
Immt
r 0.35 (VEH -0.23, PAS 0.60)
p 0.25 pooled, 0.027 within
Suclg1
r 0.13 (VEH 0.11, PAS 0.86)
p 0.25 pooled, 0.015 within
Ndufa7
r 0.21 (VEH 0.70, PAS 0.52)
p 0.25 pooled, 0.01 within
Clip1
r 0.38 (VEH 0.78, PAS 0.71)
p 0.26 pooled, 0.049 within
Mlec
r -0.22 (VEH -0.22, PAS -0.75)
p 0.26 pooled, 0.048 within
Hspa4
r 0.41 (VEH 0.40, PAS 0.67)
p 0.27 pooled, 0.0064 within
Ndufs2
r 0.40 (VEH 0.42, PAS 0.83)
p 0.27 pooled, 0.042 within
Immt (mito)
r 0.36 (VEH -0.24, PAS 0.62)
p 0.29 pooled, 0.011 within
Eif2s2
r 0.12 (VEH 0.15, PAS 0.99)
p 0.3 pooled, 7.4e-05 within
Aqp1
r -0.16 (VEH -0.85, PAS -0.26)
p 0.3 pooled, 0.044 within
Tkt
r -0.29 (VEH -0.13, PAS -0.56)
p 0.32 pooled, 0.018 within
Rpl15
r 0.21 (VEH 0.35, PAS 0.80)
p 0.33 pooled, 0.0056 within
Fam162a (mito)
r 0.44 (VEH 0.50, PAS 0.62)
p 0.33 pooled, 0.045 within
Feature value
Phenotype
VEH_F
PAS_F

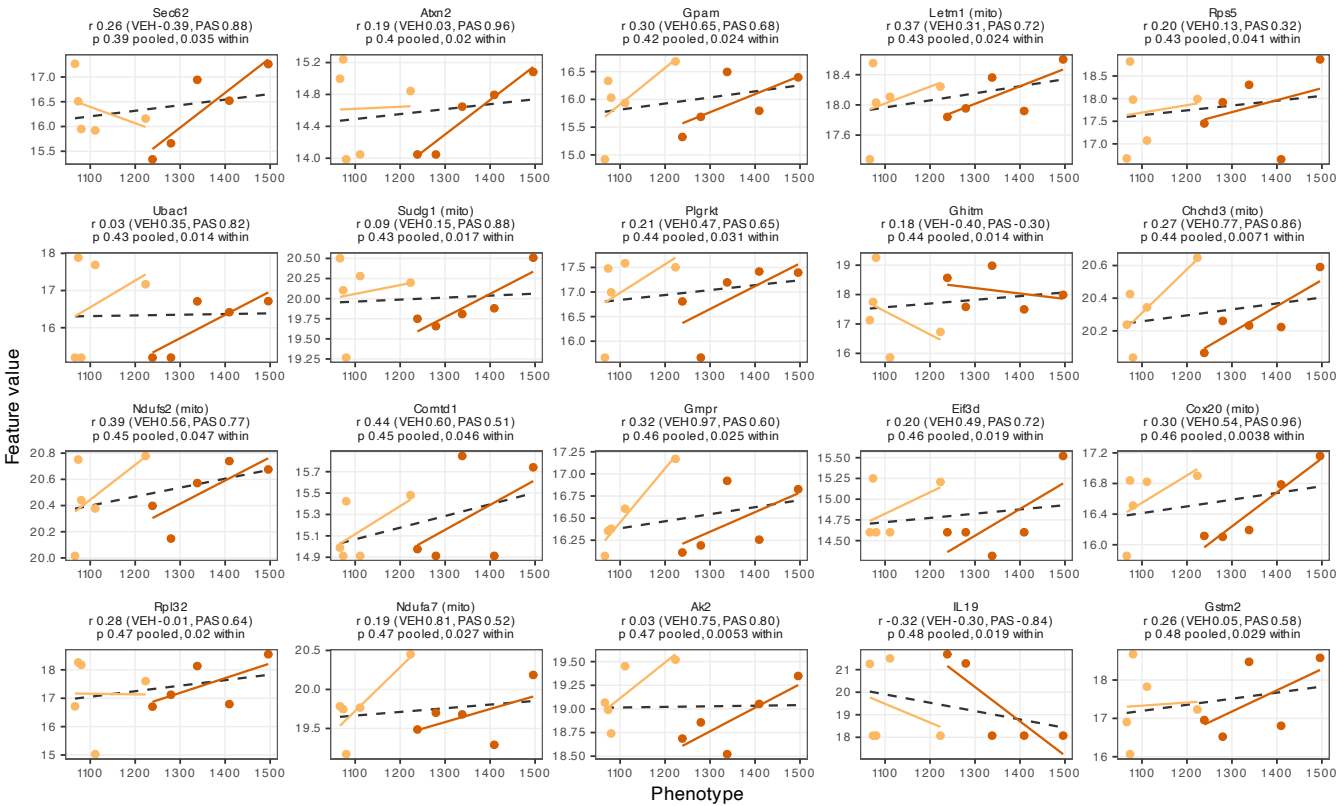
Female: Proteins associated with CSA, p ≤ 0.05 in either model (n = 328)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



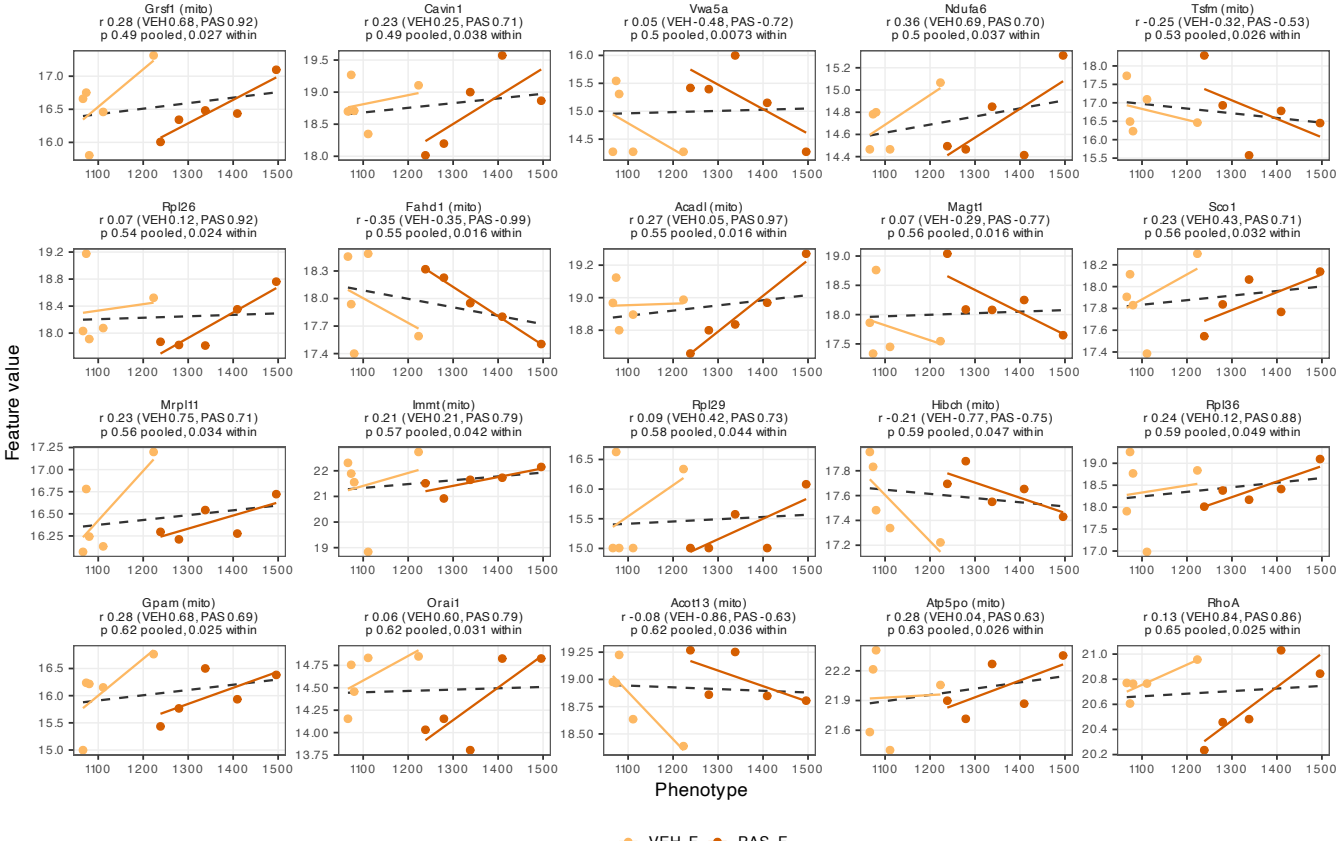
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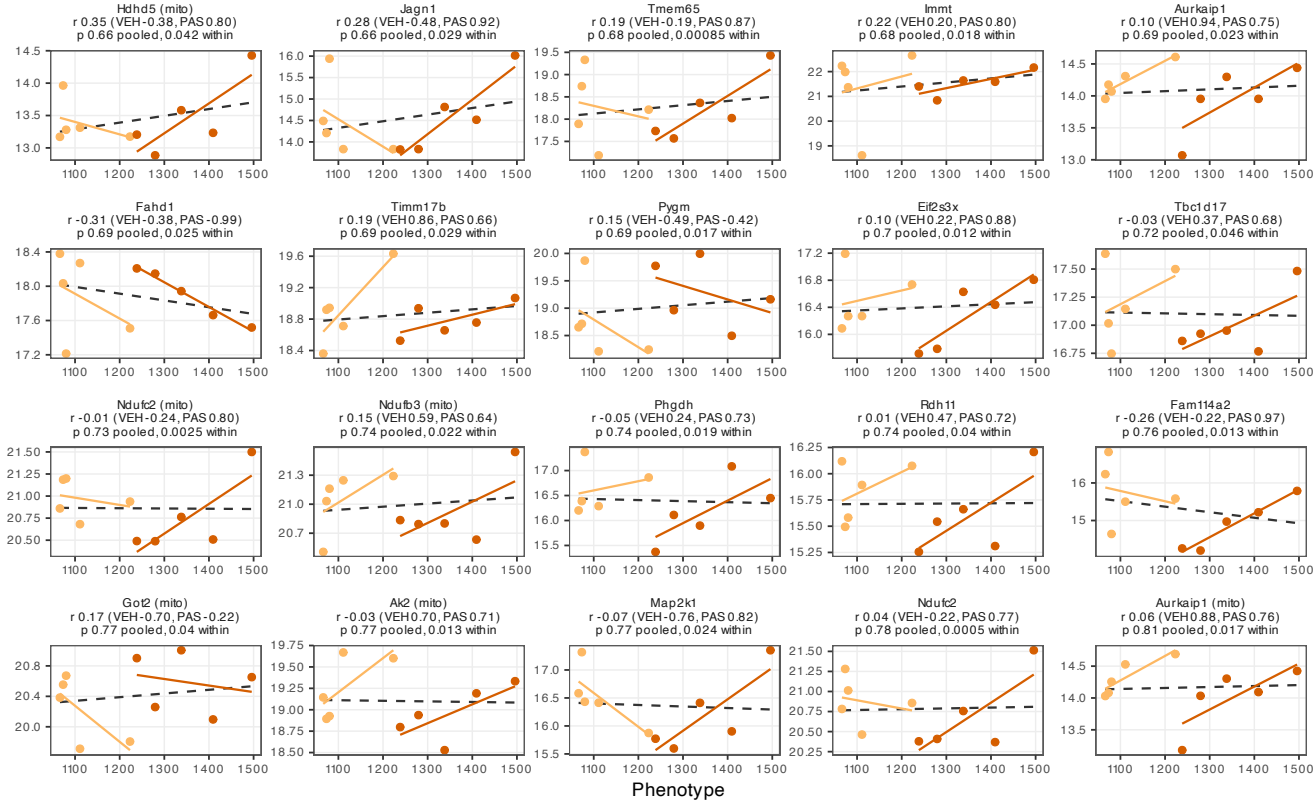
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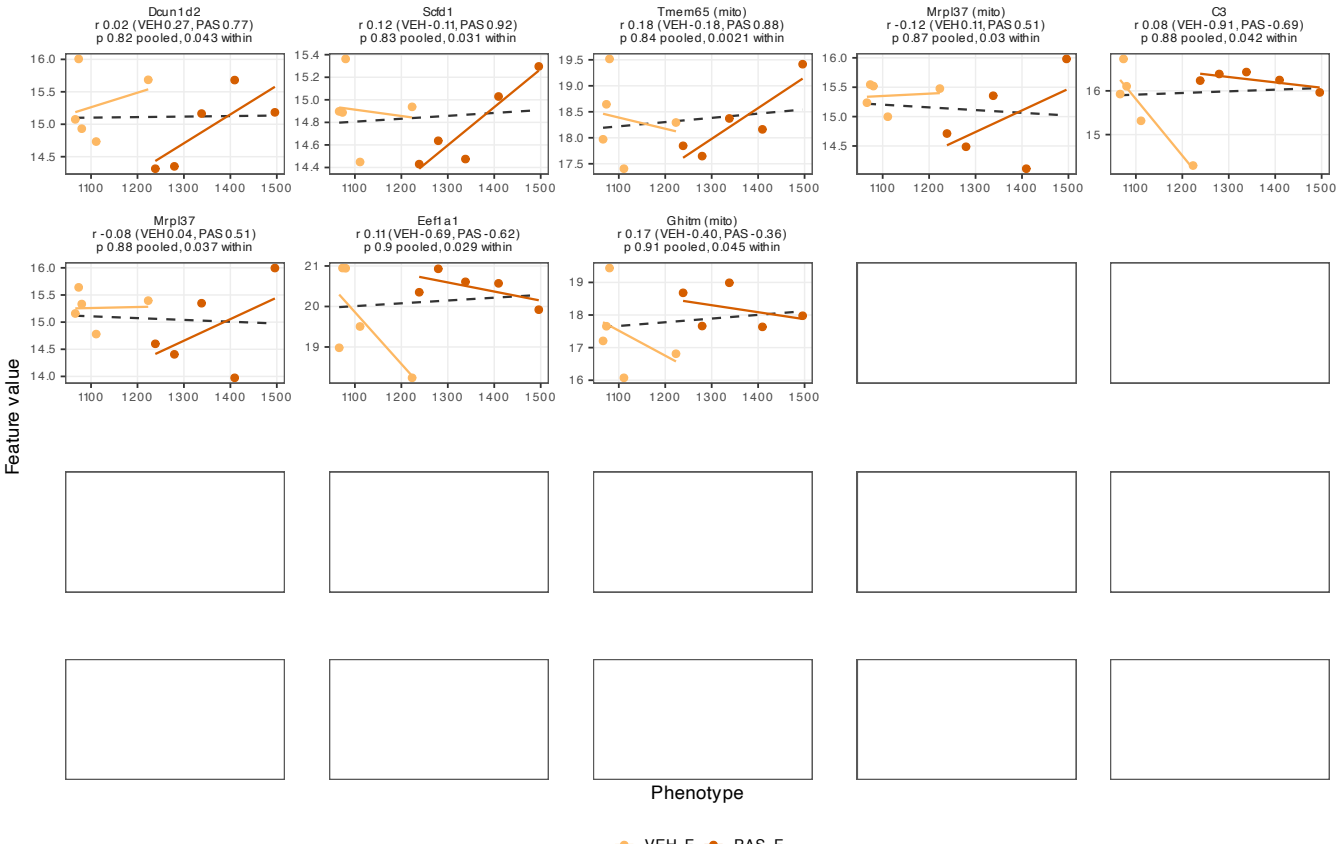
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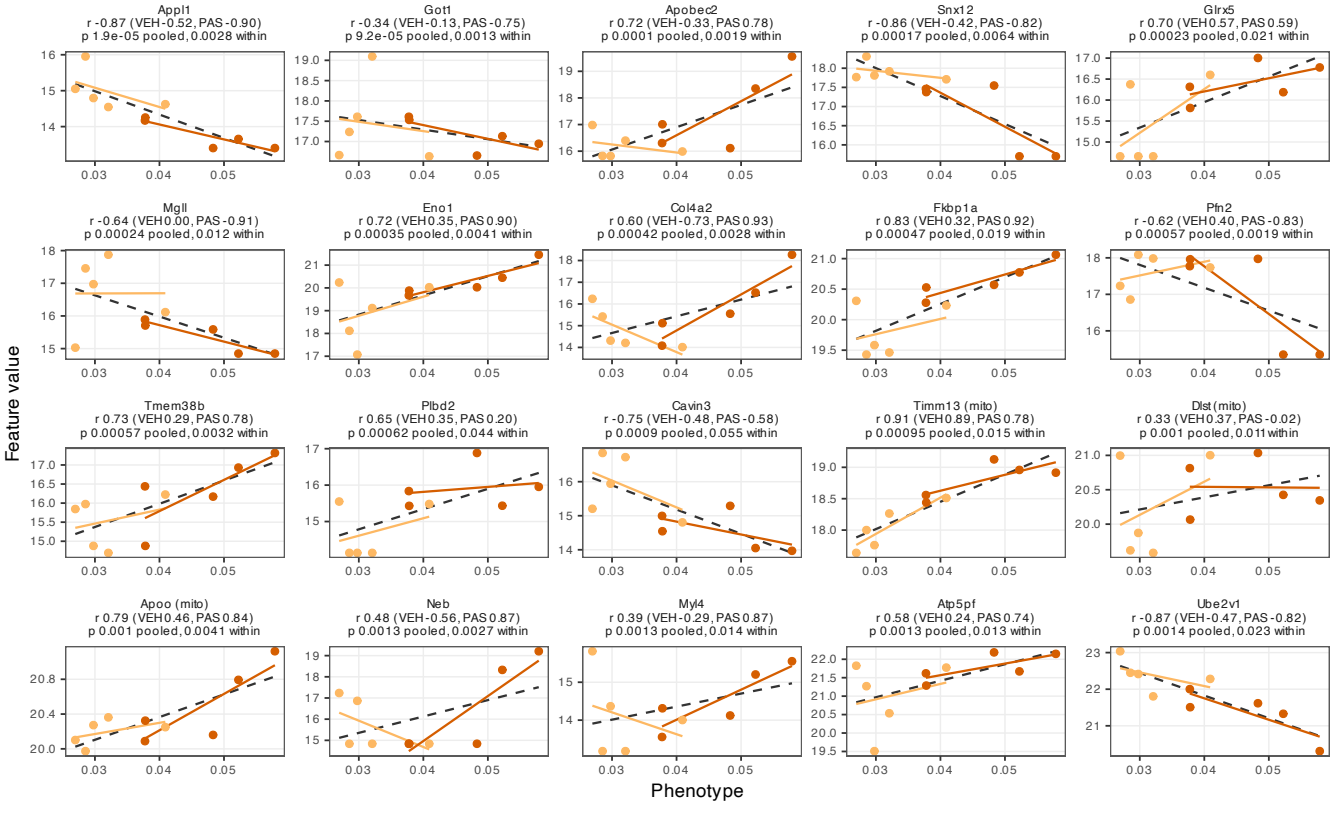
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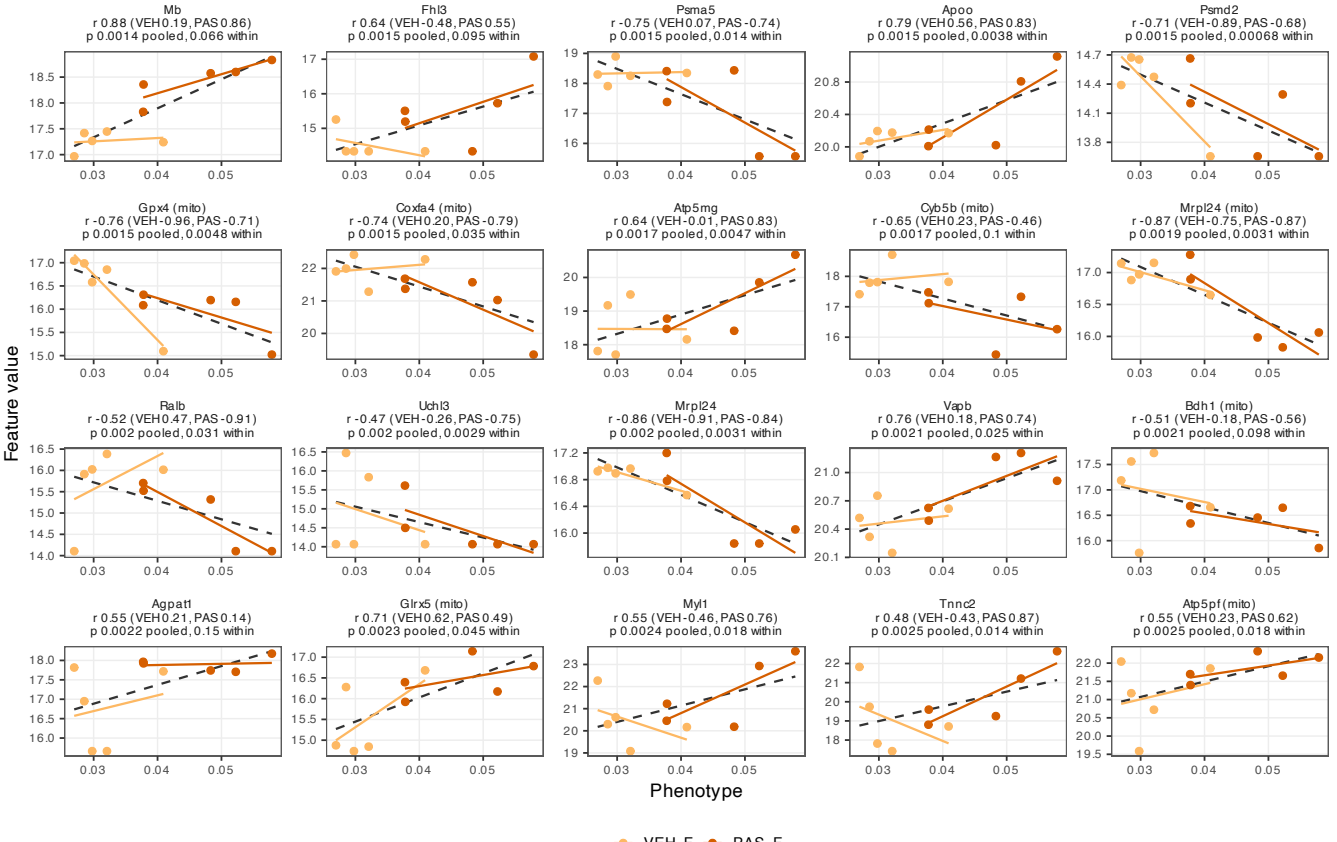
Female: Proteins associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 360)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



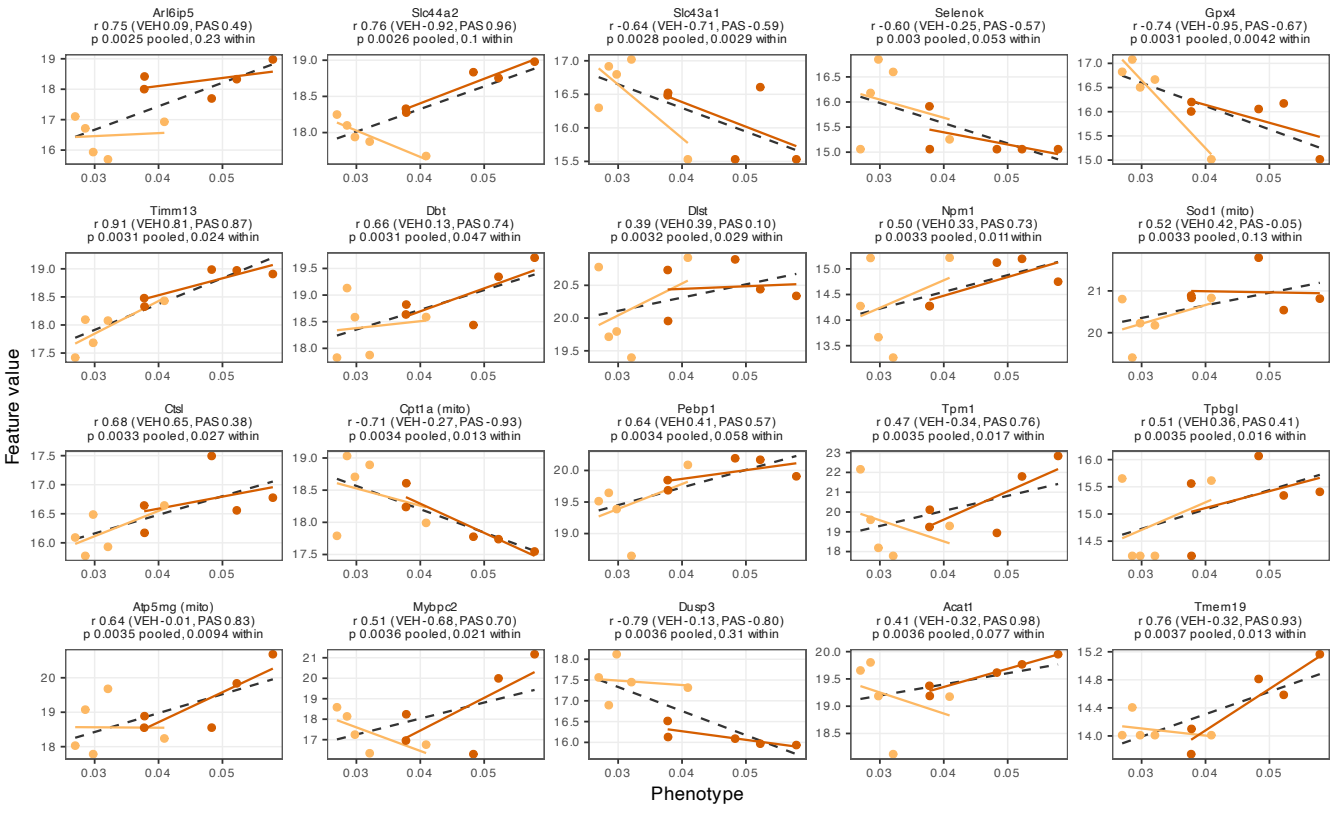
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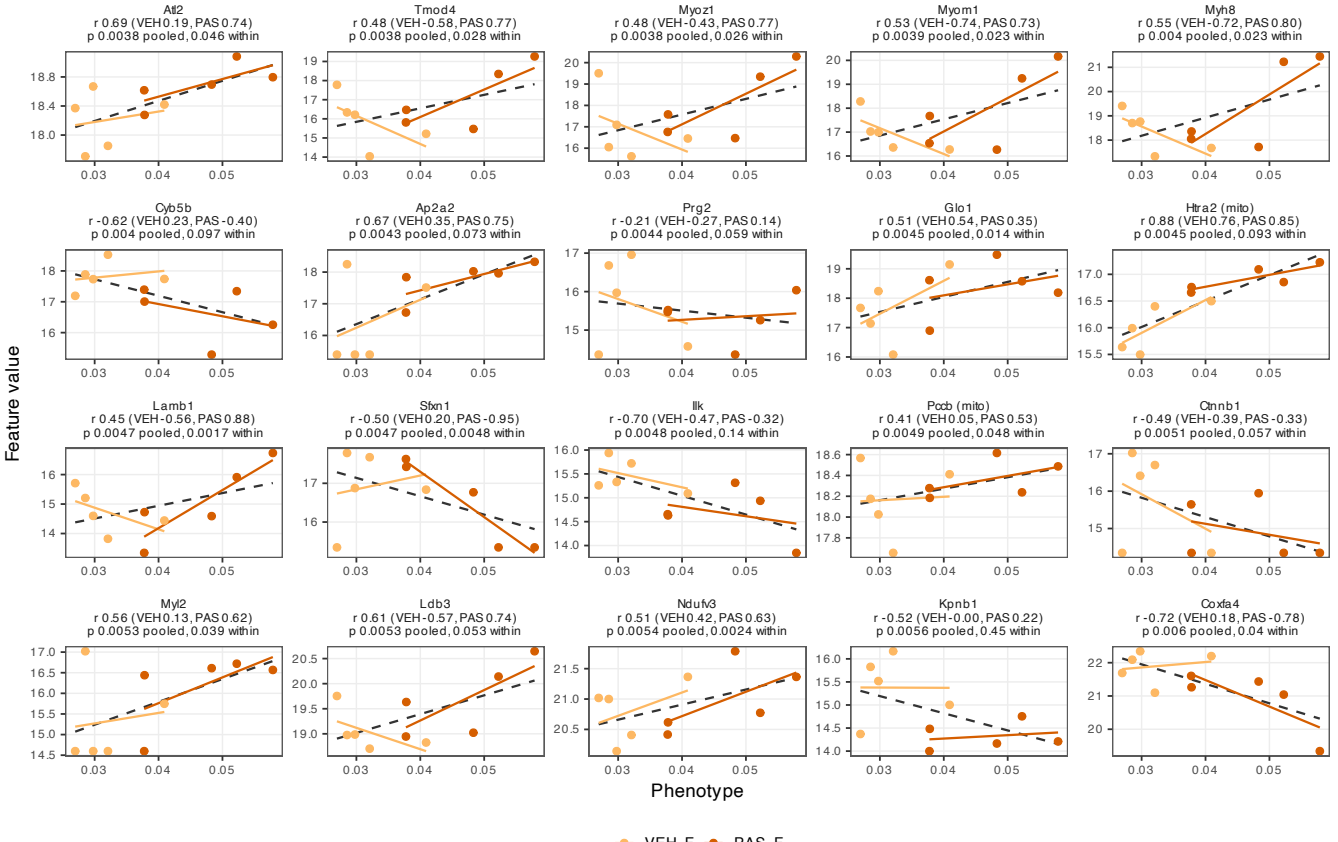
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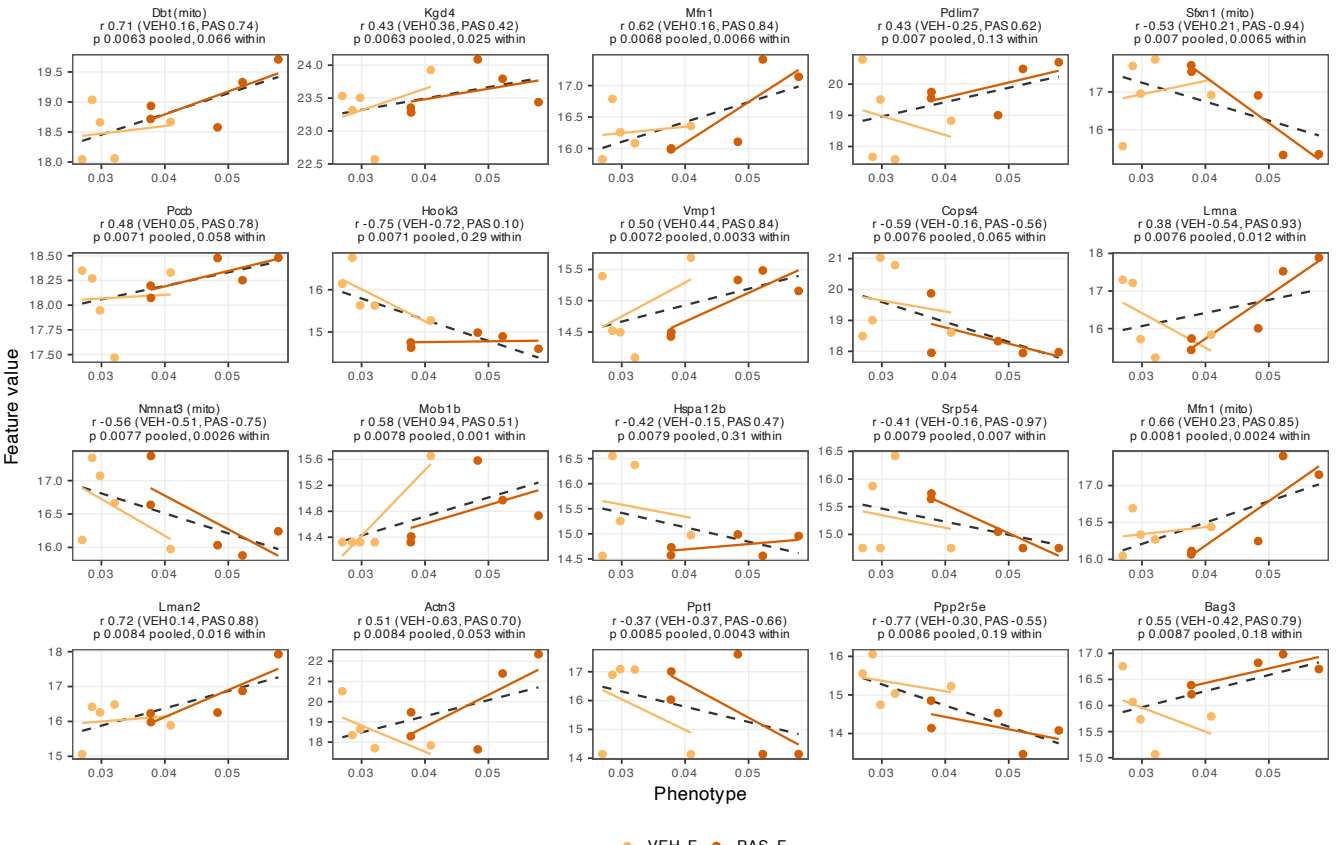
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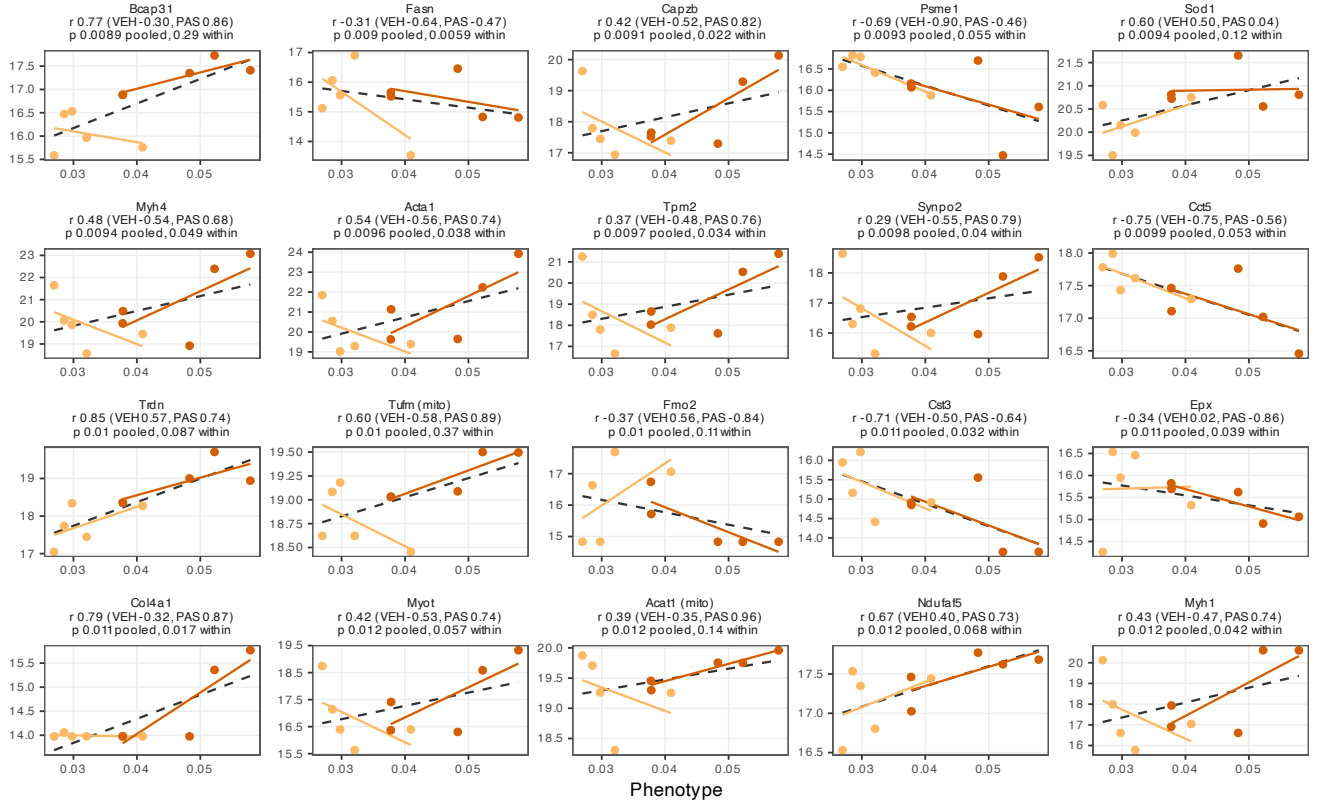
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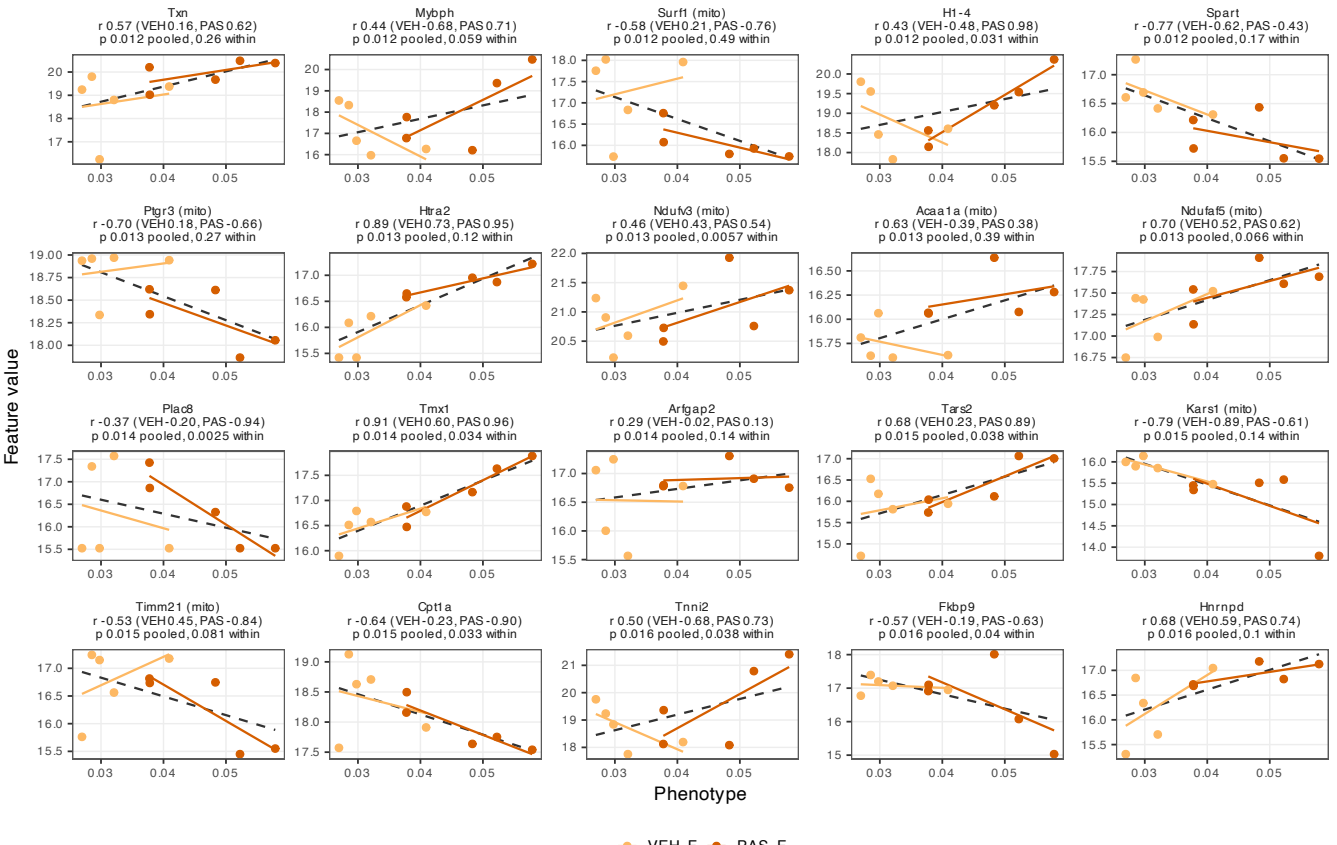
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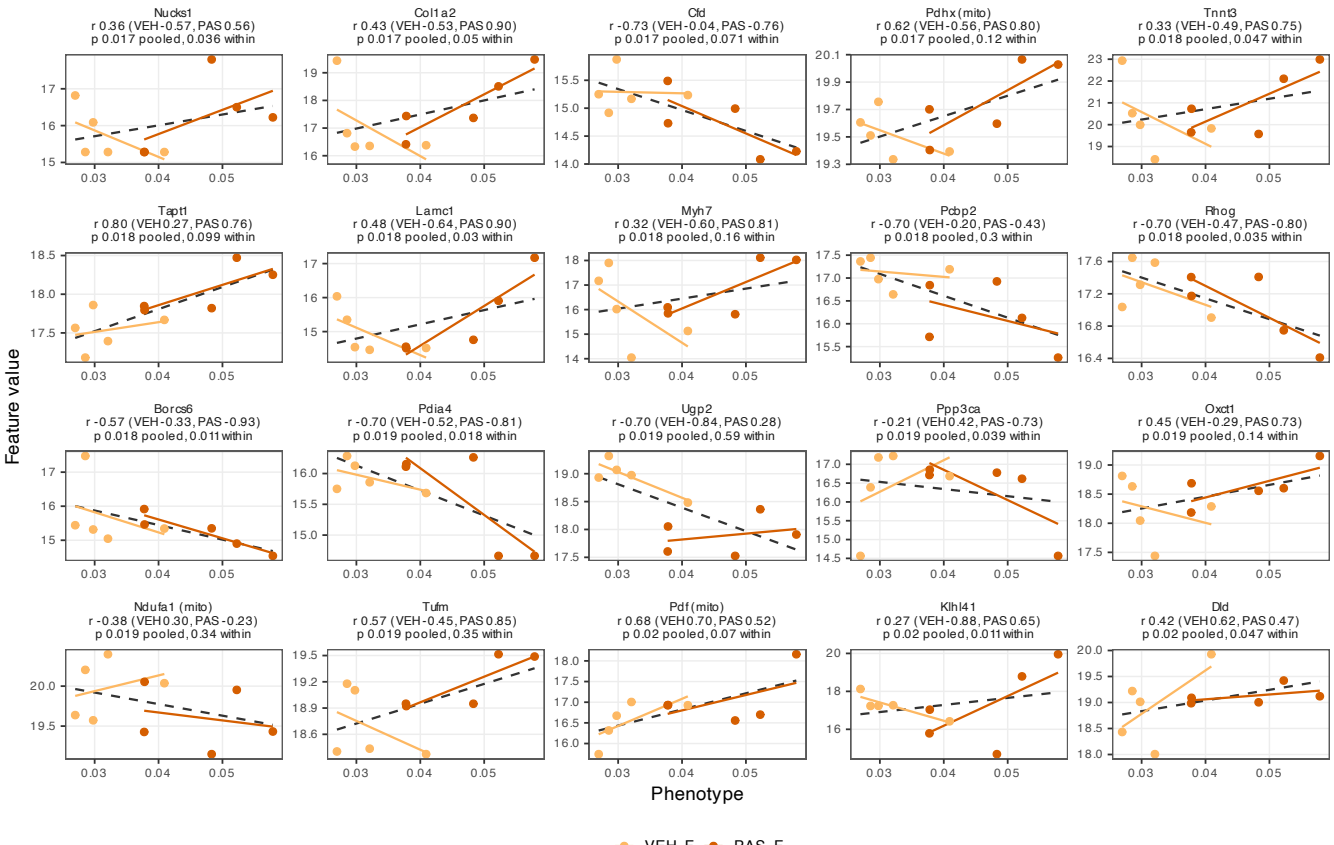
Female: Proteins associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 360)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



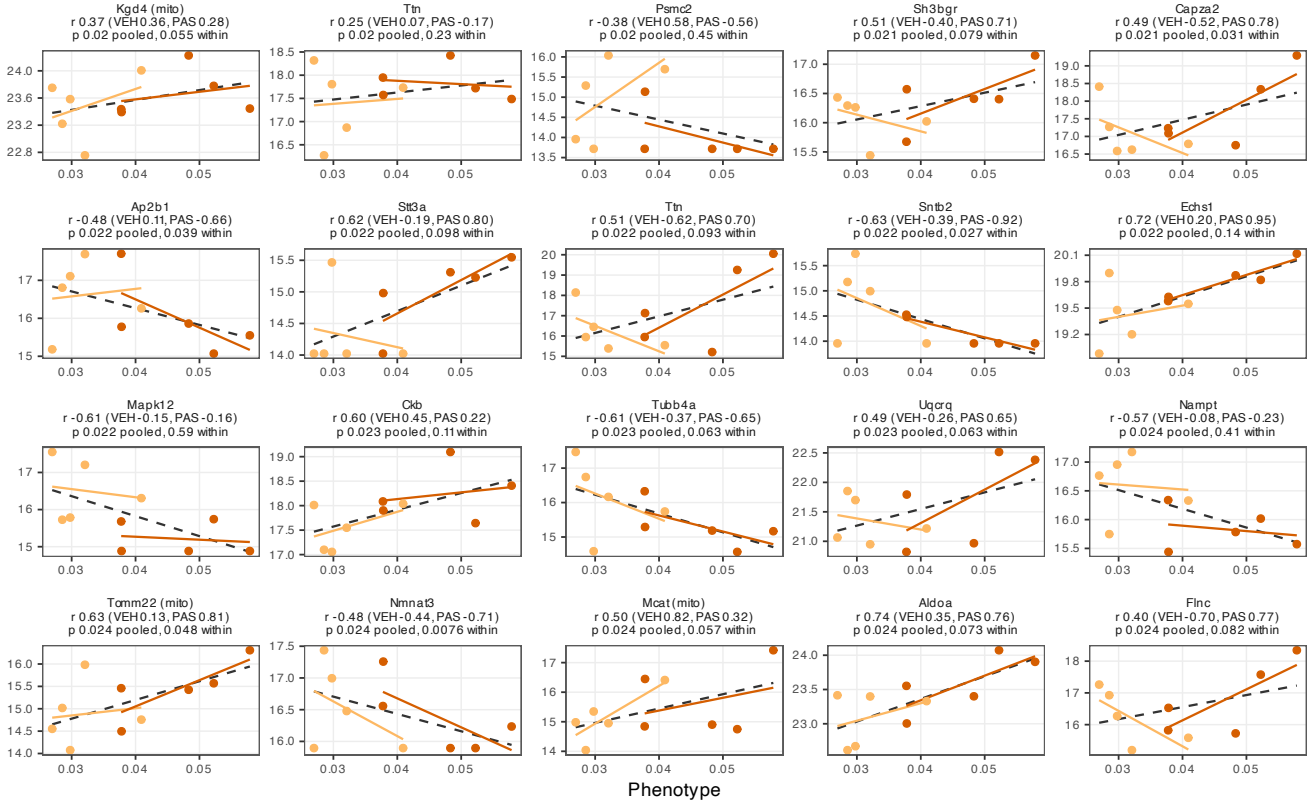
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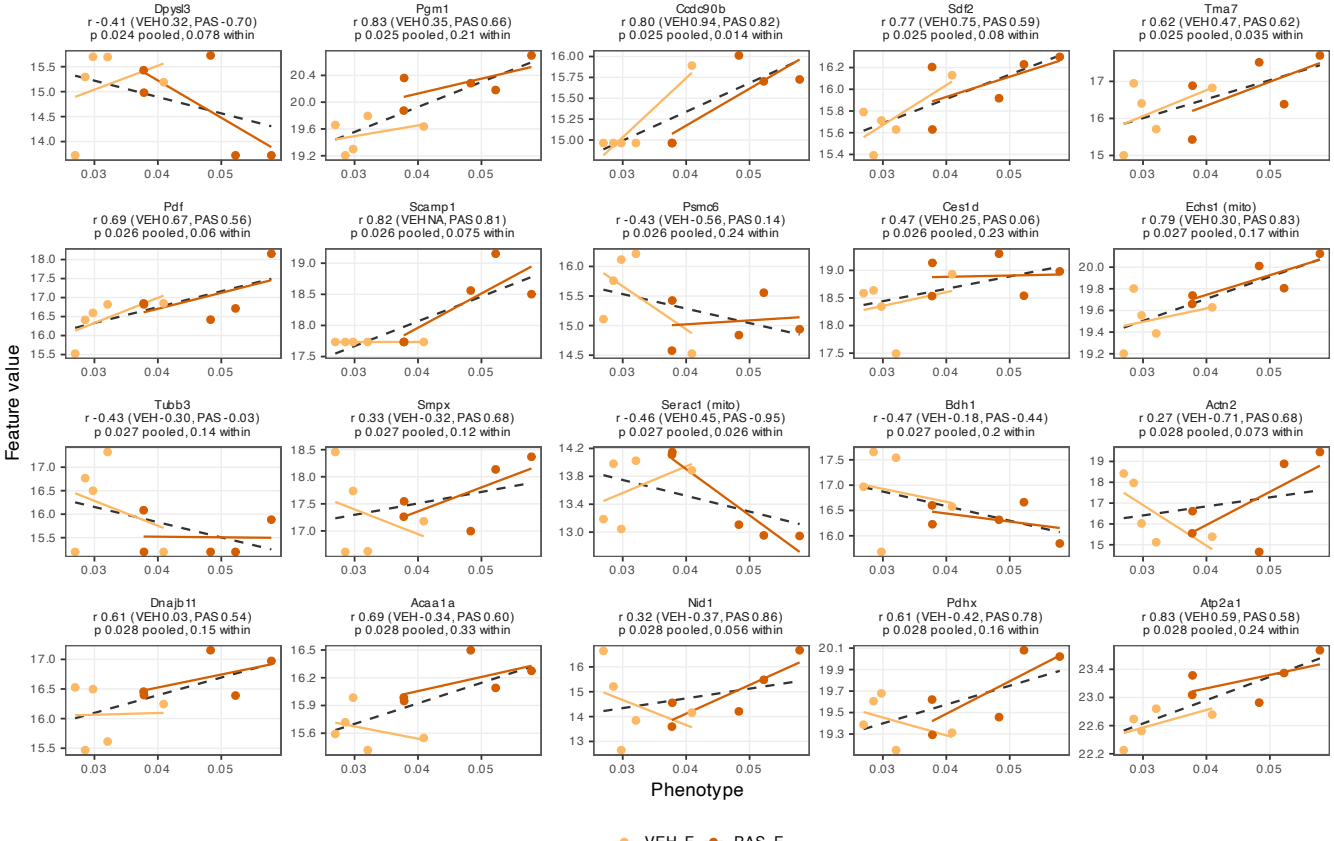
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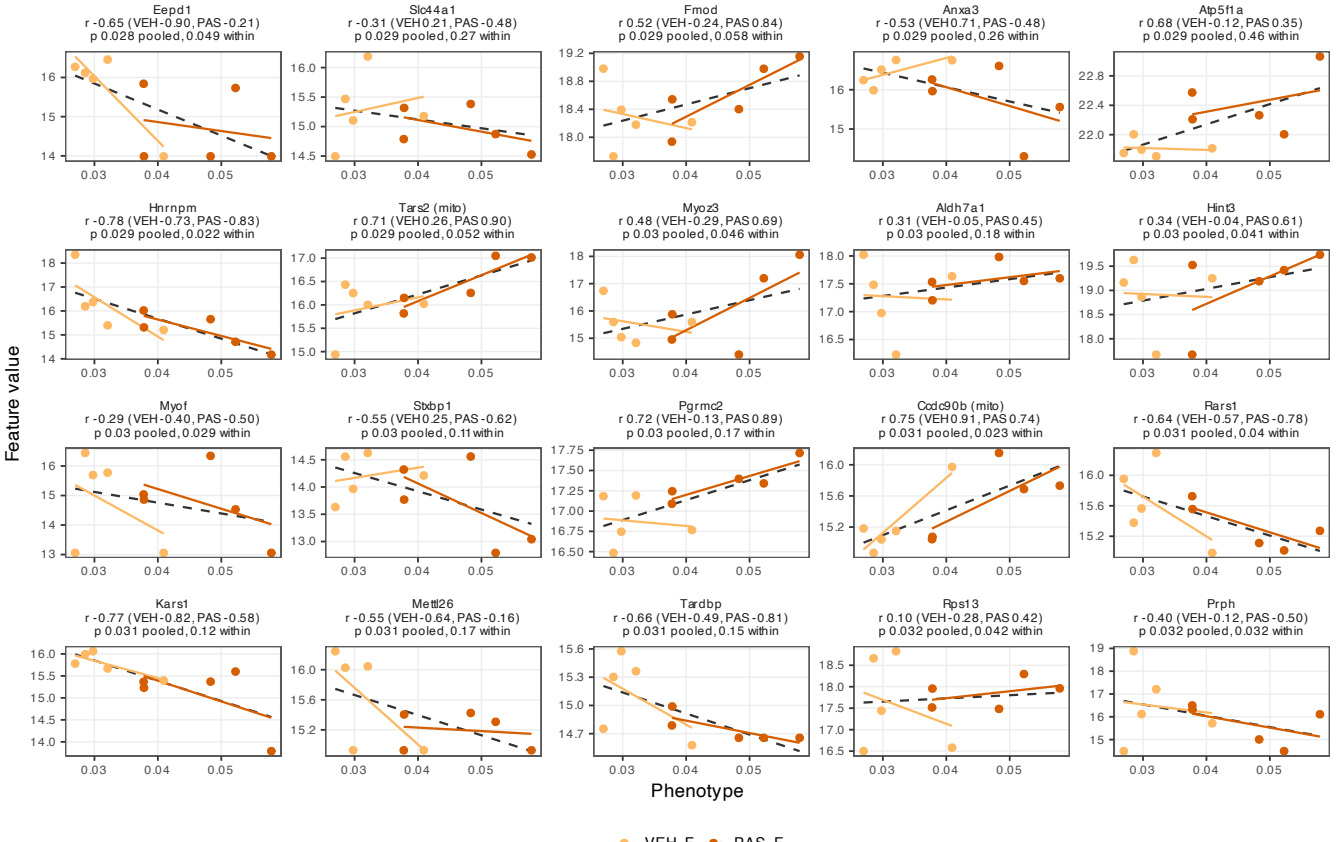
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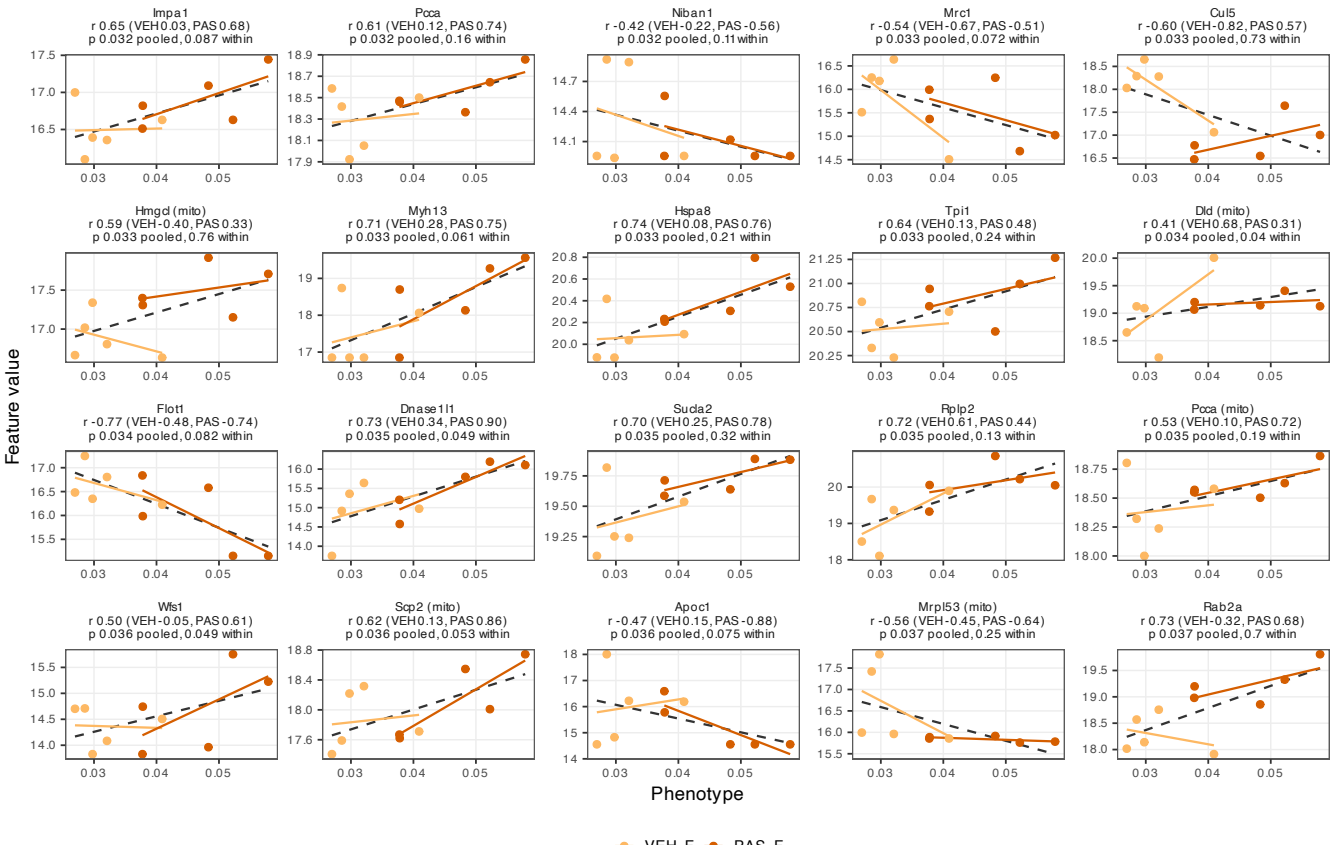
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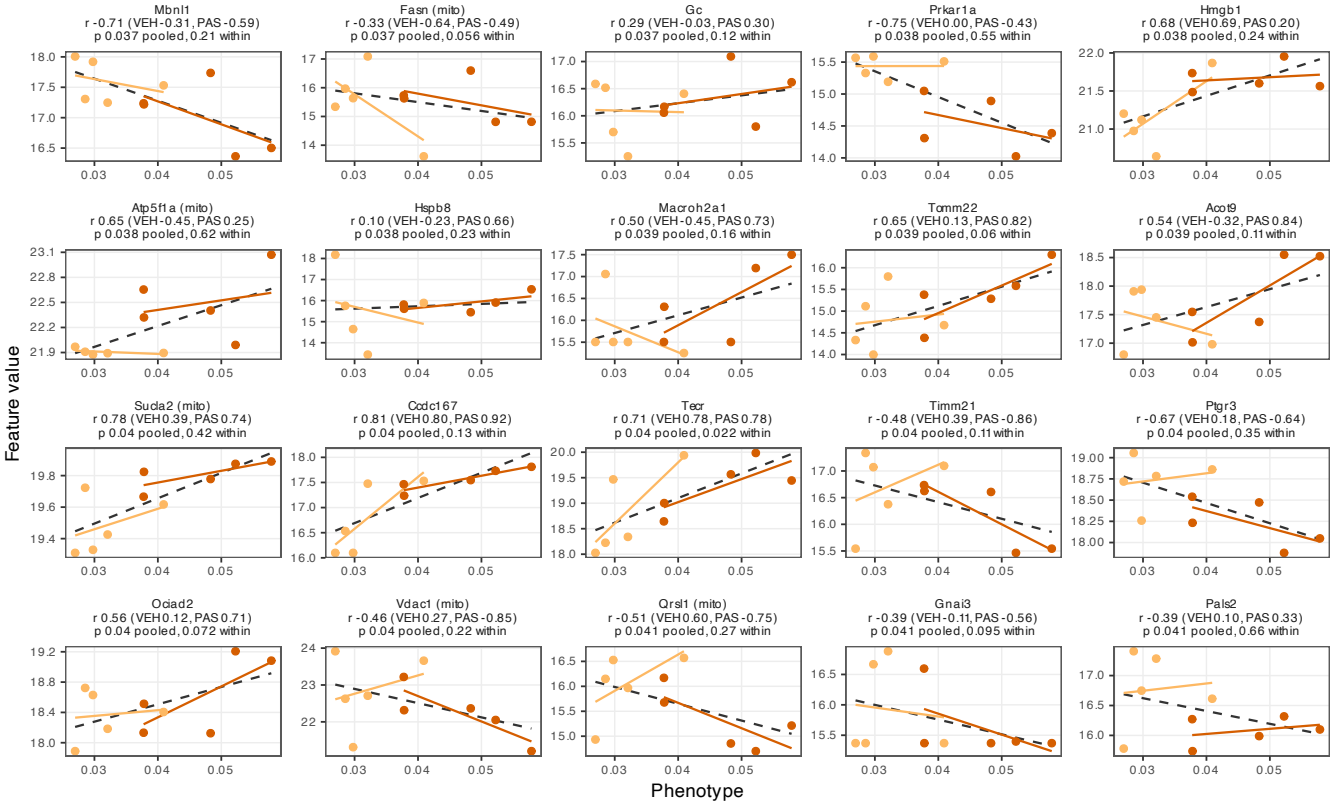
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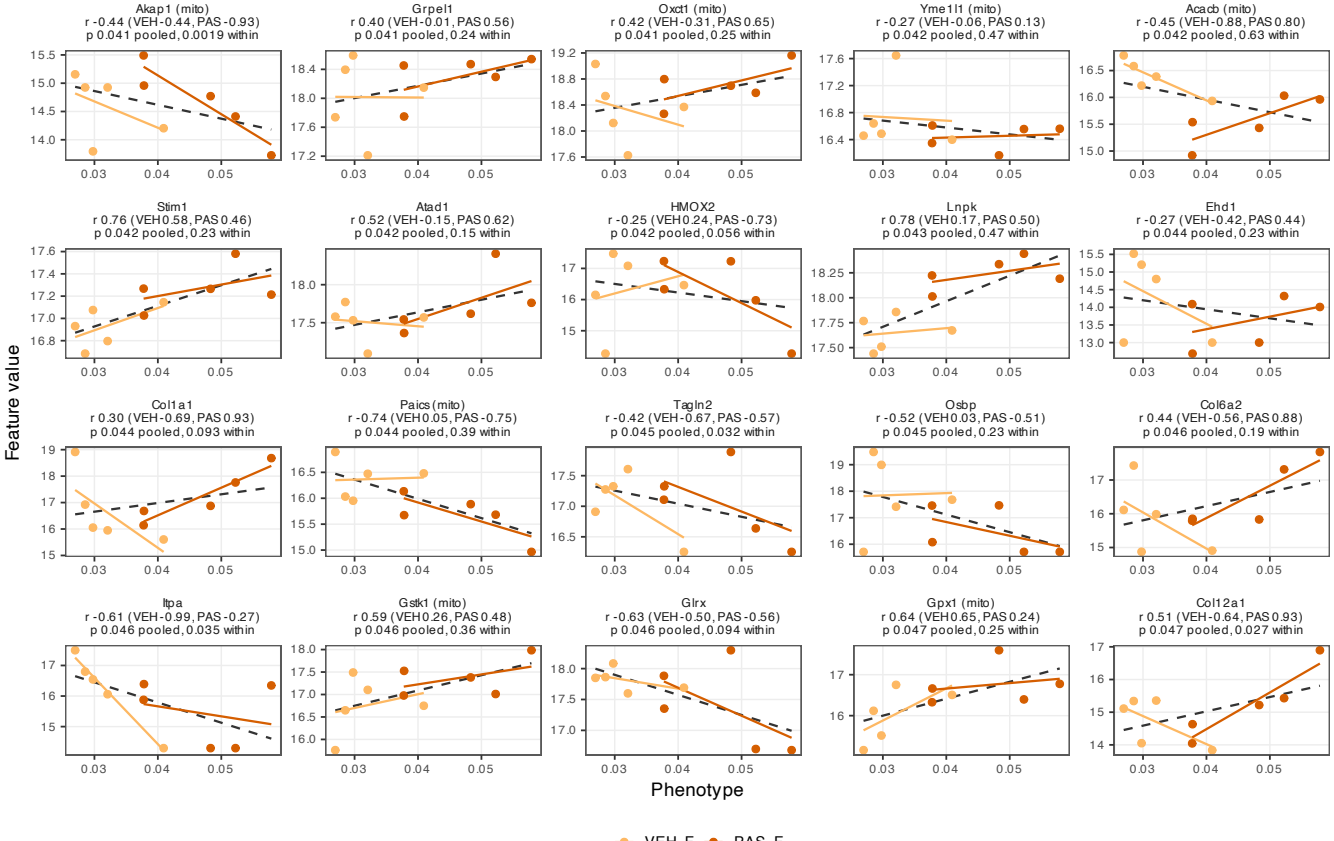
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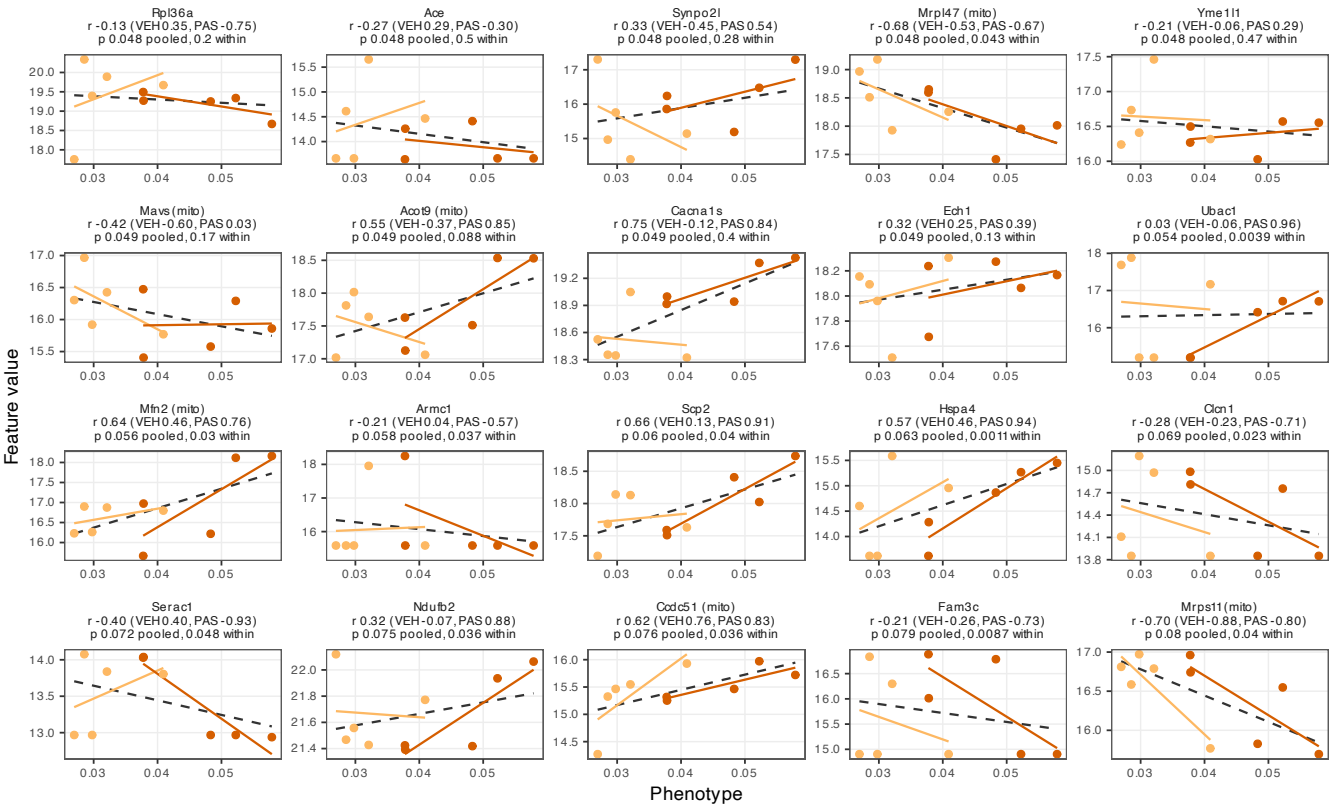
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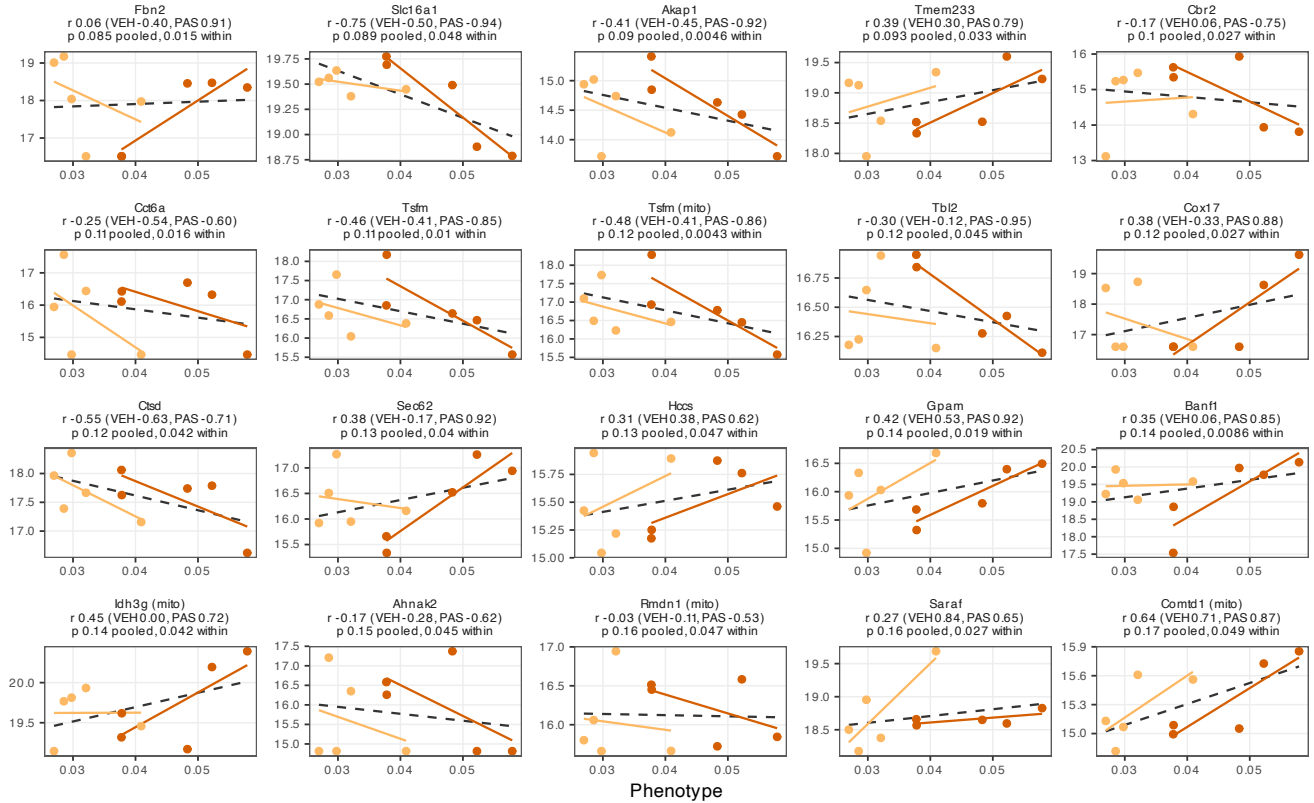
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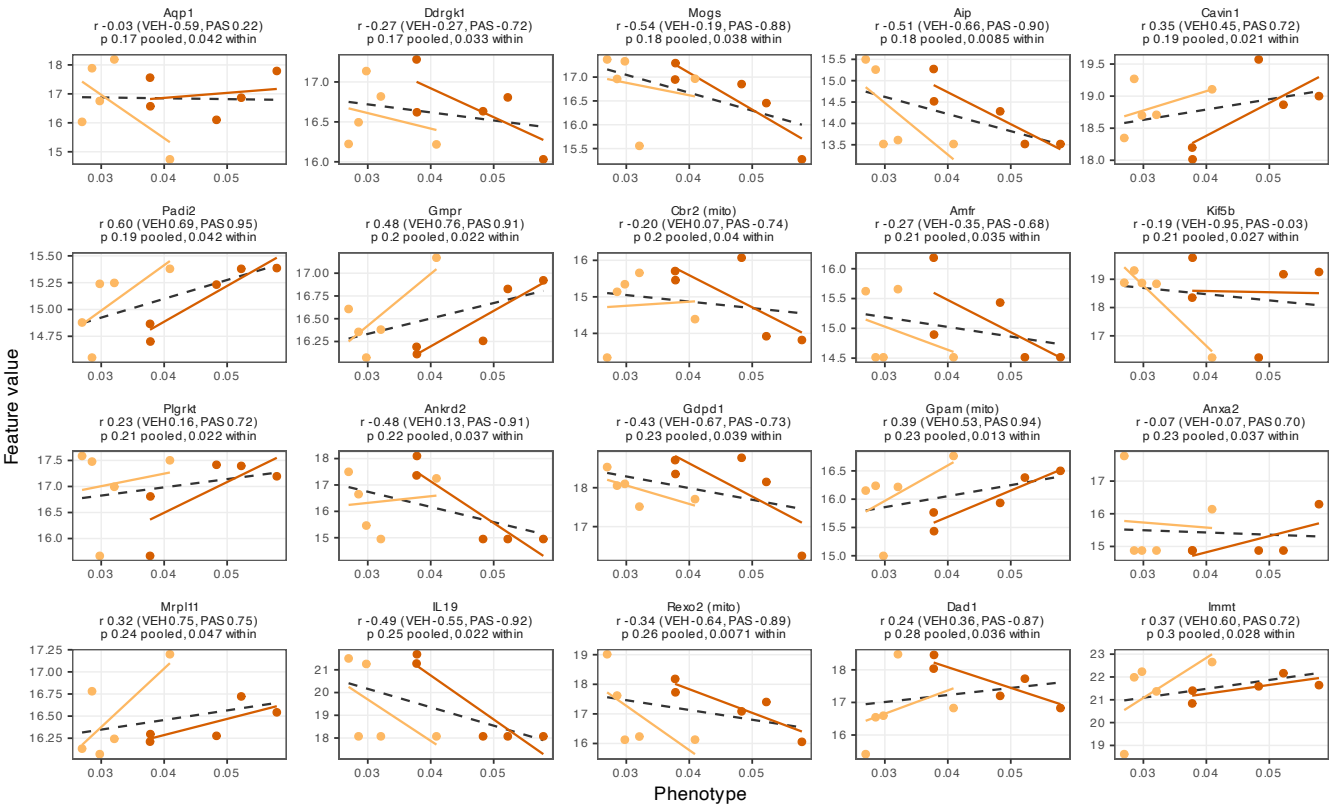
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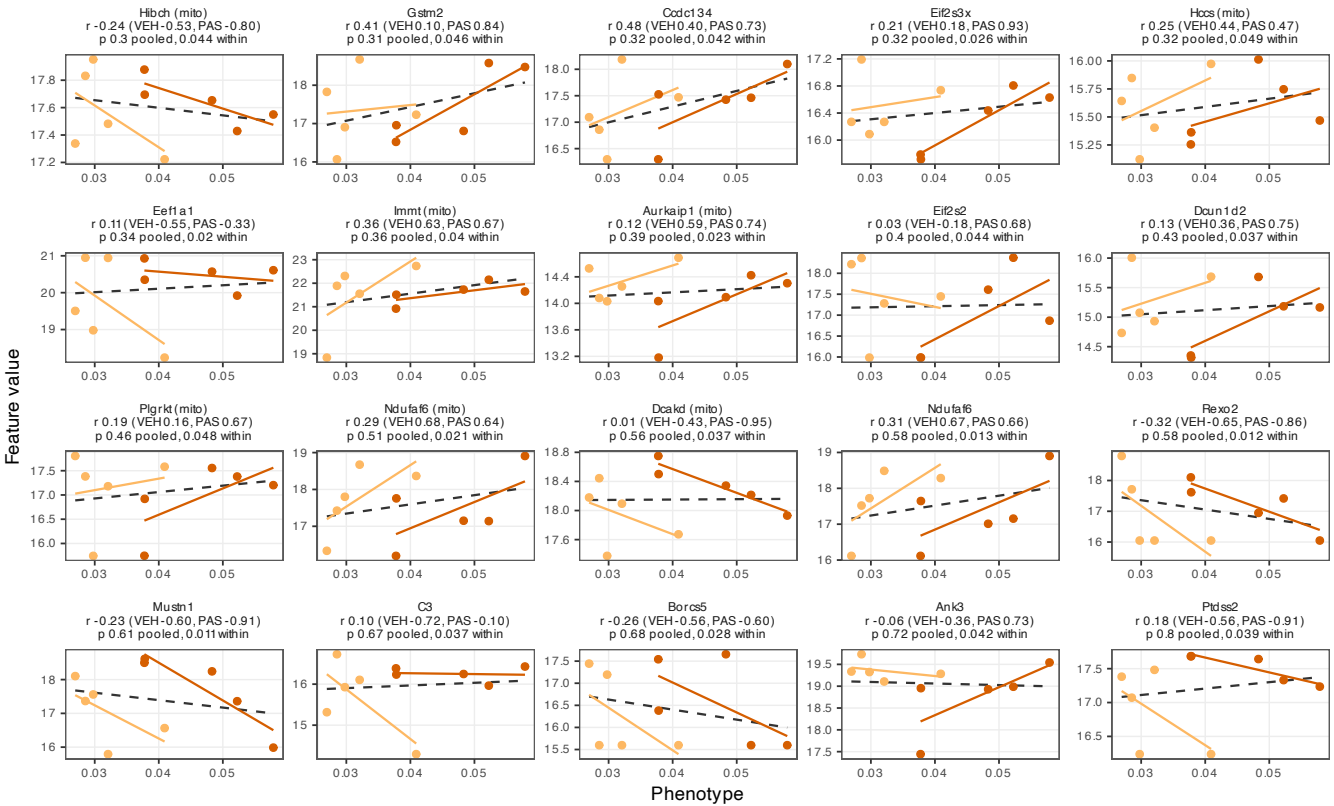
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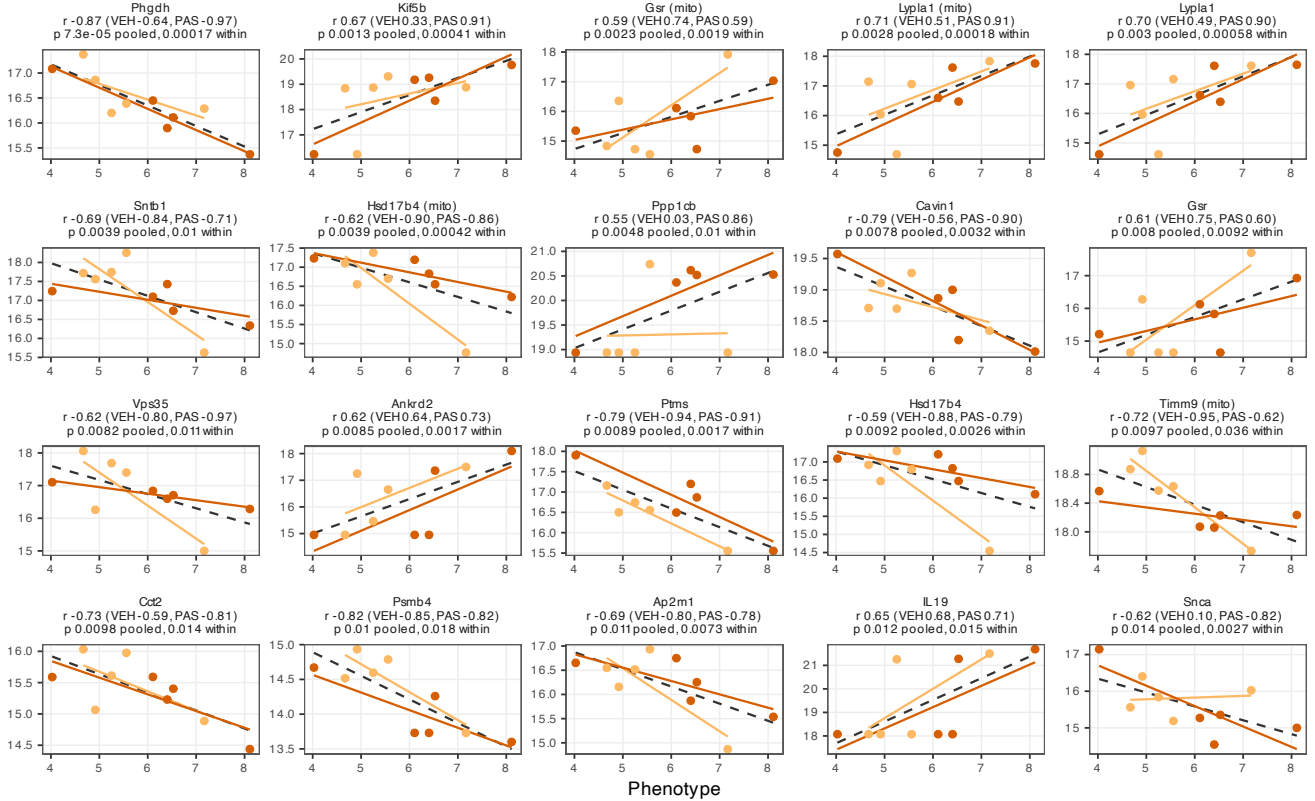
Female: Proteins associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 360)

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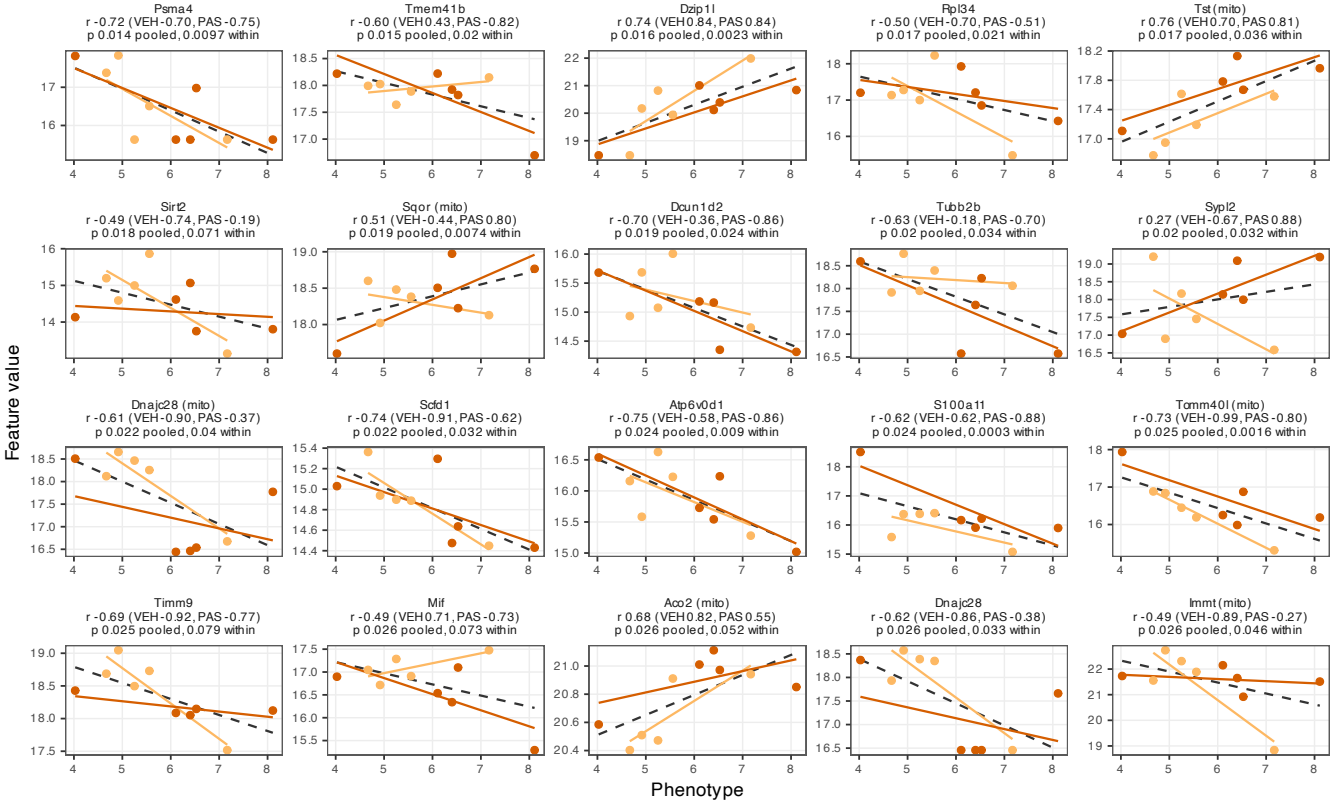
Female: Proteins associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 176)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



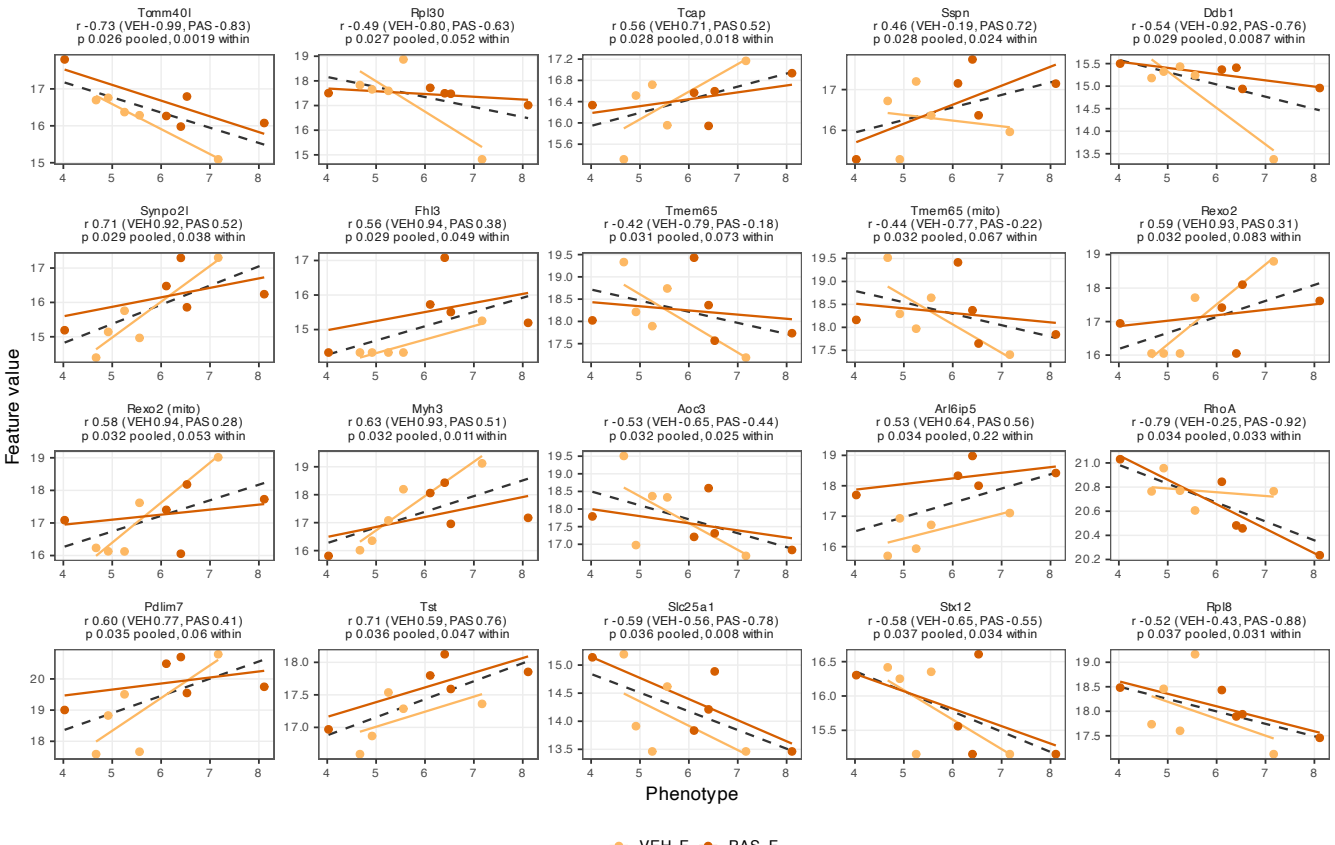
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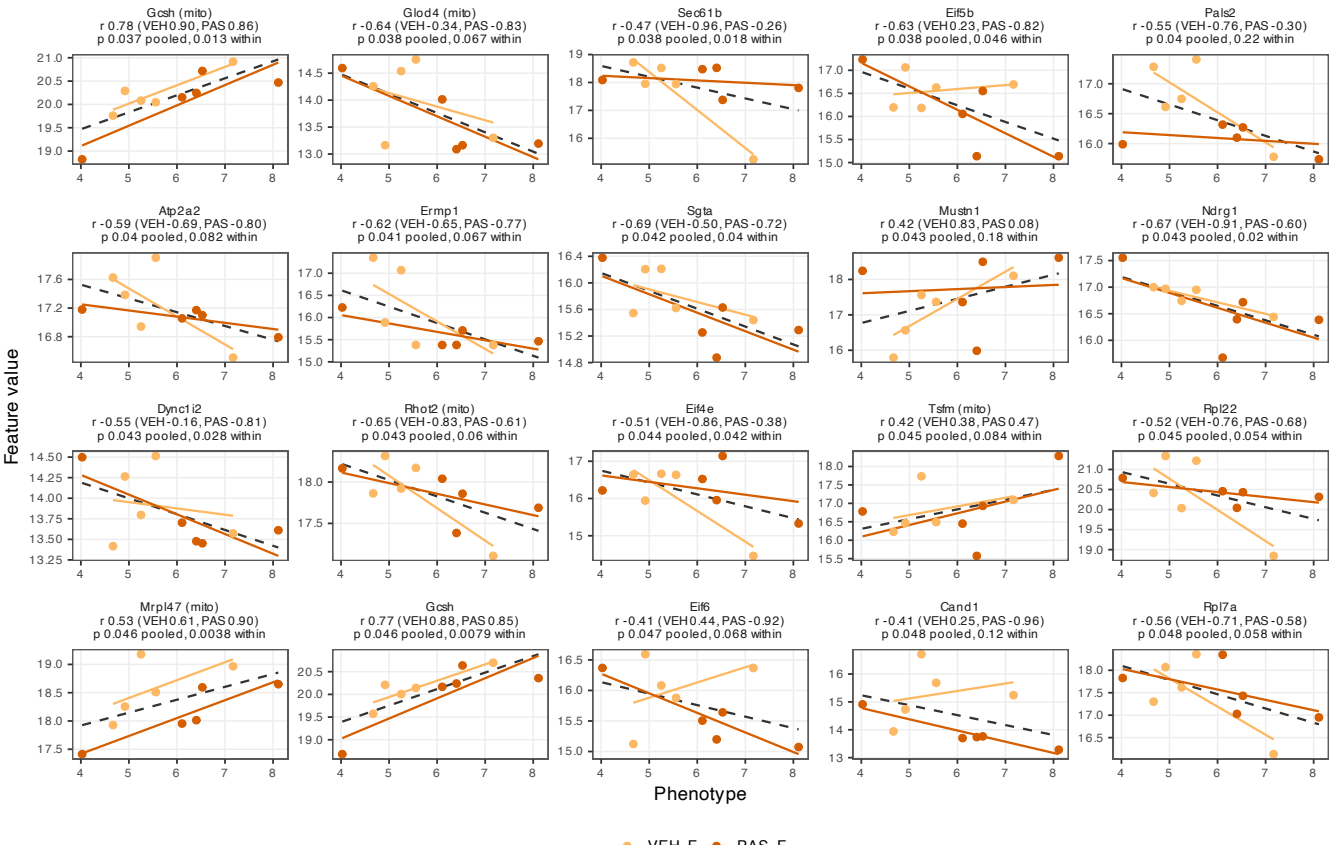
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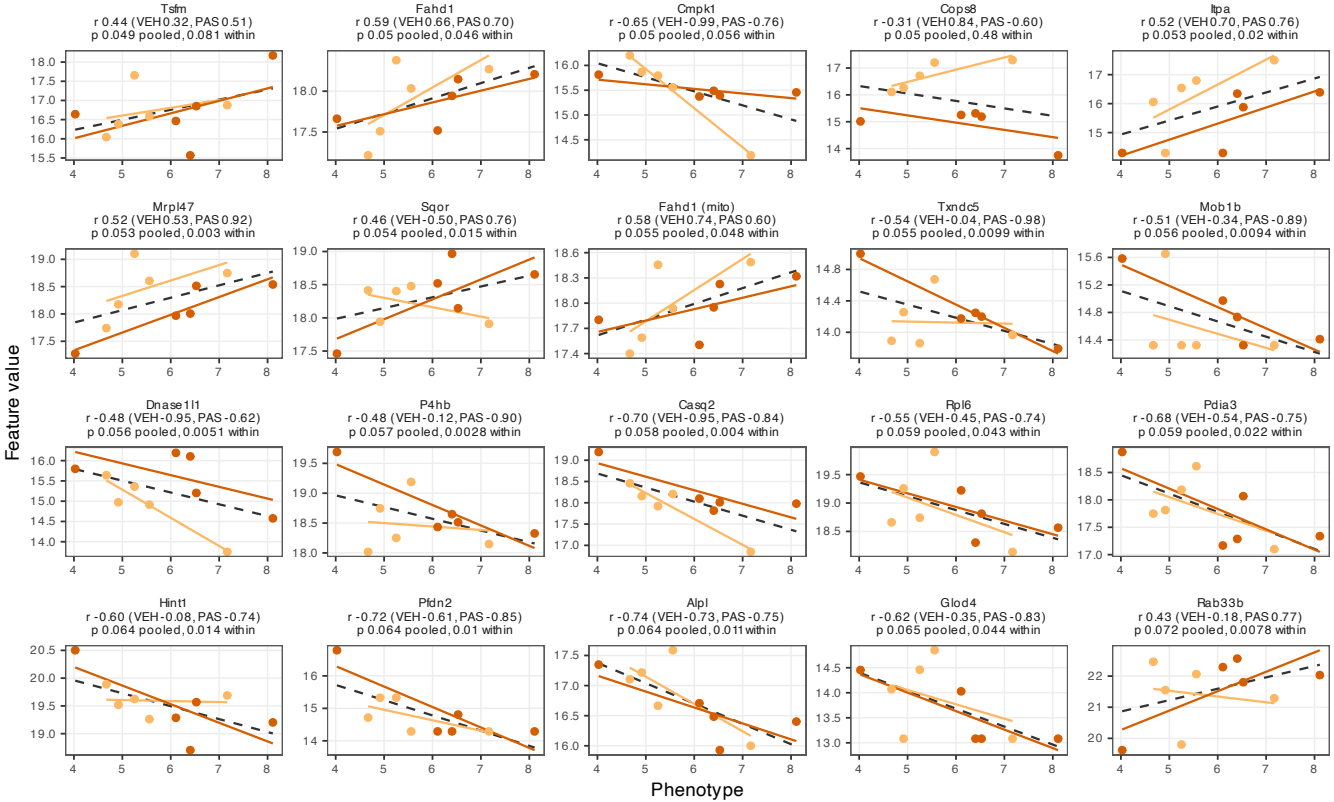
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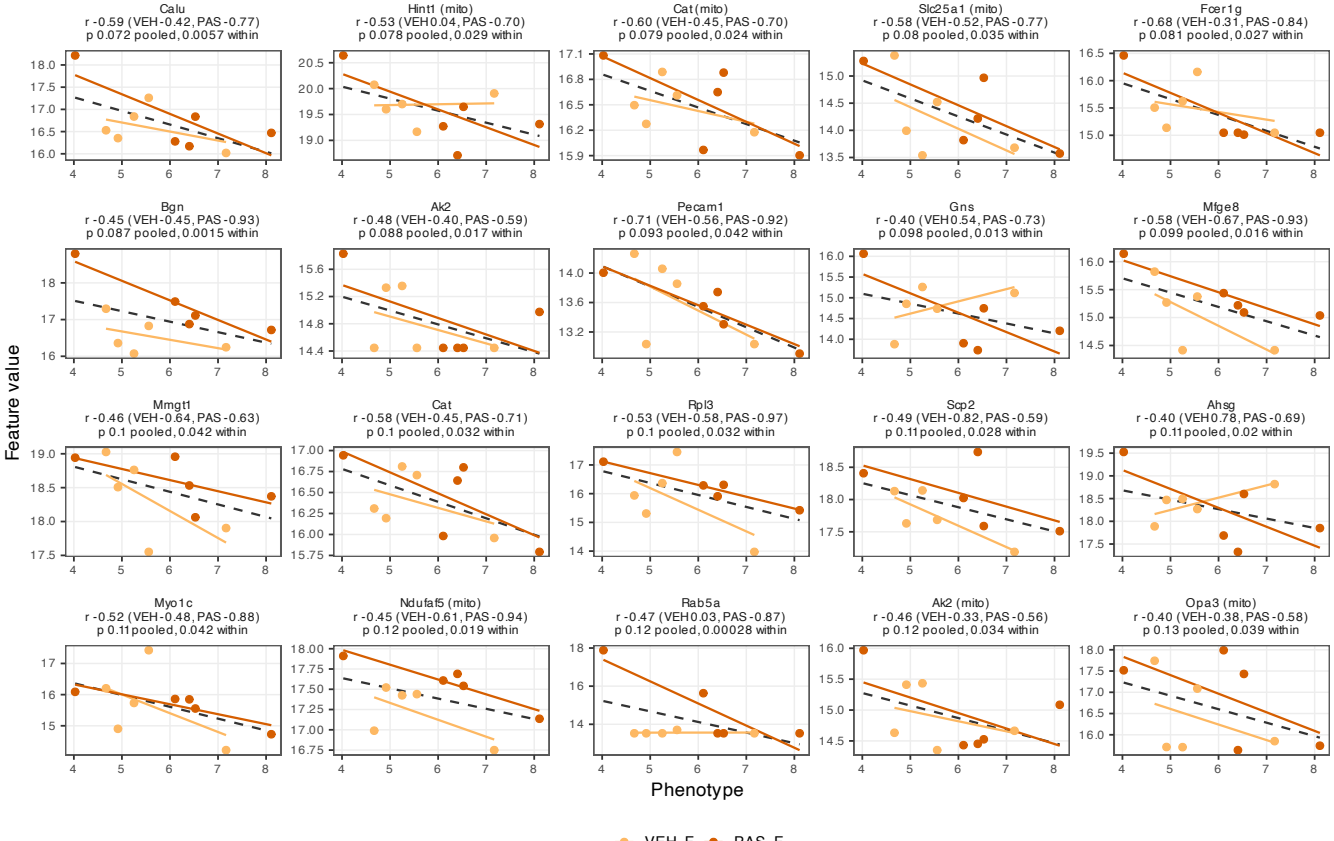
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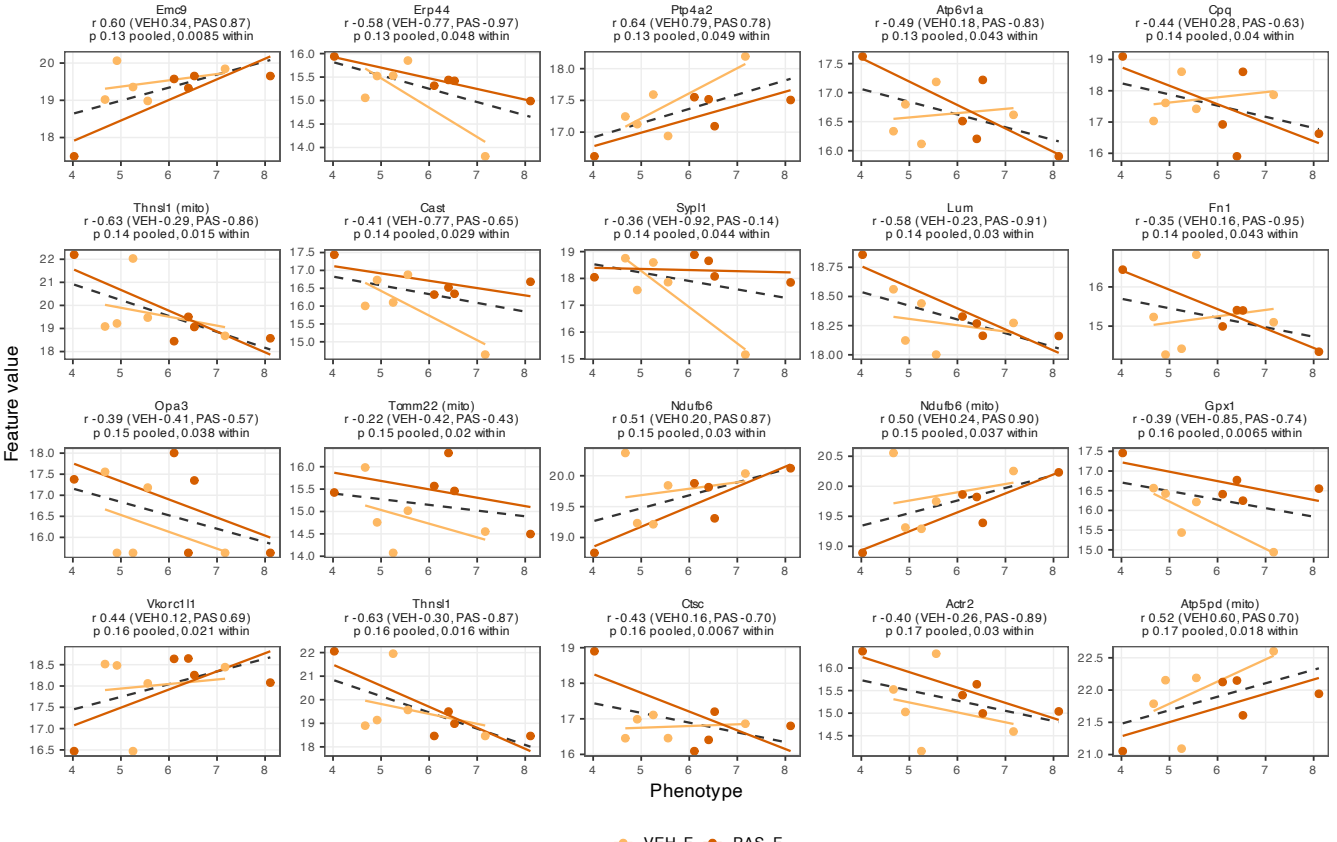
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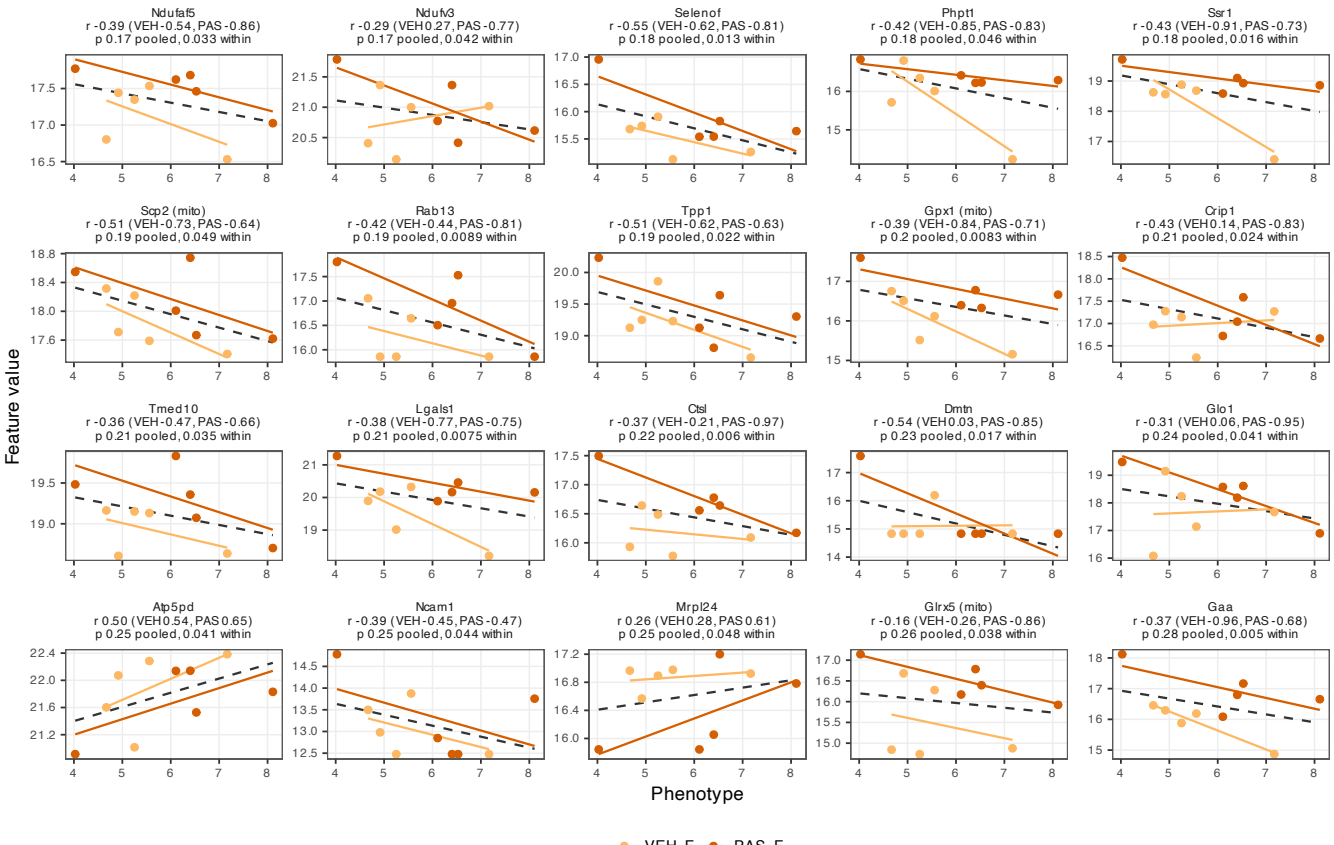
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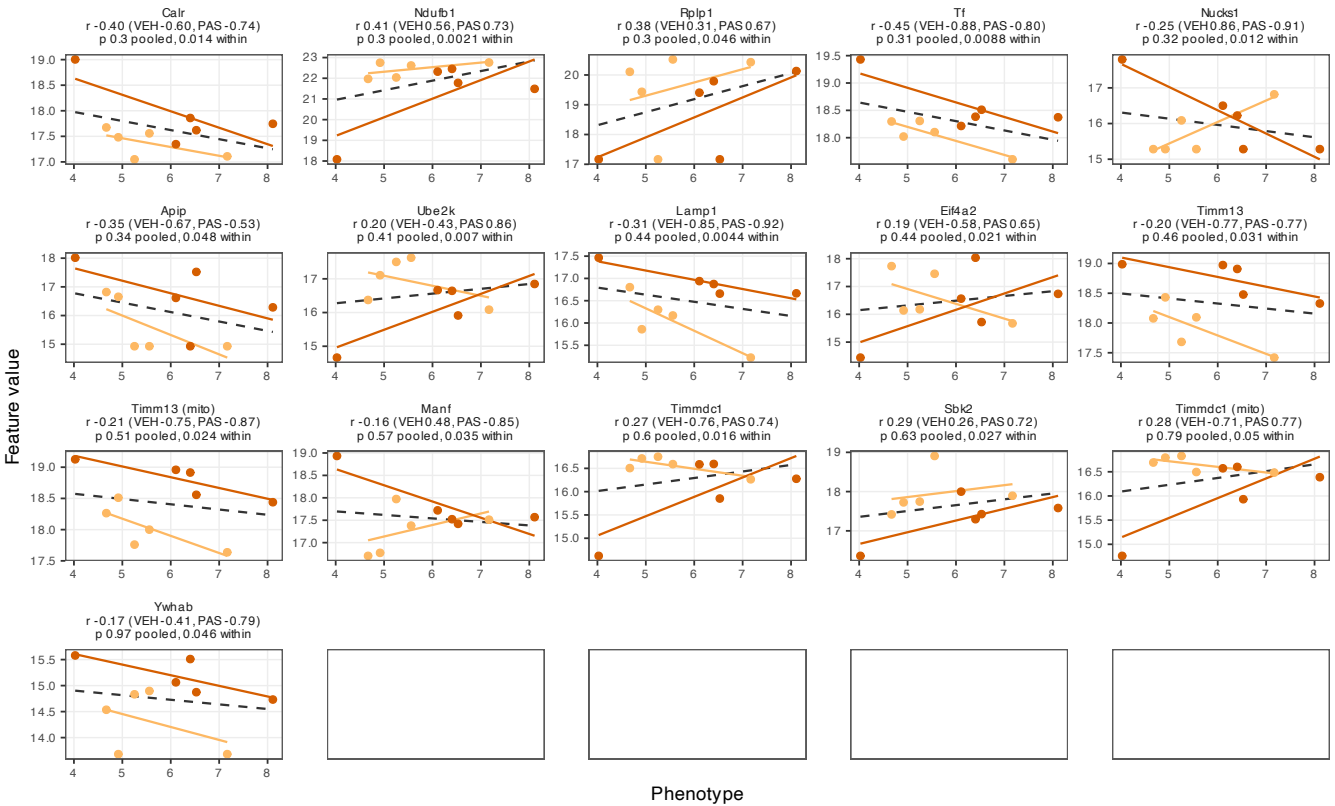
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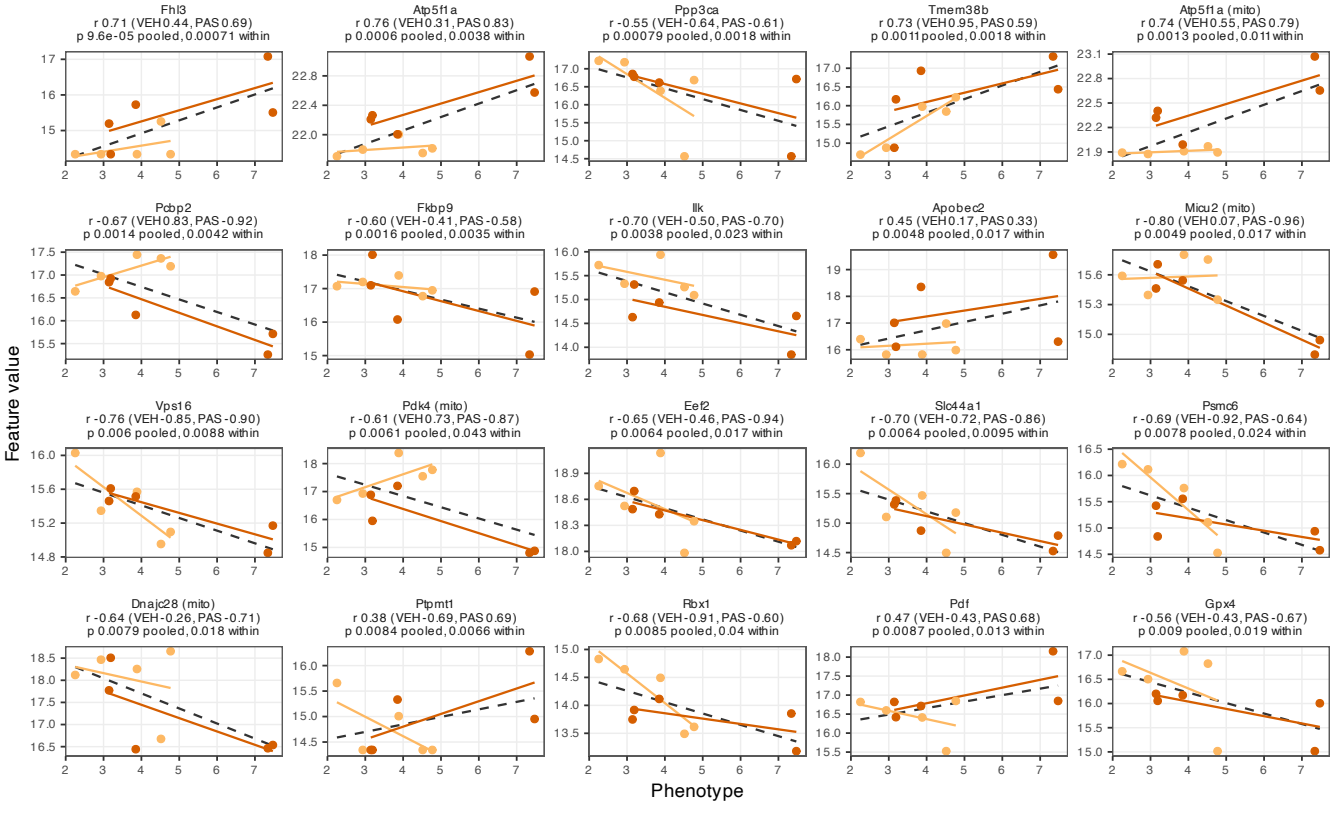
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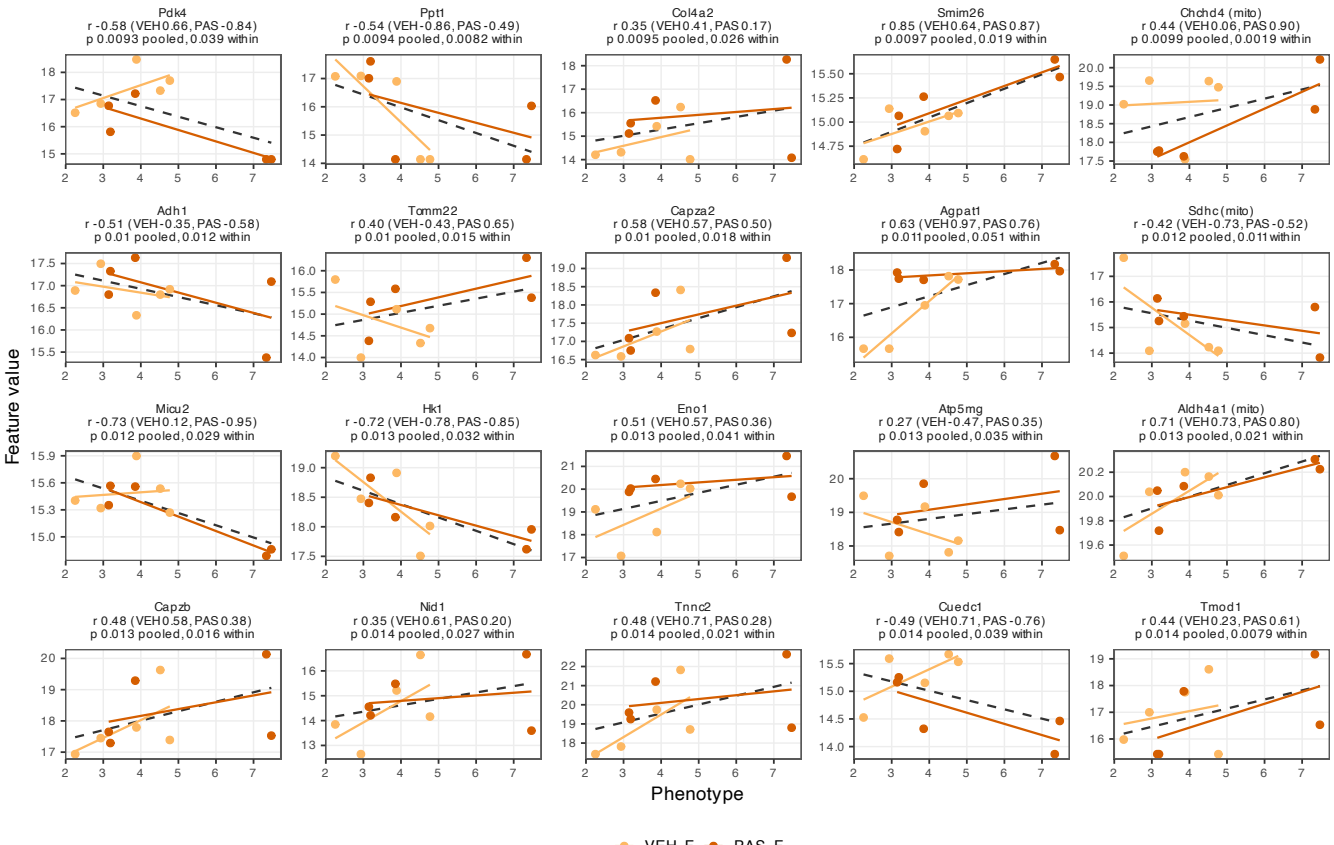
Female: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 195)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



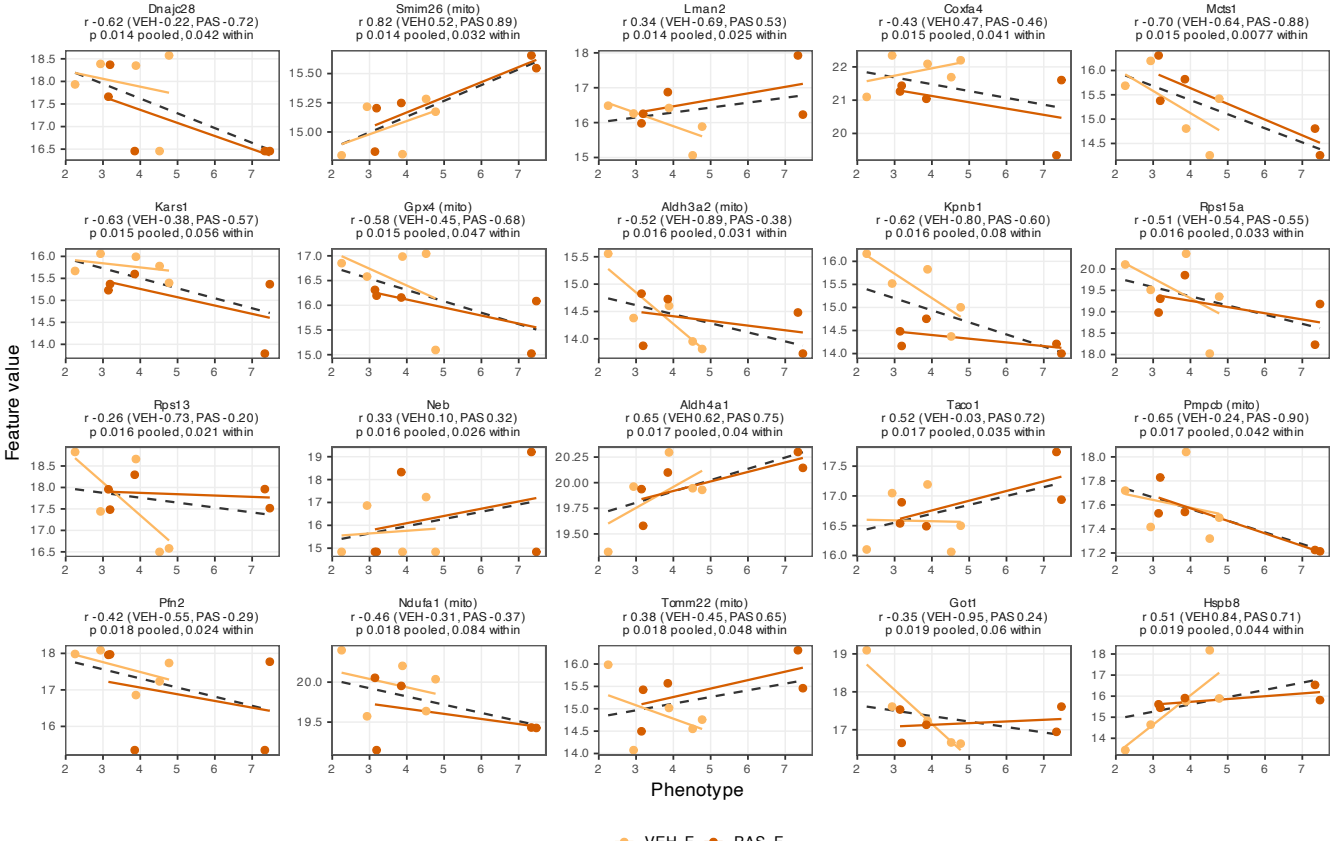
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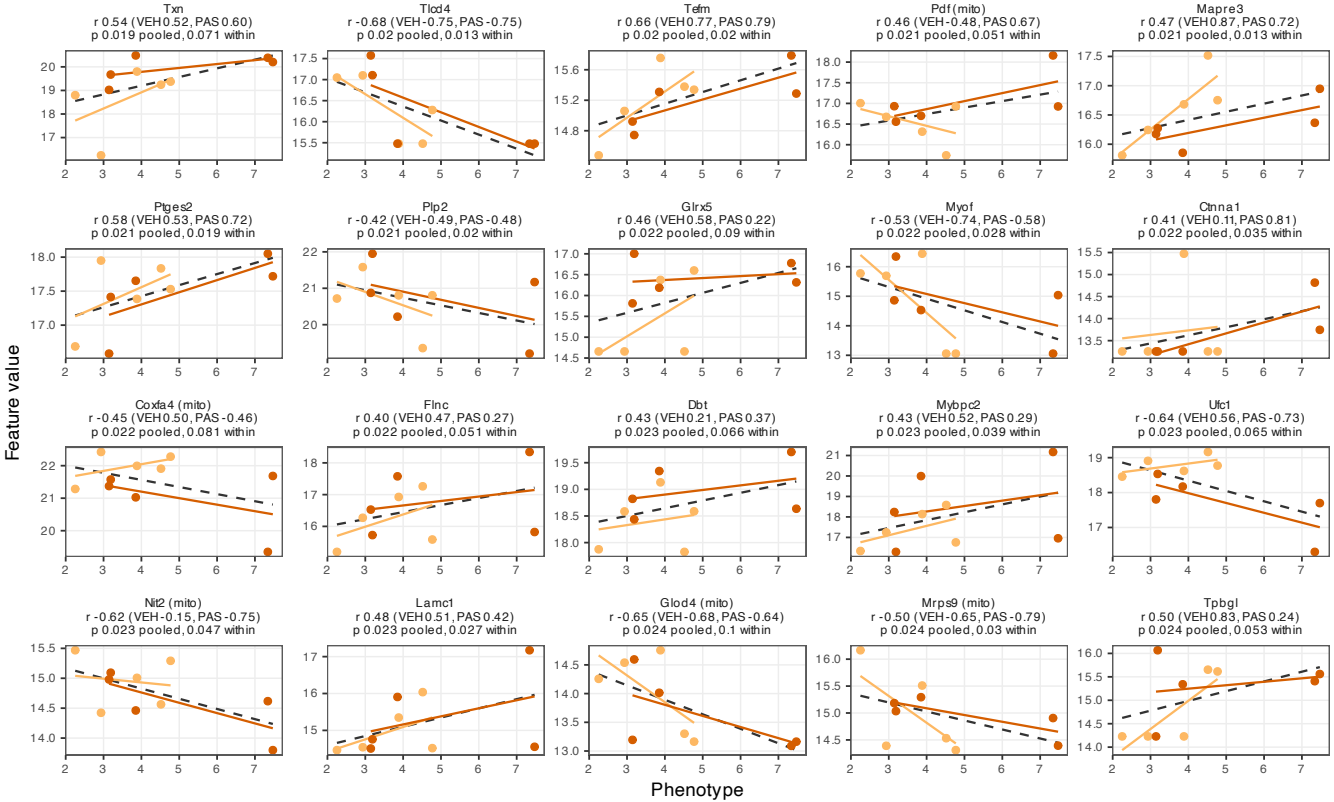
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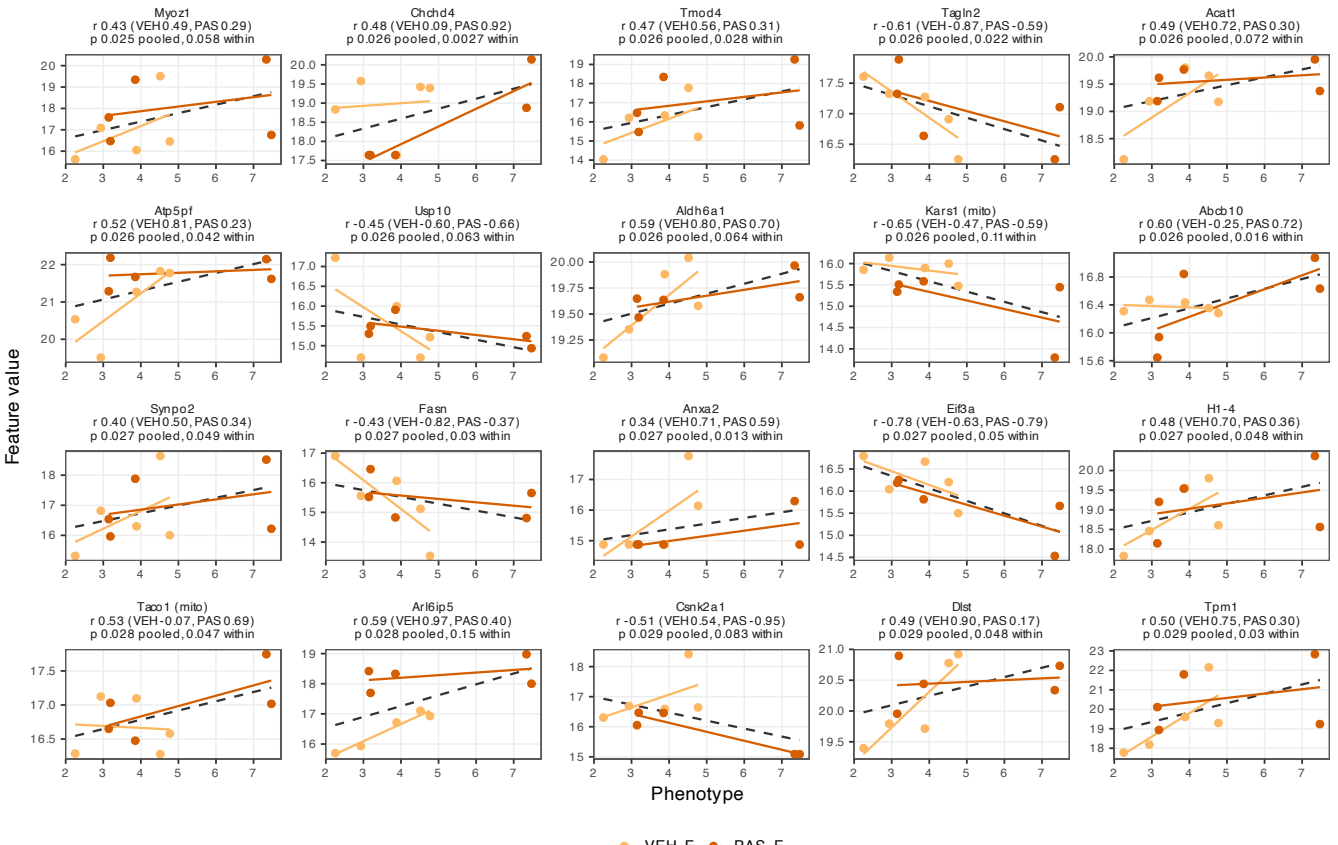
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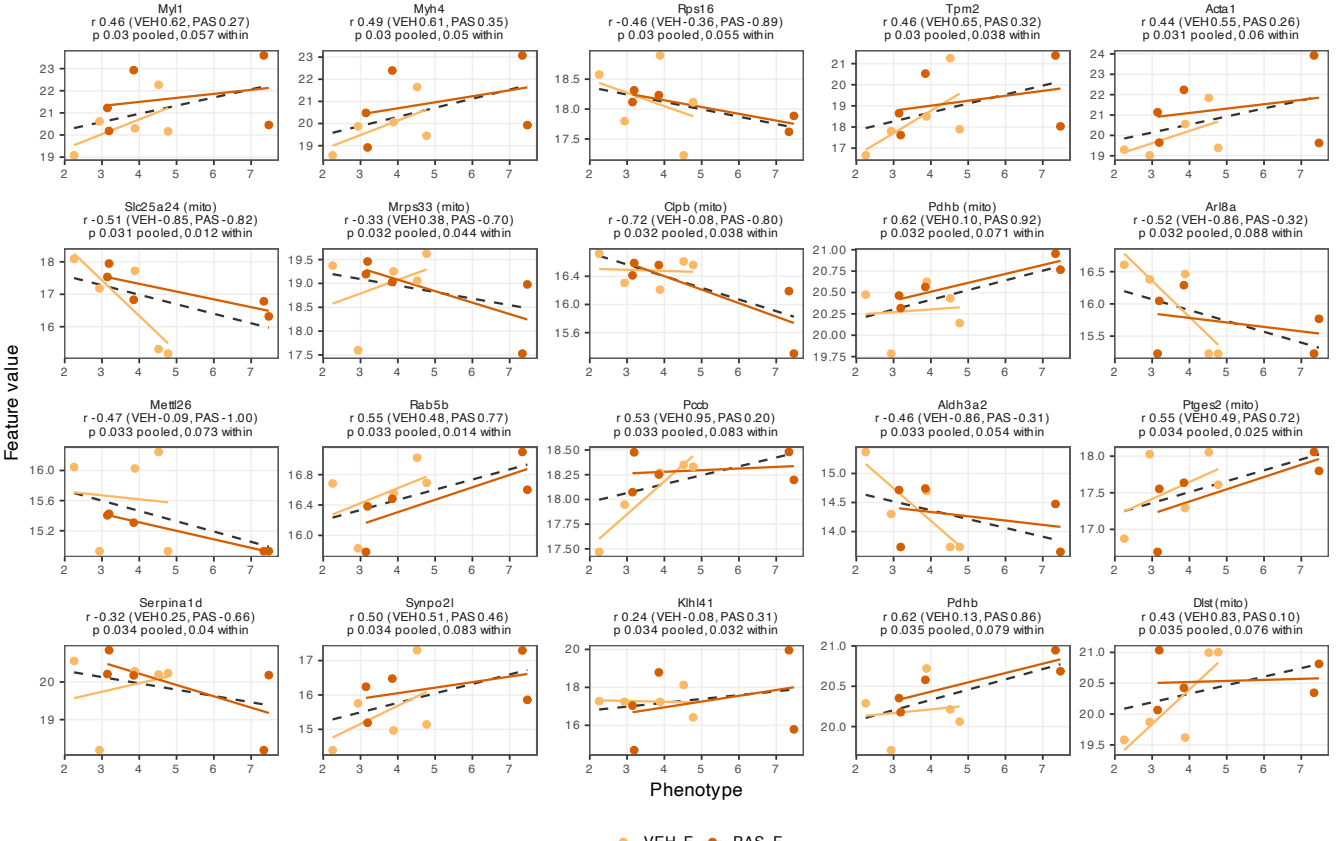
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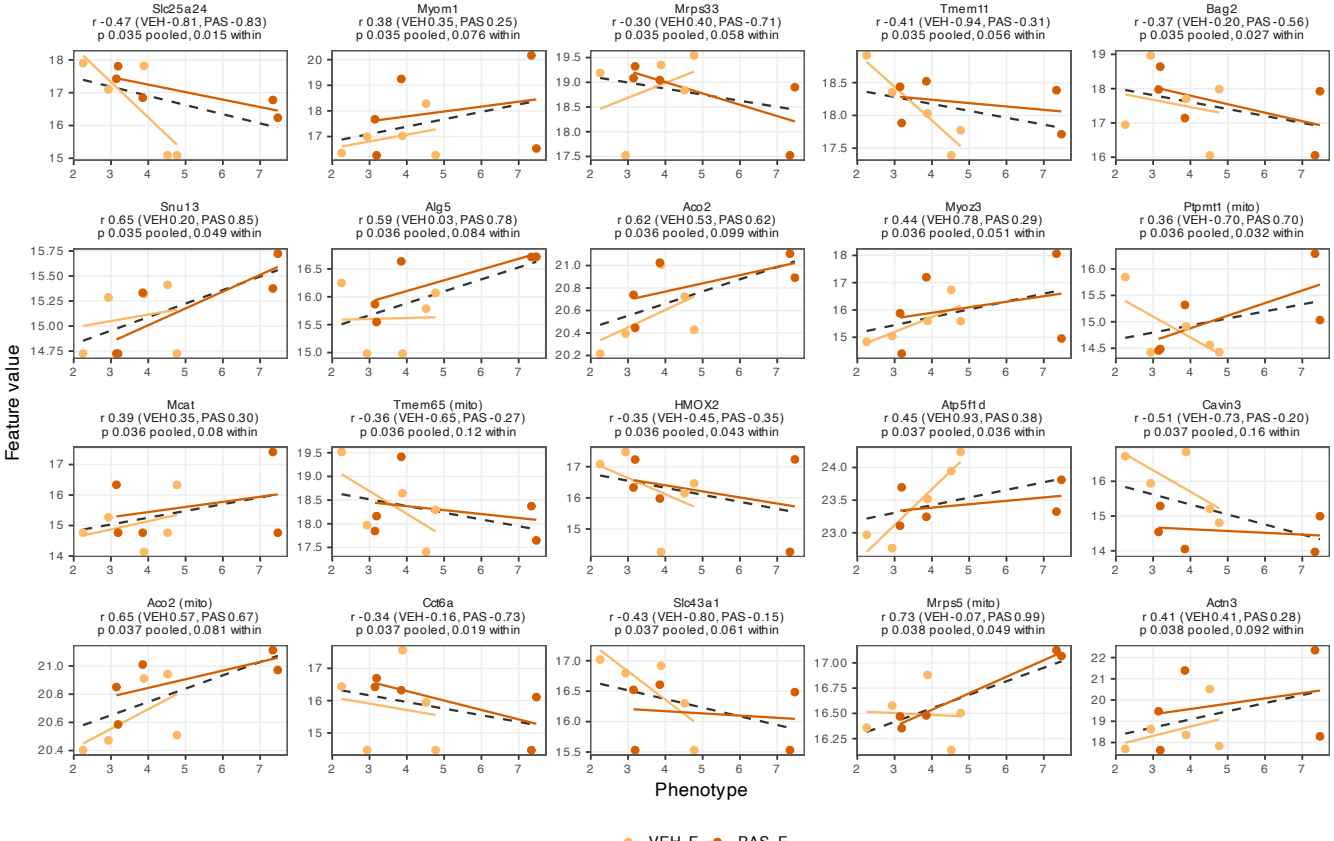
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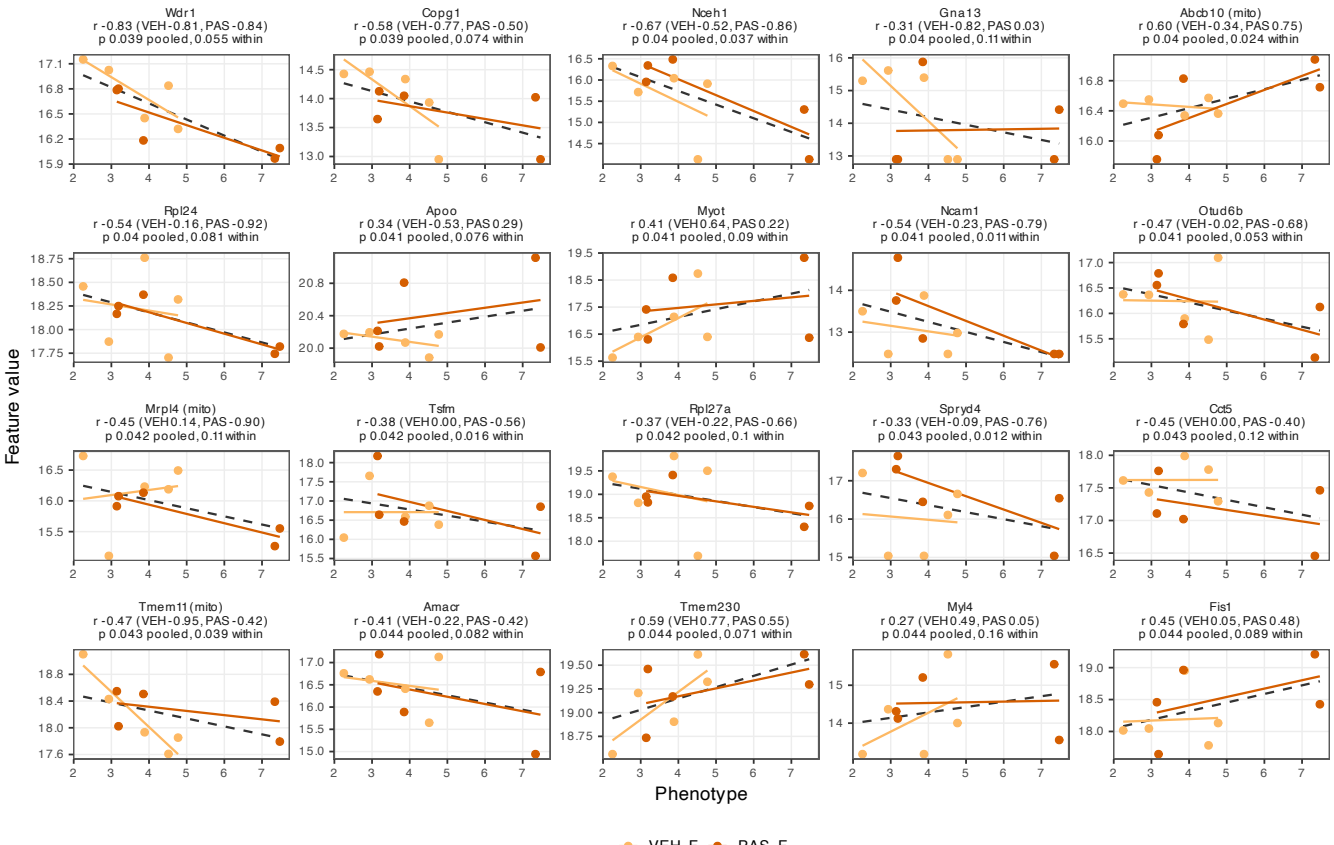
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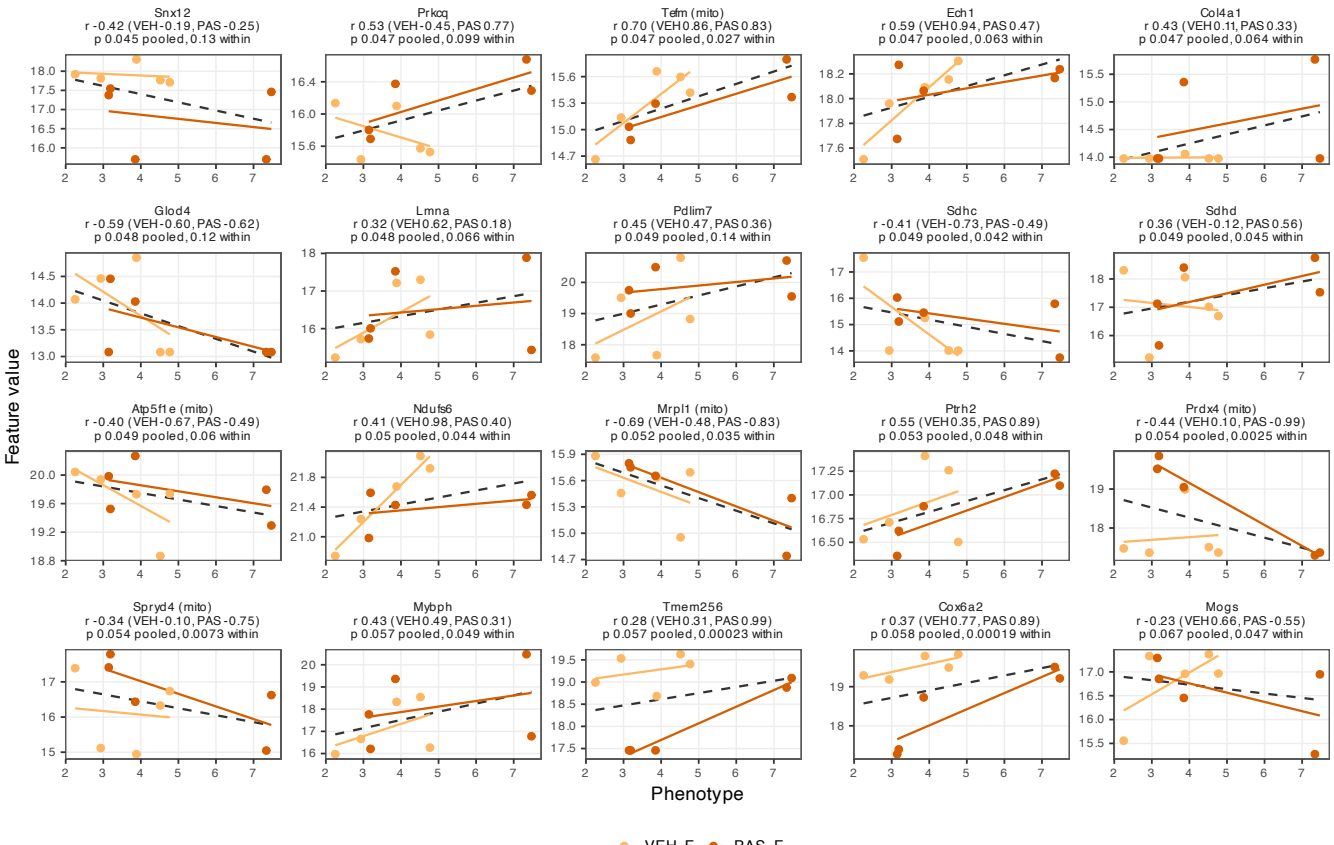
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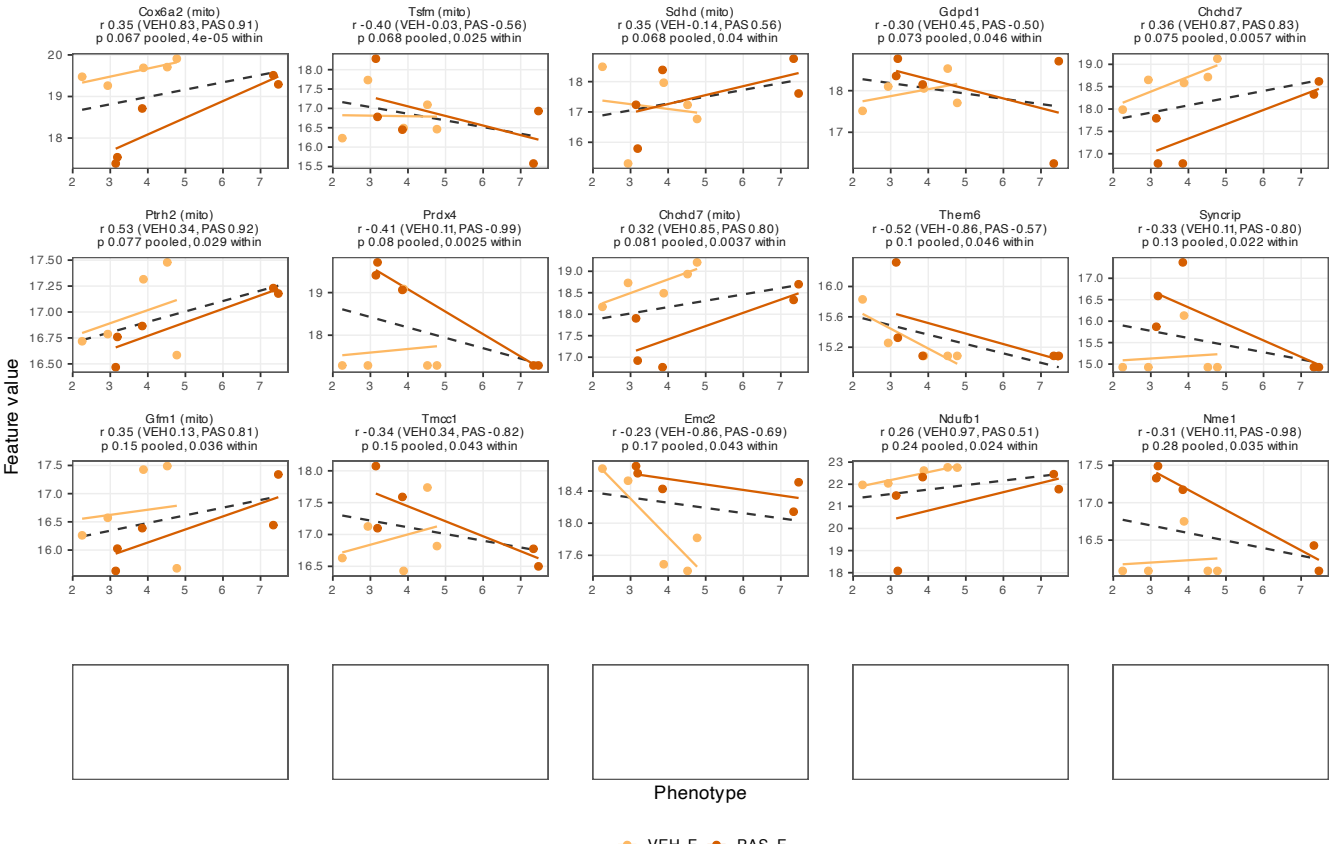
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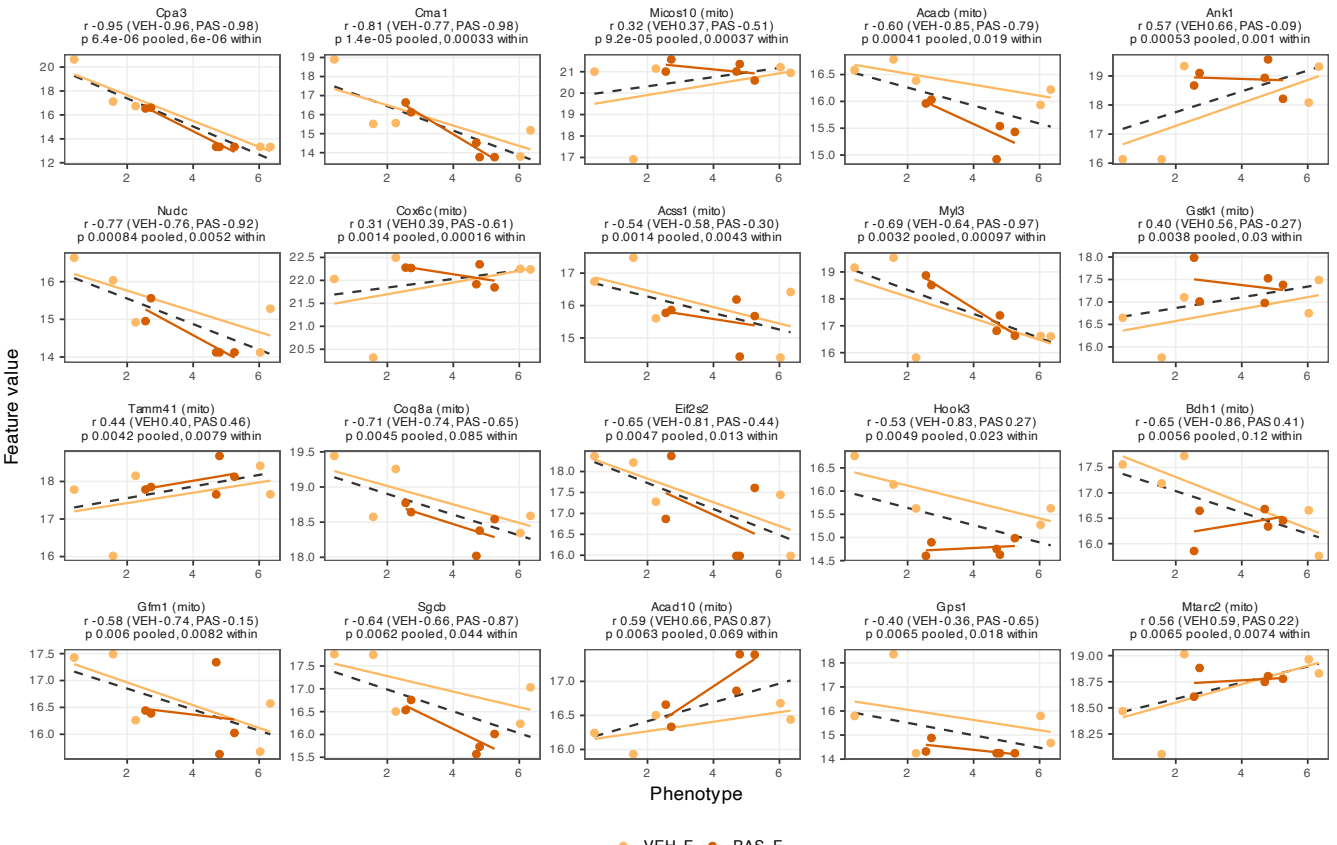
Female: Proteins associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 195)

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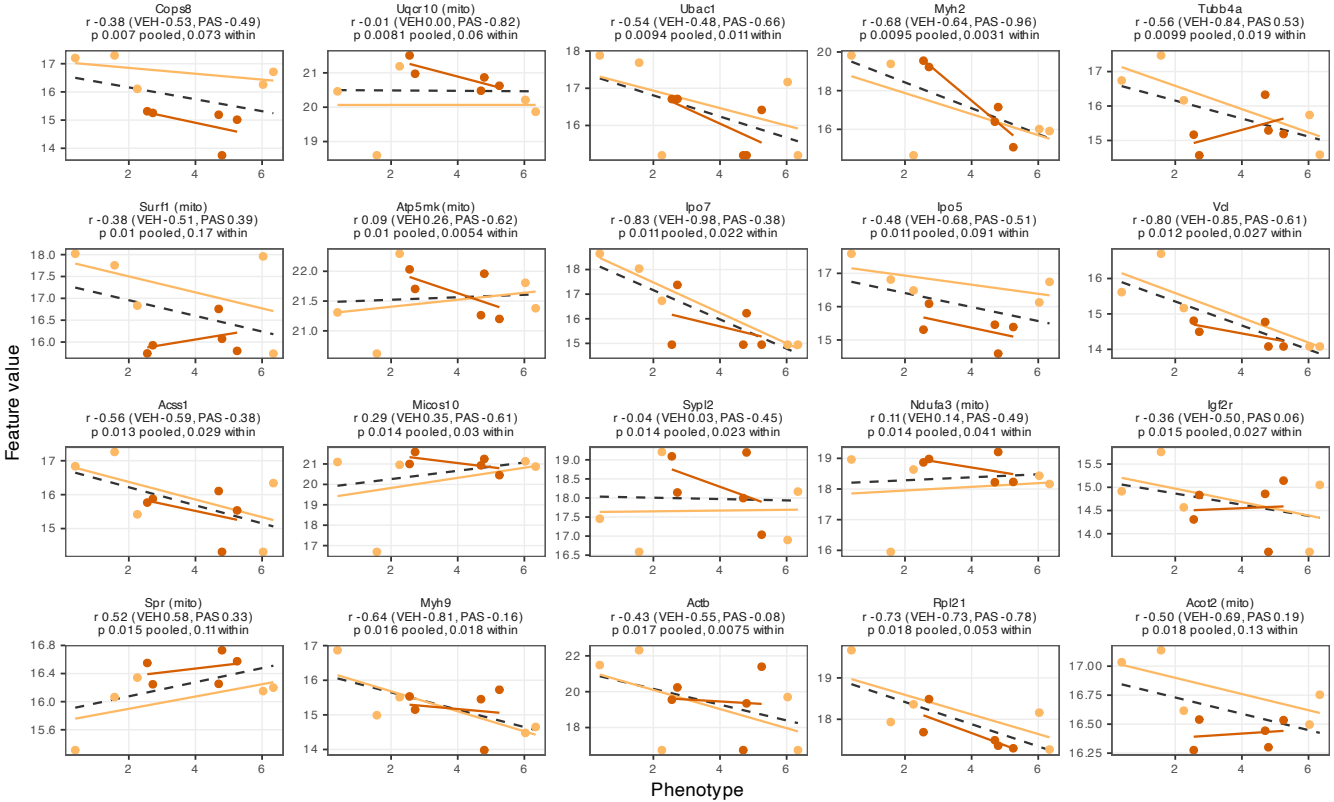
Female: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 189)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



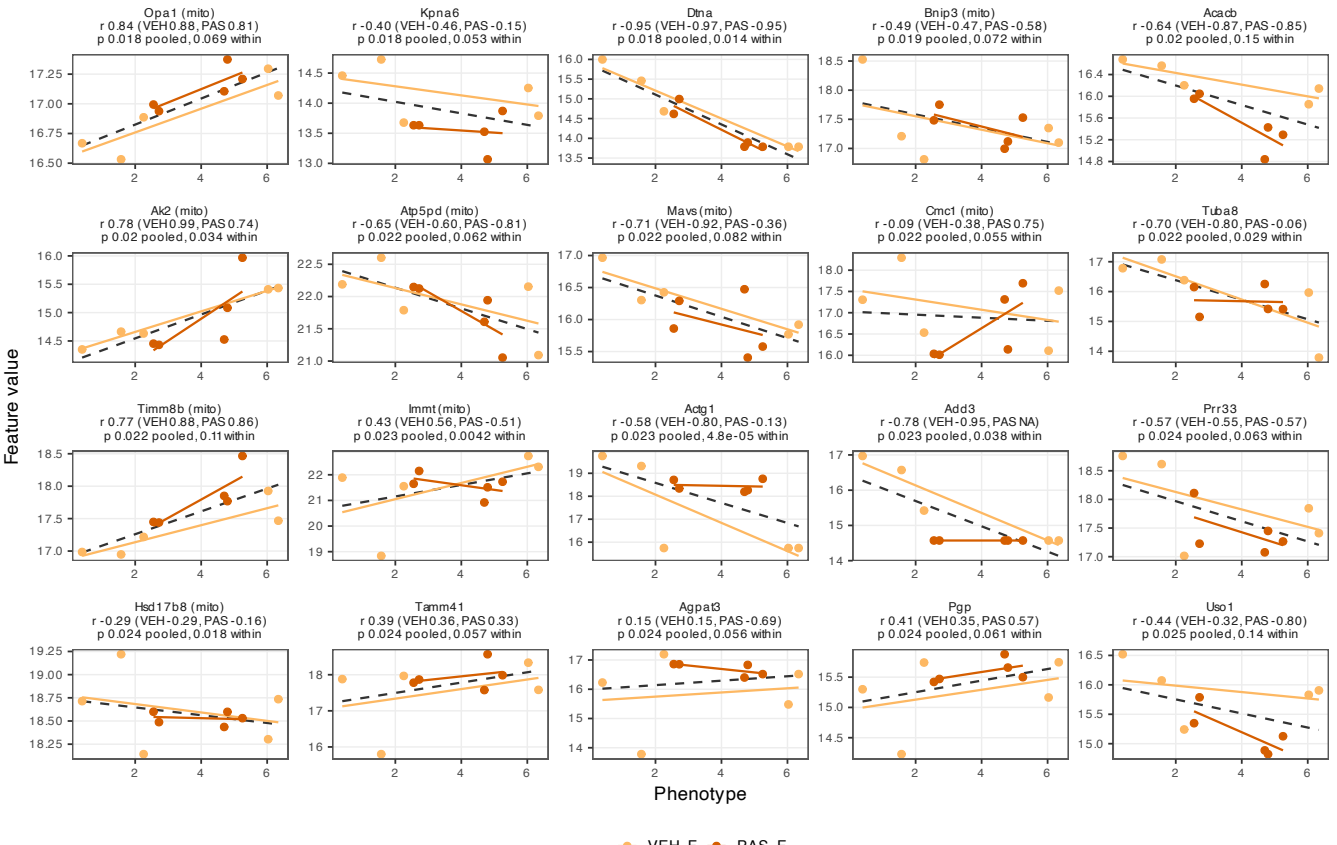
Female: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 189)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
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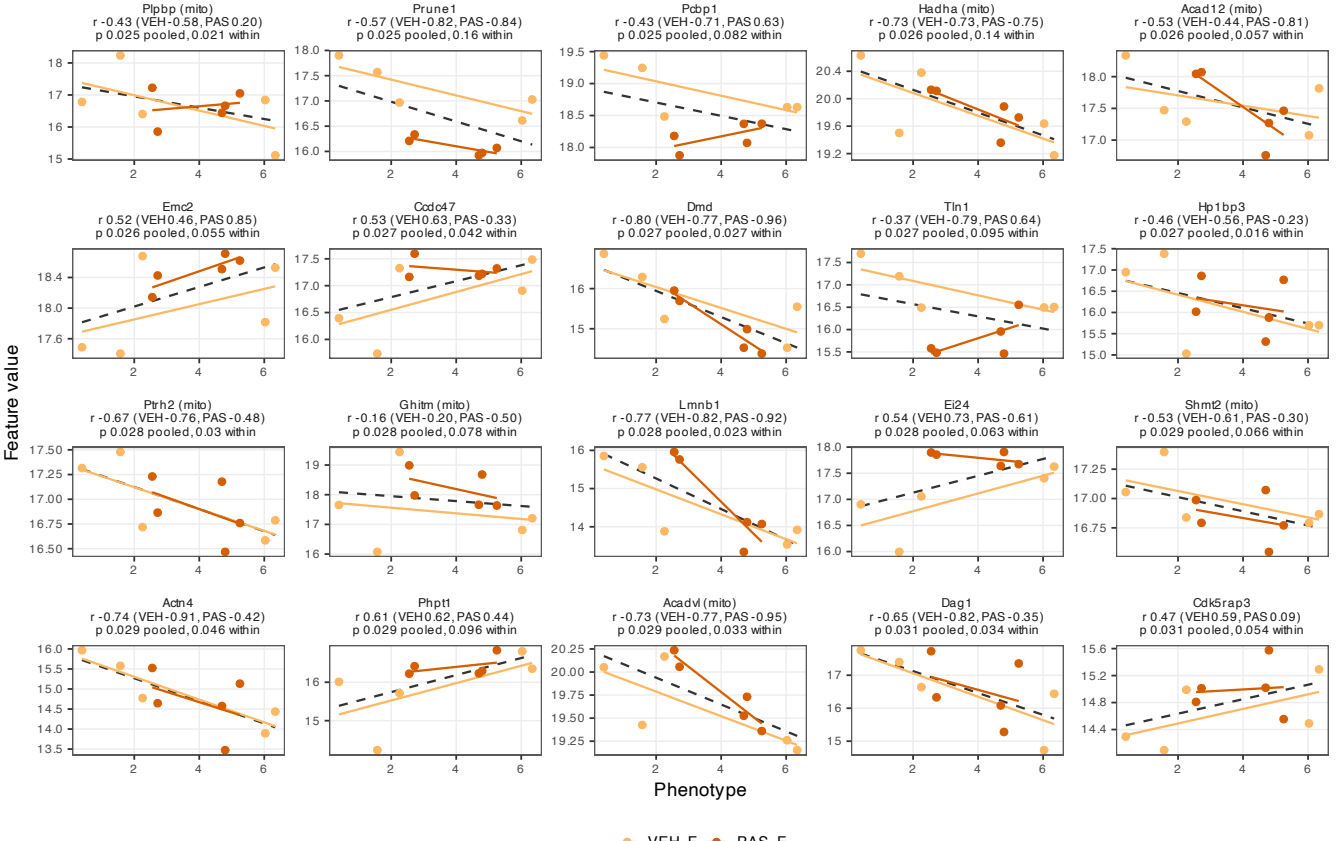
Female: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 189)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
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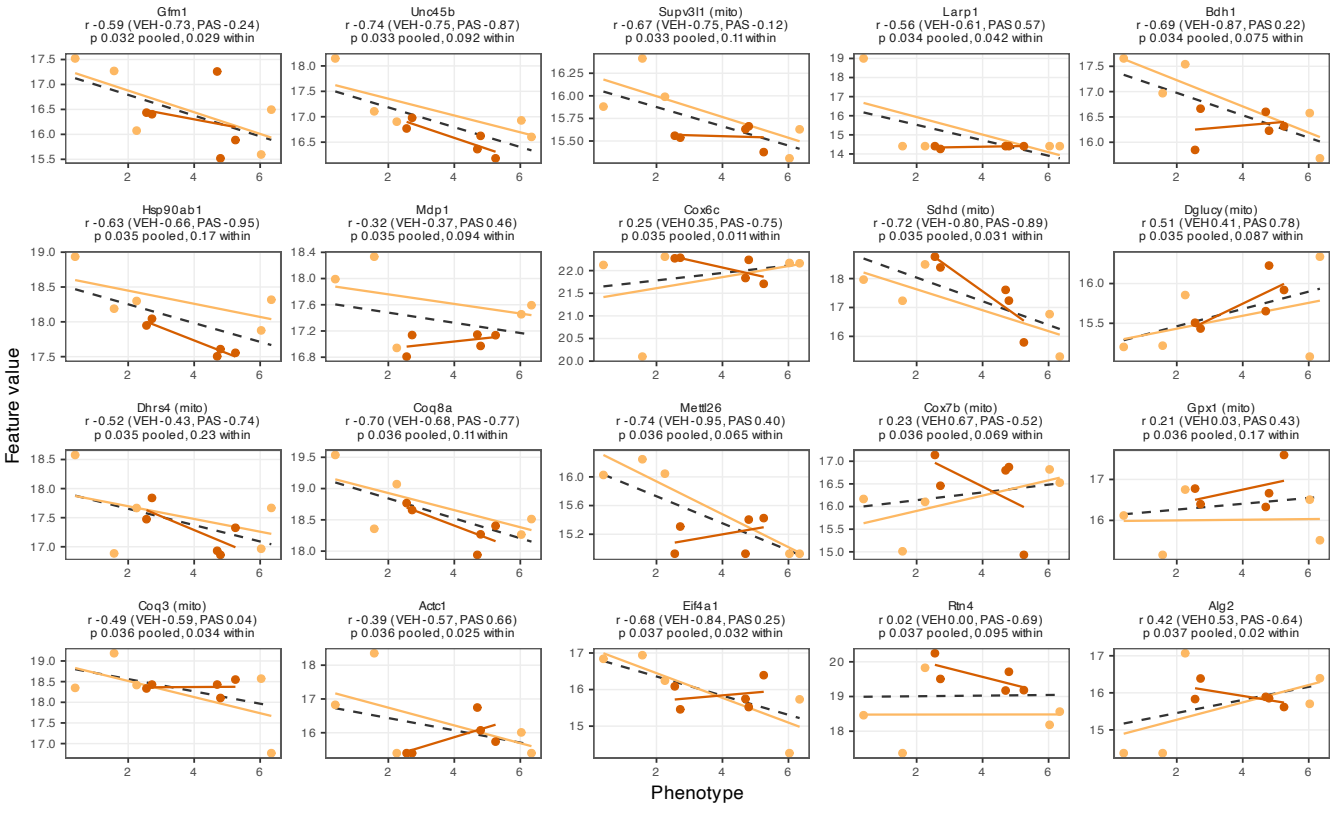
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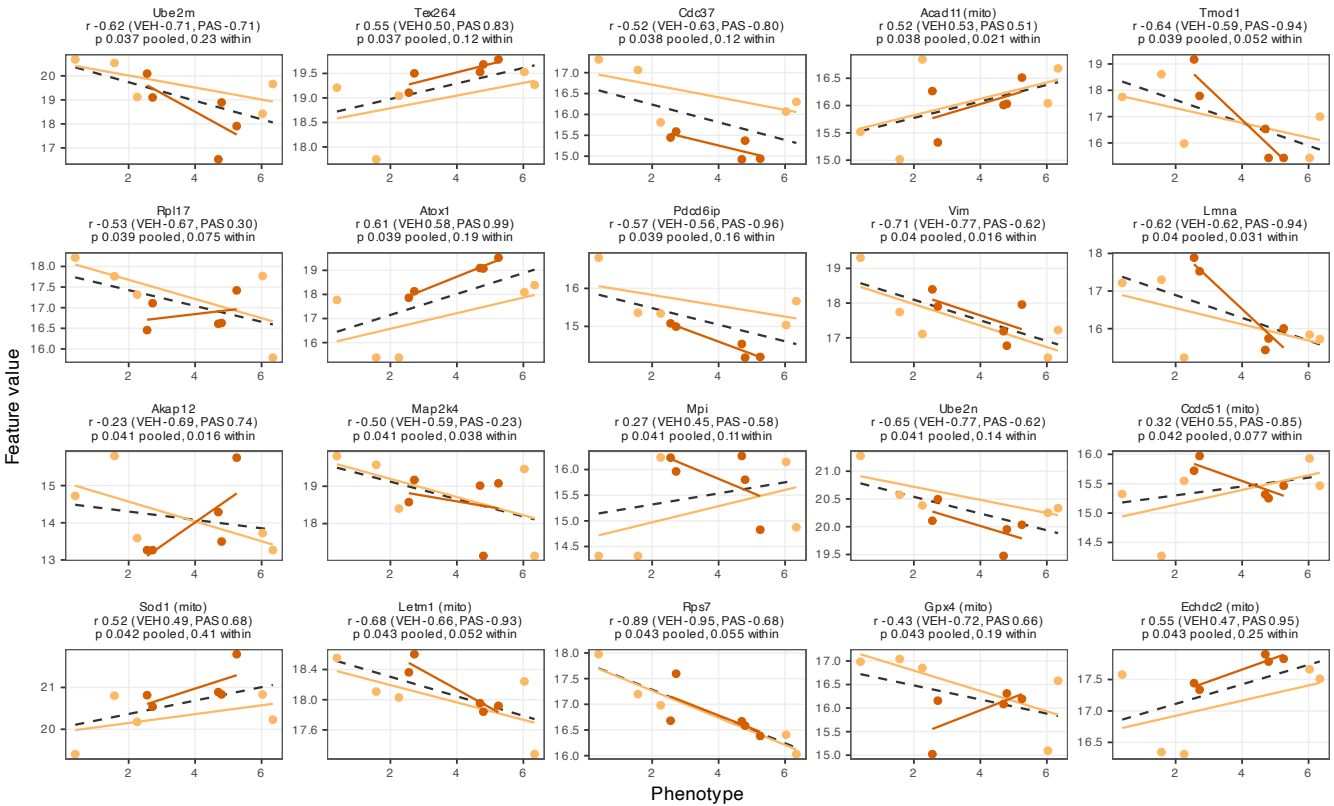
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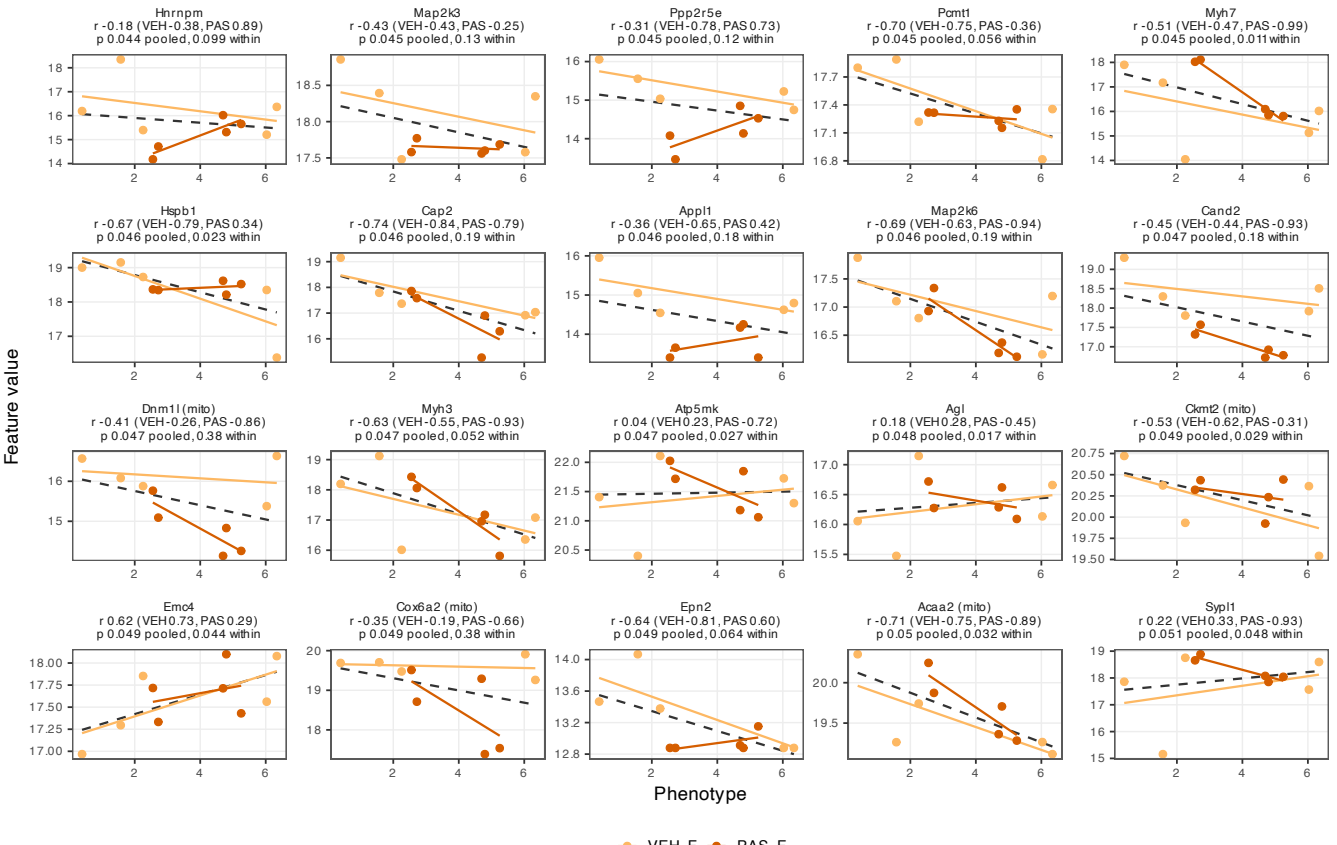
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Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



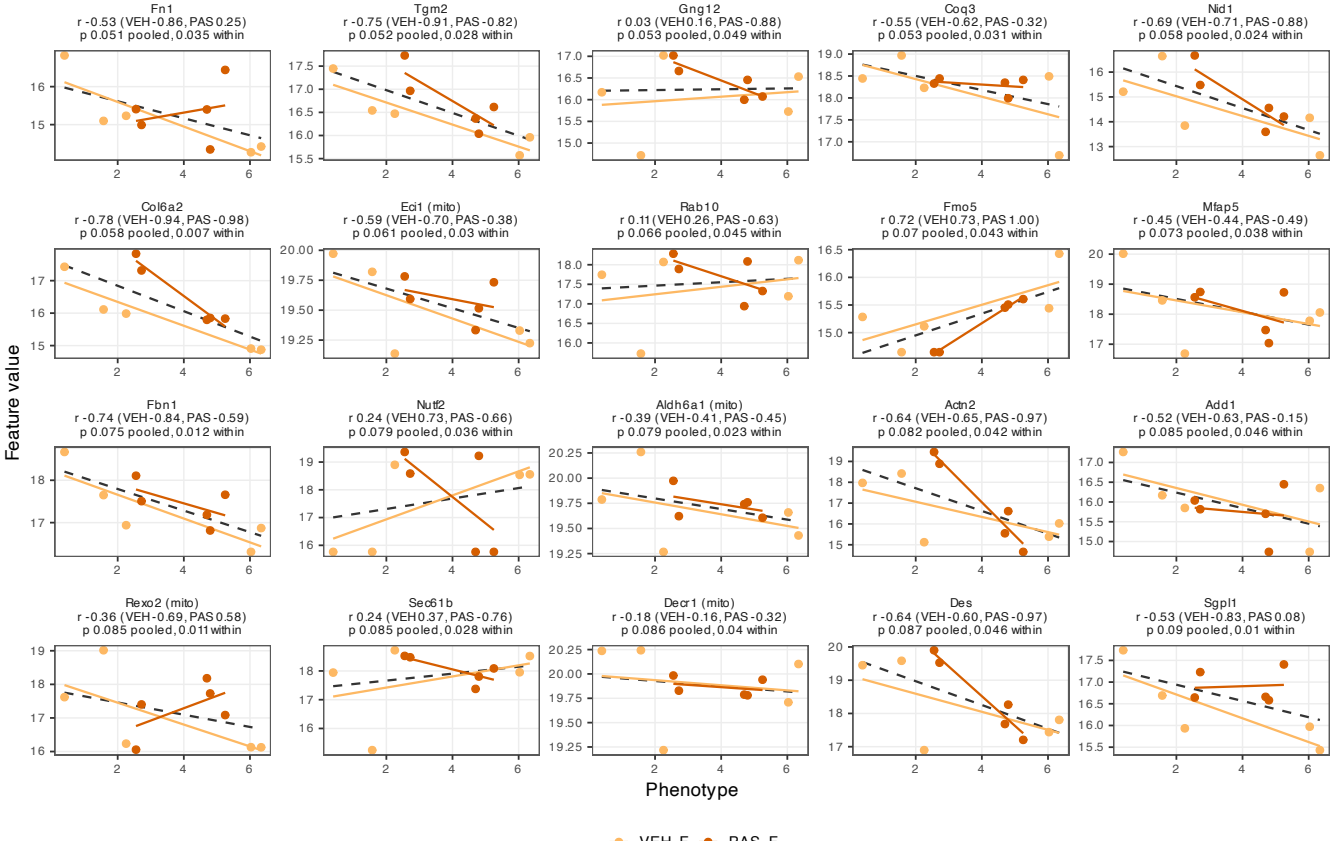
Female: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 189)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



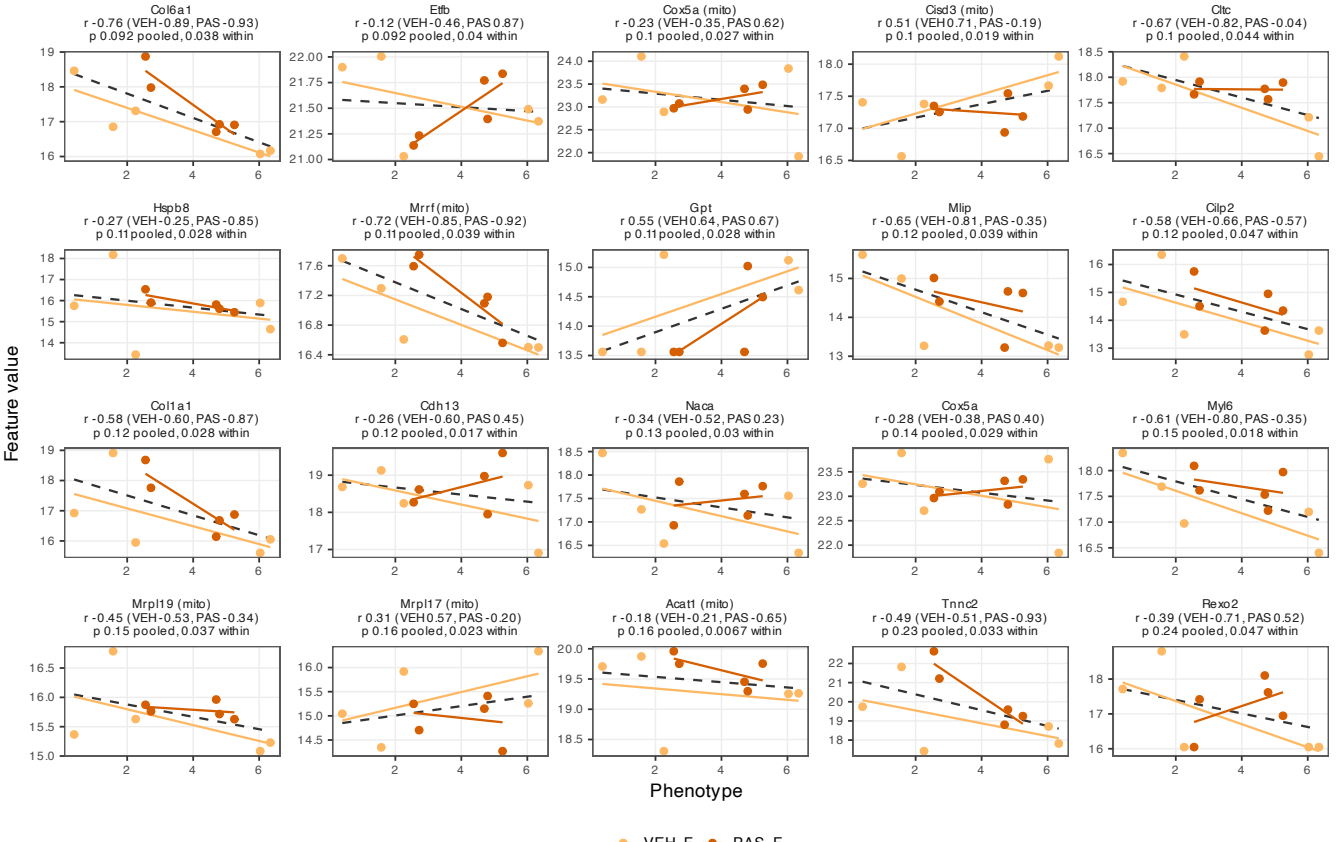
Female: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 189)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



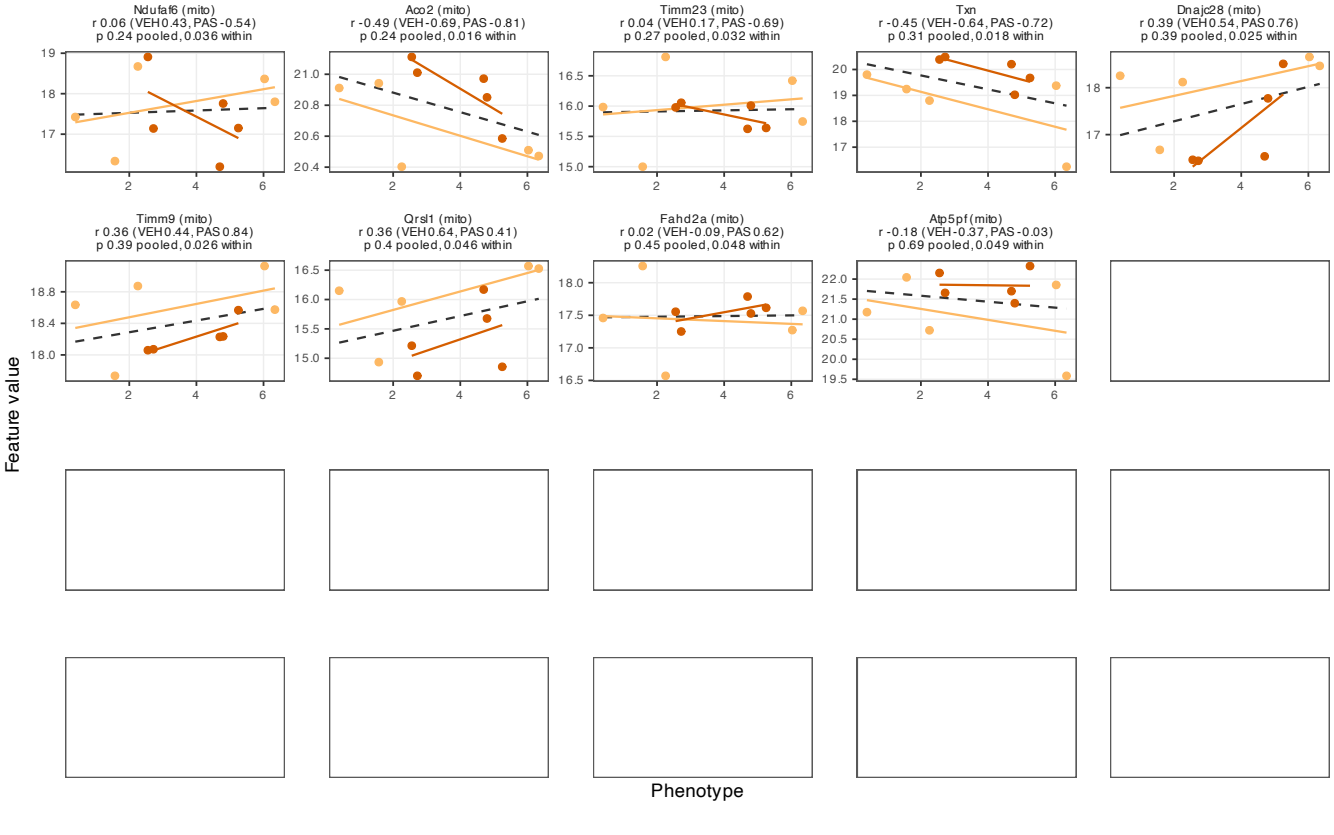
Female: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 189)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



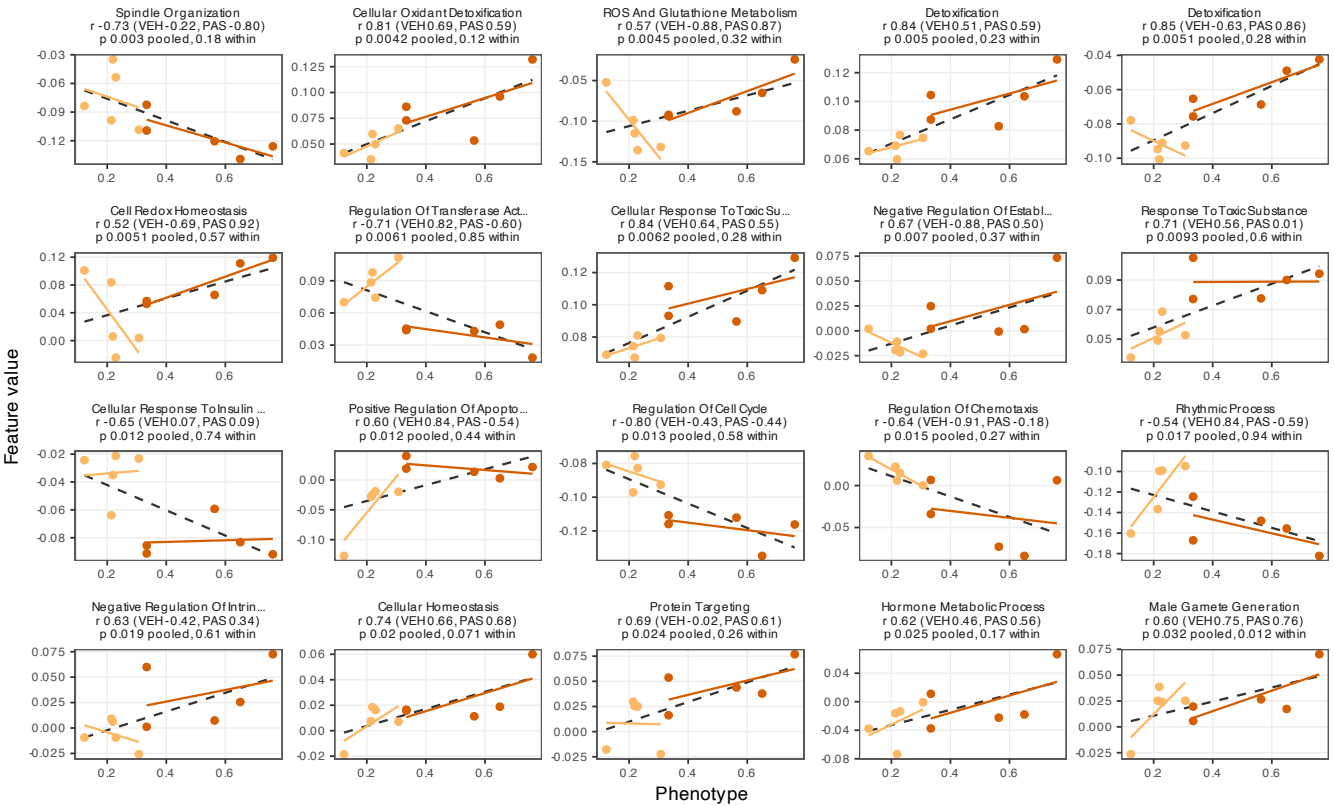
Female: Proteins associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 189)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



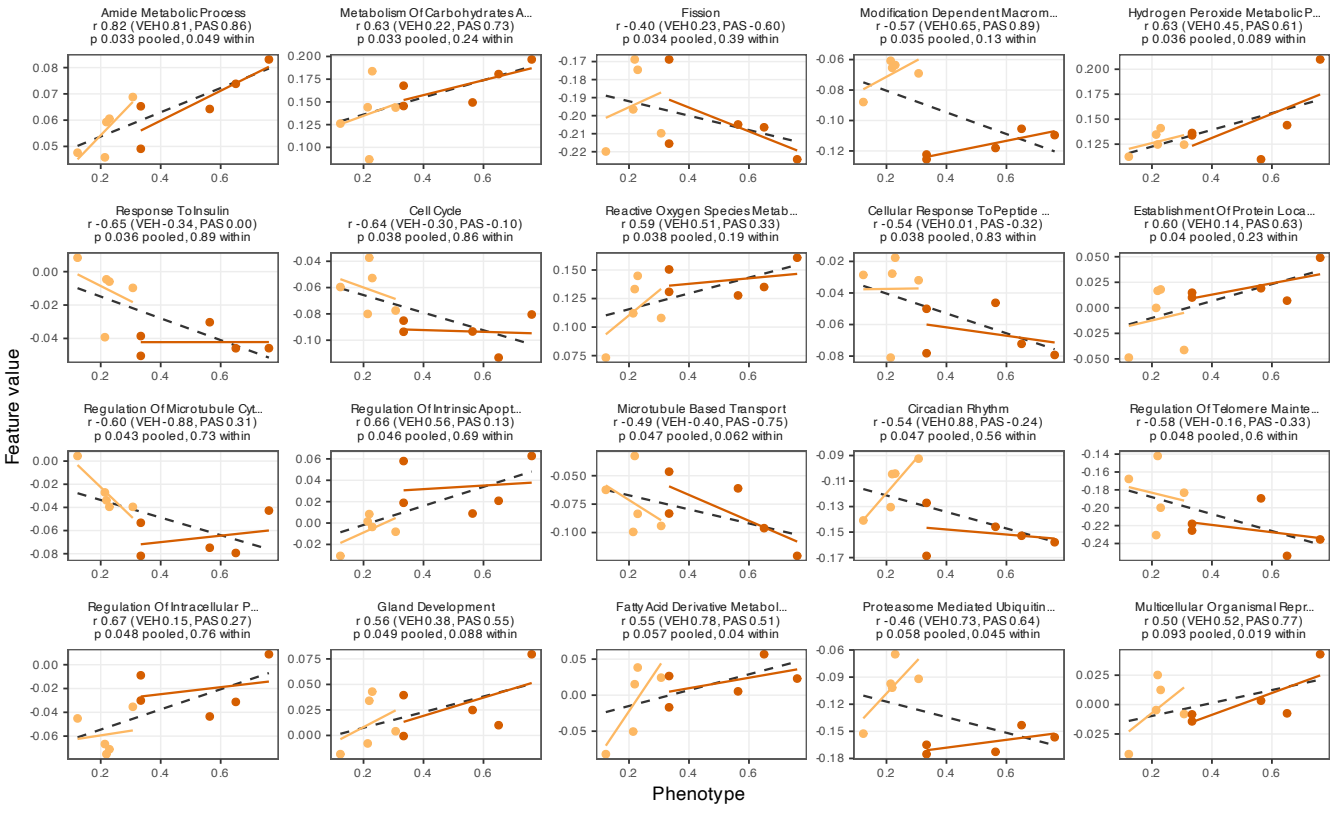
Female: Pathways associated with Capillary density, p ≤ 0.05 in either model (n = 51)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



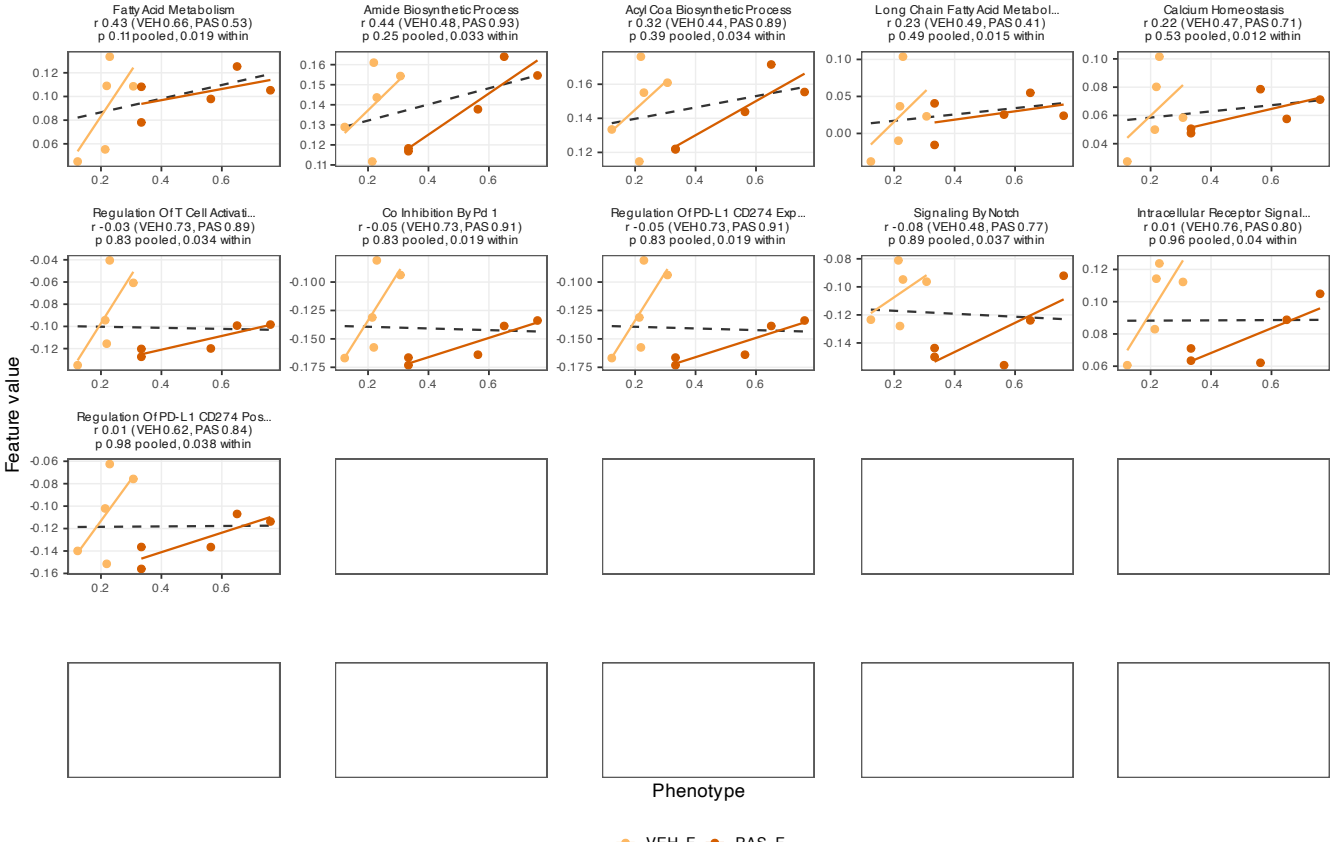
Female: Pathways associated with Capillary density, p ≤ 0.05 in either model (n = 51)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



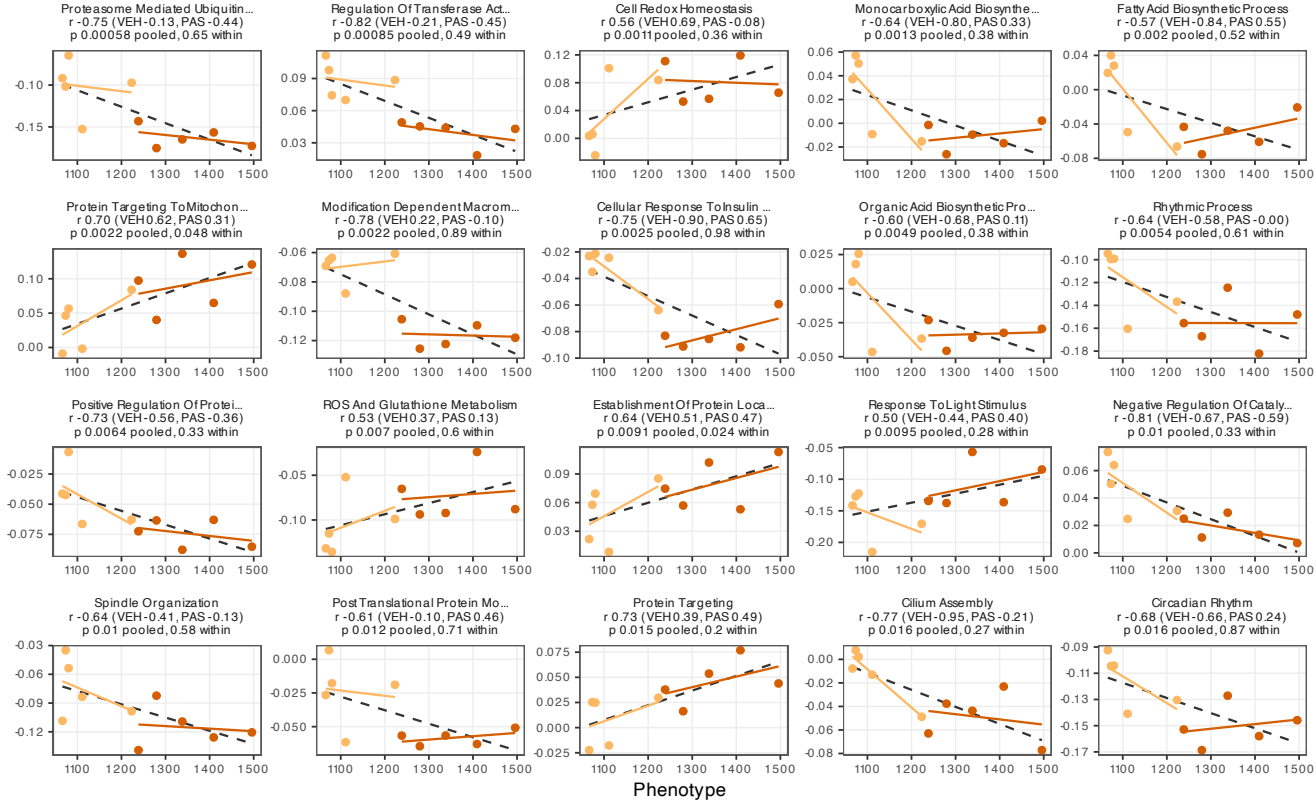
Female: Pathways associated with Capillary density, p ≤ 0.05 in either model (n = 51)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



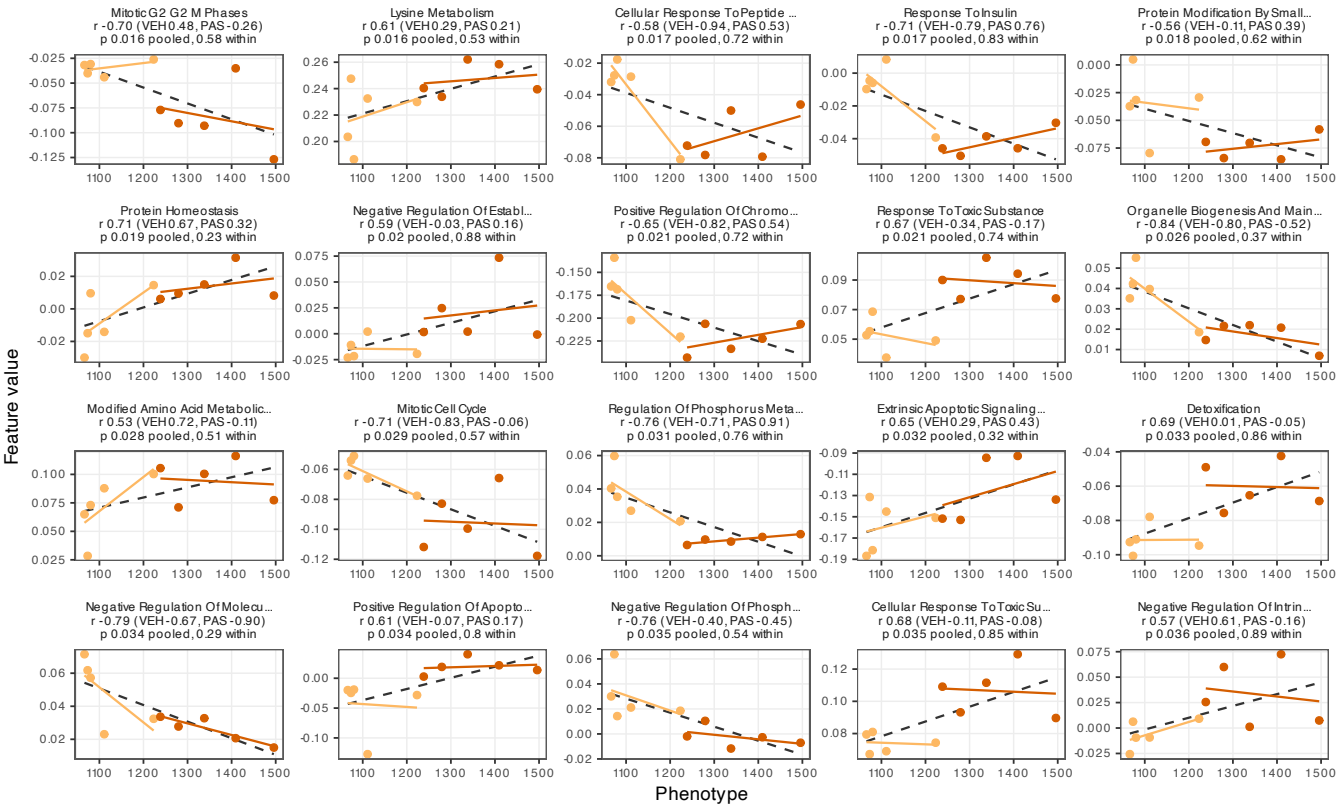
Female: Pathways associated with CSA, p ≤ 0.05 in either model (n = 59)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



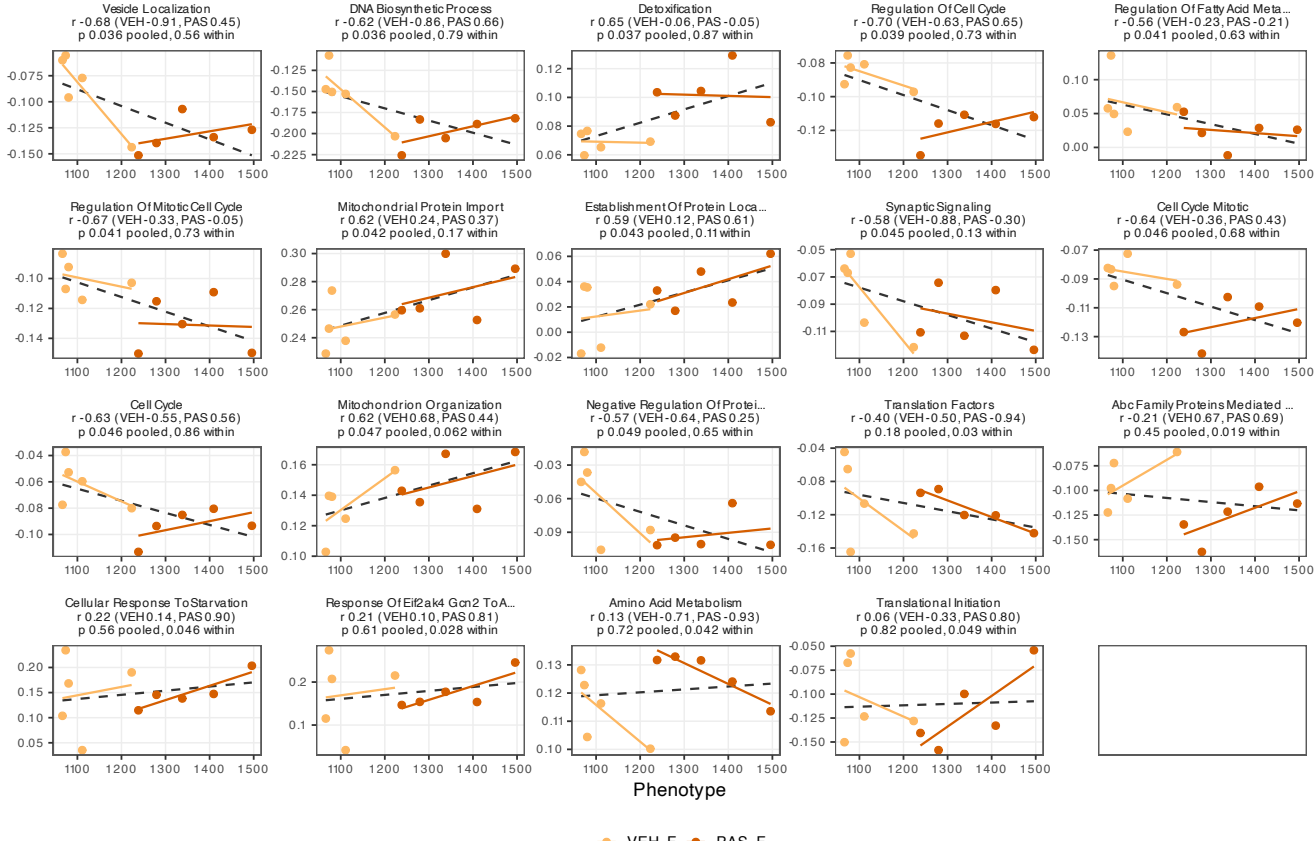
Female: Pathways associated with CSA, p ≤ 0.05 in either model (n = 59)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



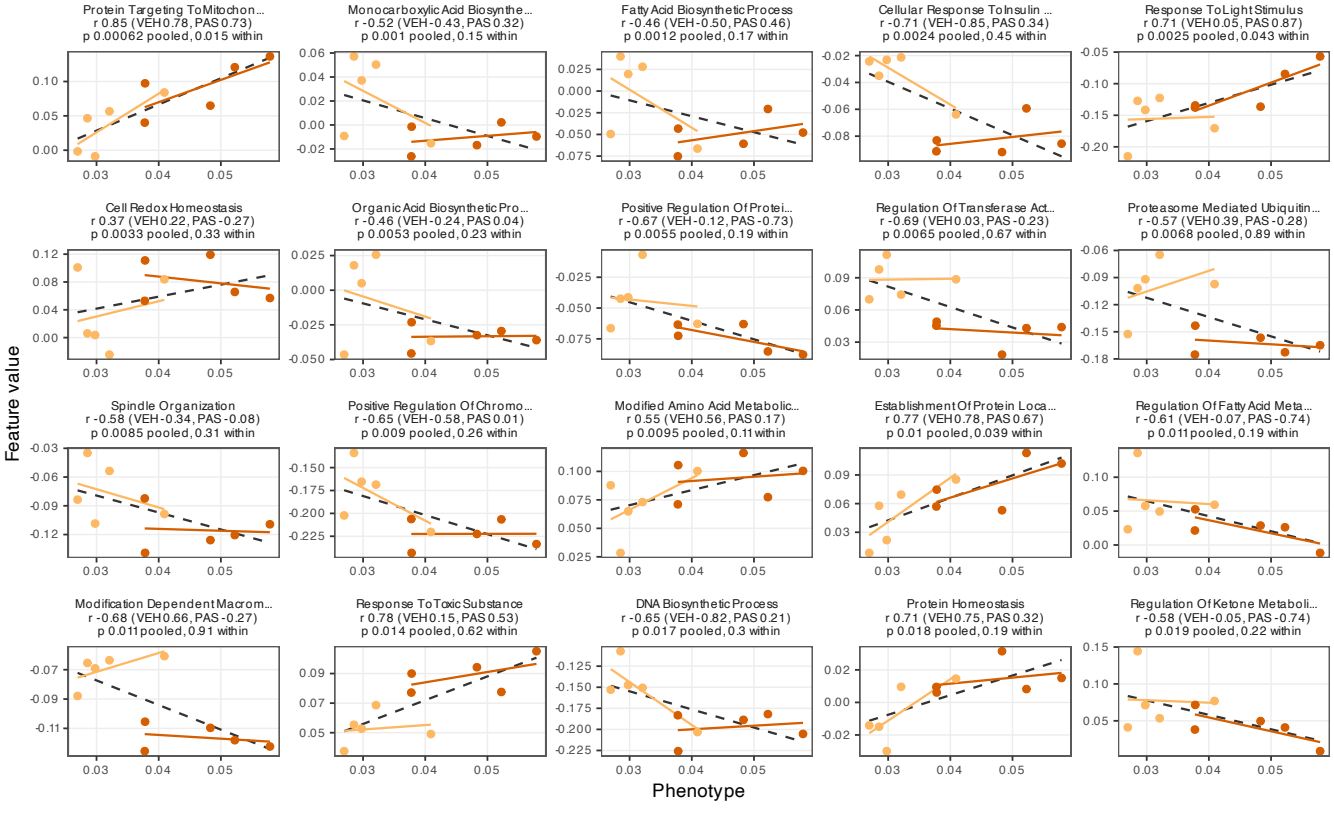
Female: Pathways associated with CSA, p ≤ 0.05 in either model (n = 59)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



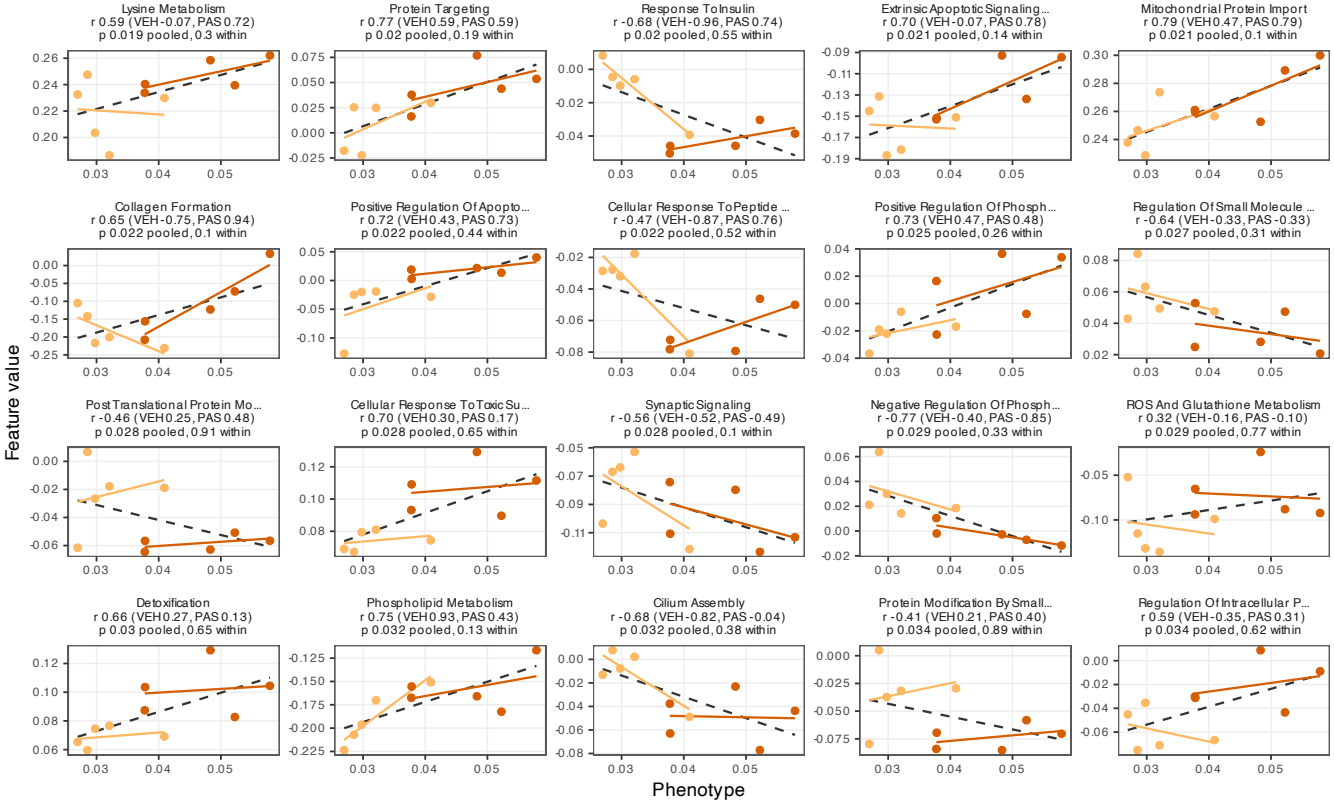
Female: Pathways associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 51)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



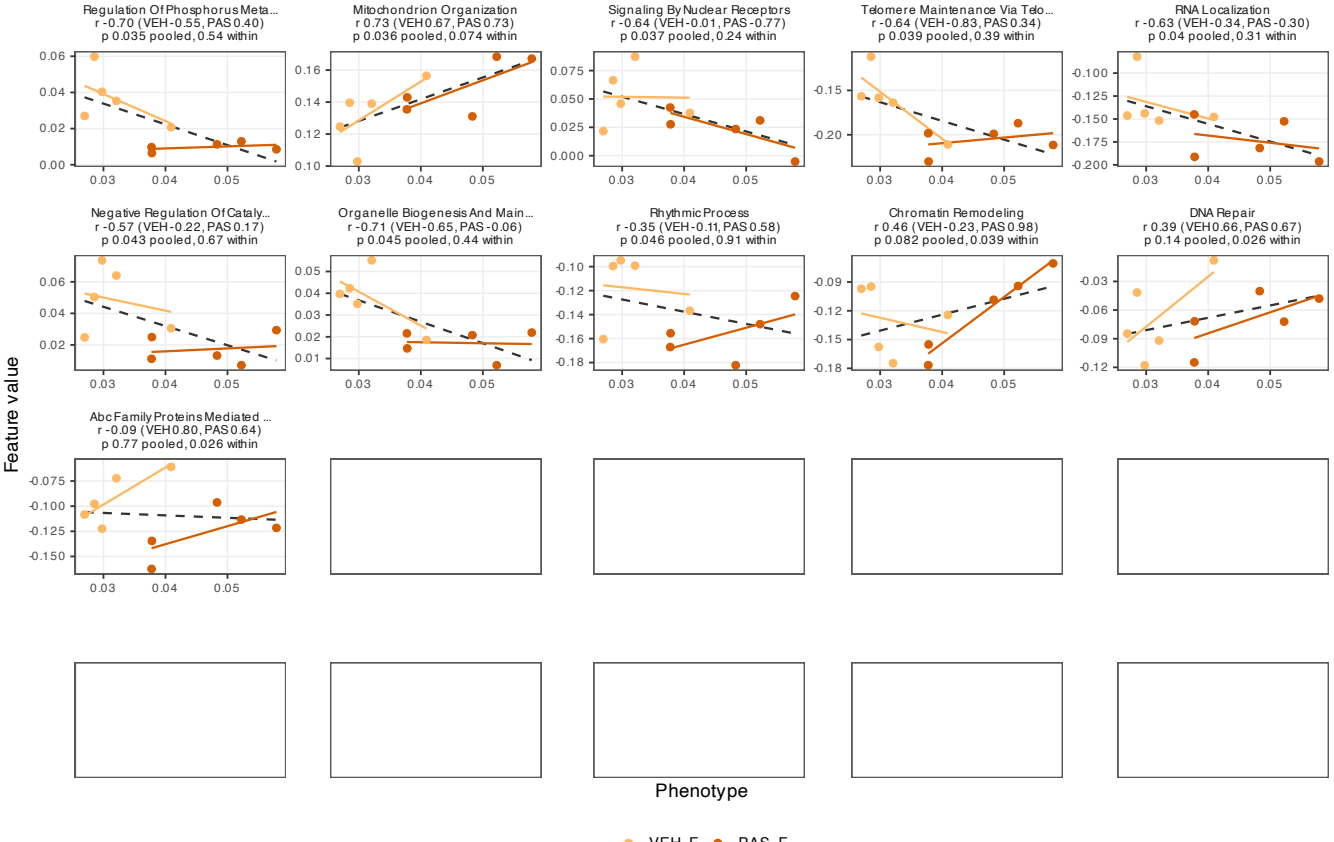
Female: Pathways associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 51)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



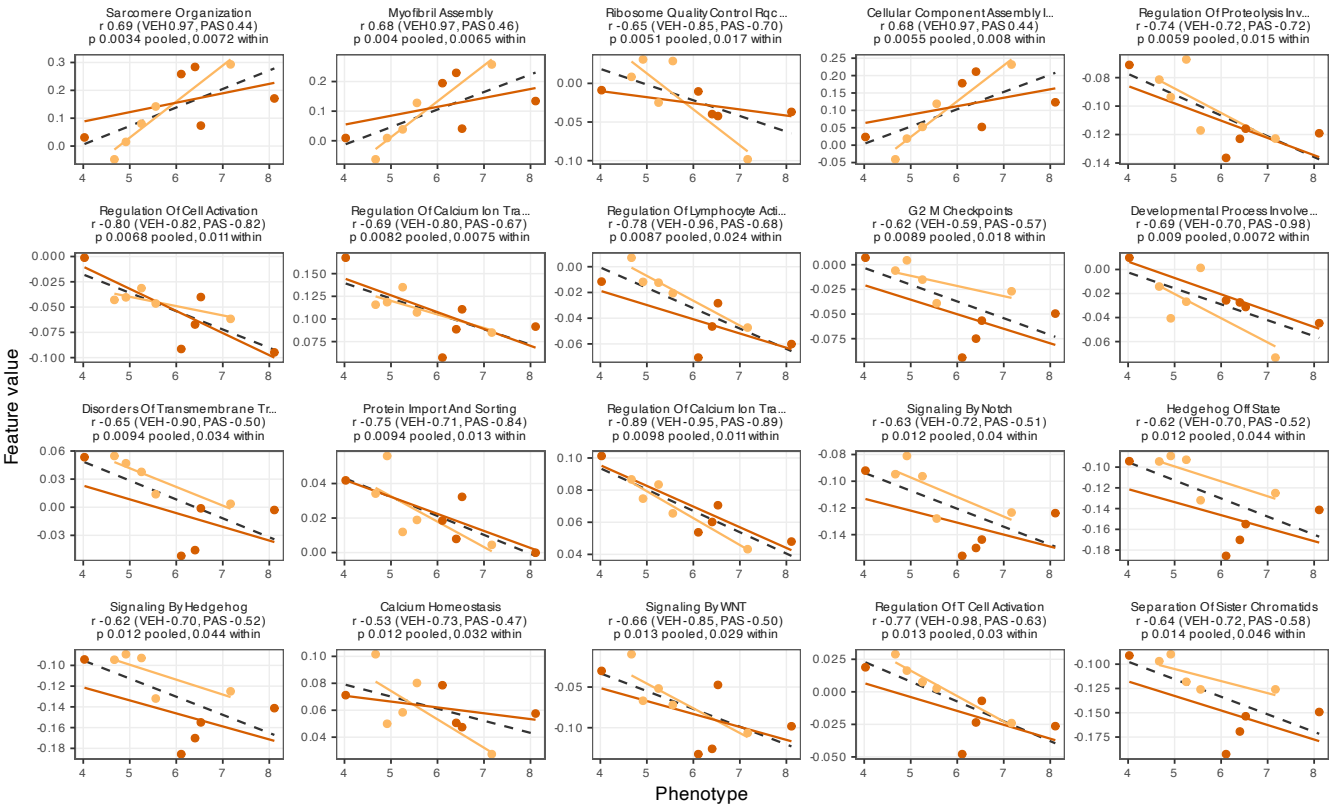
Female: Pathways associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 51)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



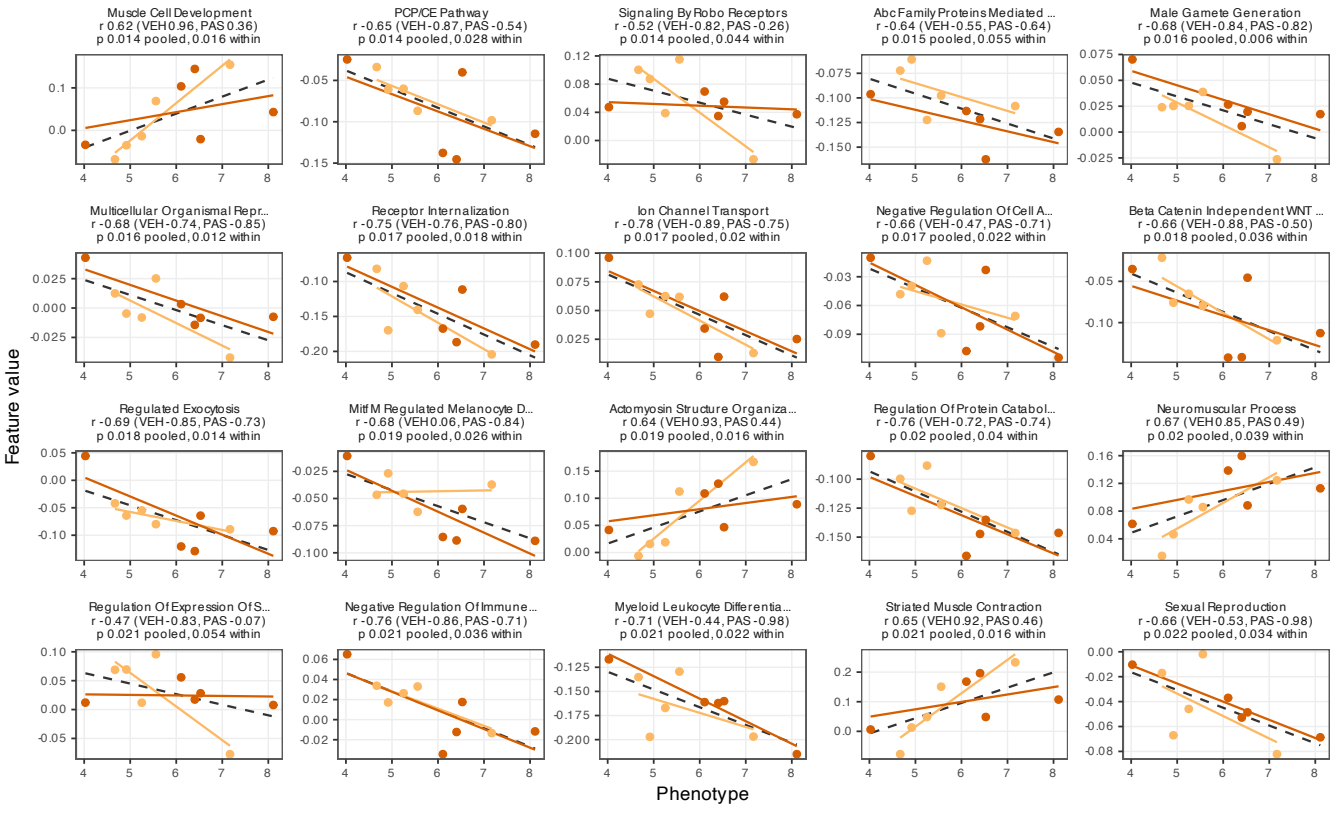
Female: Pathways associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 102)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



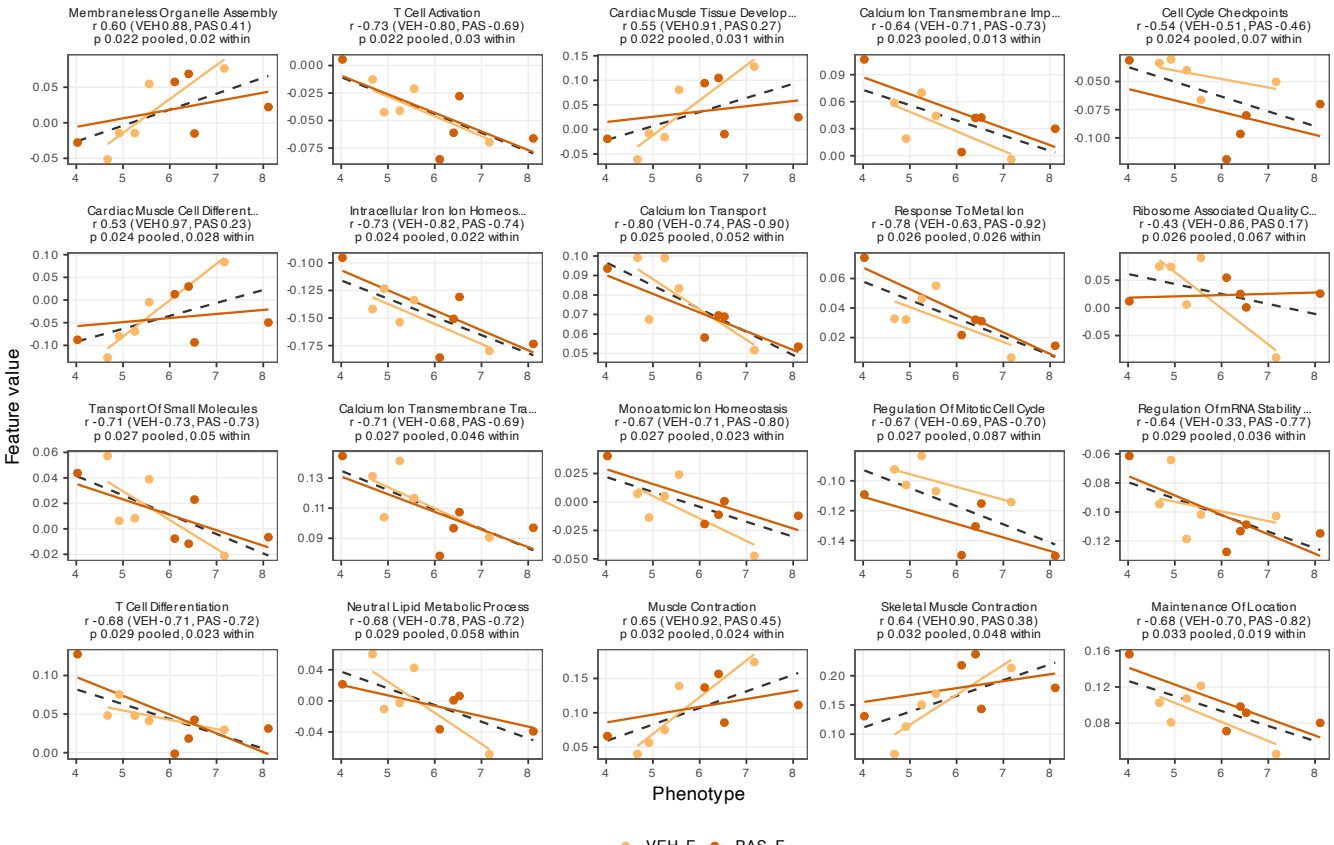
Female: Pathways associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 102)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



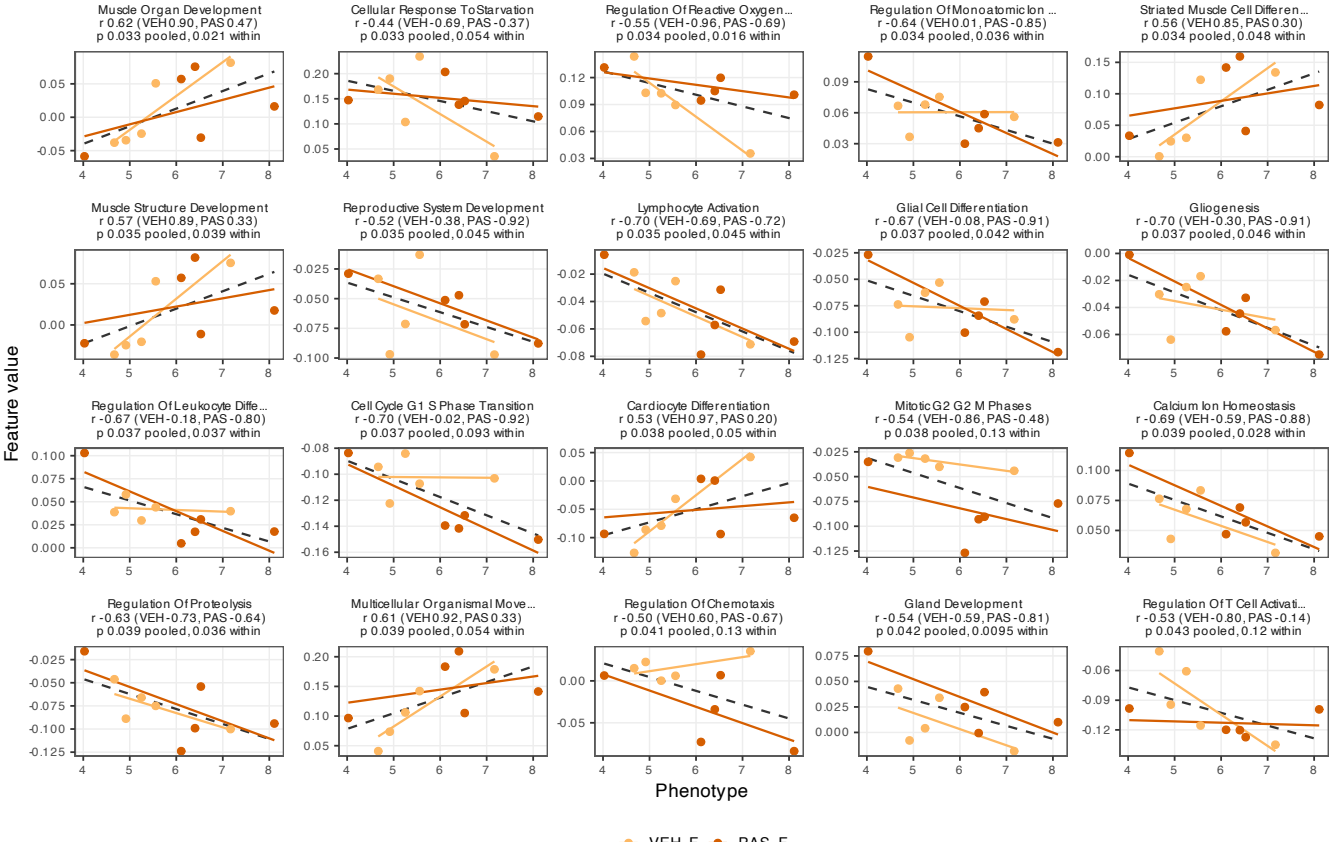
Female: Pathways associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 102)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



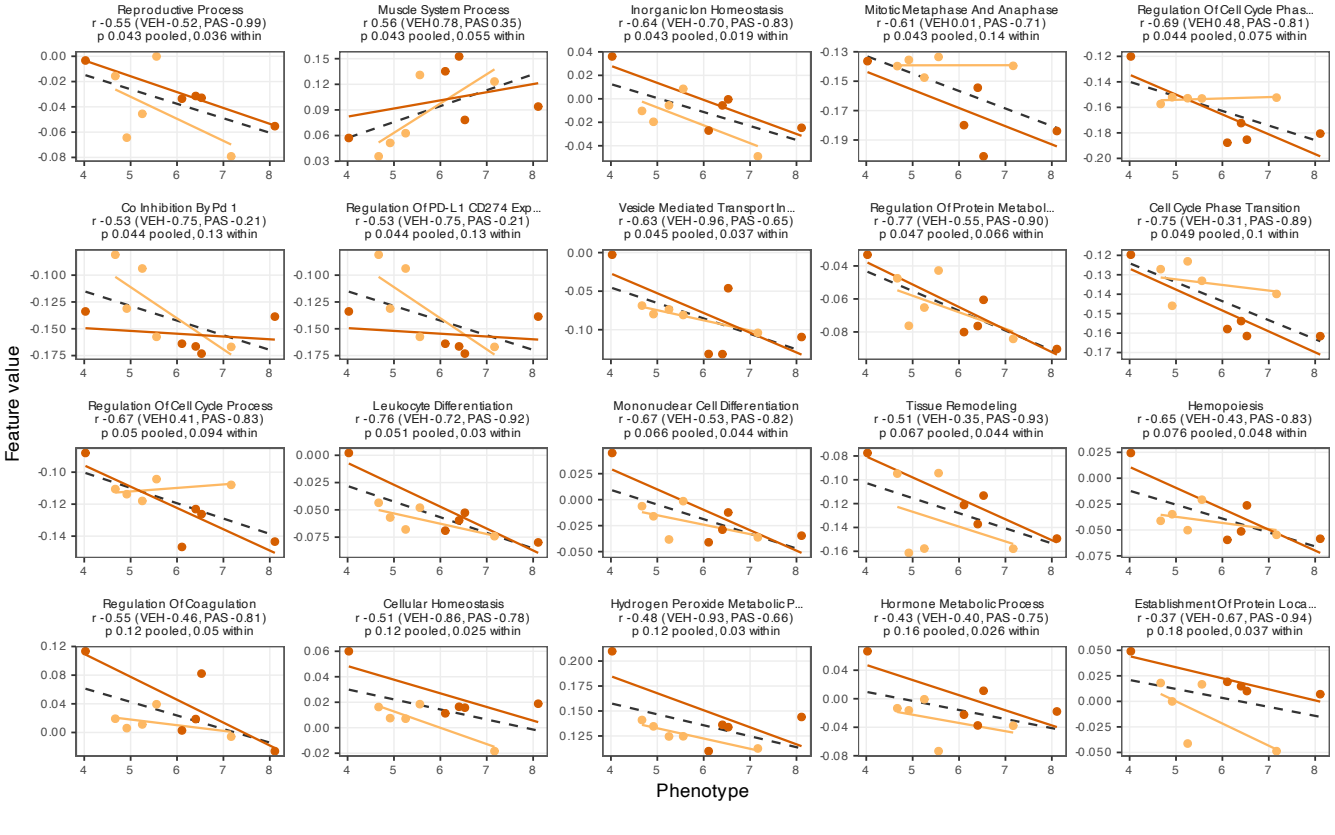
Female: Pathways associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 102)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



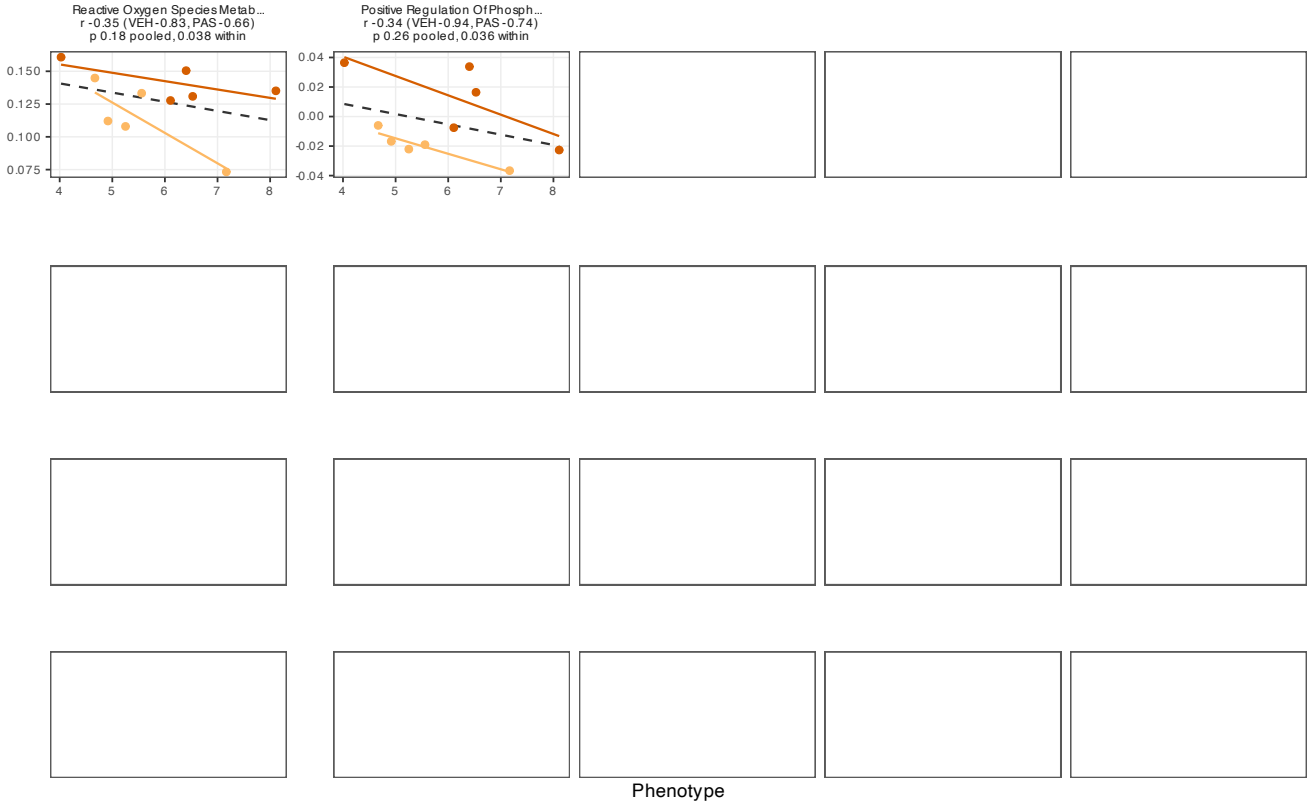
Female: Pathways associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 102)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



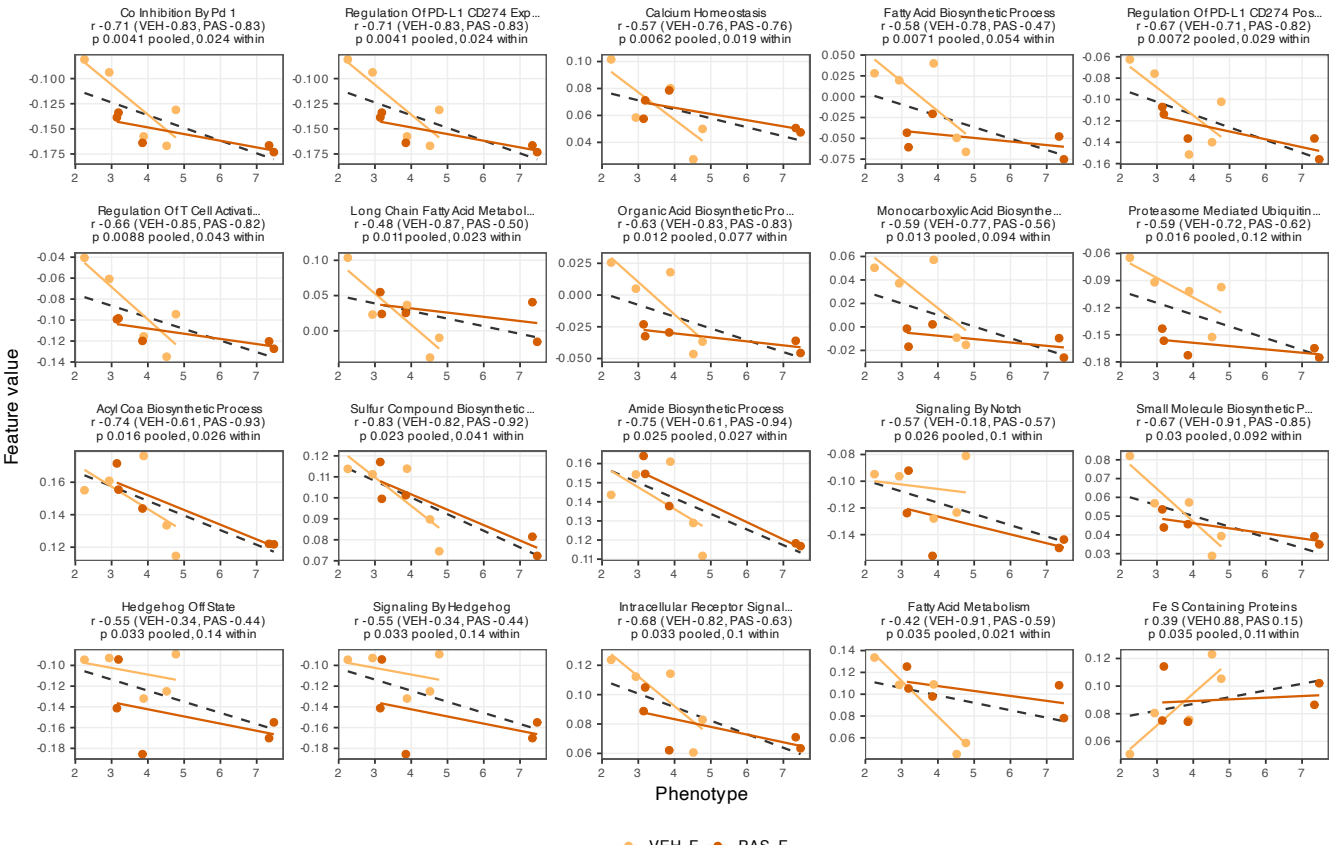
Female: Pathways associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 102)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



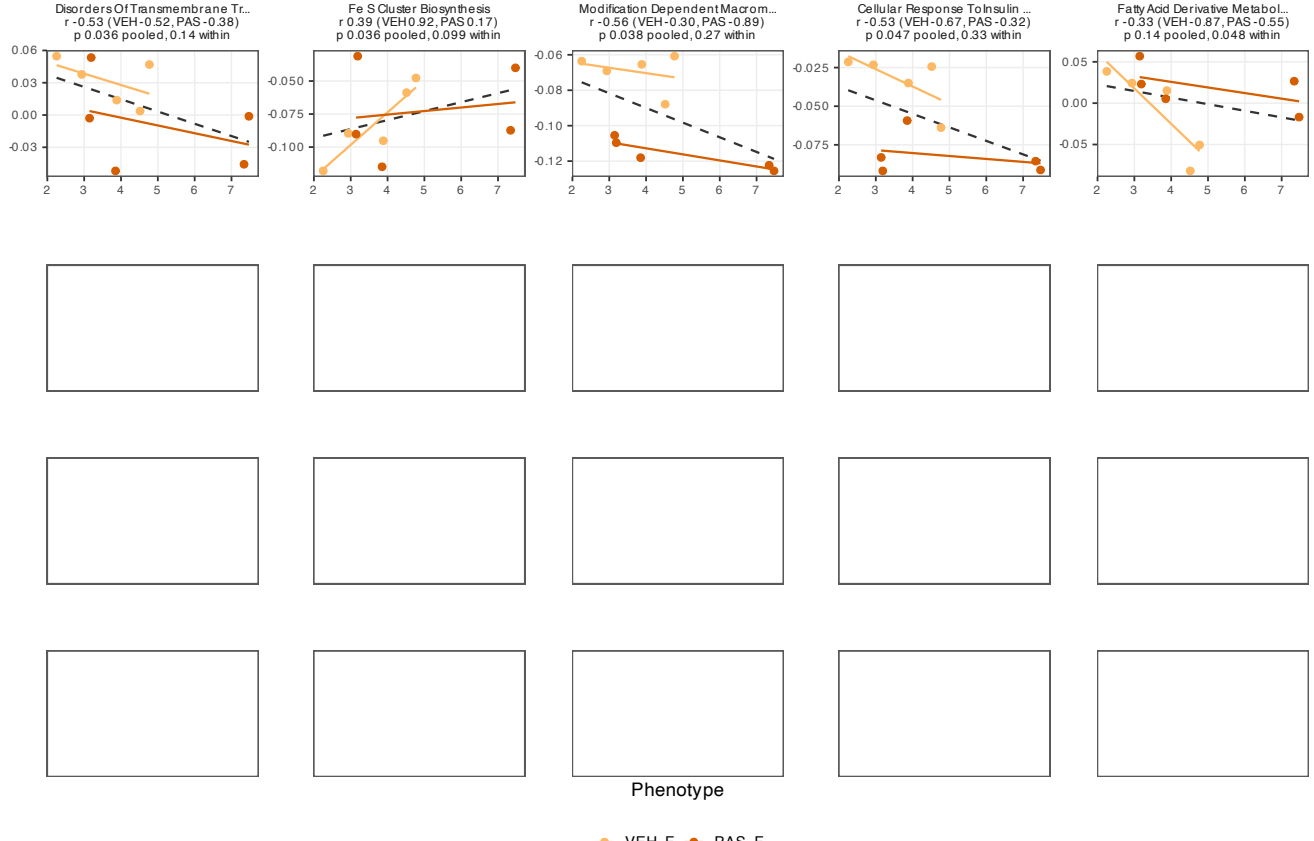
Female: Pathways associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 25)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



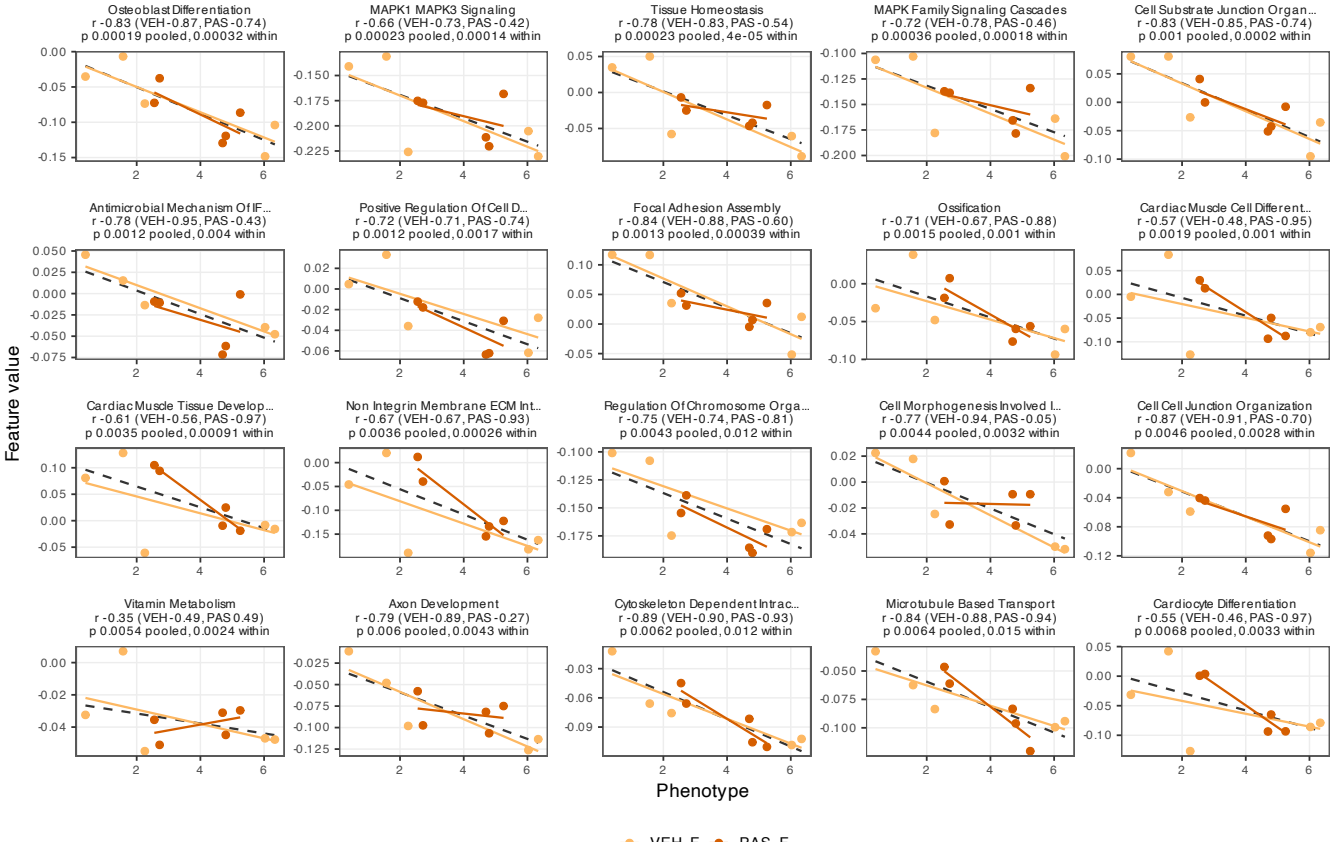
Female: Pathways associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 25)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



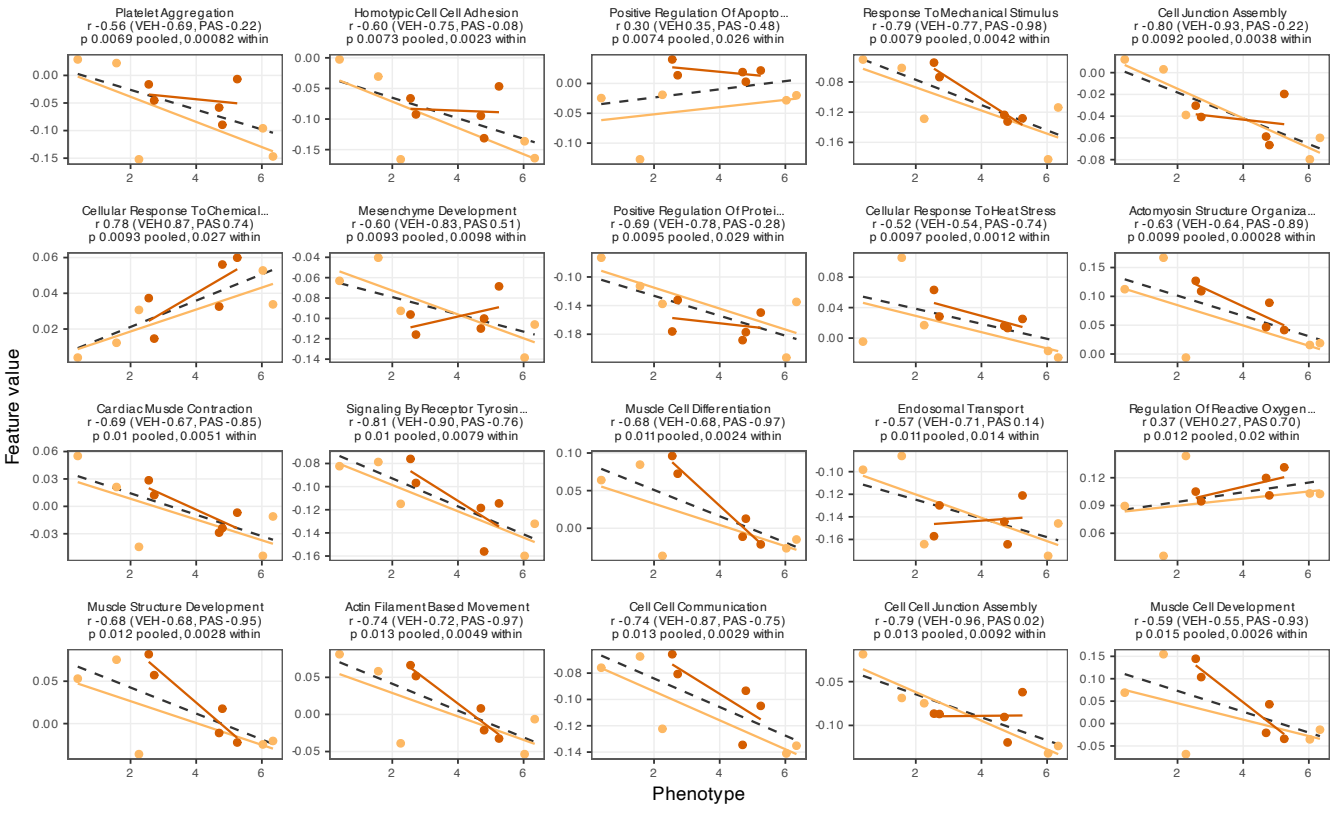
Female: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 149)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



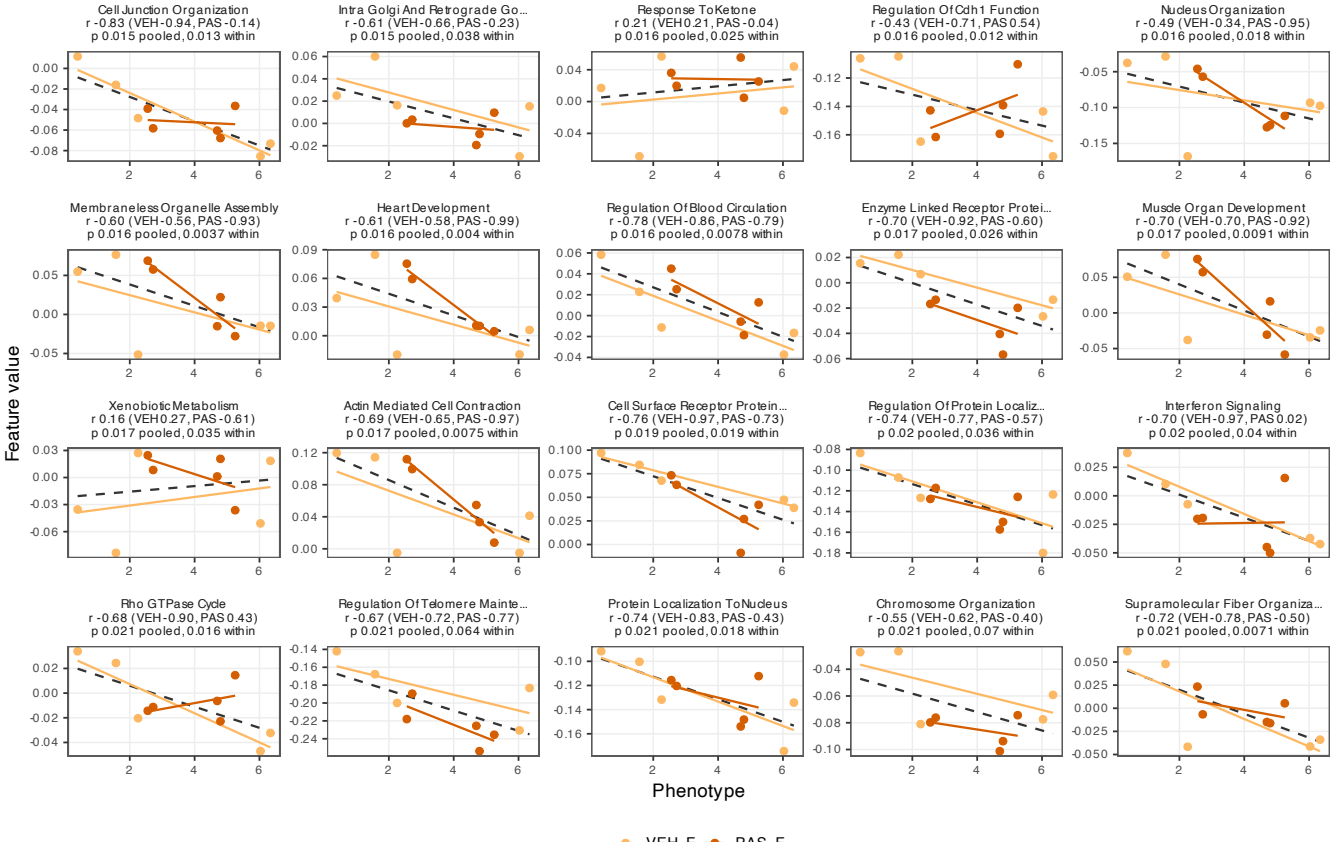
Female: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 149)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



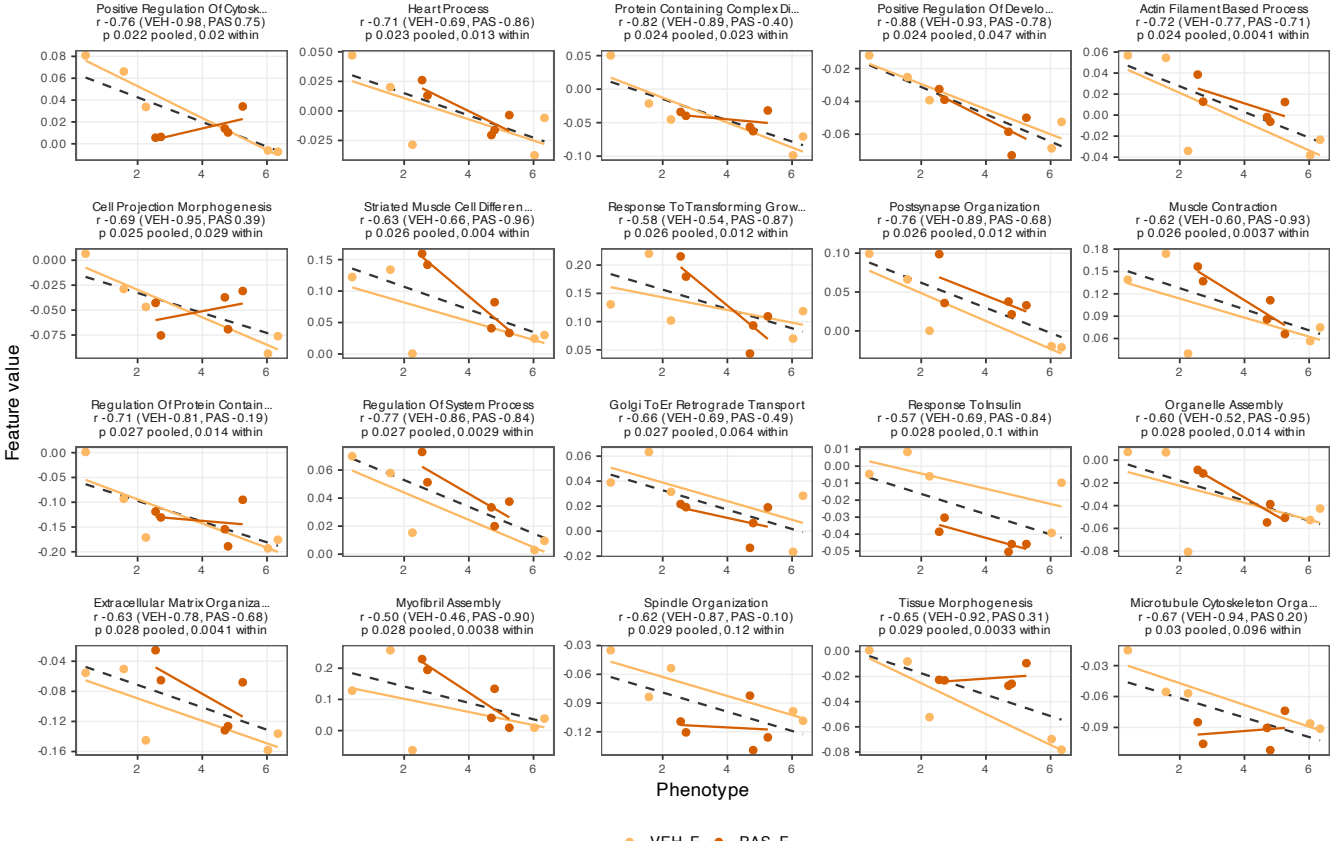
Female: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 149)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



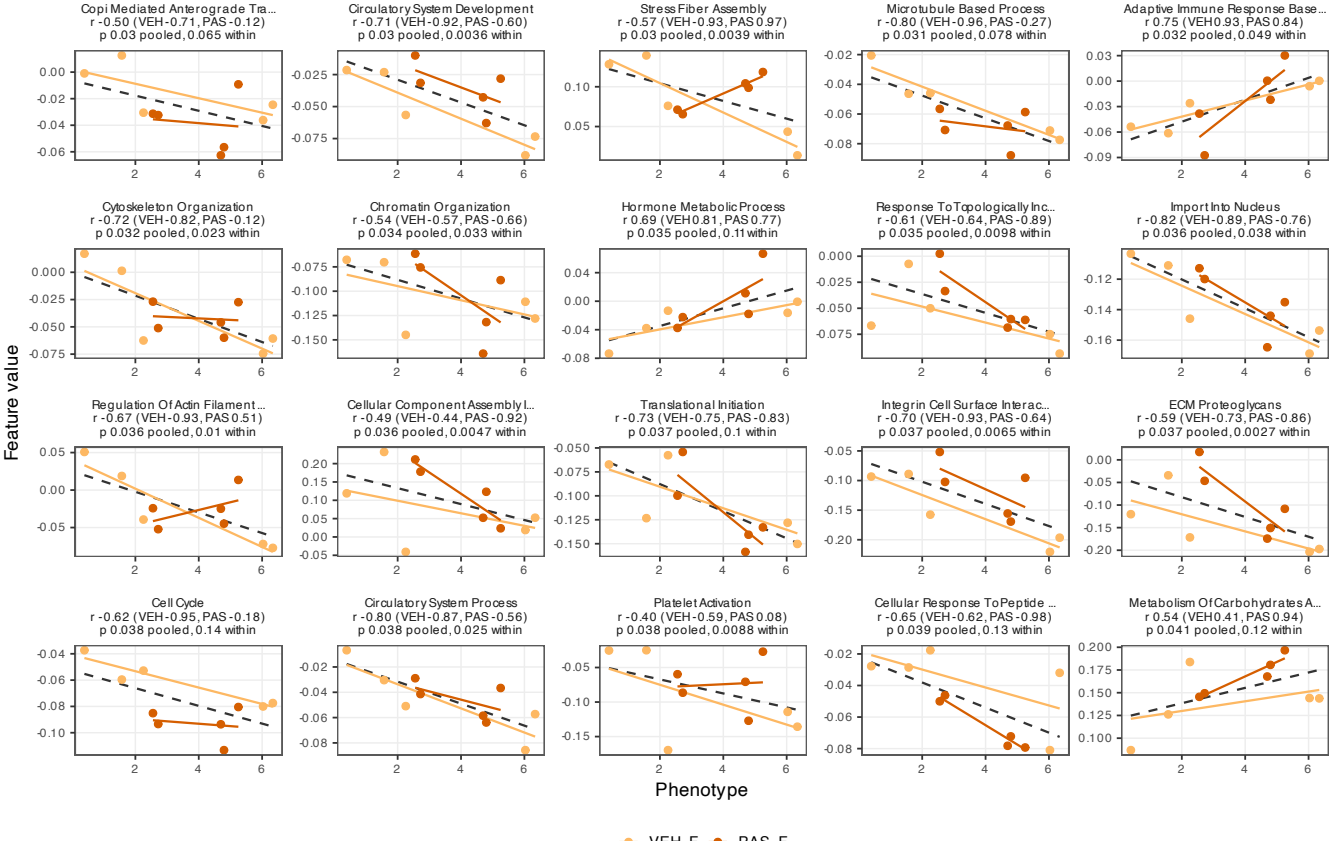
Female: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 149)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



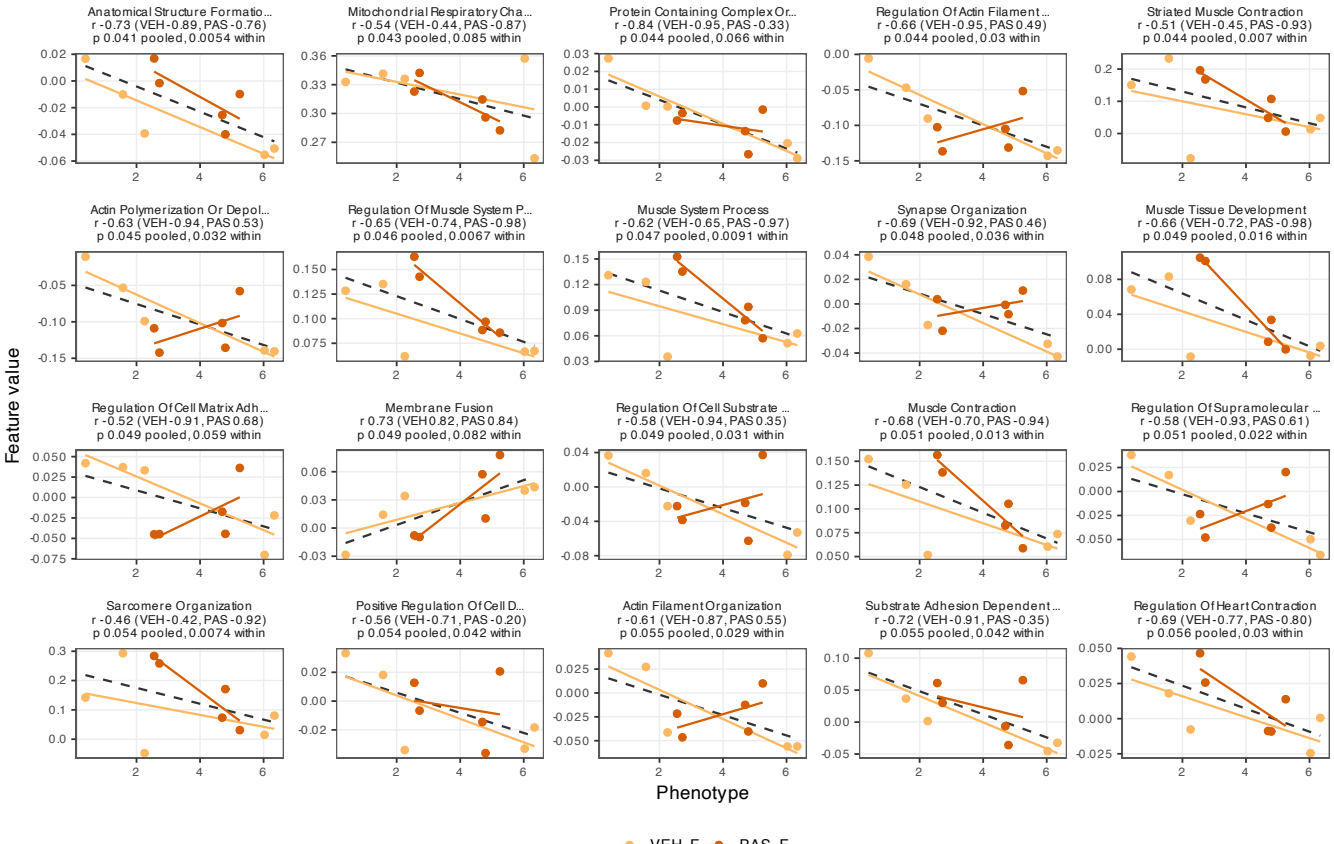
Female: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 149)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



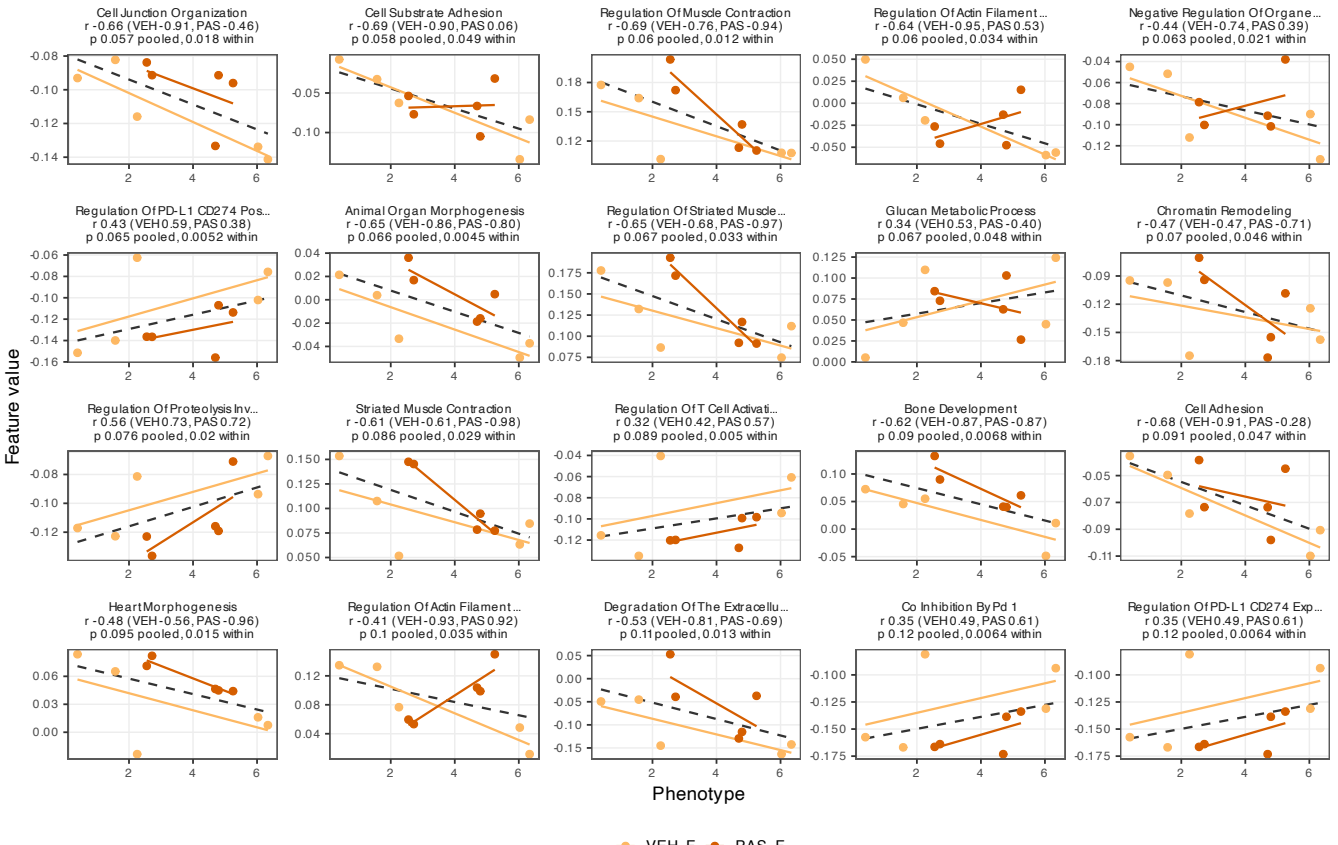
Female: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 149)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



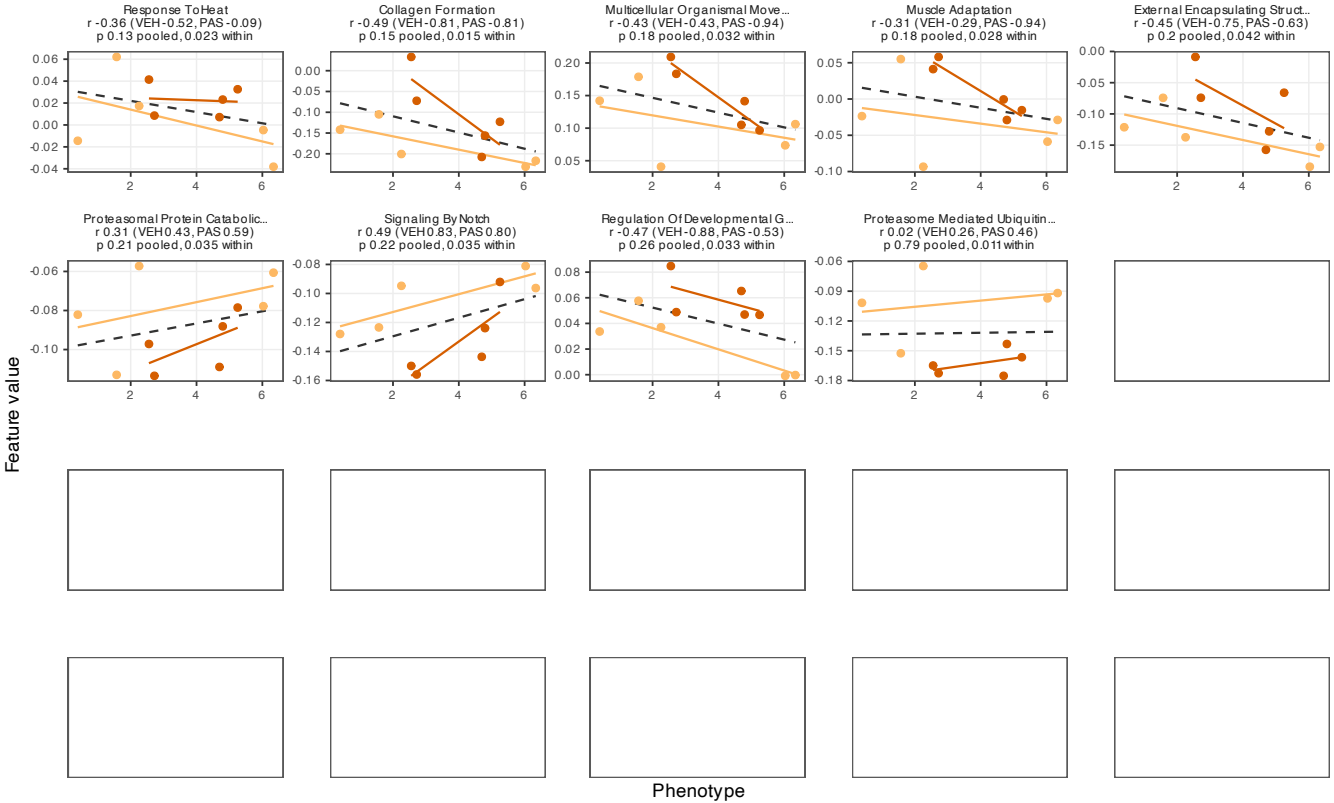
Female: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 149)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



Female: Pathways associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 149)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



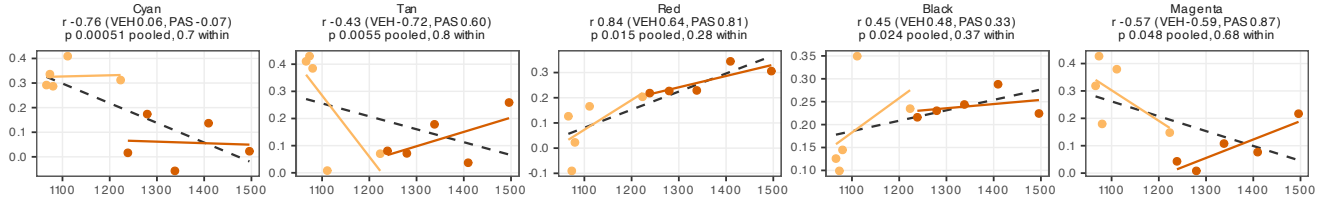
Female: Modules associated with Capillary density, p ≤ 0.05 in either model (n = 1)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



Female: Modules associated with CSA, p ≤ 0.05 in either model (n = 5)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



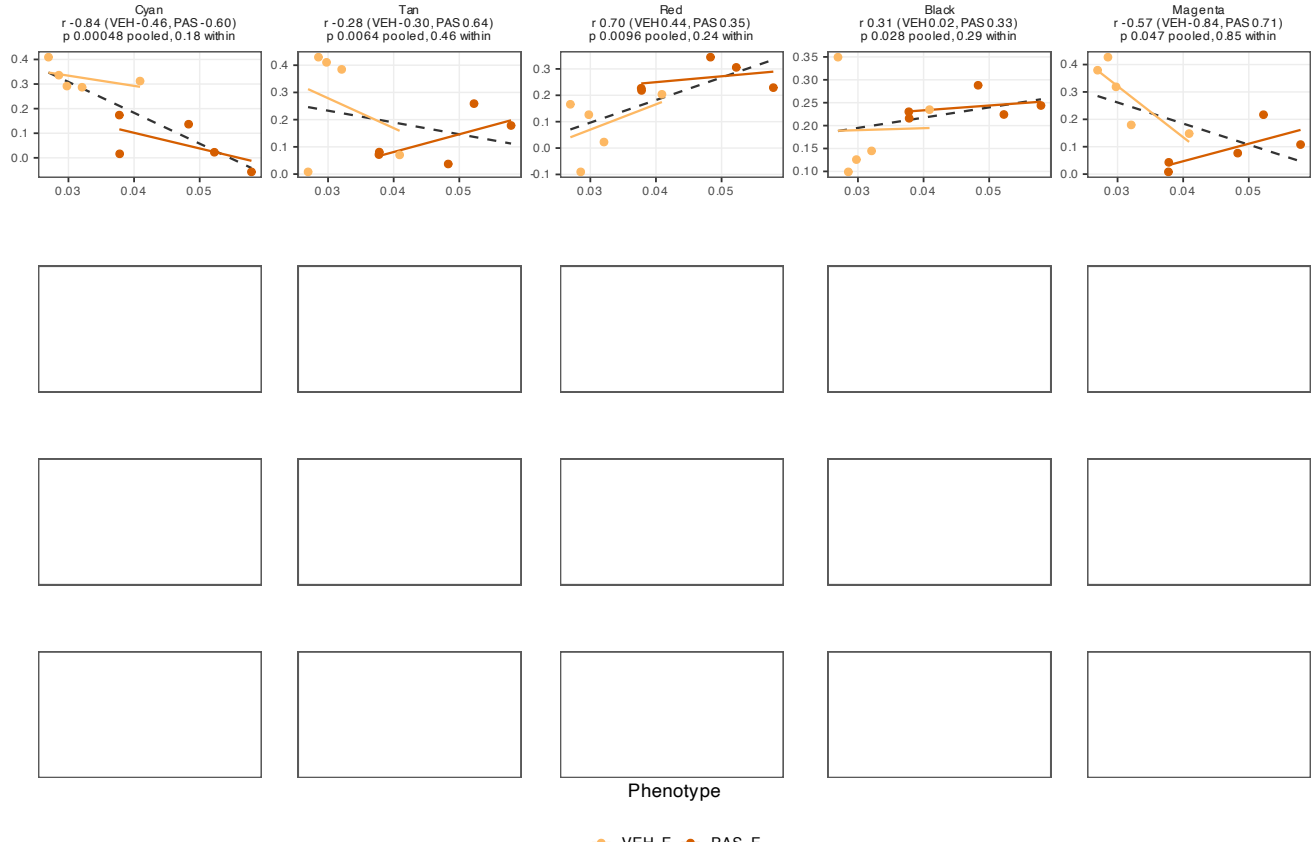
Feature value

Phenotype

VEH_F PAS_F

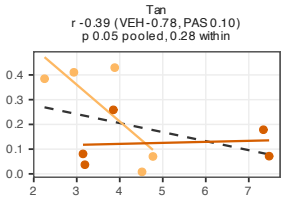
Female: Modules associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 5)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



Female: Modules associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 1)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



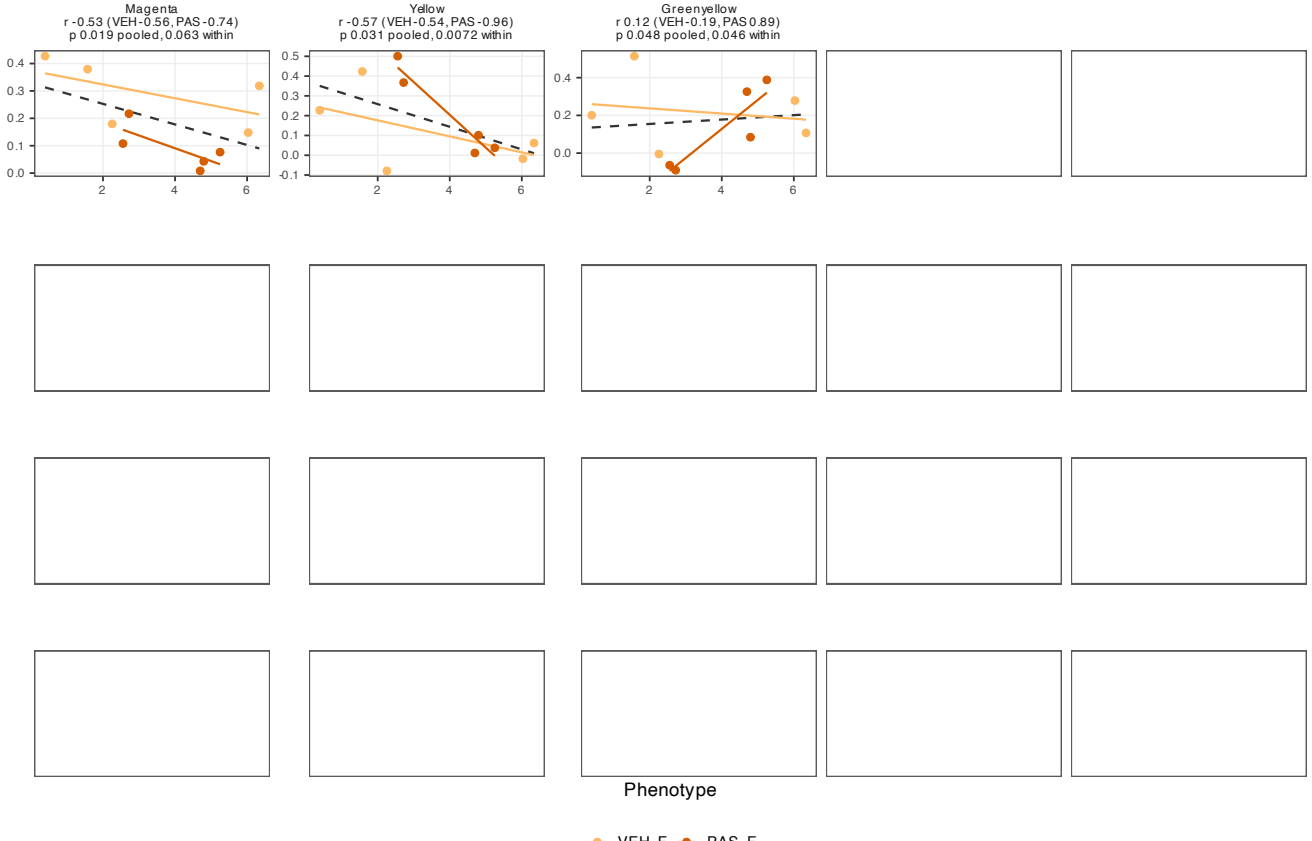
Feature value

Phenotype

VEH_F PAS_F

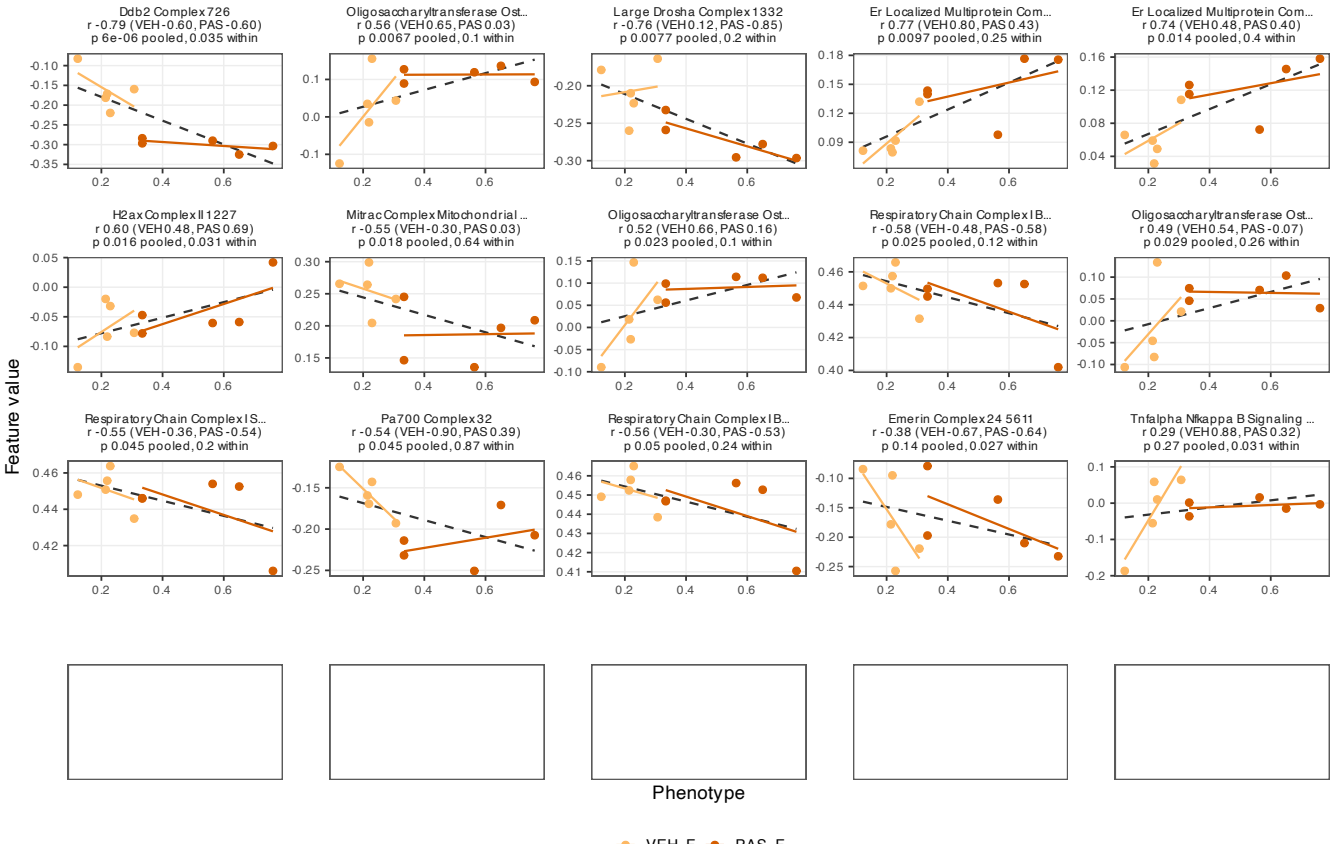
Female: Modules associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 3)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



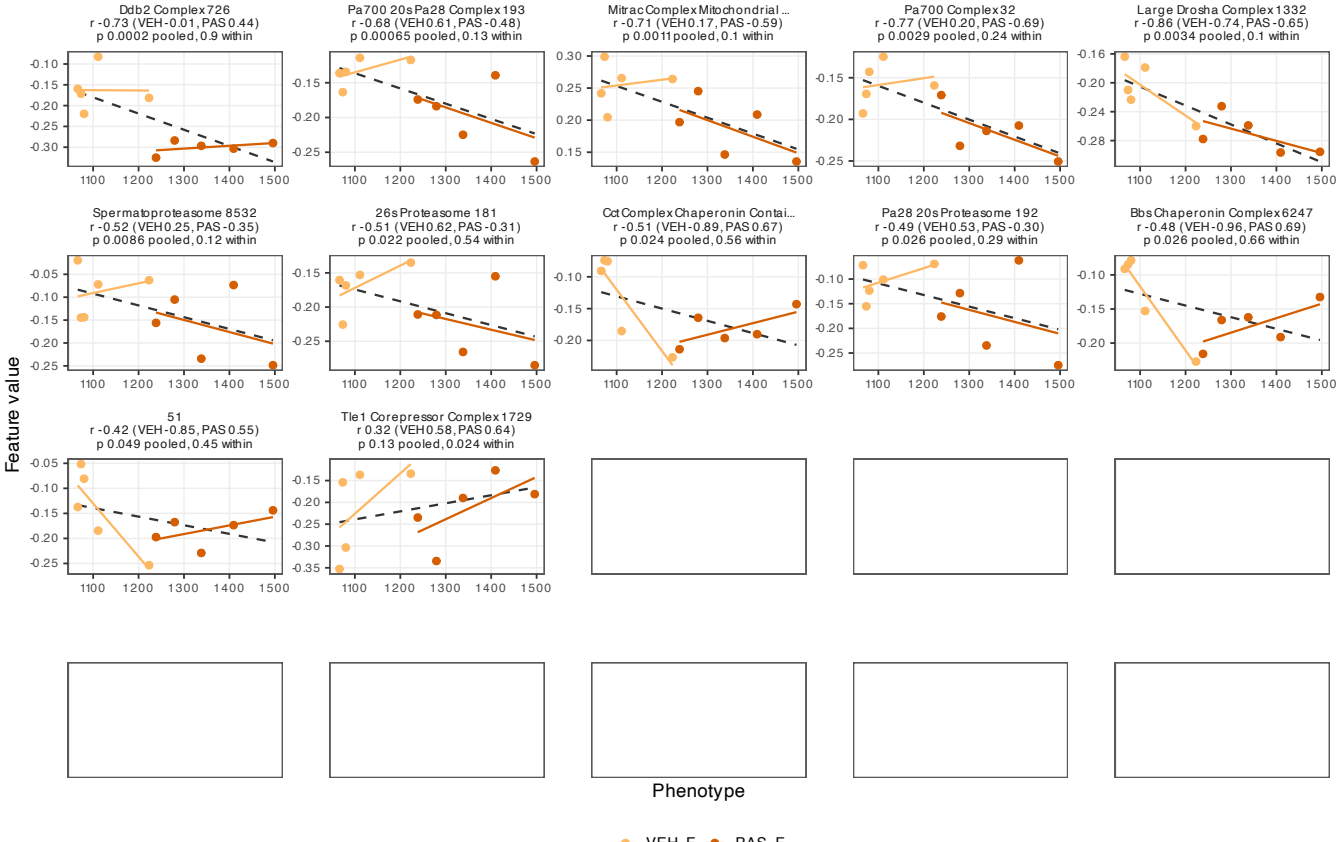
Female: Complexes associated with Capillary density, p ≤ 0.05 in either model (n = 15)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



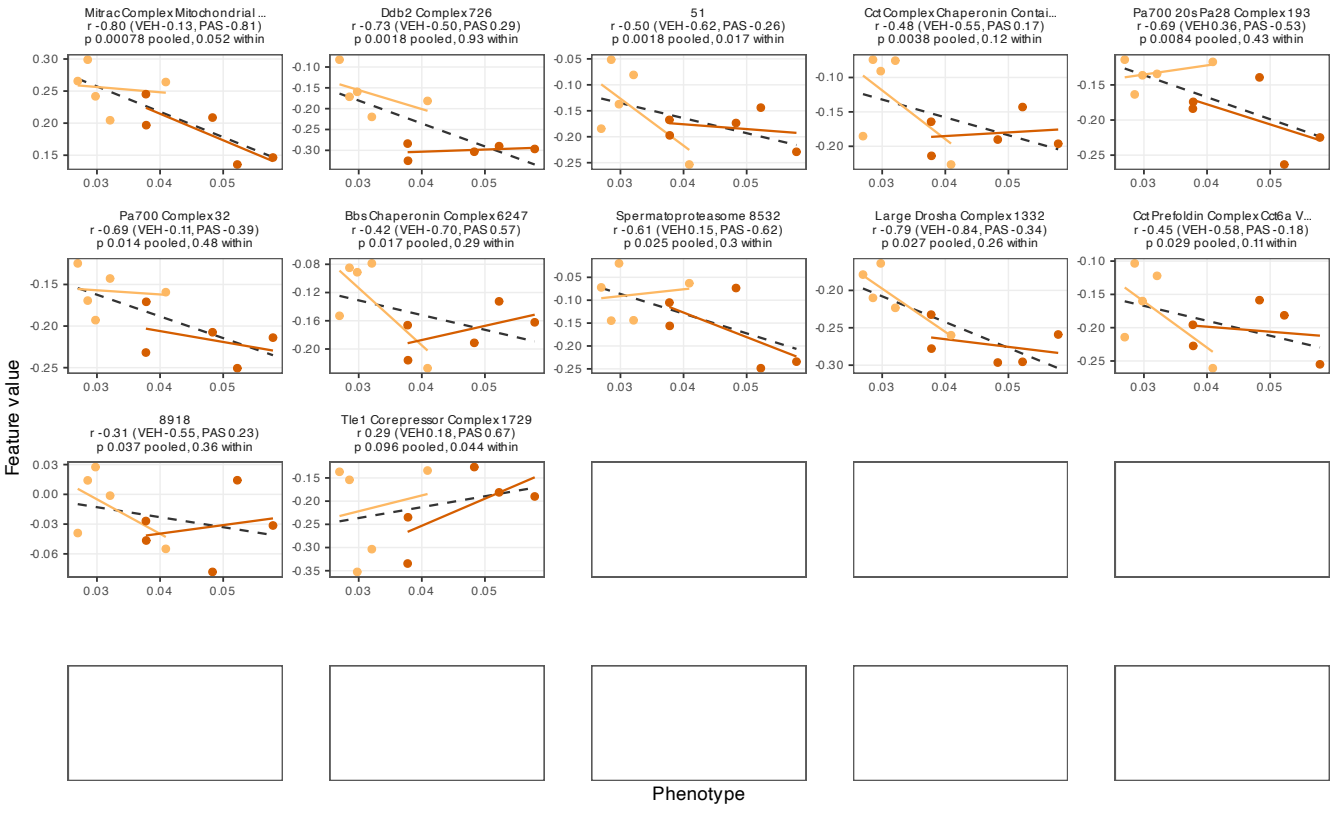
Female: Complexes associated with CSA, p ≤ 0.05 in either model (n = 12)

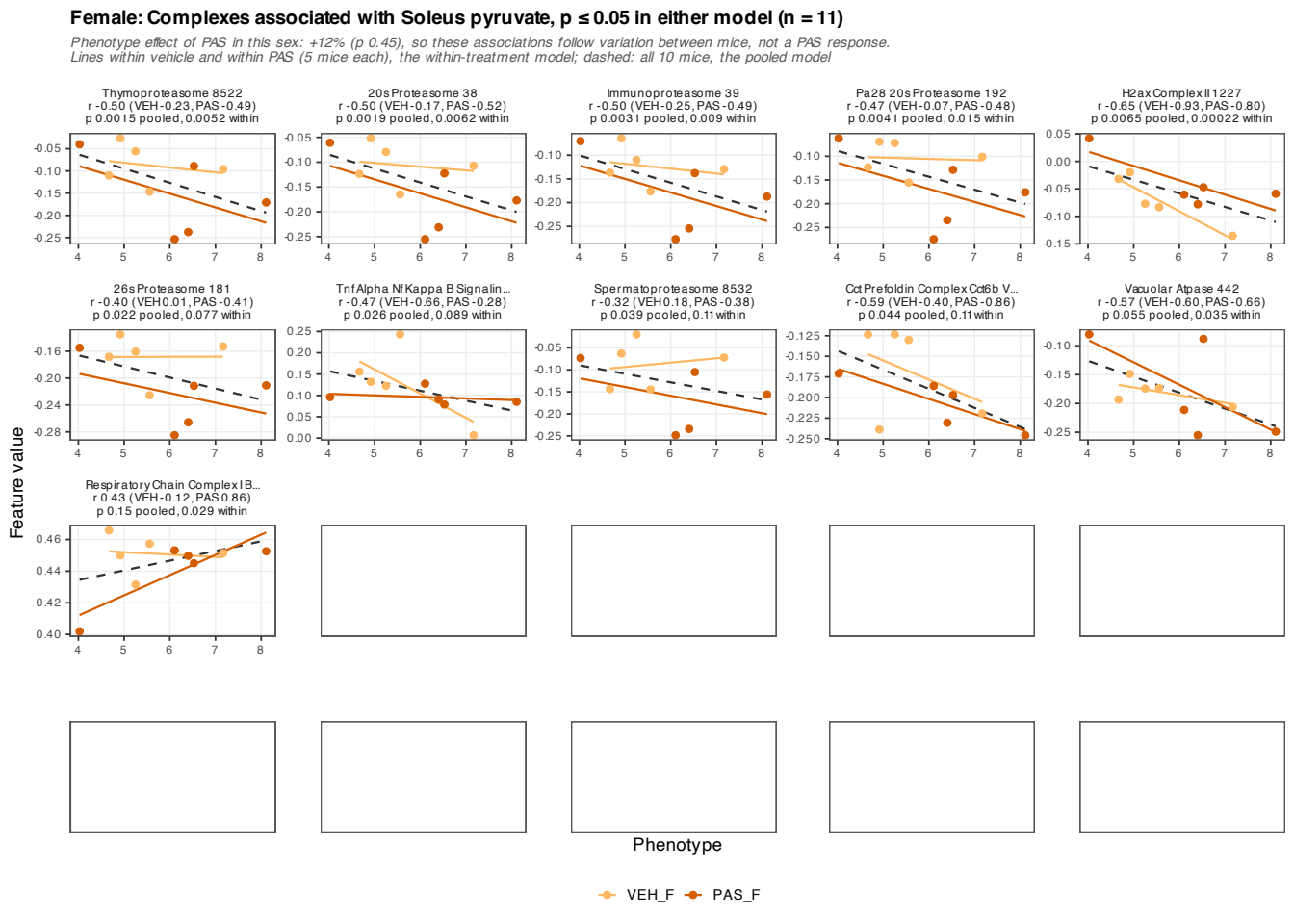
Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



Female: Complexes associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 12)

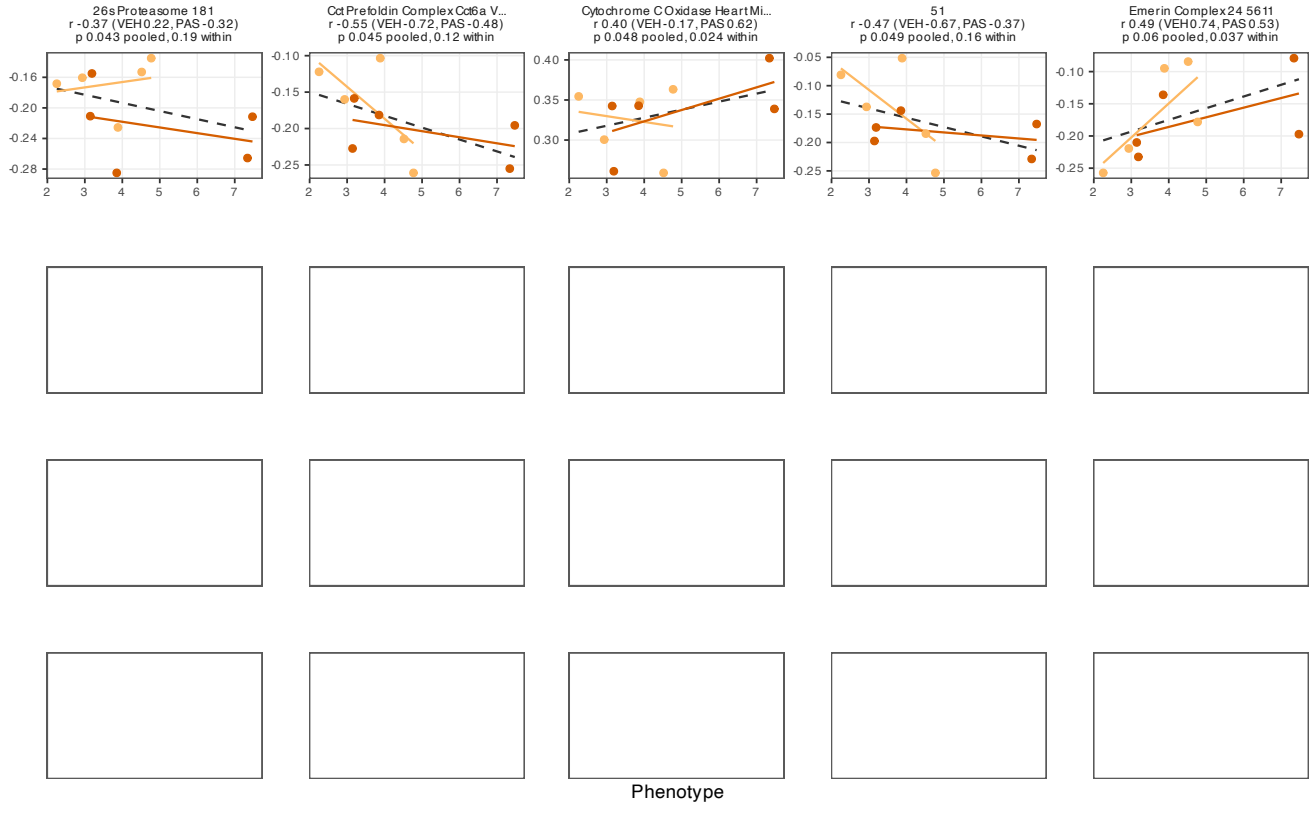
Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model




Female: Complexes associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 11)
Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model
Thymoproteasome 8522
r -0.50 (VEH -0.23, PAS -0.49)
p 0.0015 pooled, 0.0052 within
20s Proteasome 38
r -0.50 (VEH -0.17, PAS -0.52)
p 0.0019 pooled, 0.0062 within
Immunoproteasome 39
r -0.50 (VEH -0.25, PAS -0.49)
p 0.0031 pooled, 0.009 within
Pa28 20s Proteasome 192
r -0.47 (VEH -0.07, PAS -0.48)
p 0.0041 pooled, 0.015 within
H2ax Complex II 1227
r -0.65 (VEH -0.93, PAS -0.80)
p 0.0065 pooled, 0.00022 within
26s Proteasome 181
r -0.40 (VEH 0.01, PAS -0.41)
p 0.022 pooled, 0.077 within
Tnf Alpha Nf Kappa B Signalin…
r -0.47 (VEH -0.66, PAS -0.28)
p 0.026 pooled, 0.089 within
Spermatoproteasome 8532
r -0.32 (VEH 0.18, PAS -0.38)
p 0.039 pooled, 0.11 within
Cct Prefoldin Complex Cct6b V…
r -0.59 (VEH -0.40, PAS -0.86)
p 0.044 pooled, 0.11 within
Vacuolar Atpase 442
r -0.57 (VEH -0.60, PAS -0.66)
p 0.055 pooled, 0.035 within
Respiratory Chain Complex I B…
r 0.43 (VEH -0.12, PAS 0.86)
p 0.15 pooled, 0.029 within
Feature value
Phenotype
VEH_F
PAS_F

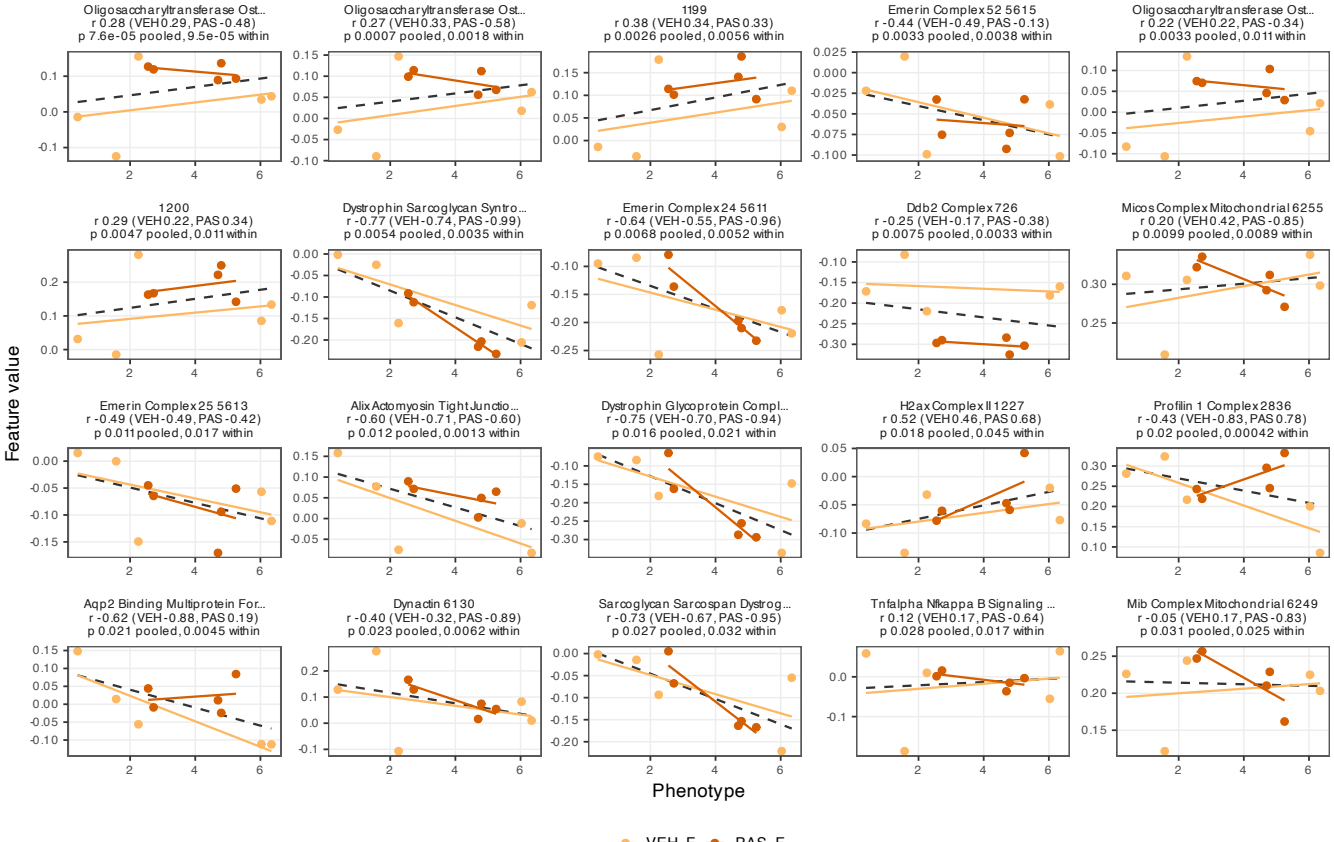
Female: Complexes associated with Plantaris pyruvate, p ≤ 0.05 in either model (n = 5)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



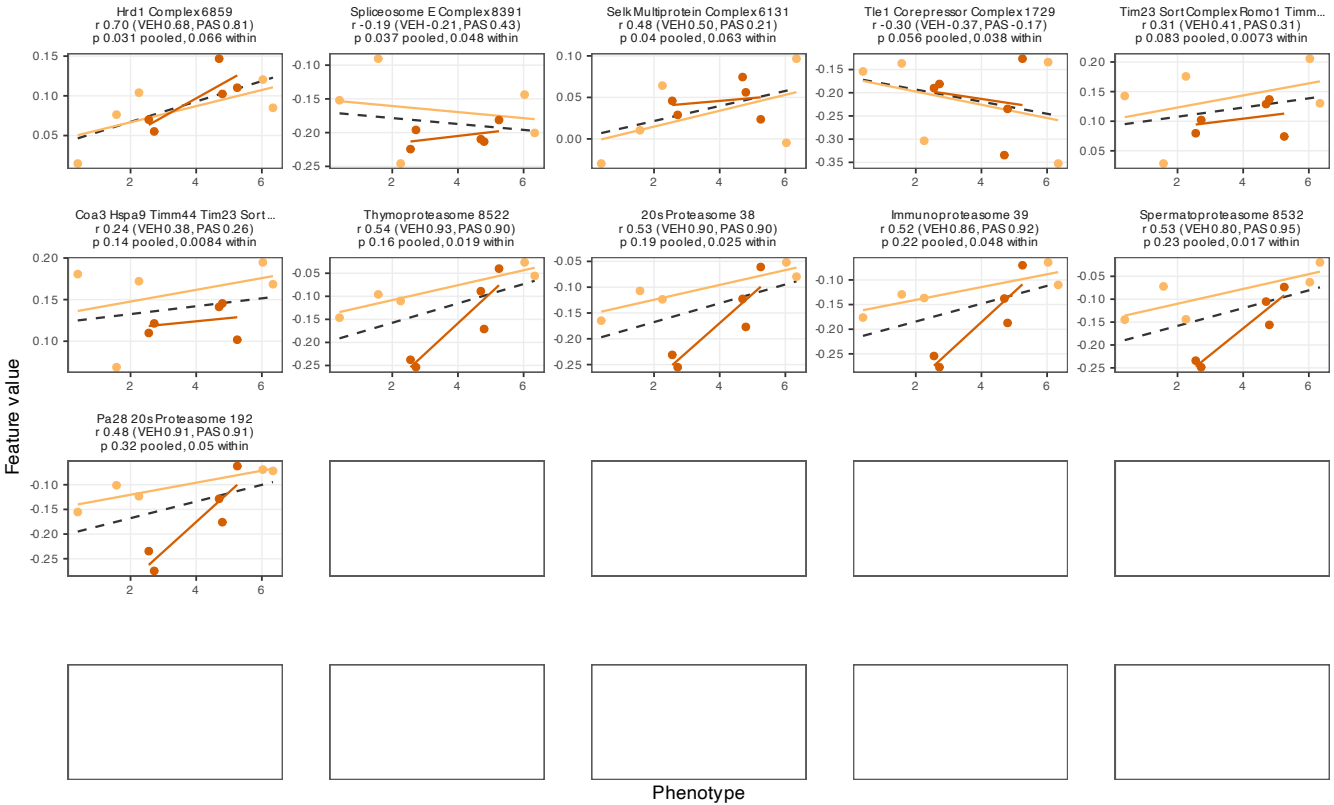
Female: Complexes associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 31)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



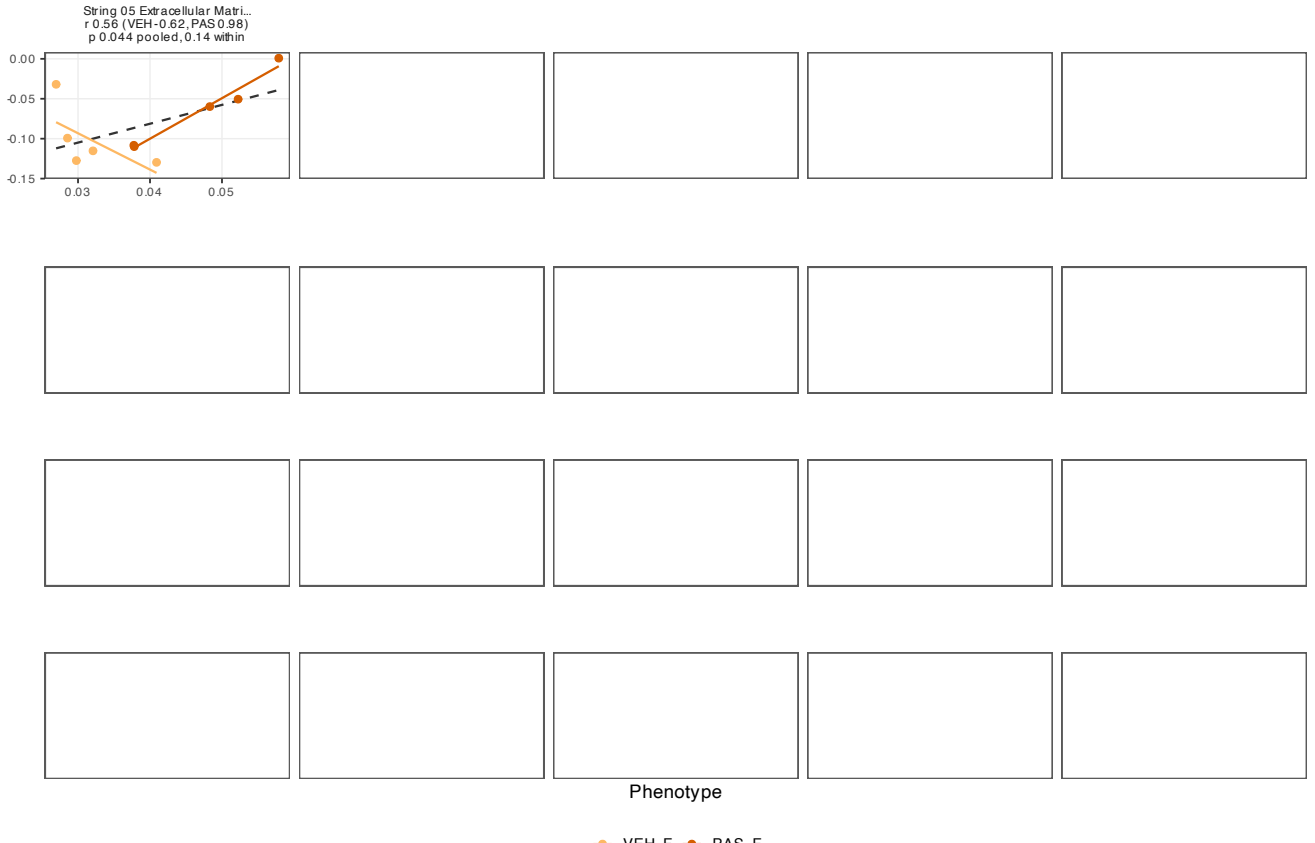
Female: Complexes associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 31)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



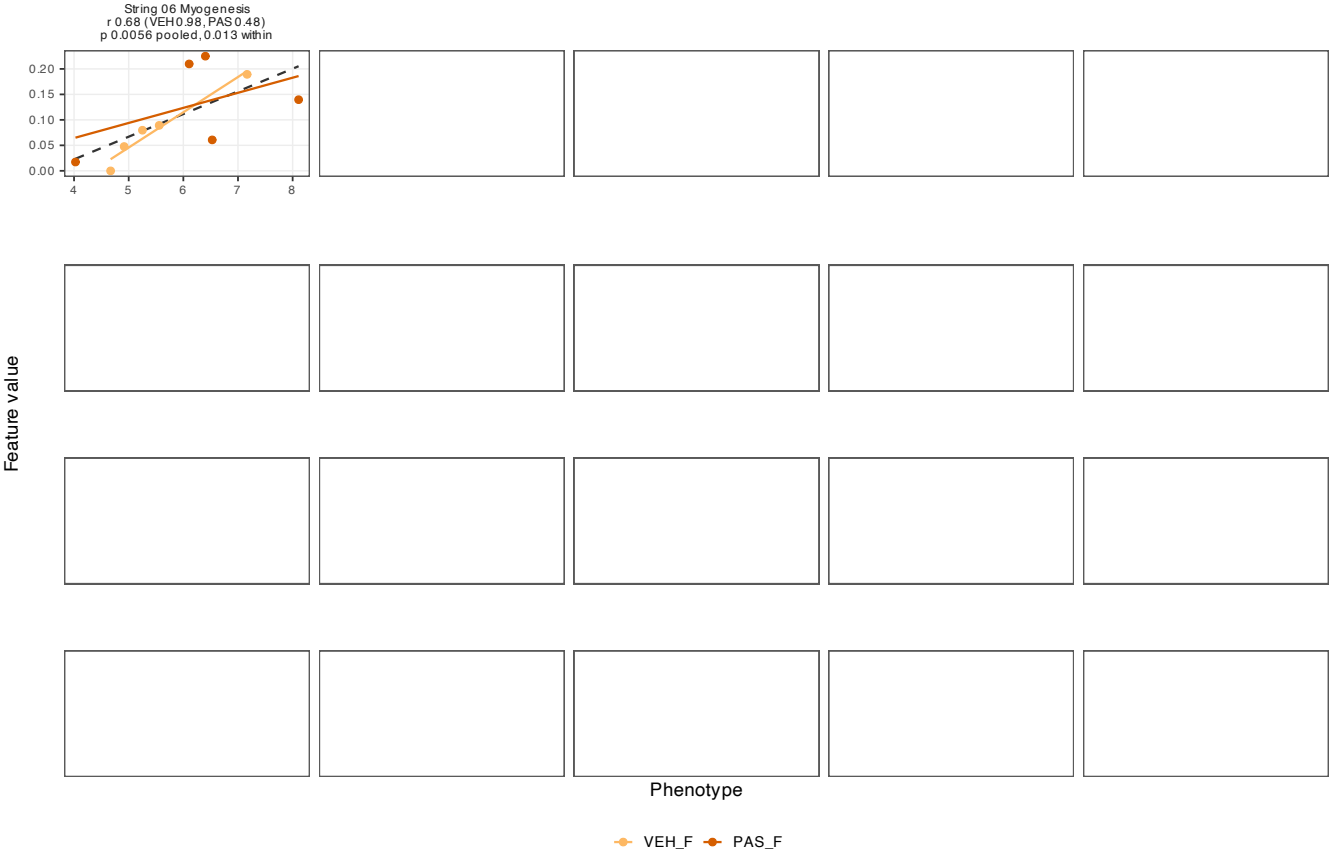
Female: Clusters associated with Satellite cells / fibre, p ≤ 0.05 in either model (n = 1)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



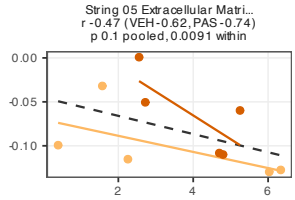
Female: Clusters associated with Soleus pyruvate, p ≤ 0.05 in either model (n = 1)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



Female: Clusters associated with Soleus octanoyl, p ≤ 0.05 in either model (n = 1)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each), the within-treatment model; dashed: all 10 mice, the pooled model



Feature value

Phenotype

VEH_F PAS_F